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UNIVERSITY OF CALIFORNIA

Los Angeles

From Minos to Midas: The Organization of Textile Production in the Aegean and in Anatolia

A dissertation submitted in partial satisfaction of the requirements for the degree Doctor of Philosophy in Archaeology

by

Richard Brendan Burke

1998

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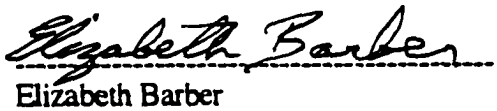
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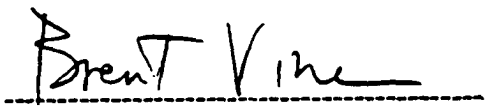
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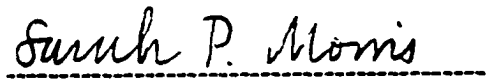
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Elizabeth Carter



Brent Vine



Sarah Morris, Committee Chair

University of California, Los Angeles

1998

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For my parents,

my brothers, and my sister.

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Burke, B. (1997). The Organization of textile production in Bronze Age Crete, TEXNH: Craftsmen, Craftswomen and Craftsmanship in the Aegean Bronze Age. Proceedings of the 6th International Aegean Conference, Temple University, 18-21 April 1996, Aegaeum 16 (eds.) R. Laffineur and P. Betancourt, 413-424.

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-----. (December, 1996). Textile Production at Gordion and the Phrygian Economy. Paper presented at the 1996 Annual Archaeological Institute of America Meeting in New York.

-----. (April, 1996). Texts and Textile Tools: the Production of Cloth in the Aegean Bronze Age. Paper presented at the 6th International Aegean Conference TEXNH: Craftsmen, Craftswomen, and Craftsmanship in the Aegean Bronze Age, at Temple University, Philadelphia, Pennsylvania.

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ABSTRACT OF THE DISSERTATION

From Minos to Midas: The Organization of Textile Production
in the Aegean and in Anatolia

by

Richard Brendan Burke
Doctor of Philosophy in Archaeology
University of California, Los Angeles, 1998
Professor Sarah Morris, Chair

The economic, political, and social implications of a well-developed, centrally controlled textile industry in three cultural regions and chronological periods are discussed in this dissertation: the Minoans in the early to mid-second millennium B.C., the Mycenaeans in the Late Bronze Age (*ca.* 1400-1150 B.C.), and the Phrygians of the Anatolian Iron Age (*ca.* 1100-700 B.C.). The organization and economic structure of these three distinct societies are compared in terms of one industry, textile production, to better understand early political economies. The study of cloth manufacture in early complex societies explores the relationship between social organization and technology. Within complex societies elites controlled craft activities, including the many phases of textile production, to maintain and finance social, religious, and military institutions.

Evidence for centralized textile production is found in administrative documents and workshop debris at palatial centers of the Aegean and Anatolia. This dissertation examines Minoan evidence related to cloth production, including early administrative documents and

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craft production residues. By mapping changes over time in Minoan textile production we can compare this information to overall social and political developments on Crete from the Neolithic through the Late Bronze Age.

The Mycenaean evidence for textile production is primarily based on the inscribed clay documents in the Linear B script. Aspects of production recorded by the palace scribes demonstrate that Mycenaean cloth was both a medium of exchange and a prestige good that played a vital role in Late Bronze Age palace economies. While the Aegean has good textual evidence for cloth production, there are few artifactual indicators from Mycenaean centers demonstrating textile manufacture.

In Iron Age Anatolia the economic and political organization of the Phrygian kingdom centered at Gordion is comparable to kingdoms of the Bronze Age Aegean. Although no administrative documents have yet been found at Gordion, an Iron Age destruction level has preserved the best archaeological evidence for controlled textile production within a regional center. Thousands of loom weights, spindle whorls and other tools of cloth production have been found in their primary-use contexts. The analysis of this equipment shows various types of tools clustered in certain parts of the industrial quarter of Gordion for attached, specialized production. Phrygian textiles played an important role in supplying a military force, providing objects of elite-gift exchange, and were also a valuable item of trade in Early Iron Age Anatolia. This study demonstrates that textile production can inform us about larger issues of cultural complexity and social organization in early states.

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From Minos to Midas: The Organization of Textile Production in the Aegean and in Anatolia

Chapter One - Introduction

This dissertation investigates the production and exchange of textiles in complex economies during the Bronze and Early Iron Ages in the Aegean and Anatolia. The study of cloth production explores the relationship between social organization and technology in developing states. Within complex societies, elites control craft activities, including the many phases of textile production, to maintain and finance social, religious, and military institutions. The economic, political, and social implications of a well-developed, centrally controlled textile industry in three cultural regions and chronological periods are discussed: the Minoans in the early to mid-second millennium B.C., the Mycenaeans in the Late Bronze Age (*ca.* 1500-1150 B.C.), and the Phrygians of the Anatolian Iron Age (*ca.* 1100-700 B.C.). The organization and economic structure of these three distinct societies are compared in terms of one industry, textile production, to better understand early political economies.

As with other craft goods in early state economies, the production and exchange of textiles required specialization and some degree of centralization. The primary focus of inquiry here is to characterize the organization of this industry in complex economic systems administered by central places. These centers show evidence of social stratification, inter-regional exchange, and organized bureaucracies, suggesting a high

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degree of regional power. The aim here is to understand the means by which a state, such as the Minoan, Mycenaean or Phrygian kingdom, financed itself with textile production.[1]

A working hypothesis of this project is that aspects of the textile industry are particularly suited to palace control.[2] Textile production is labor intensive and involves many different processes. Cloth requires the mobilization of agricultural resources for raw materials, preparation of these raw materials, skilled manufacturing, storage, and distribution, all of which are best administered under the direct control of a regional center or palace. The concentration of archaeological research at palatial centers of the Aegean and in Anatolia makes the cloth industry well suited to the study of changes in administration and organization.

Nature of the Evidence

The field of textile studies is multifaceted and multidisciplinary, taking account of all the various types, uses and roles of cloth in societies around the world. As two organizers of a recent conference on textiles state, 'On a world-wide scale, complex moral and ethical issues related to dominance and autonomy, opulence and poverty, political legitimacy and succession, and gender and sexuality, find ready expression through cloth'.[3] This dissertation confines itself to the material record of organized textile

---

[1]For general theory about early economies see K. Polanyi, "The Economy as Instituted Process," Trade and Market in the Early Empires (eds.) K. Polanyi, V. Arensberg and P. Pearson (1957) 243-70; C. Renfrew, "Trade as Action at a Distance: Questions of Integration and Communication," Ancient Civilization and Trade (eds.) J. Sabloff and C.C. Lamberg-Karlovsky (1975) 3-59; T. Earle and E. Brumfiel, "Specialization, Exchange and Complex Societies: An Introduction," in Specialization, Exchange and Complex Societies (eds.) T. Earle and E. Brumfiel (1987) 1-9.

[2]As Stein and Blackman state, 'In this perspective, it becomes essential to understand the degree to which elites and centralized institutions of complex societies controlled the raw materials, labor, and output of craft specialists.' G. Stein and M. Blackman, "Specialized Craft Production in Mesopotamia," in B. Isaac (ed.) Research in Economic Anthropology 14 (1993) 30.

[3]A. Weiner and J. Schneider, Cloth and Human Experience (1989) xi.

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production indicated by two main sources of information: excavated craft-residues and administrative documents. Loom weights, spindle whorls, knives, needles, dye vats, etc. are artifactual indicators of craft activity described in some detail in the following chapters. Culturally distinctive features of some of these tools found in concentrated deposits suggest a significant degree of regulated production.[4] In addition, textual records document the acquisition of raw materials and the production and distribution of finished textile goods. Other sources of evidence include representations of garments and cloth in ancient works of art, particularly wall-paintings, and preserved fragments of ancient cloth.[5] While data from cloth fragments and painted depictions are highly informative, they do not readily relate to aspects of economic organization. These sources are more properly the domain of art historians, fiber specialists, and weavers, and are not included in any great detail in this research. Rather than catalogue every known loom weight and spindle whorl, this dissertation primarily focuses on cogent examples of textile tool assemblages and written documents from the various centers of the Minoans, Mycenaeans and Phrygians (when available), to better understand economic organization.

---

[4]Certain types of textile tools, for example, may be regionally and chronologically restricted, as is shown in Chapter Two. The average warp-weighted loom uses about twenty to fifty weights. Some loom weight deposits, such as the four hundred (plus) from the Loom-Weight Basement at Knossos, the 450 from Room D15 of the West House at Akrotiri, and the hundreds of loom weights found in the Early Phrygian destruction level at Gordion suggest that substantial deposits of certain tools locate centralized production beyond the household level; A. Evans, Palace of Minos vol. 1 (1921) 253; S. Marinatos, Thera IV (1971) plate 50; Ι. Τζαχίλη, Υφαντική και Υφάντρες στο Προϊστορικό Αιγαίο 2000 - 1000 π.Χ. (1997) 183-193.

[5]The remains of cloth in the Aegean are fairly scanty, while artistic representations of costumes are fairly common. See catalogues and analysis of these sources in E. Barber, Prehistoric Textiles: The Development of Cloth in the Neolithic and Bronze Ages, with Special Reference to the Aegean (1991) 174 n. 12 and 311-357.

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Linen and Wool

The two primary fibers exploited by Aegean and Anatolian spinners and weavers were linen and wool. Recent research indicates that textile technology predates agriculture, pottery, and metallurgy by millennia. Twisted fibers approximately 25,000 years old have been found at the Upper Paleolithic site of Pavlov in the Czech Republic.[6] The earliest evidence for domesticated flax (*Linum usitatissimum*), which yields linen, is indicated by charred linseeds at Preceramic Neolithic A sites of the Near East and from scraps of linen cloth from Nahal Hemar in the Judean desert.[7] Domesticated flax is the earliest of the oil and fiber crops in Anatolia, widely cultivated before the eighth millennium B.C.[8] Not surprisingly, from the sedentary Neolithic we also have clay impressions of textiles that are the earliest evidence for loom-woven cloth, most certainly linen, at 7th millennium Jarmo.[9] Neolithic linen is also known from Çatal Höyük and Çayönü in Anatolia.[10] While linen often occurs in Dynastic Egypt it does not appear in the archaeological record until the fifth millennium B.C.[11]

The cultivation of flax is one of the most labor-intensive chores in agriculture. With proper timing the harvest of flax can yield two benefits: ripe seeds for linseed oil and stalks

---

[6]M. Adovasio, O. Soffer, and B. Klíma, "Upper Palaeolithic Fibre Technology from Pavlov I, Czech Republic," Antiquity 70 (1996) 526-534.

[7]E. Barber (1991) 11-15; J. McCorriston, "The Fiber Revolution: Textile Extensification, Alienation, and Social Stratification in Ancient Mesopotamia," Current Anthropology 38.4 (1997) 519; W. van Zeist and J. Bakker-Heeres, "Evidence for Linseed Cultivation Before 6000 B.C.," Journal of Archaeological Science (1975) 215-219; T. Schick, "Cordage, Basketry, and Fabrics," Atiquot 18 (1988) 31-43.

[8]M. Nesbitt, "Plants and People in Ancient Anatolia," Biblical Archaeologist 58.2 (1995) 75.

[9]J. Adovasio, "Notes on the Textile and Basketry Impressions from Jarmo," in Prehistoric Archaeology along the Zagros Flanks (eds.) L. Braidwood, *et al.* (1983) 425-426.

[10]J. McCorriston (1997) 519 .

[11]Pieces of coarse linen were found in the Faiyum along with spindle whorls and flax seeds. See E. Barber (1991) 10.

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with fibers still pliable enough for fine linen cloth. While we know linseed oil most commonly as paint thinner, with proper processing it can be a calorie-rich food source. Well-watered, easily drained agricultural land is needed for flax. Strong manual labor is also required first for weeding and then pulling the stalks of flax out of the ground whole. The stalks are then retted, *i.e.* left to rot, either in standing water or spread out on a roof to collect dew. After 10-14 days, the fibers are left to dry and then beaten and combed to break the pith away to reach the usable fibers in the center. At this point the flax is now ready for spinning.

Thousands of years after flax was domesticated, sheep were selectively bred for their woolly fibers.[12] The earliest preserved wool dates to the 4th millennium, from Upper Egypt.[13] Other evidence for woolly sheep is found in animal figurines and Mesopotamian texts tallying records of herds. Prior to the fourth millennium B.C. most sheep had fairly short hair of coarse fibers not well-suited to spinning. They were most probably raised for their meat. Palaeozoologists studying the age and sex ratios of sheep bones have been able to suggest whether certain sheep were used primarily for meat or wool. Male and female ratios of sheep are helpful in determining the primary use of the sheep, since females yield better wool than males, although the highest quality wool comes from young neutered males called wethers. Butcher marks are an obvious indicator of meat use. Finally, if the sheep bones appear more robust over time, it is probable that heavier musculature developed on a new breed of sheep able to carry a heavier (*i.e.* woollier) coat.[14] The

---

[12]A. Sherratt, "The Secondary Products Revolution of Animals in the Old World," World Archaeology 15 (1983) 90-104; M. Ryder, Sheep and Man (1983); *Idem*, "Sheep and Goat Industry with Particular Reference to Textile Fibre and Milk Production," Bulletin on Sumerian Agriculture 7 (1993) 9-32; E. Barber (1991) 20-30.

[13]E. Barber (1991) 25.

[14]J. McCorriston (1997) 521; S. Payne, "Kill-off Patterns in Sheep and Goats," Anatolian Studies 23 (1973) 281-303.

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fleecier sheep were purposefully bred for use in textile production, a shift in agricultural patterns that had significant social ramifications in Mesopotamia, according to McCorriston.[15] This theory postulates that 'It was the shift to wool, a fiber more efficient to produce than linen, that catalyzed the initial development of large textile workshops and the labor class that eventually was attached to them.'[16]

Methodology

Various sources are used in this investigation of cloth manufacture. In some areas of the world, archaeologists have approached the study of textiles as socio-cultural anthropologists, building theoretical models from ethnographic parallels and comparing these observations to the archaeological record.[17] Some scholars, particularly in the Near East where records are abundant, have looked exclusively at textual documents related to textile production,[18] while others have relied on a combination of textual and archaeological

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[15]See J. McCorriston (1997) 517-549.

[16]J. McCorriston (1997) 518.

[17]M. Parsons, "The Distribution of Late Postclassic Spindle Whorls in the Valley of Mexico", American Antiquity 40 (1975) 209-215; M. Smith and K. Hirth, "The Development of Prehispanic Cotton-Spinning Technology in Western Morelos, Mexico," JFA 15 (1988) 349-358; J. Parsons and M. Parsons, Maguey Utilization in Highland Central Mexico: An Archaeological Ethnography. (1990); E. Brumfiel, "Weaving and Cooking: Women's Production in Aztec Mexico" in Engendering Archaeology: Women and Prehistory (eds.) J. Gero and M. Coney (1991) 224-251; C. Costin, "Craft Specialization: Issues in Defining, Documenting, and Explaining the Organization of Production," Archaeological Method and Theory (ed.) M. Stiffer 3 (1991) 1-56; *Eadem*, "Textiles, Women, and Political Economy in Late Prehispanic Peru," Research in Economic Anthropology 14 (1993) 3-28.

[18]Th. Jacobsen, "On the Textile Industry of Ur under Ibbi-Sin," Toward the Image of Tammuz and Other Essays on Mesopotamian History and Culture (ed.) W. Moran (1970) 216-230; H. Waetzoldt, Untersuchungen zur neusumerischen Textilindustrie (1972); S. Dalley, "Old Babylonian Trade in Textiles at Tell al Rimah," Iraq 39 (1977) 155-59; E. Sollberger, Administrative Texts Chiefly Concerning Textiles. Archivi Reali di Ebla Testi VIII (1986); K. Szarzynska, "Records of Garments and Cloth in Archaic Uruk/Warka," Altorientalische Forschungen 15 (1988) 220-230.

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evidence for studying textile production within complex economies.[19]

Ideally, one would like to have archaeological evidence from an excavated political and economic center that includes workshop facilities preserving textile equipment. In addition, one would like administrative records from this same site that describe the raw materials, their sources, the production targets, the ultimate destination of the finished products, the objects traded in return, and the organization of the labor used to make these goods. In the Aegean and Anatolia, however, the evidence is not quite so clear. As it is, the data sets of textile production vary in the Aegean and in Anatolia. The primary sources of evidence are the tools associated with cloth production and inscribed documents recording the administration of this industry. These data are preserved to varying degrees at palatial centers of the Minoans, Mycenaeans, and Phrygians.

Minoan Evidence

During the Neolithic period at Knossos, cloth production was a major communal activity of the fourth millennium. The foundations of Palatial Crete, however, do not begin until the third millennium, towards the end of the Early Minoan period. The major question regarding the origins of the Minoan palace concerns the rate of economic and social development in the palatial system. Was it a protracted, progressive process, as advocated by gradualists, or was it a swift quantum leap forward brought about by sudden stimuli new to the island?[20] Related to this is the issue of contact with pre-existent Near Eastern states that show similar organizational features to the Minoans. As Cherry has noted,

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[19] For example, in the New World, see J. Murra and C. Morris, "Dynastic Oral Tradition, Administrative Records and Archaeology in the Andes," World Archaeology 7.3 (1976) 270-79; J. Murra, "Cloth and Its Function in the Inca State," Cloth and Human Experience (eds.) A. Weiner and J. Schneider (1989) 275-302. In Mesopotamia, see G. Stein and J. Blackman (1993) 29-59; J. McCorriston (1997) 517-549.

[20] J. Cherry, "Evolution, Revolution and Origins of Complex Societies," Minoan Society (eds.) O. Krzyszkowska and L. Nixon (1983) 33-45.

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'what is beyond doubt in terms of Minoan development is that major changes in the scale and nature of socio-political integration on Crete took place in the twentieth century BC (1950-1930), signaled by the construction of political, religious, and economic central places, or "Palaces".'[21] The multi-component aspects of emergent complexity can not be addressed in great detail in this study, but by looking at Minoan textile production as one more feature of the Minoan palatial system, we will gain further evidence for interpreting its overall development.

Destruction and rebuilding mark the end of the Old and beginning of the New Palace period on Crete. Because the New Palaces were built over the Old, the earliest phases of development are obscured. What we do see, however, is that Minoan palaces share architectural parallels in overall concepts. They are large, multi-storied buildings, with rooms built around open, central, porticoed courtyards facing north. On their western side is often a courtyard. In terms of construction, all palaces have large, sawn ashlar blocks, downward-tapering columns, pier and door partitions, light wells, lustral basins, monumental stairways, major public rooms on upper floors. Similarly, they seem to have multiple functions, with storage areas, craft workshops, and ritual installations (particularly along the west side of the central court). They also functioned as administrative centers complete with archive rooms. Minoan hieroglyphs and Linear A inscriptions provide some information about early administration of agriculture, including aspects of textile production.

Preserved within many Minoan palaces and settlements are the tools of textile manufacture, primarily loom weights for weaving textiles. These weights vary throughout the Minoan periods and in different regions of Crete. Some weights show inscribed

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[21] J. Cherry, "Polities and Palaces: Some Problems in Minoan State Formation," Peer Polity Interaction (eds.) C. Renfrew and J. Cherry (1986) 19-45. These issues of Minoan state formation will be explored more fully in Chapter two.

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markings and some seem to occur only in certain archaeological contexts. In marked contrast to the numbers of preserved Minoan loom weights is the relative absence of Minoan spindle whorls used for spinning thread. In many parts of Crete, for example, the proportion of excavated spindle whorls to loom weights is very low, suggesting that spinning was either done without whorls or dispersed throughout the island of Crete.[22] Some have suggested that the Minoans may have used perishable materials for the whorls at the end of their spindles. While this is possible, it may also be the case that archaeologists have not been looking in the area where most of the spinning took place, in the domestic settlements of working Minoans. With all the attention in Minoan archaeology devoted to the palaces, the activity of spinning may be one of the tasks not located in major centers. Most palaces were abandoned or destroyed in 1450, Late Minoan I B. This was followed by the Mycenaean incursion into Crete where the mainland Greeks established their control at the palace of Knossos.

Mycenaean Evidence

After the collapse of the Minoan palace system in the Late Bronze Age, Mycenaean culture plays a dominating role in the economic history of the Aegean. Clay tablets inscribed in the Linear B script record economic transactions at Mycenaean palaces in the earliest form of written Greek. A large percentage of these tablets refer in some way to a well-organized textile industry, recording many different types of cloth. Tablets record large amounts of raw textile material, wool and linen, assessed as tribute from surrounding towns and villages under the administration of the palaces. The documents also provide information about textile personnel, including food rations for dependent workers in this

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[22]Excavations at the Palace of Knossos, for example, report no spindle whorls from Minoan levels. This absence is striking in comparison to the thousands of whorls reported from various levels at Troy, many on display in the Istanbul Museum. Compare also the many whorls from Bronze Age Aphrodisias; M. Joukowsky, *et al.* Prehistoric Aphrodisias (1986) 373-381.

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industry. The tablets suggest a high degree of specialization for the craftsmen and craftswomen listed, recording specific occupations such as spinners, weavers, and fullers. Second-order administrators are also known from the Mycenaean tablets. The status of these individuals has puzzled scholars for some time. Their role in the textile industry is explored in Chapter three.

The Linear B studies by John Killen and others describe how the palace scribes recorded the many stages of textile production in the Late Bronze Age. Linear B tablets are our best source of information for the organization of labor in Bronze Age Greece. From such records we see that Mycenaean textile production was subject to the central authority of the palace. These philological investigations clarify theories regarding the organization of the Late Bronze Age economy in general but often do not take full advantage of the archaeological evidence.

Artistic representation of Mycenaean cloth is known from various media: frescoes, sculpture, and vase painting. Decorative patterns and weaving technology described in the tablets is probably reflected in the various artistic images of Mycenaeans. In addition, while Mycenaean loom weights are not as plentiful at palace centers as they are in the Minoan and Phrygian centers, there are archaeological indicators of weaving at or near the Mycenaean palaces. Because the archaeological evidence for Mycenaean weaving is scanty, it is possible that Mycenaean weaving was done primarily on a different type of loom, such as the horizontal ground loom, which would leave no archaeological trace. It is also possible that the Mycenaeans excelled at band-weaving technology which would also leave little archaeological evidence.[23] Finally, as is the case with Minoan spindle whorls, it may be that we should not expect loom weights at major Mycenaean centers, where most

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[23]Narrow strips, either plaited or woven, are discussed in E. Barber, "Minoan Women and the Challenges of Weaving for Home, Trade, and Shrine," TEXNH (1997) 515-21, especially pl. CXCIIIc-d.

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excavations have been conducted. Textiles were more probably woven in outlying Mycenaean settlements but administered by palatial centers, such as Pylos and Knossos.

Phrygian Evidence

After the Late Bronze Age collapse that ended the Mycenaean civilization, the palatial economic system ends in Greece. In Iron Age Anatolia, however, strong regional centers that fit the criteria for palaces emerge. Throughout most of the first millennium B.C. the Phrygian capital of Gordion was one of the strongest economic and political centers on the Anatolian plateau. In its early phase a destruction level preserved the best archaeological evidence for textile production on a scale comparable to that described in the Linear B tablets and that demonstrated by finds from Minoan Crete.[24] While the destruction level at Gordion dates approximately 450 years after the Mycenaean age, to about 700 B.C., the palace economy of the Phrygians may have been similarly organized to palaces of the Bronze Age Aegean.

Regarding Phrygian textiles, we have substantial amounts of cloth preserved from burials around the site and from the citadel itself. Some of the fragments were highly burnt but they all reveal sophisticated and diverse textile technologies. Analyses of these fragments provide descriptions of the weave structure and suggest identifications for the fibers, most probably wool and linen. Rather than a detailed study of the Gordion textile fragments, the analysis here focuses on the tools of cloth manufacture at the capital and how this reflects organized production within the Phrygian kingdom.

Not only have substantial textile remains been recovered and studied from the Gordion excavations, but also artifactual evidence for the large-scale textile production

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[24]For a summary of the chronology and architecture from Rodney Young's excavations at Gordion, see K. DeVries, "The Gordion Excavation Seasons of 1969-1971 and Subsequent Research," AJA 94 (1990) 371-406.

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within the palace has been found. This dissertation presents a systematic study of the tools of textile manufacture located in the workshop areas within the fortified walls at Gordion.[25] While no textual records of palace administration from Gordion have yet been found, we can infer from the architecture at Gordion and from historical references to King Midas that Phrygian society was fairly complex, with a well-established economy, partly based on textile production. Thousands of spindle whorls and loom weights were found in primary depositional contexts within workshop buildings. Measurements of their weight and distribution are presented to get a detailed picture of organized cloth manufacture at a major center.

Specialization and Exchange

The explicit research objective of this dissertation is to understand the role of textile industries in complex societies. Two general research questions are addressed: how was textile production within a kingdom organized and what was the purpose of organized cloth manufacture? Although there are differences in time and place between centers in the Aegean and in Anatolia, textiles were a major source of wealth for these capitals. The examination of one industry in detail will better inform us about the general structure of the production system, from the acquisition of raw materials to the final distribution of finished goods. Once the production of a certain commodity is understood it is possible to place that product within the context of the overall political economy and answer the question, 'What were textiles used for?' Information can also be compared to other industries, such as food production, pottery manufacture, and metallurgy, to confirm or refute suggestions about social complexity. Data from other cultural regions, such as those in the New World

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[25]The excavation notebooks from R. Young's excavation are stored in the Gordion archives of the University Museum, University of Pennsylvania. The artifacts themselves are in storage at the site museum of Gordion in Yassıhöyük.

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and Mesopotamia, demonstrate that cloth production played a vital role in the economy, society and ideology of many early states.

As Stein and Blackman state, '...the organizational context of craft production reflects both the institutional structure of the state and the nature of the state's interaction with the broader, more heterogeneous society in which it functions.'[26] For investigating the organization of craft activities, the context, concentration, scale, and intensity of production require examination.[27] Specialized craft production tends to be localized within regional centers. Concentrated deposits of craft residues, such as raw materials, tools of manufacture, craft debris, unfinished and finished goods, are all indicative of specialized production. Administrative documents facilitate this investigation. The consumption of textiles is more difficult to study than production but is addressed when evidence is available.

Of interest here is specialized production, *i.e.* production of surplus cloth for exchange. This kind of organized production transforms raw material (wool and linen) and component elements (dyes, perfumed oil, decorations) into usable objects at the disposal of the political economy. This is distinguished from non-specialized production in that specialists receive raw materials from the palace. The distribution and standardization of certain tools of manufacture concentrated within the palace complex indicate specialized production. Finally, the palace directs consumption of the finished product, for military, ideological or political purposes, to acquire other prestige goods and to maintain their position of power.

Craft specialization in this study is defined as repeated, surplus production of one type of good (cloth) by attached, dependent specialists for exchange directed by a central

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[26]G. Stein and M. Blackman (1993) 30-31.

[27]See C. Costin (1991) 1-56.

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authority (the palace). All phases of production, from the acquisition and preparation of raw materials, to the subsistence of the personnel and the distribution of the final product, is administered by a centralized regional authority, or palace. Differences in the distribution of textile tools reflect different levels of specialization and complexity. Many of the eight types of specialized production proposed by Costin find expression in Minoan, Mycenaean, and Phrygian examples included within this research.[28] These types of specialized production are:

1) Autonomous specialiststs producing for unrestricted local consumption,

2) Dispersed workshops producing for unrestricted local consumption,

3) Community specialization by autonomous individual or household-based production units, aggregated within a single community to produce for unrestricted regional consumption,

4) Larger, nucleated workshops aggregated within a single community, producing for unrestricted regional consumption,

5) Dispersed part-time corvée laborers producing for elite or government institutions within household or local community settings,

6) Individual retainers anad individual artisans usually working full-time for elite patrons or governments,

7) Nucleated part-time corvée laborers recruited by the government for special purposes,

8) Large-scale retainer workshops with full-time artisans working for an elite patron or government institution within a segregated, highly specialized, setting or facility.

These types of specialization are applied to the cases examined in this dissertation to show the variability in productive strategies of textile manufacture and how this reflects social organization. These criteria do not distinguish social positions for the craft specialists. In

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[28]C. Costin (1991) 1-56 and see below, Chapter Five.

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some cases it is likely that these workers were not free laborers but were more likely slaves or prisoners captured in war and employed by the state.

Staple and Wealth Finance

A theoretical model of staple and wealth finance in a redistributive economy directs analysis of textile data, both artifactual and textual. This model heuristically distinguishes two types of commodities: staple and wealth goods.[29] Of primary concern in the staple and wealth finance system is the mobilization and distribution of labor and goods from the general population to state institutions and personnel. The model allows us to map the relationship between the evolution of social complexity and increasing energy capture.

Staple finance involves what we generally think of as the redistribution, or mobilization, of agricultural resources.[30] Goods, such as livestock, grain, ceramics, and perhaps textiles, collected by the state as a share of commoner produce in large store houses, are used to support non-food producers. Staple finance items are 'collected by the state as a share of commoner produce, as a specified levy, or as produce from land worked with corvée labor.'[31] Products manufactured and used by a majority of the population functioned as staple goods within the palace economy. The storage and general necessity of staple items in daily life are of key importance in understanding this finance model. The major drawback of this system is the high cost of transporting and storing bulky subsistence goods. For this reason staple finance is often replaced by a wealth finance

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[29]K. Polanyi, Primitive, Archaic, and Modern Economies: Essays of Karl Polanyi (ed.) G. Dalton (1968) 321; T. D'Altroy and T. Earle, "Staple Finance, Wealth Finance and Storage in the Inka Political Economy," Current Anthropology 26 (1985) 187-197; E. Brumfiel and T. Earle (1987) 6. See also J. Bennet, "Outside in the Distance: the Economic Geography," Texts, Tablets and Scribes. Minos Suppl. 10 (ed.) J.-P. Olivier and T. Palaima (1988) 33; G. Walberg, "Minoan Economy: an Alternative Model," Politeia. Aegaeum 12 (eds.) R. Laffineur and W.-D. Niemeier (1995) 157-161 and the discussion following.

[30]T. D'Altroy and Earle (1985) 188.

[31]T. D'Altroy and Earle (1985) 188.

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system.[32] The connection between staple goods, such as food, with textile production is evident in the earliest Minoan palaces, equipped with many storage facilities, and at Gordion, where food processing occurs along with textile manufacture. Much of the other evidence from these centers stresses the use of cloth in a wealth finance system.

Wealth finance involves the production of goods, which have some standard of value and are used by the state possibly as some form of payment to craft specialists, administrators, or other elites.[33] Specialists and second-order administrators are able to draw from the central places their own rations and are granted access to provincial land, labor, and tribute.[34] These goods may be converted into staples or exchanged for other wealth goods. Elizabeth Brumfiel has studied tribute demands during the late Aztec period in Mexico and demonstrated that state textiles originated from commoner tribute and were distributed to some urban dwellers for service to the state. This cloth would then be exchanged at market for food. The farmers who produced the food in exchange for textiles would then send the cloth back to the state as tribute.[35] Here textiles function as a kind of currency or wealth commodity. The evidence presented in this dissertation demonstrates that the Minoans, Mycenaeans, and Phrygians, primarily employed the wealth finance system with regard to textiles. Standardized tools in centralized workshops, under the

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[32]T. Earle and E. Brumfiel (1987) 4.

[33]T. D'Altroy and Earle (1985) 188. For a Mesopotamian example of this, see Th. Jacobsen, "On the Textile Industry of Ur under Ibbi-Sin," Toward the Image of Tammuz and Other Essays on Mesopotamian History and Culture (ed.) W. Moran (1970) 216-230. Wool from the bookkeeper in charge of the copper pays the positive balance on his account in wool rather than in silver, using textiles as currency.

[34]For a fully integrated discussion of the staple and wealth finance model as it applies to the Inca, see T. Levine, (ed.) Inka Storage Systems (1992).

[35]E. Brumfiel, "Weaving and Cooking in Aztec Mexico," Engendering Archaeology: Women and Prehistory (eds.) J. Gero and M. Conkey (1991) 224-251.

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administration of political elites, produced cloth that had significant value, as an item of exchange, prestige, and wealth.

Other examples of wealth goods include objects of precious metal, ivory, and worked stone, often found within palatial structures of the Minoan, Mycenaean, and Phrygian kingdoms. Production of these goods required a great deal of capital, such as imported raw materials of high value, fairly elaborate installations, like kilns or furnaces, and highly skilled crafts-people. In the Aegean and in Anatolia, textiles, including murex-dyed purple cloth, were probably traded in exchange for imported raw materials such as ivory, tin, and gold. Crafts-people and their industries were dependent upon the palatial infrastructure for sustained production and also for the storage, distribution, and consumption of their finished wealth product.

Textiles are an unusual class of product in that they can be either a high value wealth good, produced by attached specialists, or a low-grade object of daily use produced independently by the general populace. Cloth is considered an agricultural product yet it is also a craft product of specialized workers. The varying degrees of production and consumption of textiles has significant socio-political implications. It should be stressed, however, that the staple/wealth finance model is a heuristic construct and is employed in this dissertation only for explanatory purposes. The focus here is to better understand the role of one multi-faceted commodity, textiles, within early states of the Aegean and Anatolia.

Summary

This study focuses on the economic aspects of textile production, combining textual, artifactual, and theoretical research, to better understand the significance of cloth manufacture in complex political economies of the Bronze and Iron Ages. The second chapter begins with a survey of textile production on Minoan Crete from the Neolithic

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through the Late Bronze Age. Aspects of textile production, such as dyeing, administration and weaving are included to better understand the changing nature of Minoan cultural complexity into a wealth finance system. The third chapter focuses on Mycenaean cloth production as revealed by the Linear B documents. The approach to these texts will be more archaeological than philological, incorporating contextual data to expand our understanding of Aegean craft production. The Linear B tablets from Mycenaean palaces provide the best evidence for the many stages of craft production activities in Late Bronze Age of mainland Greece and on Crete.

In contrast to the Aegean centers, the Phrygian site of Gordion lacks textual evidence for any craft activity yet provides the best archaeological evidence for the large-scale production of textiles. Chapter four analyses the large amounts of textile tools preserved in original workshop contexts within the citadel mound of the Phrygian capital. Many cultural features of the Aegean, including the use of the warp-weighted loom and the redistributive palace economy, are paralleled in Anatolia.[36] For these reasons, the dissertation bridges the Aegean and Anatolia from the critical periods of the Middle Bronze to the Early Iron Age. The final chapter concludes with a summary of specialized textile production examined in this dissertation. We see the degree of workshop variability reflected in the types of sites studied. Reference is also made to comparative examples of cloth production in other regional centers around the world. These data are useful in explaining some of the gaps observed in the Aegean and Anatolian evidence. This

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[36]People of the Aegean and Anatolia used the warp-weighted loom for the production of the majority of their textiles, beginning as early as the Middle Neolithic period. This technology gave them advantages over the ground looms of Egypt and the Near East. See M. Hoffmann, The Warp-Weighted Loom: Studies in the History and Technology of an Ancient Implement (1964) 21, 314; G. Crowfoot, "Of the Warp-Weighted Loom," BSA 37 (1936/37) 36-47; J. Carington Smith, Spinning, Weaving and Textile Manufacture in Prehistoric Crete Ph. D. thesis, University of Tasmania, Hobart, (1975) 97-99. Many of the women textile workers listed in the Linear B tablets have ethnic descriptions locating their origin in Anatolia, at places such as Knidos (*ki-ni-di-ja*), Miletus (*mi-ra-ti-ja*), and Halikarnassos (Zephyra, *ze-pu$_2$-ra$_3$*). See J. Chadwick, "The Women of Pylos," Texts, Tablets and Scribes, Minos Suppl. 10 (eds.) J.-P. Olivier and T.G. Palaima (1989) 43-96. See also Chapter Three.

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dissertation demonstrates that organized textile production reflects a wealth-based economy for the Minoans, Mycenaeans, and Phrygians.

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Chapter Two - The Organization of Textile Production on Minoan Crete

Although the actual remains of wool and linen cloth in the prehistoric Aegean are few, we are fairly well informed about textile production from a variety of sources. These include craft production residues and administrative documents from Minoan centers. As textiles were a major source of wealth in Minoan palace economies of the second millennium B.C., changes in the organization of production should reflect socio-economic and political developments. Does the archaeological record for textile production reflect what we know, or think we know, about Minoan Crete and the origins of Minoan complexity? While some may question whether loom weights and spindle whorls will explain the 'emergence of civilisation' on Crete, it appears likely that organized textile production did contribute to the development of the Minoan state. By examining the archaeological indicators of cloth manufacture on Crete before the Mycenaean incursion we should be able to map changes over time in the complexity of early Minoan society.

The emergence of complex society on Minoan Crete is demonstrated by the Minoan palace. In both its architectural and political form, the palace emerges at the beginning of the second millennium B.C. Eventhough textile production on Crete can be traced back to the middle Neolithic period, it is not until the second millennium that significant changes in production technology are observed during the earliest phases of the Minoan palaces. These changes include specialized spinning, weaving, and dyeing, and some evidence for an international exchange in textiles.

In this chapter, textile technology is analyzed in relation to emerging complexity on Crete from the Neolithic to the Late Bronze Age. After describing the types of tools

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associated with cloth production on Crete, Part I of this chapter presents significant examples of textile production from Prepalatial Crete. Part II focuses on Palatial Crete before the Mycenaean period and compares this to Part I. Rather than catalogue every reported Minoan loom weight or spindle whorl, I examine certain deposits of various Minoan tool types to see how changes in technology reflect social organization in Minoan society. Administrative documents are also included when they refer to cloth production.

Minoan Textile Equipment

There are basically two major types of archaeological indicator for textile production on Minoan Crete: loom weights and spindle whorls.[1] These tools are most often clay weights of various shapes and sizes used to add tension to thread and are found at many sites. Spindle whorls are usually small, fired clay objects of various shapes and weights.[2] They are attached as a weight at the bottom of a long slender rod called a spindle. Raw wool is drawn out and attached to this rod and then spun manually. Spun thread then winds itself around the narrow spindle. All weaving technologies depend upon some system of tension-spun thread. In the Aegean this was achieved with the dropped, whorl-weighted spindle.

Minoan spindle whorls have been a source of quiet consternation among archaeologists for some time. From literary references and ethnographic analogies we might speculate that women of all classes in Bronze Age society spun thread. Spinning is quite labor intensive yet can be done almost anywhere at anytime. It does not require the physical or mental attachment that the loom needs. The spinner must find one steady

[1]Other artifacts associated with textile production include weft bobbins, needles, dye vats, fiber-wetting bowls, and decks of tablets for tablet weaving. See M. Hoffmann (1964) Chapters 1 and 5; J. Carington Smith (1975); E. Barber (1991) Chapters 2, 3, 6, 12, 14-16; and I. Τζαχίλη (1997).

[2]I. Τζαχίλη (1997) 105-123; E. Barber (1991) 51-68, 299-310.

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rhythm of rotation and coordinate it with her own movements throughout the day.[3] Barber estimates that it took seven to eight hours to spin thread that could be woven in one hour.[4] Because of this, one would expect to find many whorls throughout the archaeological record on Crete. This, however, is not the case. On Crete, the proportion of excavated spindle whorls to loom weights is very low in palatial contexts. Excavations at the Palace of Knossos, excluding the Prepalatial settlement, report no spindle whorls.[5] The sites that do yield spinning equipment are fairly small, isolated settlements such as Debla, Kastellos, Sklavokampos and Myrtos.[6] As Carington Smith states, 'The most interesting thing about Minoan whorls is that there are so few of them ...Their numbers would seldom have been adequate to supply all the thread needed by the island's many looms at any one time'.[7] Tzachili gives a total count of whorls from Crete: 48 from the Early Minoan period, 15 whorls from Middle Minoan sites, and 11 dating to the Late Minoan period.[8] In striking

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[3]I. Τζαχίλη (1997) 126.

[4]E. Barber, "Minoan Women and the Challenges of Weaving for Home, Trade, and Shrine," TEXNH. Craftsmen, Craftswomen, and Craftsmanship in the Aegean Bronze Age. Proceedings of the 6th International Aegean Conference, Temple University, 18-21 April, 1996. Aegaeum 16 (eds.) R. Laffineur and P. Betancourt (1997) 515.

[5]This absence is striking in comparison to the thousands of whorls reported from various levels at Troy and Aphrodisias. For Troy, see H. Schliemann, Troy and Its Remains (1976) pls. 22-51; Ilios: The City and Country of the Trojans (1880) 327, 361; W. Dörpfeld, Troja und Ilion (1902) 340; C. Blegen, *et al.* Troy vol. I Princeton (1950) 19, 24, 29, 37, 50, 104, 216, 219, 337-8, 350-3, 369; vol. II Princeton (1951) 15, 74, 85, 116, 169-172, 233f, 261; vol. III Princeton (1953) 30-3, 232, 272, 315; vol. IV Princeton (1958) 17-19, 52, 82, 151, 199, 250-1. And see K. Balfanz, "Bronzezeitliche Spinnwirtel aus Troia," Studia Troica V (1995) 117-144. For Aphrodisias, see M. Joukowsky, *et al.* Prehistoric Aphrodisias vol. I (1986) 373-381. See E. Barber (1991) Chapter 14, 'The Weight Chase,' for a discussion of the importance and resilience of the spindle whorl in the archaeological record, including examples at Troy and Aphrodisias.

[6]J. Carington Smith (1975) 268.

[7]J. Carington Smith (1975) 263.

[8]I. Τζαχίλη (1997) 126.

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contrast, after the palatial period, in Sub-Minoan Crete, a site such as Karphi yielded 110 spindle whorls.[9]

In Tzachili's study of weavers and weaving in the Bronze Age, she suggests three possible explanations for the absence of Minoan spindle whorls, none of which is mutually exclusive.[10] 1) Most Minoan whorls may have been made of similarly perishable material as the spindle, *e.g.* wood. 2) Fibers could have been spun by a different technique from the whorl-weighted drop-spindle. 3) Finally, tasks such as spinning might have become highly specialized in an organized Minoan textile industry and occurred far away from excavated centers with only the spun thread sent on to weaving workshops. More than likely the spinning of wool and linen during the Old and New Palace periods occurred in outlying areas. Spun thread would then be sent on to the weavers located at the palaces.

As with spun thread, woven textiles can be made in a variety of ways.[11] Horizontal ground-looms and two-beam vertical looms are known in different parts of the Mediterranean region. On Crete and most of Bronze Age Greece, however, it is generally assumed that the warp-weighted, vertical loom was used most often, based on pictograms in Aegean scripts and on later representations of weaving scenes in Archaic and Classical Greece. Suspended clay loom weights gave tension to the warp threads hanging vertically at the bottom of a standing loom. The weights were attached in a variety of ways. They were either tied directly to bunches of the warp threads, attached to a cord which was fastened to the warp, or they were tied to a thin bar which had warp threads connected to it.[12]

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[9]J. Carington Smith (1975) 263-272.

[10]I. Τζαχίλη (1997) 125-129.

[11]I. Τζαχίλη (1997) 145-171; E. Barber (1991) 79-125.

[12]J. Carington Smith (1975) 275-279; E. Barber (1991) 104-105; I. Tzachili, "All Important Yet Elusive: Looking for Evidence of Cloth-Making at Akrotiri," Thera and the Aegean World vol. III.1 (1990) 380-389,

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There are three basic shapes of Minoan loom weight (see fig. 2.1 a-c):[13]

The first is a rectangular clay weight with four holes in each corner (fig. 2.1 a). These have been found as early as the Neolithic period at Knossos. A variant of this cube type occurs in the Middle Minoan period concentrated in Eastern Crete. Most often four small suspension holes, one in each corner, occur along the longitudinal axes, showing thread wear.[14] Over one hundred of these 'block' shaped weights were found at Palaikastro.[15] It is of course possible that these objects were not at all connected with weaving since it is difficult to reconstruct the use of the four holes in each corner. Some of these weights are, however, stamped in a similar manner to other known loom weights and they do appear fairly standardized in form and weight like other known loom weights.[16]

Next, melon-shaped or spherical weights occur exclusively on Crete during the MM to LM II period (figure 2.1 b). None have been found outside of Crete, even at sites such as Kythera or Thera. The weights seem to be found in areas with religious associations, particularly around the area of Knossos.[17] These tend to weigh more than the other loom weights, around 150 to 250 grams.

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fig. 9. The attachment of these weights was first illustrated in J. Hazzidakis, Les Villas Minoennes de Tylissos (1934) 105, Pl. 30.1. The weights, however, were identified as 'clay axe and club heads' for ritual purposes. Some loom weights in the Classical period also seem to have used a wooden rod inserted horizontally through the top of the weights. One of the looms on the famous aryballos in the Corinth museum illustrating the weaving contest between Arachne and Athena shows loom weights attached to horizontal bars. Two loom weights found at Nemea show this rod. See B. K. McLauchlin, "New Evidence on the Mechanics of Loom Weights," AJA 85 (1981) 79-81.

[13]All illustrations are at the end of each chapter.

[14]J. Carington Smith (1975) 293-296.

[15] R.W. Hutchinson, "Unpublished Objects from Palaikastro and Praisos," BSA 40 (1939-40) 47-49.

[16]A great many of these stamped weights are on display in the Hagios Nikolaos Museum.

[17]See J. Sakellarakis and E. Sapouna-Sakellaraki, Archanes (1991) 87, fig. 62. Another sizeable group of melon weights comes from LM II levels of Room N in the Unexplored Mansion; M. Popham, *et al.*, The Minoan Unexplored Mansion. BSA Suppl. vol. 17 (1984). This type of weight is discussed in detail below, pp. 70-78.

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The third type, the circular or discoid weight, is the most typical Minoan loom weight (fig. 2.1 c). This weight appears on Crete from EM II through LM III. Carington Smith states that the Minoan discoid weight is just as culturally distinctive of the Minoans as is the Double Ax or the Horns of Consecration.[18] In Early Bronze Age Anatolia, at Troy I, II, IV, and at Aphrodisias, a similar disc-shaped loom weight is common. Since Aegean examples are known from the same time periods, Carington Smith suggests that both types are derived from a common source.[19] The weights have holes in the upper half and often a slightly flattened and grooved top side. In the Old Palace at Knossos Evans found in the Loom Weight Basement of the East wing over 400 of these discoid loom weights. A similarly large number was found at Akrotiri as well.[20] I was able to weigh and study all of the surviving Loom Weight Basement weights (45 in number) at the Stratigraphical Museum. The weights are fairly standardized, with an average weight of 155 grams. The total of 400 loom weights suggests ten to fifteen working looms if we assume each loom used about 20-30 weights for plain weeave (more if pattern weaving).[21]

Large deposits of Minoan loom weights are known from many different centers, including Minoanized sites of the Aegean such as Akrotiri, Ialysos, Iasos and Ayia Irini.[22]

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[18]J. Carington Smith (1975) 275.

[19]See J. Carington Smith, "Spinning and Weaving Equipment" in Excavations at Nichoria in Southwestern Messenia, Vol. II: The Bronze Age Occupation (eds.) W. McDonald and N. Wilkie (1992) 687-91.

[20]The textile equipment from Akrotiri has been thoroughly studied and published by I. Tzachili (1990); (1997) 183-193.

[21]E. Barber (1991) 104. See further in Part II.

[22]For a thorough analysis of the loom weights from Akrotiri, see. I. Tzachili (1990) 407-419; (1997) 183-193. MM II pottery was found with spindle whorls and loom weights on the acropolis at Ialysos. See M. Benzi, "Evidence for MM Settlement on the Acropolis at Ialysos," in The Minoan Thalassocracy: Myth and Reality. Proceedings of the Third International Symposium at the Swedish Institute in Athens, 31 May - 5 June, 1982 (eds.) R. Hägg and N. Marinatos (1984) 93-105. Discoid loom weights were found at Iasos. See R. Laviosa, "The Minoan Thalassocracy, Iasos," in The Minoan Thalassocracy (1984) 183-85. For textile

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J. Davis' study of the distribution of loom weights throughout the settlement of Ayia Irini on Keos is a very good example of the cultural implications suggested by large quantities of excavated textile equipment. Middle Bronze Age discoid loom weights found on Keos may have closer parallels in Anatolia than Crete, although both types may be descended from the same original source.[23] During the Middle Bronze Age terra-cotta spools are also prevalent on Keos. These objects are about five to seven centimeters in height with a pinched waist and are often catalogued with other textile equipment at many sites throughout the Aegean, such as Zygouries and Eutresis.[24] The function of these objects has long puzzled scholars.[25] Barber suggests that they may have been used to give tension to threads while being twisted into a belt-cord.[26] Carington Smith associates them with a warping creel or spool rack, used to warp a horizontal loom.[27] If they are associated with a horizontal loom, Davis remarks that two fundamentally different weaving technologies are present at Ayia Irini in the Middle Bronze Age, the warp-weighted loom and the horizontal loom.[28]

By the LC I period, on Keos, it is the Minoan discoid type of loom weight that dominates the textile assemblages. Over 185 clay loom weights are found in LC I levels, heavily concentrated in the north central part of the site. Davis speculates that this concentration of standardized clay weights (rather than stone) indicates specialized textiles woven with Minoan technology for trade within the Aegean sphere. This warp-weighted loom

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equipment on Keos, see J. Davis (1984) and W. Cummer and E. Schofield, Keos III: Ayia Irini, House A (1984) pl. vii.

[23]J. Carington Smith (1992) 688.

[24]C. Blegen, Zygouries (1928) 190-1, fig. 179.4-5; H. Goldman, Excavations at Eutresis in Boeotia (1931) 193; J. Davis (1984) 162-163.

[25]See E. Barber (1991) 107 for discussion and bibliography of Aegean spools.

[26]See E. Barber (1997) 515-519, especially fig. CXCII.c.

[27]In modern Greek, the term is *diastra* or *klouvi*. J. Carington Smith (1975) 400-404.

[28]J. Davis (1984) 163.

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had an advantage over the other loom types. It provided increased control over warp thread tension and permitted cloths of greater width.[29] By understanding the distribution of loom weights through time and space at sites like Ayia Irini we can greatly increase our understanding of cultural interactions and economic development in the Bronze Age Aegean.

Part I - Prepalatial Crete

Neolithic Knossos

The history of textile production on Crete begins, as most subjects Minoan, at Knossos.[30] Excavated by J. Evans and the British School at Athens from 1957-1960 and 1969-70, the earliest, largest, and longest-lived Neolithic settlement known on Crete was built on the promontory junction of the Vlykhia and Kairatos streams at Knossos.[31] This aceramic hamlet of the 7th millennium B.C. had houses of mud-brick on stone foundations built on the south side of the knoll. The houses covered *ca.* 5 hectares and had a population estimated at 1000 individuals. The earliest settlement was built atop a rather sloping mound that was later leveled for the MM II West Court. The site was ideally located near the sea, fresh water, and fertile land. Sheep, goat, pigs, dog, and small cattle

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[29]J. Davis (1984) 161-63. The advantages of a warp-weighted loom over a horizontal loom are discussed by M. Hoffmann (1964) 21, 314; G. Crowfoot (1936/37) 36-47; J. Carington Smith (1975) 97-99; A. Sherratt and S. Sherratt, "From Luxuries to Commodities: The Nature of Mediterranean Bronze Age Trading Systems," Bronze Age Trade in the Mediterranean. SIMA vol. 90 (ed.) N. Gale (1991) 359.

[30]J. Evans, "Excavations in the Neolithic Settlement at Knossos, 1957-60," BSA 59 (1964) 132-240; "Knossos Neolithic, Part II," BSA 63 (1968) 239-276; "The Early Millennia: Continuity and Change in a Farming Settlement," in Knossos: A Labyrinth of Hisory. Papers Presented in Honour of Sinclair Hood (eds.) D. Evely, H. Hughes-Brock, and N. Momigliano (1994) 1-20. See also T. Strasser, Neolithic Settlement and Land Use on Crete, Ph. D. Dissertation, Indiana University (1992) 65-73.

[31]Other Early Neolithic sites on Crete are the Gerani cave and Lera cave. Watrous summarizes evidence that might suggest an even earlier habitation of Crete, perhaps as early as the 12th millennium; L. V. Watrous, "Review of Aegean Prehistory III: Crete from Earliest Prehistory through the Protopalatial Period," AJA 98 (1994) 699-700.

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were domesticates here as well as legumes, emmer, and one-corn wheat. None of these foods is unusual for this period.

No wild goats or sheep, cattle, pigs or wild wheat have been found on Crete, suggesting that original settlers of the island planned a colony and transported at least 15-20 animals of each species for a good genetic base overseas.[32] Before humans reached the island, herbivores, such as deer, pygmy elephants and pygmy hippopotami thrived in the predator-free environment of Crete. Once humans arrived on the island these 'pygmy' species became extinct, suggesting that there was competition for the limited island resources.[33] The animals that are found at Neolithic Knossos, sheep/goat, pigs, dogs and small cattle, are found in contemporary levels at Çatal Höyük. Breadwheat (*Triticum aestivum*) is also found in the palaeobotanical record, the only parallel of which is found at Çatal Hüyük. In addition, the mud-brick architecture of Neolithic Knossos finds parallels at Hacılar and Aşıklı Höyük. Taken together, this evidence suggests an eastern origin for the first inhabitants of Crete, probably from southwestern Anatolia.[34]

The Neolithic settlement at Knossos lasted at least 3500 years. Ten occupation phases were discerned (strata I-X) by J. Evans, the primary excavator. For simplicity, the site has also been classified into four ceramic phases:[35]

1) Early Neolithic I (EN I). This period had fired mud-brick architecture and was the longest lasting Neolithic period (1500-2000 years).

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[32]C. Groves, "Feral Mammals of the Mediterranean Islands: Documents of Early Domestication," The Walking Larder (ed.) J. Clutton-Brock (1990) 46-58 (*non vidi*).

[33]See E. Lax and T. Strasser, "Early Holocene Extinctions on Crete: The Search for the Cause," JMA 5 (1992) 203-224.

[34]See J. Evans (1994) 1-20. See also pp. 83 and following below.

[35]J. Evans (1994). See also A. Furness, "The Neolithic Pottery of Knossos," BSA 48 (1953) 94-134.

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2) EN II. This level lies four meters above EN I and dates to the 5th millennium, lasting several hundred years.

3) MN. Tests under the Central Court revealed two large square multi-room buildings dating to late 5th millennium. They seem to have been occupied for only a few hundred years.

4) LN. Also from the Central Court excavations were three building levels dating to the Late Neolithic period.

According to the excavator there was steady, uninterrupted growth of the settlement throughout the Neolithic period. Broodbank and others, however, see a less uniform rate of growing complexity and identify a shift in material culture in the transition from Early to Middle Neolithic.[36] These changes are reflected in mortuary practices, styles of pottery decoration, and the quantity of figurines. How do changes in the textile equipment from Neolithic Knossos reflect changing complexity?

Over 150 clay spindle whorls are reported from the latest three Neolithic periods (III-I) of the Central Court excavations, indicating the spinning of raw fiber into usable thread. Numerous clay loom weights of various shapes and sizes, 4-holed cuboid or 2-holed half ovals, were also found, demonstrating that weaving and spinning equipment were located together.[37] Uncorrected radio-carbon dates suggest that the first appearance of textile equipment dates to the Middle Neolithic, in approximately the first quarter of the 4th millennium B.C.[38] Stratum III (Middle Neolithic) had 25 clay whorls, 2 stone whorls,

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[36]C. Broodbank, "The Neolithic Labyrinth: Social Change at Knossos before the Bronze Age," JMA 5.1 (1992) 39-75. See also commentary by T. Whitelaw, "Lost in the Labyrinth? Comments on Broodbank's 'Social change at Knossos before the Bronze Age'" JMA 5.2 (1992) 225-32 and J. Bintliff, "Structuralism and the Minoan Myth," Antiquity 58 (1984) 33-38.

[37]J. Evans (1968) 233-235, fig. 56, pls. 56 and 57.

[38]J. Evans (1968) 272. See also discussion by J. Carington Smith (1975) 182-84.

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88 sherd-discs, of which six were pierced,[39] two possible weft-bobbins[40] and two sets of clay loom weights, with 7 and 13 weights in each set.[41] Stratum II had nearly 100 spindle whorls, three of which were decorated.[42] The Late Neolithic (Stratum I) was greatly disturbed by the later Knossos builders but it did reveal a group of loom weights, 36 spindle whorls and 2 weft-bobbins.

The earliest loom weights from Neolithic Knossos are box-shaped or cuboid, with four holes in each corner, some showing evidence of thread wear. Of the Stratum III weights, the first set were made of creamy white clay, ranging in weight from 350 to 635 grams.[43] The other set of 13 was less well-preserved and lighter, weighing approximately 200-300 grams.[44] Whether or not these weights were used on warp-weighted looms is difficult to answer with certainty, although the form of the weight does recur in the Middle Minoan period associated with textile equipment (discussed further on pp. 81f.).

The archaeological literature seems to assume that the textiles woven at Knossos were wool.[45] In the early fourth millennium this is possible but it would be close to the earliest known exploitation of woolly sheep. One should at least be aware of the long

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[39]See E. Barber (1991) 54 n. 10 for the occurrence elsewhere in the Aegean of pierced discs, probably used as spindle whorls.

[40]One is of clay, the other of bone. Weft bobbins would have thread wrapped around them which was then fed through the warp threads. According to Barber, these are often erroneously referred to as shuttles. See E. Barber (1991) 84-85 n. 3 for the definition of a true shuttle, a rather late invention in the Mediterranean region (*ca.* 10th century A.D.). In modern Greek, the term is σαΐτα; I. Τζαχίλη (1997) 175-77, εικ. 86.

[41]J. Evans (1964) 180-82, 234-35, fig. 56.1, 61.13, pl. 56.1-2, pl. 57.2 no. 1-7, pl. 60, 1, no. 5; E. Barber (1991) 100, 387.

[42]J. Evans (1964) 182-188.

[43]E. Barber (1991) 100; J. Carington Smith (1975) 185.

[44]E. Barber (1991) 100.

[45]See, for example, the Aegean Prehistory notes online by J. Rutter discussing the model proposed by van Andel and Runnels.

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tradition of flax and linen textiles and the likelihood of their use. Preliminary faunal analysis ran contrary to what was expected of a growing wool-based industry at Neolithic Knossos.[46] Broodbank states, 'The appearance of spinning and weaving technology at exactly the time when the sheep and goat component goes into sharpest decline represents a suspicious mismatch between two types of data.'[47] After re-analysis, however, the faunal remains from Knossos do seem to indicate that sheep and goat were the main species exploited, probably in part for their wool. Whitelaw reports that the re-analysis of faunal remains by Winder conclusively shows that any changes in the relative exploitation of different species is a result of taphonomic (*i.e.* depositional), not cultural, processes.[48]

Towards the end of the Neolithic period, in stratum I at Knossos, the cuboid weights seem to be replaced by a taller, heavier weight, pear or pyramidal in shape with a suspension hole at the top. This type, Carington Smith reports, concurrently appears in northern Greece.[49] Interestingly, the older cuboid type of weight reappears in Crete during the Middle Minoan period, but almost exclusively at the eastern end of the island, only as far west as Mallia (also spelt Malia). Carington Smith suggests that perhaps the users of the cuboid weights were the earliest inhabitants of Crete and that toward the end of the Neolithic, they were pushed eastwards by the newly arrived 'Minoans'.[50] It may be that

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[46]M. Jarman and H. Jarman, "The Fauna and Economy of Neolithic Knossos," BSA 63 (1968) 241-64; T. Strasser (1992) 70-73; and C. Broodbank (1992) 39-75.

[47]Broodbank (1992) 62.

[48]T. Whitelaw (1992) 232 and N. Winder, Faunal Analysis: Studies in the Analysis and Intepretation of Animal Bones from Large, Multi-phase Archaeological Excavations Unpublished Ph. D. Dissertation, Southampton University (1986) (*non vidi*).

[49]J. Carington Smith (1975) 154-156.

[50]Carington Smith cautions, however, that the four-holed cuboid weight of the Neolithic differs in terms of which axis is pierced, compared to the Middle Minoan weights. The short side is pierced in Neolithic weights versus the long axis of the Minoan examples. These later weights are also lighter and have smaller holes, perhaps for finer thread, compared to the Neolithic weights; J. Carington Smith (1975) 293-296.

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the weights were used in specialized weaving, perhaps with four different color threads through the holes, like tablet weaving.[51] The bulkiness of the clay weights, however, makes the tablet weaving comparison difficult to imagine.

Textile equipment first appears in the Middle Neolithic and in such significant concentrations that this suggests a change in social complexity and organization between the Early and Middle Neolithic.[52] While Broodbank's analysis uses archaeological theory and is more statistically rigorous than J. Evans', the nature of the data and the tremendous rebuilding that occurs at Knossos in the Bronze Age prevents his theory from being satisfactorily proven. The appearance of textile equipment, it can be said, does indicate a rather marked cultural change at Knossos from the Early to Middle Neolithic period. The Neolithic excavations at Knossos uncovered only a small portion of the total extent of the site. While the exact nature of the habitation at Neolithic Knossos is debated, the changing pattern of textile equipment found in such a small area suggests a shift in the degree of craft specialization. The evidence suggests that in the Middle Neolithic period there were dispersed workshops producing for unrestricted consumption. By the Later Neolithic, however, there may have been community specialization where autonomous individuals or household-units aggregated to produce cloth for regional consumption.[53] The large amount of production equipment and the limited area excavated suggests that these textile tools were shared by the community at large.

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[51] J. Carington Smith (1975) 185-186. See also A. Henshall, "Textiles and Weaving Appliances in Prehistoric Britain," Proceedings of the Prehistoric Society 16 (1950) 130, 148 and E. S. van Reesma, Contributions to Early History of Textile Technics (1926).

[52] C. Broodbank (1992) 39-75.

[53] Following C. Costin (1990) 1-56. For a discussion of archaeological indicators of craft production in general, see M. Tosi, "The Notion of Craft Specialization and its Representation in the Archaeological Record of Early States in the Turanian Basin," in Marxist Perspectives in Archaeology (ed.) M. Spriggs (1984) 22-52.

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Early Minoan Knossos

In 1904 Arthur Evans' assistant and record keeper, Duncan Mackenzie, dug test trenches below the West Court at Knossos and drew the North section. To Evans, who may not have even been present at Knossos at the time, this section showed a clear, uninterrupted Neolithic to Middle Minoan chronology of pottery with three successive floor levels of EM pottery. Evans' scheme based on this section was delivered at an International Conference in Athens in the Spring of 1905 and presented the tripartite division logic that has guided many Minoan archaeologists' dating ever since.[54] L. Palmer reports that in the autumn of the same year (1905), Mackenzie wrote to Evans with special reference to the EM phases and said, 'As regards an Early Minoan series, I cannot understand how you came to imagine such a series either from the West Square Section or anywhere else at Knossos'.[55] None of the pottery was from a floor deposit because the workmen had dug right through the floor. The stratigraphy presented by Evans was adapted from an ideal sketch of a section in Mackenzie's notebook. Evans disregarded this. He looked to the Egyptian three-part chronology system as a model, (Old, Middle and New Kingdom) as well as Darwinian evolutionary theory of growth, maturity and decay.[56]

Doro Levi dug Early Minoan sections at the palace of Phaistos that he believed were in conflict with the influential West Section of Knossos. The EM levels he had showed only a short transition period between the Neolithic and the Middle Minoan, not the 1000 years Evans indicated. While there have been some modifications to the time spans of the periods, Evans' overall schema has remained virtually unchanged in Minoan

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[54]A. Evans, Essai de Classification des Epoques de la Civilisation Minoenne (1906).

[55]Letter from the Evans archives in the Ashmolean Museum, Oxford, quoted in L. Palmer, A New Guide to the Palace of Knossos (1969) 33.

[56]The Egyptian system was based on king lists with dates of succession. Evans cited Egyptian imports found at Knossos to support his chronological framework in the Middle and Late Minoan periods.

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archaeology.[57] Rather than later ceramic finds revising the chronology, they have been either confined by it or used to validate the Knossos sequence.[58] In his *magnum opus*, Evans notes that all chronological demarcations are arbitrary, no distinct compartments exist for Knossos and he expects that opinions on his divisions will differ constantly.[59]

Stepping out of the shadow of Evans, J. Cherry has argued that the origins of palatial society on Crete were the result of swift, punctuated changes in social organization.[60] Looking at the foundations of palatial Crete in the Prepalatial period of the third millennium B.C., Cherry sees the development of social complexity as a swift quantum leap forward rather than a protracted, progressive process. He also notes the importance of contact with pre-existing Near Eastern states in the formation of what we know of as the Minoan palace.[61] This revolutionary notion challenges the traditional

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[57]See the full critique in F. Hentschel, The Basis for the Standard Chronology of the Late Bronze Age at Knossos, Ph. D. Dissertation, Yale University (1982).

[58]Often, excavated Minoan pottery will be chronologically delineated at one site on Crete and then correlated with the accepted ceramic periods at Knossos. For example, the ceramic sequence at Kommos identifies 'Early LM IA' pottery there corresponding to the 'Beginning of Transitional MM III/LM IA' at Knossos and Kommos 'Advanced LM IA' corresponding to the 'Remainder of Transitional MM IIIB/LM IA and an interval of unknown duration' at Knossos; J. Shaw, A. Van de Moortel, P. Day and V. Kilikoglou, "A LM IA Pottery Kiln at Kommos, Crete," TEXNH. Aegaeum 16 (1997) 325, n. 10. A good attempt at clarity is found in L. Bernini, "Ceramics of the Early Neo-Palatial Period at Palaikastro," BSA 90 (1995) 55-82.

[59]A. Evans The Palace of Minos I (1921) 30. Nearly one hundred years after Evans first presented his chronological system, many problems in Minoan dating still seem to come down to stratigraphic versus stylistic sequences of ceramics. By shifting the emphasis from ceramics and looking at many other features of Minoan society, including aspects of textile production and the organization of labor, we are better equipped to explain changes over time on Crete.

[60]J. Cherry, "Evolution, Revolution, and the Origins of Complex Societies," in Minoan Society (eds.) O. Krzyszkowska and L. Nixon (1983) 33-45; "Polities and Palaces: Some Problems in Minoan State Formation," Peer Polity Interaction and Socio-Political Change (eds.) J. Cherry and C. Renfrew (1986) 19-45.

[61]For other views on the prepalatial period, see K. Branigan, Pre-Palatial: The Foundations of Palatial Crete (1988); "Social Transformation and the Rise of the State in Crete," in Politeia: Society and State in the Aegean Bronze Age. Proceedings of the 5th International Aegean Conference. Aegaeum 14 (1995) 33-40; P. Halstead, "From Determinism to Uncertainty: Social Storage and the Rise of the Minoan Palace," Economic Anthropology. BAR 96 (1981) 187-213; D. Wilson, "Knossos Before the Palaces: an Overview

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concept of progressive evolution: growth, maturity, then decline, based on Evans and Darwinism of the late nineteenth and early twentieth centuries.

Archaeological evidence indicates that from EM I to EM II, a period of about 1200 years, all of Crete had a scatter of fairly simple farmhouses.[62] To some, evidence from EM II and EM III Crete shows the early signs of palatial society.[63] The codified form of the Old Palaces first occurs in MM I B at Knossos and elsewhere on Crete. Something external has stimulated the system between EM II and MM I B. What this was is difficult to say but it cannot simply be explained as a gradual change over time. During the Early Minoan II A period the first Cycladic imports on Crete are found at Knossos and the first Minoan colony is established on Kythera.[64] Over 30 EC II jars and sauce-boats, an obsidian bowl, and examples of Urfirnis ware, that may have come from mainland via Keos or Melos, were found at Knossos. Also, in EM II A levels a Hippopotamus tusk was found at Knossos suggesting early foreign contact between Minoans and Egyptians. As Watrous notes, 'In contrast (to the imports) the list of EM II exports in the Aegean is relatively short.'[65] Perhaps textiles were Minoan exports, the evidence for which no longer survives.

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of the Early Bronze Age (EM I-EM II)," in Knossos: A Labyrinth of History (1994) Chapter 2; L. V. Watrous (1994) 695-753; S. Manning Before Daidalos. The Origins of Complex Society and the Genesis of the State on Crete Ph. D. Dissertation, University of Cambridge (1995) (*non vidi*).

[62]L. V. Watrous (1994) 697 for chronology of EM I- II.

[63]The French excavators at Mallia have suggested that they have the early palatial features by EM III. See O. Pelon, "Fonction politique dans un palais minoen," Minoan Society (1983) 251-257; J.-Cl. Poursat, "Le debut de l'époque protopalatiale à Mallia," Eilapini (1987) 461-466. J. Soles, "Social Ranking in Prepalatial Cemeteries," Problems in Greek Prehistory (ed.) E. French and K. Wardle (1988) 49-62 suggests that the prepalatial cemeteries at Mochlos, Gournia, and Mallia show signs of social and economic rankng, yet the differences between individuals are minimal, according to L. V. Watrous (1994) 713.

[64]D. Wilson (1994); J. Coldstream and G. Huxley, "The Minoans of Kythera," Minoan Thalassocracy (1984) 106-108; L.V. Watrous (1994) 711.

[65]L.V. Watrous (1994) 711.

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In terms of textile production, D. Wilson reports that the West Court Houses at Knossos, excavated in 1969-70, yielded spindle whorls and loom weights of Early Minoan II A date.[66] The fabric of the whorls was fine, gray to grayish brown, usually smoothed, not burnished, with a brown to black surface. The clay was not well fired and crumbled once the whorls broke. Some examples from the West Court Houses may be Neolithic since a few resemble Neolithic types discussed above. All the spindle whorls but one are plain; one has four rows of punctuated single or double dots dividing the upper surface into four equal parts. The biconical whorl was the most common at Knossos, ranging from 3.2 to 4.9 cm. in diameter, while at Myrtos, discussed below, the most common form of spindle whorl was the cylindrical type.[67] The loom weights are dome shaped or cylindrical and of coarse, made of unbaked or poorly fired clay. It is difficult to say how much textile production was going on at Prepalatial Knossos but we can be certain that the earliest Minoans were spinning and weaving.

Myrtos Fournou Korifi

Elsewhere on Crete we find another Prepalatial settlement contemporary with the sketchy remains from Knossos. The Early Minoan II A-B site of Myrtos (Fournou Korifi) is located on the southeastern coast between Ierapetra and Arvi. Dating to about 2600-2170 B.C. the site is significant because of its central role in debates regarding the origins of complexity on Crete. It is also of great interest because of the substantial data found there related to textile production. I will argue that the two issues are not unrelated.

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[66]D. Wilson, The Early Minoan II A West Court Houses at Knossos Ph.D. Dissertation, University of Cincinnati (1984) 214-219.

[67]Comparative examples of other early whorls on Crete are found at Kastellos (Lasithi), 'sub-Neolithic period' in J. Pendlebury and M. Money-Coutts, "Excavations in the Plain of Lasithi II," BSA 38 (1937-1938) 55, no. 46; at EM I Arkalokhori, in J. Hazzidakis (1912-13) 42, fig. 4m; at Debla, Phase I = EM I in P. Warren and I. Tzedakis (1974) 330, fig. 21.

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The complete excavation and thorough publication of Myrtos Fournou Korifi by Peter Warren provides a great deal of evidence for varying interpretations of Prepalatial Crete.[68] Originally, Warren interpreted Myrtos as a single, integrated community with specialized areas for economic, political, and ritual purposes.[69] Branigan interpreted the one hundred-room site as an integrated whole, a mansion, or proto-palace, for a local chief.[70] Whitelaw's reinvestigation of EM II A - B Myrtos indicates that the social organization shows no signs of complexity beyond that of a large nuclear family, with perhaps six separate households.[71] Given such varying interpretations of the same evidence, we can look specifically at the textile production material from Myrtos to see if this better informs us of Early Minoan social organization.

The textile equipment from Myrtos has been well published by Warren, with cautionary remarks about over-interpretation.[72] Taking full account of the textile equipment, we see that Myrtos yielded four spindle whorls from EM II A levels, and 28 from EM II B or mixed contexts. The majority of the whorls were cylindrical in shape. Fifteen of them were painted.[73] Discoid loom weights, with one to three holes pierced

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[68]P. Warren, Myrtos. An Early Bronze Age Settlement in Crete. BSA Suppl. vol. 7 (1972).

[69]P. Warren (1972) 267. For a reassessment of his own views, see P. Warren, "The Genesis of the Minoan Palace," in The Function of the Minoan Palaces. Proceedings of the Fourth International Symposium at the Swedish Institute in Athens, 10-16 June, 1984 (eds.) R. Hägg and N. Marinatos (1987) 47-64.

[70]K. Branigan, The Foundations of Palatial Crete (1970) 47-48.

[71]T. Whitelaw, "The Settlement at Fournou Korifi Myrtos and Aspects of Early Minoan Social Organization," in Minoan Society (eds.) O. Krzyszkowska and L. Nixon (1983) 323-345. Myrtos Fournou Korifi has been further re-investigated by D. Sanders, in Domestic Architecture and the Use of Space (ed.) S. Kent (1990) and by C. Ten Wolde, AJA (1992) 333, although neither of these studies specifically addresses textile production.

[72]P. Warren (1972) 262-263. See also P. Warren, "A Textile Town 4500 Years Ago?" Illustrated London News 252 (February 17, 1968) 25-7.

[73]J. Carington Smith (1975) 260. Comparable spindle whorls are reported in D. Blackman and K. Branigan, "An EM Tholos Tomb at Ayia Kyriaki," BSA 77 (1982) 18.

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through the top and a grooved top edge, were also found in similar quantities to the spindle whorls, about thirty of them. Some of these weights show thread-wear marks on the suspension holes. The presence of both loom weights and spindle whorls in the same area, some in the same room, such as rooms 59 and 81, is similar to the distribution of finds at Neolithic Knossos.[74]

In addition to the loom weights and spindle whorls, several tubs and channels were located in rooms 8, 59, and 81 suggesting textile dye installations at Myrtos. Room 8 was located near a hearth that could have been used to heat certain dyes. Burnt pieces of oak may have either been used as fuel or were possibly the remains of a loom. Residue analysis of the tub in room 59 showed animal lipids. This would be expected if the tub was used for washing and dyeing wool.[75] The site of Myrtos lies close to the area of southeastern Crete where the earliest traces of murex dyeing occurred a few centuries later. Warren reports that very few shells were found in the settlement and discounts the possibility that purple dyeing occurred at Myrtos.[76] One would not expect to find molluscs shells from purple dye production in the settlement but, more likely, they would be down closer to the shore. Perhaps future surveys in the area of Myrtos will reveal some beds of murex shells used for dyeing.

Minoan spinning bowls were also found at Myrtos in areas associated with textile production (fig. 2.2). Warren recognized from wear marks under the internal loop-handle that these vessels were associated with thread making.[77] Spinning bowls have been found

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[74]The faunal remains, not surprisingly, show sheep and goat dominant at Myrtos suggesting that wool, rather than linen, was the main textile being produced. M. Jarman, "The Fauna", in P. Warren (1972) Appendix VI, 318-320.

[75]D. Bowyer, "Chromatography Analyses for Lipids in Clay," Appendix X, in P. Warren (1972) 330-331; E. Barber (1991) 240.

[76]P. Warren (1972) 263. See below, pp. 50-58, for more on purple dyeing.

[77]P. Warren (1972) 209.

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elsewhere on Crete dating to the Early and Middle Minoan periods at Drakones,[78] Phaistos, Palaikastro,[79] Kommos, and Archanes.[80] Examples outside of Crete are known from Messenia, Palestine, and Egypt.[81] The examples from Myrtos are the earliest known spinning bowls. E. Barber has suggested the possibility that the Egyptians adopted the idea of fiber-wetting bowls from the Minoans.[82] She also suggests that some of the textiles produced at Myrtos could have been elaborate pattern textiles for export partly on the basis of 10th Dynasty wall-paintings in Egypt showing the typically Minoan double-heart spiral pattern.[83] The proximity of Myrtos to Egypt, compared to many other Minoan sites, the absence of local wool in Egypt, the growing number of EM II imports on Crete, and the long history of international textile trade, suggests that a Minoan origin for the spinning bowl in Egypt is possible.

Another EM II A-B settlement contemporary with Myrtos is Vasiliki. It consisted of two to four houses and in some of the rooms were found deposits of loom weights. The inhabitants here had access to foreign materials, copper and obsidian, and probably made bronze tools in molds.[84] As at Myrtos, the craft residues suggest that specialization at

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[78]S. Xanthoudides, The Vaulted Tombs of the Messara (1924) 78.

[79]R. Bosanquet and R. Dawkins, The Unpublished Objects from the Palaikastro Excavations 1902-1906 BSA Suppl. 1 (1923) pl. 6c.

[80]A. Λεμπέση, "Ανασκαφικαί έρευναι εις Ανατολική Κρήτην." Πρακτικά (1970) 269, pl. 378d.

[81]Known as διμιτεῖς in modern Greek, see S. Marinatos, "Ανασκαφή εν Πύλω." Πρακτικά (1964) 87-88; J. Carington Smith (1992) 686-7, pl. 11-34 ; E. Barber (1991) 74-76; I. Τζαχίλη (1997) 124-125; T. Dothan, "Spinning Bowls," Israel Exploration Journal 13 (1963) 97-112.

[82]E. Barber (1991) 74-76. Other possible interpretations include independent inventions of the spinning bowl on Crete and Egypt, earlier spinning bowls not yet found in Egypt, or that these bowls were not really used for spinning fibers.

[83]E. Barber (1994) 109; *Eadem* (1997) 516 and plate CXCIII a and b.

[84]L. V. Watrous (1994) 709.

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Vasiliki was aggregated within household units, possibly for exchange within Crete and perhaps even to Egypt.

Spinning and weaving crafts were both located at Myrtos Fournou Korifi, Vasiliki, and Early Minoan Knossos. This point is highly significant in terms of craft specialization. The two activities occurring together do not suggest task specialization in these early settlements. It is of course possible that at Prepalatial sites older and younger people spun while the more skilled and able wove, but probably at the household level. With the rise of the Minoan palaces the archaeological distribution of textile equipment changes; no longer are spinning and weaving tools found together as they were at Prepalatial sites. The textile equipment at these EM II sites reflects the levels of social organization we reconstruct for these settlements based on other archaeological data such as settlement density and ceramic analysis.[85] The sites are not major economic centers but the quantity of finds related to textile manufacture indicate that these people may have been producing a small surplus of cloth for some kind of exchange. As Warren states, 'It also remains unknown whether textiles were produced at Myrtos for home use only, or were exported. We may feel that the latter is possible, in view of the numbers of finds connected with textile manufacture in so small a settlement.'[86] The presence of Melian obsidian, metals that were most likely imported to Crete, and the growing evidence for specialized pottery production at Myrtos, suggest growing complexity, part of which was based on a Minoan textile industry.[87]

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[85]For an analysis of the evidence for ranked EM II society on Crete, all disproven by Watrous, see L. V. Watrous (1994) 712-717.

[86]P. Warren (1972) 263.

[87]See T. Whitelaw, *et al.*, "Ceramic Traditions at EM IIB Myrtos, Fournou Korifi," TEXNH. Aegaeum 16 (1997) 265-274, and P. Day, *et al.* "Reassessing Specialization in Prepalatial Cretan Ceramic Production," TEXNH. Aegaeum 16 (1997) 276-290.

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Part II - Palatial Crete

Models of Cultural Complexity in the Aegean

Agricultural surplus has often been assumed as a pre-requisite for complex society. The major exponent of this processual model in the Aegean is Colin Renfrew and his seminal work, The Emergence of Civilisation (1972). Briefly, Renfrew's redistributive model suggests that by the end of the Neolithic period, Aegean farmers mainly exploited the 'Mediterranean triad', cultivation based on wheat, olives, and the grape. This diversification led to local specialization in agricultural production, based mainly on the geographical predisposition of the region. Lower and flatter land was suited to cereals while more mountainous terrain with thinner soil cover was used for cultivating olives and vines. Successful agriculture of this kind led to both increased productivity and an increase in population. With this success, Renfrew posits that 'redistributive chiefs' emerged at the local level to administer the specialized products from different farmers and to coordinate storage facilities prior to redistribution. Supporting Renfrew, J. Bintliff states that 'the prehistoric cultures of Greece were fundamentally agricultural, pastoral and piscatorial.... Minoan and Mycenaean civilizations were in all essentials constructed upon the redistribution of local foodstuffs rather than commerce and mercenary service.'[88] Under the patronage of these local chiefs, craft specialization led to a positive feedback loop where the production of prestige goods demanded increased agricultural production and at the same time reinforced the interdependence of the specialized farmers on the controlling elites.[89]

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[88]J. Bintliff, Natural Environment and Human Settlement in Prehistoric Greece. BAR Suppl. 28 (1977) 115-116.

[89]C. Renfrew (1972) 340-45, 483-485.

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Renfrew's theory posits an indigenous development of 'civilisation' in the Aegean without any 'diffusion' from the Near East.[90] The international age of exchange between the Aegean, Near East and Egypt takes off around 2000 B.C., at the same that there is clear evidence for palaces on Crete.[91] Architectural and administrative similarities of large centers in the Near East and on Crete suggest that there was some foreign inspiration if not influence upon the earliest Minoans.[92] Renfrew's model also does not fully account for the initial stimulus that changed Neolithic and Early Minoan society into the complex civilization we call Minoan. It assumes basically altruistic motives for the 'redistributive chiefs' that are probably unrealistic.[93] It is also unlikely that Early Bronze Age farmers would willingly take the risk of specializing in one crop with the expectation that trade with their neighbors would supply other essentials. As an adjustment to Renfrew's model, Gamble has suggested that early farmers were forcefully made to live in large nucleated settlements by a powerful elite, removing them from their farmsteads and coercing them into producing specialized crops.[94] Yet this model also does not fully explain how the 'elites' first developed.

Rather than an altruistic or coercive theory, P. Halstead and O'Shea's 'social storage' model takes a mutually beneficial approach to explaining surplus and emergent

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[90]C. Renfrew (1972) 15-44. For a critique of this and other views regarding Near Eastern influence on Greek culture, see S. Morris, "Greece and the Levant," JMA 3.1 (1990) 57-66.

[91]L. V. Watrous (1994) 736; E. Cline, "Tinker, Tailor, Soldier, Sailor: Minoans and Mycenaeans Abroad," Politeia. Aegaeum 14 (1995) 265-283.

[92]J. Weingarten, "The Sealing Structure of Karahöyük and Some Administrative Links with Phaistos on Crete," Oriens Antiquus 29 (1990) 63-95.

[93]See comments by J. Rutter (1997).

[94]See C. Gamble, "Animal Husbandry, Population and Urbanism," in An Island Polity: The Archaeology and Exploitation of Melos (eds.) C. Renfrew and M. Wagstaff (1982) 161-171.

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complexity.[95] In Greece, where climate and topography can differ within fairly short distances, stored surplus crops can be shared with another group suffering a short-term deficit. This is a way of insuring a stable food supply in case the situations are reversed later. In addition, any unused surplus could be fed to animals increasing their secondary products, such as wool, hides, and milk. Evidence in support of this model includes the additional storage areas of MM IB Knossos and Phaistos. Storage magazines and circular pit structures, known as *koulouras*, are added during the MM II period in the area of the West Courts at these palace centers.[96] Administrative records and archives focusing primarily on agricultural goods also support this model of agricultural surplus. This evidence suggests a staple finance system for the earliest palaces on Crete. Perhaps specialized food producers sent their surplus to regional centers where this food was disbursed to non-food producers or those in need. The contemporary evidence for the production of craft goods, including textiles (see discussion below) in the Old Palaces, indicates that both a staple finance and wealth finance system were in operation. An accumulation of wealth is possible in this model if an unused surplus in food accrues over-time. When another group experiences a shortage then they could 'contribute' some kind of prestige good or craft item to the more stable party in exchange for food.[97] More often than not people who had restricted access to good crop lands compensated for it by diverting their attention to craft production, such as textile manufacture.[98] This labor, freed from agriculture, would divise strategies for better goods. This quickly fosters an

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[95]P. Halstead and J. O'Shea, "A Friend in Need is a Friend Indeed: Social Storage and the Orgins of Social Ranking," Ranking, Resource, and Exchange (eds.) C. Renfrew and S. Shennan (1982) 92-99.

[96]See L. V. Watrous (1994) 736-753.

[97]P. Halstead, "Agriculture in the Bronze Age Aegean: Towards a Model of a Palatial Economy," Agriculture in Ancient Greece. Proceedings of the Seventh International Symposium at the Swedish Institute at Athens, 16-17 May 1990 (ed.) B. Wells (1992) 105-116.

[98]J. McCorriston (1997) 525.

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accumulation of wealth and an increase in specialized production for exchange. Ceramic studies such as T. Whitelaw and P. Day in the Early Minoan period are also revealing fairly sophisticated craft technologies during this formative period.[99]

The emphasis, however, on agriculture for the origins of complexity in the 'social storage' model might be misplaced. Recently, others have focused attention on long distance commercial trade rather than the primacy of agriculture in their models for emergent Aegean complexity.[100] By the mid-third millennium, advances in sailing technology, such as the long-boat commonly shown inscribed on Cycladic pottery, and the invention of the sail in the Aegean allowed for long-distance exchange along well-established trade networks.[101] Increases in agricultural productivity are a response to this growing trade. This would in turn lead to a greater emphasis on secondary products like wool and textiles. Relatively contemporary advances in metallurgy increase the exchange of metallic raw materials and finished goods that further enhance the development of trade networks. Exchange centers located along these networks would quickly accumulate wealth in prestige goods that would empower emerging elites. In this case one would think that islanders of the Cyclades would be the first elites since they were both middlemen in the trade networks and had valuable resources such as obsidian, silver, lead and emery.[102] Yet this is not the case.

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[99]See T. Whitelaw, *et al.* (1997) 265-274 and P. Day, *et al.* (1997) 276-290.

[100]C. Runnels and Tj. van Andel, "Trade and the Origins of Agriculture in the Eastern Mediterranean," JMA 1 (1988) 83-109. See also debate of the model in JMA 2 (1989) 139-56, 297-302; A. and S. Sherratt (1991) 351-382. See also the analysis of this model by J. Rutter in Aegean Prehistory Notes Online, Dartmouth College (1997).

[101]See L. Casson, Ships and Seamanship in the Ancient World (1971).

[102]J. Rutter (1997).

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It is on Crete that foreign trade increases the wealth of emerging elites and brings about the first palaces in the Aegean.[103] The appearance of mountain shrines built just at the end of the Prepalatial period on Crete demonstrates the importance of pastoralism.[104] At these sites the main dedications are informal figurines of women and sheep, strongly indicating the importance of a wool-based economy for the first palaces. Long distance trade supported by textile exports and abundantly fertile land each contributed to the rise of social complexity on Crete, taking the form of the Old Palaces at Knossos, Phaistos, Mallia, and Zakros. While Younger and Rehak have recently summarized the evidence that the Old palaces generally functioned as the heads of agricultural states, I think it is the production and exchange of prestige goods that first allowed such centers, and their elites, to develop.[105] The one problem with the role of agriculture in models of early state formation is the great difficulties in transport, storage, distribution, and administration of bulk produce. As Earle and Brumfiel, among others, have demonstrated, the specialization in subsistence goods is largely unrelated to political development.[106]

References to the Aegean traders and goods in international exchange first occur in the early second millennium B.C. Mari tablet ARMT 21.432: 4-12 lists clothes and leather shoes sent from Kaptara, presumably Crete.[107] By the Late Bronze Age, international

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[103]For this model of long-distance trade and emergent complexity in the New world, see W. Rathje, "Praise the Gods and Pass the Metates: A Hypothesis of the development of Lowland Rainforest Civilization in Middle America," Contemporary Archaeology (ed.) M. Leone, (1972) 365-392.

[104] G. Cadogan, "Why was Crete Different?" The End of the Early Bronze Age (ed.) G. Cadogan (1986) 161.

[105]P. Rehak and J. Younger, "Review of Aegean Prehistory VI: The Neopalatial, Final Palatial, and Post-Palatial Periods on Crete," AJA 101 (1998) 97.

[106]See essays and introduction in Specialization, Exchange and Complex Societies (eds.) T. Earle and E. Brumfiel (1987), especially 6-7, 64-85.

[107]B. Knapp "Spices, Drugs, Grain and Grog: Organic Goods in Bronze Age East Mediterranean Trade," in Bronze Age Trade in the Mediterranean, SIMA vol. 90 (ed.) N. Gale (1991) 42. See also, E. Cline (1995) 265-283.

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prestige giving was common. From Amarna letter 34, lines 16-24, we read that the King of Alashiya (possibly Cyprus) sent 100 talents of copper to his 'brother' the Egyptian pharaoh expecting in return a gold decorated bed, a chariot with gold, two horses, 14 (pieces?) of ebony, 17 jars of good oil, 54 items of linen, and, finally, eight pieces of 'the royal linen'.[108] It is highly significant that half of all the elaborate goods listed in this letter represent some kind of textile. Another el-Amarna letter records an official of Byblos complaining to an official in Egypt that there is neither wool nor blue-or red-purple garments to give as tribute, implying that the Egyptians had requested some from the Levantines.[109] Knapp also refers to an unpublished document from Mari that lists a variety of goods from Crete, including 'elaborately-decorated weapons, a *katappu*-container, a large vase, and perhaps some fabric.'[110]

Purple-dyed woolen textiles were probably among the first goods exchanged by Minoans overseas.[111] Written sources tell us that purple-dyed textiles achieved a status and value comparable to, if not exceeding, that of precious metals. During the 12th Dynasty in Egyptian art, we see a new, heavier cloak on seated Egyptian figures, possibly a Minoan import.[112] Good evidence for contact between the Aegean and Egypt exists from 12th Dynasty 'weaver's waste' reported by Sir Flinders Petrie at Kahun. 'A handful of weaver's waste...mainly made up of blue worsted ends, and blue wool, with some red and

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[108]B. Knapp (1991) 21 and 38.

[109]B. Knapp (1991) 38.

[110]B. Knapp (1991) 38. Further in this article Knapp also describes the later references in Hittite and Amarna documents which list the trade in linen and wool, the latter occasionally dyed purple.

[111]Minoan purple is discussed in more detail below, pp. 49f.

[112]Information kindly supplied to me by V. Golovina, visiting Eastern European Mellon scholar at the ASCSA, 1997.

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some green ends' and a 'lump of red dyed wool, not yet spun,' were found.[113] As Barber points out, what is so significant about this find is that at this time there is no reason to believe the Egyptians were dyeing or weaving wool.[114] The workers of the wool were possibly immigrants, perhaps from Crete, or at least they were working with imported materials. The wool itself is probably non-Egyptian since Egyptian representations of sheep during this period show that hairy species rather than the woolly kind were prevalent.

In the first half of the second millennium B.C. there is good evidence for Minoan contact with Egypt, particularly in the Eighteenth dynasty.[115] Five tombs dating to the reigns of Hatshepsut and Thutmose III (1504-1450 B.C.) show Aegean ambassadors bearing gifts and wearing Minoan and Mycenaean style clothing. The three earliest tomb paintings show Minoan figures wearing loincloths, cinchbelts and codpieces in the typical red, white, and blue Minoan color scheme. The fourth tomb was built for the Vizier Rekhmire and shows 16 Aegean figures bearing gifts on its walls. These figures are identified as princes of Keftiu, 'from the isles in the midst of the Great Green (sea)'.[116] At some point in the mid-fifteenth century Rekhmire repainted the costumes of only these figures, changing the Minoan-style loincloths into Mycenaean-style kilts. This period coincides with major destructions in Neopalatial Crete in LM IB and may mark the beginning of Mycenaean participation (domination?) in overseas trade. The final tomb is

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[113]W.M.F. Petrie, F. Griffith, and P. Newberry, Kahun, Gurob, and Hawara (1890) 28 (*non vidi*) cited by E. Barber (1991) 224.

[114]E. Barber (1991) 224, 351.

[115]B. Kemp and R. Merrillees, Minoan Pottery in Second Millennium Egypt (1980) 245.

[116]E. Barber (1991) 330-351 and P. Rehak, "Aegean Breechcloths, Kilts, and the Keftiu Paintings," AJA 100 (1996) 35-51 discuss the significance of Aegean ambassadors and textiles in Egyptian art. See also S. Wachsmann, Aegeans in the Thebans Tombs OLA 20 (1987) and M. Heltzer (1988) 167-169. The generalized term 'Aegeans' seems to be the best identity for these figures. See. A. and S. Sherratt (1991) 368; See S. Morris (1992) 94 for the generalized Aegean identity of the *kptr*.

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that of Menkheperraseneb, son of Rekhmire. The scenes from this tomb most likely copied and rearranged the tribute scene found in Rekhmire's tomb but shows the ambassadors bringing new gifts to the Egyptian court, including Aegean (Mycenaean?) textiles.[117] While the origin of the 'Aegeans' in the Theban tombs is a debated issue, the ambassadors from 'The Isles in the midst of the great Green sea' most likely traded with the Egyptians for resources they lacked, probably precious metals. We might assume that higher status was achieved and greater wealth acquired by the Aegean elites controlling these ventures. In addition to the material exchange between the Aegean and the Near East, we can also assume an exchange of ideas involving craft technologies, rulership and economic administration.[118]

It is also likely that the Minoans were in regular contact with the rulers at Mari and other metal traders of the Near East by the early second millennium, trading agricultural goods, manufactured metal tools, vessels (with oil, wine, perfume?), and dyed textiles.[119] A reference at Mari to a Kaptorite merchant receiving tin in exchange for western made goods at Ugarit is fairly clear.[120]

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[117]E. Barber (1991) 335.

[118]For one recent view of Near Eastern influence on the Minoan palace, see L. V. Watrous (1994) 734-753.

[119]In the first millennium B.C. we have good evidence for the exchange of textiles and metals. Two Neo-Babylonian tablets of the mid-sixth century B.C. from Uruk are discussed in L. Oppenheim's, "Essay on Overland Trade in the First Millennium B.C." Journal of Cuneiform Studies 21 (1967) 236-254. These economic documents record transactions of copper, tin, iron, purple dyes and mordants, and blue-purple linen and wool. They are remarkable for the light they shed on international trade of the period. A merchant receives his payment, or share, listed at the end of each tablet. In both cases the men receive blue-purple textiles, one of linen, the other of wool. While no point of origin is given for these textiles, all of the items are said to be from the West.

[120]See A. and S. Sherratt (1991) 368.

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The Old Palaces

The Minoan palace is defined as a central building complex within a settlement showing specific architectural features and specialized functions. The most prominent feature of all Minoan palaces is an open, central, porticoed court with an orientation within 15 degrees of North. This orientation is probably to allow light into the western cult rooms. In addition, most Minoan palaces have a west court, storage facilities, workshops, ritual areas, evidence for administration and perhaps public reception spaces.[121] As was mentioned before, other features include an overall large plan with many stories, large sawn ashlar blocks, downward-tapering columns, pier and door partitions, light wells, lustral basins, monumental stairways, and major public rooms on the upper floors. These are the highest order sites in the Minoan cultural koine. In addition to the architectural criteria, the political function of the palace is important to consider. Broadly speaking, three functions, not mutually exclusive, are suggested for the Minoan palace: an elite residence, a redistribution center, and a religious complex. Of chief concern to an understanding of the textile industry is the Minoan palace as a redistribution center.

The main function of these centers was probably to facilitate the production of valuable trade goods (wealth finance) and to coordinate the redistribution of agricultural goods (staple finance).[122] The term 'redistribution' is potent and often misunderstood in relation to the Bronze Age palaces of the Aegean. Redistribution is perhaps better understood as the mobilization of raw materials and products. Goods, such as livestock, grain, wine, oil, and some textiles, collected by the state as a share of commoner produce

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[121]For other criteria and definitions, see articles in R. Hägg and N. Marinatos (eds.) The Function of the Minoan Palaces (1987).

[122]According to Rehak and Younger, 'an important role of the palaces was the systematic manufacture of wine, beginning in the Protopalatial period.' P. Rehak and J. Younger (1998) 102.

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in large store houses, are mobilized to support non-food producers who are employed by or dependent upon the kingdom.

The Old Palaces begin early in the second millennium. After thousands of years, the Neolithic and succeeding Early Minoan settlements at Knossos are leveled and replaced by a new building form, the Minoan palace. Large-scale structures are also built over Early Minoan remains at Phaistos, Mallia, and Zakros. Polychrome pottery of MM IA date found in the destruction debris predate these first palace. It is therefore MM I B that marks a fairly sudden and dramatic shift in social organization, *ca.* 1930 B.C., that lasts through MM II B, *ca* 1750 B.C., delimiting the Old Palace Period on Crete. The cycle of destruction and rebuilding continues, however, and several fires destroy the Old Palaces in the MM II B period at sites throughout Crete, including Knossos, Phaistos, Mallia, Monastiraki, Apolodoulou, and Galatas.[123] The cause(s) of these calamities are not completely understood.

Minoan Purple

Explaining the emergence of the first Minoan palaces is difficult given the evidence from Prepalatial and Protopalatial Crete.[124] One factor, however, that is certain to have contributed to this major social, political and economic change is greater exploitation of natural resources on Crete that allowed for the production of marketable commodities desired by foreign partners. By the early second millennium B.C. there is a new development in cloth production technology that may have had a profound impact on the incipient Minoan economy, including the emergence of the Minoan palace. Archaeological evidence from eastern Crete

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[123]P. Rehak and J. Younger (1998) 100; J. Driessen and I. Schoep, "The Architect and the Scribe: Political Implications of Architectural and Administrative Changes on MM II- LM IIIA Crete," Politeia. Aegaeum 14 (1995) 649-664.

[124]See G. Cadogan (1986) 153-173. Much of the Old Palaces is obscured by the New Palaces built over them.

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suggests that the purple-dye industry from sea-snails originated here in the Mediterranean region. The earliest known deposits of murex shells on Crete, in quantities substantial enough to indicate dye-extraction, occur on the small island of Kouphonisi, at Palaikastro, and at Kommos.[125] Pottery found at these sites suggests a date within the early to middle MM period, *ca.* 20th-18th c. B.C.

Dye installations are often identified as wine presses in archaeological literature. Their multiple use is not unlikely, depending perhaps on the agricultural season.[126] Water channels and spouted tubs found at Kato Zakros suggest dyeing of textiles in Hogarth's House I. Platon reports that House J also contained a dye vat and numerous loom weights. Nine rectangular troughs were found coated with plaster in an industrial annex of the palace.[127] I have also wondered whether the 'cistern' in the eastern wing of the palace at Zakros might be related to wool washing or dyeing. The only parallel for this hydraulic feature is found in the 'Caravansarai' at Knossos. Rather than see the Caravansarai as a rest-stop or a place for travellers to wash their feet, I suggest that it is a place for washing textiles, or perhaps dyeing them.[128] The sophisticated drainage system at Knossos might also be related to dyeing and textile production.[129] Evidence from Vathypetro near Knossos also indicates dye installations.

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[125]For the original references to Kouphonisi and Palaikastro see R. Bosanquet, "Some 'Late Minoan' Vases found in Greece," JHS 24 (1904) 317-329 and "Dikte and the Temples of Dictaean Zeus," BSA 40 (1939-40) 60-77. For recent research at Kommos, see D. Reese, 'Muricacea,' in Kommos I (1995) 258-261; D. Ruscillo, "Working Double Tides: The Marine Molluscs from Kommos, Crete," AJA 102 (1998) 392.

[126]P. Warren (1972) 139.

[127]See D. Hogarth (1900-01) 138-141; N. Platon Πρακτικά (1964) 146, pl. 144a; "Kato Zakros" in Ancient Crete (ed.) L. von Matt (1967) 167; Zakros: The Discovery of a Lost Palace of Ancient Crete (1985) 29; J. Carington Smith (1977) 63-64; E. Barber (1991) 241.

[128]See the recent article on the Caravansarai at Knossos, E. Schofield, "Wash and Brush-Up at the 'Travellers' Rest': the Caravanserai (sic) Reconsidered," Minotaur and Centaur. Studies in the Archaeology of Crete and Euboia Presented to Mervyn Popham, BAR International Series 638, (eds.) D. Evely, I. Lemos and S. Sherratt (1996) 27-33.

[129]See C. Macdonald and J. Driessen, "The Drainage System of the Domestic Quarter in the Palace at Knossos," BSA 83 (1988) 235-258.

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A spouted tub and spherical loom weights were found along with burnt wood that might be the remains of a standing loom in Room 13.[130]

In the Levant, the earliest textual reference to trade in purple dye comes from Nuzi, recording the transfer of dye from Syro-Phoenicia to the eastern Tigris, dating to the mid-second millennium B.C.[131] The earliest archaeological evidence for purple is found at Minet el-Beidha, the harbor of ancient Ugarit, tentatively dated to the fifteenth century B.C. based on textual evidence.[132] M. Dothan notes that the earliest evidence for purple workshops in southern Canaan dates to the 13th-12th centuries B.C. at Akko. Dothan suggests that new settlers, the Sherdana, brought purple dyeing to the Levant, beginning the intensified purple-dye production for which the region became so well-known.[133] Perhaps the transfer of technology is attributable to displaced Minoans arriving in the Levant.[134] Wall paintings at Tell El Dab'a, Tel Kabri, and Alalakh demonstrate the transfer of ideas and artistic skills from Crete to the Eastern Mediterranean.[135] Technologies of

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[130]See S. Marinatos, Πρακτικά (1951) 269; J. Carington Smith (1975) 63; J. Driessen and J. Sakellarakis, "The Vathypetro Complex," The Function of the Minoan 'Villa'. Proceedings of the Eighth International Symposium at the Swedish Institute at Athens, 6-8 June, 1992 (ed.) R. Hägg (1997) 75.

[131]M. Reinhold, History of Purple as a Status Symbol in Antiquity, Collection Latomus, vol. 116 (1970) 7-21 9, n. 4. See also E. Speiser, "One Hundred New Selected Nuzi Texts," ASOR 16 (1935-36) 121-122.

[132]D. Reese, "Palaikastro Shells and Bronze Age Purple-Dye Production in the Mediterranean Basin," BSA 82 (1987) 205. The earliest date for purple dye based on scientific analysis of dye residues is found at Sarepta, dating to the thirteenth c. B.C.; P. McGovern and R. Michel, "Royal Purple and the Pre-Phoenician Dye Industry of Lebanon," MASCA Journal 3 (1984) 67-70. S. Dalley infers purple dyeing from murex snails at Mari from the later Ugaritic evidence, although the term for purple-dyed fabrics in the Mari tablets cannot yet be isolated and there is no archaeological evidence for such textiles there; S. Dalley, Mari and Karana: Two Old Babylonian Cities (1984) 54.

[133]M. Dothan, "The Significance of Some Artisans' Workshops along the Canaanite Coast," in Society and Economy in the Eastern Mediterranean (c. 1500-1000 B.C.) (eds.) M. Heltzer and E. Lipinski (1988) 295-303.

[134]A comparable model of technology transfer is found in J. Carney, "From Hands to Tutors: African Expertise in the South Carolina Rice Economy," Agricultural History 67.3 (1993) 1-30.

[135]See W.-D. Niemeier, "Minoan Frescoes in the Eastern Mediterranean," in The Aegean and the Orient in the Second Millennium B.C.: Proceedings of the 50th Anniversary Symposium, Aegaeum 18 (eds.) E.

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textile arts would not be out of place in such transactions. While evidence for earlier purple dyeing might be revealed in the Levant, it is equally possible that there is still earlier Minoan evidence for purple dyeing in the Aegean. The area around Myrtos is one area to be explored as are some of the earliest overseas ventures by the Minoans. The Early Minoan colony on Kythera, known as '*Porphyroussa*', 'the purple-island' to Aristotle, may have been settled to acquire more sources of purple dye along with valuable mineral resources.[136]

From archaeology and ancient literary evidence, it is certain that the shell species, *murex brandaris* (fig. 2.3), *murex trunculus*, and *purpura* (a.k.a. *thais*) *haemastoma* were used for purple production. Pliny discusses the principal luxury dyes from antiquity, including purple from murex.[137] Pliny is clear that the shade of purple itself can vary depending on the species and the technique, ranging from 'black', 'purple' and 'red'. Referring to purple, he states, '*laus ei summa in colore sanguinis concreti, nigricans aspectu idemque suspectu refulgens; unde et Homero purpureus dicitur sanguis.*' (translation: Its highest glory consists in the colour of congealed blood, blackish at first glance when held up to the light; this is the origin of Homer's phrase, 'blood of purple hue'.)[138]

Cline and D. Harris-Cline, (forthcoming, *non vidi*); Abstracts of the conference proceedings online at http://ucaswww.mcm.uc.edu/classics/program.html. See also M. Shaw, "Aegean Sponsors and Artists: Reflections of their Roles in the Patterns of Distribution of Themes and Representational Conventions in the Murals," TEXNH. Aegaeum 16 (1997) 481-503, especially n. 9.

[136]References to trade and colonization on Kythera, the Purple Colony, probably in search of murex dye among other resources, can be found in S. Morris, "Hollow Lakedaimon," Harvard Studies in Classical Philology 88 (1984) 9; Daidalos and the Origins of Greek Art (1992) 134-5; discussion in The Minoan Thalassocracy (1984) 111. The early presence of Minoans on Kythera suggests that purple dyeing might date back to the third millennium. Similarly, other Phoenician colonies may have been founded in part as sources of purple. Perhaps the Tunisian site of D'Jerba, ancient Mennix, cited by Strabo xvii.3.18 and Pliny, Natural History ix.127 as the best source of dye in Africa, was an earlier Phoenician settlement.

[137]Natural History xxii, 2-3.

[138]Natural History ix, 125-142 - Loeb translation.

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We are fortunate that the nature of the purple dye industry is such that it can be located archaeologically. Systematic counts, however, of crushed shells based on a thorough understanding of the particulars of dye extraction have only been a recent development in archaeology. Often in the publications one finds references to 'heaps' or the mere presence of certain shells is noted. Distinguishing between the shells as food debris and residues of dyeing is difficult and important. For a dye industry, the shells need to be living, and therefore any locus of production must be close to the sea, probably within a few kilometers.

A thorough understanding of some traits relevant to purple-producing snails will improve our understanding of the ancient evidence. The process of extracting purple dye centers on the hypobranchial gland of the snail. Pliny explains the dye source best: '..It has the famous flower of purple, sought after for dyeing robes, in the middle of its throat: here there is a white vein of very scanty fluid from which that precious dye, suffused with a dark rose colour, is drained, but the rest of the body produces nothing. People strive to catch this fish alive, because it discharges this juice with its life.'[139]

Purple dyeing was very arduous, smelly and labor intensive lasting over a period of several days. The live sea snails were often trapped or baited in baskets. It is not unlikely that Bronze Age dyers experimented with bait to trap the purple producing snails. Recent experiments with the species known to produce purple have shown that, while muricid snails possess cephalic eyes, their main method of attaining food is from chemoreceptors, common to other gastropods.[140] Living and dead mollusks, of varying age, were used as bait in transparent boxes to observe the feeding habits of these snails. The least selective eater was the *m. trunculus*, or rock mollusc. This might explain why at ancient harbors,

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[139]Pliny Natural History ix. 126.

[140]E. Spanier and N. Karmon, "Muricid Snails and the Ancient Dye Industry," in The Royal Purple and the Biblical Blue: *Argaman* and *Tekhelet* (ed.) E. Spainer (1987) 183.

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such as Caesarea or Akko, where dead fish and other edible refuse are common, the variety of *m. trunculus* is most common. When eating prey within a hard shell, the muricid snails approach the bivalve, snail, or, most commonly in the Mediterranean, the limpet (*patella caerulea*), and then proceed to bore a hole through the shell using two organs, one of which secretes a substance, possibly hydrochloric acid, to soften the surface crystals of the calcium carbonate of the shell so that the second organ, the radula, can penetrate and extract the food.[141]

An alternative to drilling for food is perhaps related to the hypobranchial extract that makes the purple dye. Recent experiments suggest that a certain chemical from the snail makes the muscles of its prey relax and allows the predator to insert its proboscis into the soft parts of the prey.[142] It may be that other muricid snails are unaffected by this chemical. When the snails were kept in an artificial environment with no additional food supplies, the snails began to cannibalize each other using the bore method, rather than producing the chemical relaxant. It is assumed that this chemical would have had no effect on their own species. The boring is significant because examples of murex shells from archaeological excavations have shown these distinctive bore holes. Many scholars in the past have interpreted these holes as the marks left by someone manually piercing the shell to extract the dye. This interpretation seems highly difficult and unlikely given what we know of the eating habits of murex. In their natural environment, along the Israeli shore, no examples of bored *m. trunculus* were found by the researchers from Haifa University. The prevalence of bored shells from antiquity strongly suggests that these snails were actually being kept in artificial installations most likely for their purple-dye producing

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[141]E. Spanier and N. Karmon (1987) 184-5, fig. 12.

[142]For a detailed description of the experiments, see E. Spanier and N. Karmon (1987) 183-186. I present only a summary of their findings.

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abilities. The bored holes result when some snails are denied a natural food source by their captors and begin to cannibalize one another.

Once the snails were collected in the fisheries, they would then be crushed and mixed with salt as a preservative. Pliny recommends one pint of salt for every hundred pounds of flesh.[143] This liquid stew should steep for three days and then gently boil for nine more days. The pot itself will turn purple.[144] The addition of sugar or honey also helps produce a deeper dye by slowing the reactions.[145] To dye a significant amount of wool purple with murex dye required a great many snails. Reports vary on the potential amount of dye one snail will yield.[146] Recent research at the University of Haifa suggests that murex dyes are actually more powerful than previously thought. Whatever the precise yield of dye, the labor required to extract the dye and the large amounts of snails needed added to the high cost of the purple.

In a recent report D. Ruscillo has shown that murex snails were actively exploited in the MM I/II B period at Kommos, most probably to dye purple yarn for textiles early in

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[143]Pliny Natural History ix 133.

[144]What are thought to be dye residues have been analysed from a spouted vat from Sarepta showing that purple dyeing extending back into the Late Bronze Age in the Phoenician/Canaanite homeland. See P. McGovern and R. Michel (1984) 67-70.

[145]Incidentally, the use of sugars might explain why some Classical loom weights, such as those in the Athenian Agora, have the stamp MEL (for meli, 'honey') or GLYK (for sugar) on them. Alternatively, MEL could be an abbreviation of ἡ μελικήρα used by Aristotle to refer to the spawn of the murex. (HA 546.b.19). See also Chapter Three.

[146]According to H. Fouqet and H. -J. Bielig, "Biological Precursors and Genesis of Tyrian Purple," Angewänte Chemie 10 (1971) 816-817, one *m. trunculus* will produce 1.2 mg of dye, and one *m. brandaris* 0.6mg. Early scientific work on purple was conducted by W. Cole in 1684 and R. Reaumur in 1711. E. Schunck, in the 1870's, isolated 7 mg. of pure purple from 400 snails. These descriptions seem to refer to pure dye extract rather than the solution described by Pliny. Bruin reports one shell makes only 0.1 mg of purple dye; F. Bruin, "Royal Purple and the Dye Industry of the Mycenaeans and Phoenicians," in M. Mollat (ed.) Societés et compagnies de commerce en Orient dans l'Ocean Indien (1966) 82. The Jensens say 100 *brandaris* shells yield 6.8-12 mg. of dye; L. Jensen and F. Jensen, The Story of Royal Purple (1965) 17, (*non vidi*), cited by D. Reese, (1980) 83. Research into purple was proposed as a separate discipline in academics, to be known as "Porphyrology" or "Pouprologie." This enthusiastic proposal was made by the Austrian A. Dedekind at the turn of this century. Dedekind erroneously believed that the Egyptians had originated the use of purple as a status symbol and he identified many remains of purple paint, ink and textiles from Egypt that are no longer accepted by scholars. See M. Reinhold (1970) 13, n. 2.

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the second millennium.[147] Ruscillo demonstrates that various types of mollusks can be valuable chronological indicators, including those used in purple dye production which consistently date to the MM I/MM II B periods at Kommos. Roughly contemporaneous evidence for Aegean contact with the Near East and Egypt has long been known.[148]

The best-known purple dye in antiquity came from the Levantine coast, in the areas around Tyre, Sarepta and Byblos. This region was the homeland of the Canaanites in the Bronze Age, and the Phoenicians in the subsequent Iron Age. Both words, 'Canaan' and 'Phoenicia', seem to refer to the same concept, some purple-red color designation associated with the name of this region and the people therein.[149] The significant question is whether the red-purple dye gave the name to the people, or, the reverse, that the dye color was named for the people who excelled at this technology. Each word has been studied and debated by linguists for many years. When the word Canaan was found in cuneiform tablets at el-Amarna and Boğazköy, as *Ki-na-ah-ni*, *Ki-na-ah-na*, *Ki-na-ah-hi*, *Ki-na-a-ah-hi*, Lewy interpreted the word as Hurrian based on its occurrence with and without the suffix *-ni/-na*.[150] The suffixed form occurs in the Amarna letters from Tyre and Byblos as *Kn'n*, corresponding to Hebrew, Egyptian, and Phoenician.[151] At Nuzi, a predominantly Hurrian city, the related term *kinahhu* occurs in Akkadian texts and is taken

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[147]D. Ruscillo (1998) 392.

[148]H. Kantor, "The Aegean and the Orient in the Second Millennium B.C.," AJA 51 (1947) 1-103. See also D. Was, "Byblos in Minoan Texts," Minos 16 (1977) 7-11; H. Klengel, "Near Eastern Trade and the Emergence of Interaction with Crete in the Third Millennium B.C.," SMEA 24 (1980) 7-19; M. Wiener, "Trade and Rule in Palatial Crete," in The Function of the Minoan Palace (1987) 261-265; M. Heltzer, "Cretans in Egypt," Kadmos 27.2 (1988) 167-9;

[149]S. Morris (1992) 98.

[150]J. Lewy, "Influences hurrites sur Israël," RÉS, (1938) 49 (*non vidi*, cited by W. Albright, "The Role of the Canaanites in the History of Civilization," Studies in the History of Civilization (1942) 25, n. 50).

[151]W. Albright (1942) 25.

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to mean 'red purple dye'.[152] This term is interpreted by Albright also as Hurrian, the 'Akkadianized' nominative from '*kinahhi*' meaning specifically 'purple'. Albright associates this term with the root **kna* and takes **ikna* as the Hurrianized form of *uknu* in Akkadian or *igna'u* in Ugaritic. For two reasons Astour disagrees with this identification. The double *-hh-* of the transliterated Hurrian voiceless velar spirant in *kinahhi* corresponds to West Semitic *h*, not the voiced one rendered in Ugaritic by *g(g)*. Also, in the Nuzi texts, *kinahhu* is wool coloring clearly distinguished from *uqnu*, 'blue purple,' and designated a different dye.[153] As in Mycenaean, there are many different kinds of red-blue-purple dye, showing the value and significance of this color. Based on earlier occurrences of *Kn'n* in newly discovered texts from Egypt, Ugarit, and Alalakh, *Kn'n* is probably not Hurrian.

Viewing 'Canaan' as West Semitic, scholars have attempted to explain the root from fairly unsubstantiated derivatives meaning 'merchants' (Mazar)[154] or 'murex' (Albright).[155] Moscati and Astour reject these etymologies. Astour suggests that the best semantic explanation is found in Biblical Hebrew with meanings related to 'setting, bending down, or lowering oneself', in relation to the sun, perhaps. He states that Canaan refers to the West, comparable to the Latin root for Occident, *occidere*. The Akkadian term for Syria, Amurru, refers to the same general concept of the West, and becomes synonymous with Canaan in the Amarna Age and in the Bible.[156] Is it possible that instead of the Levant, the term refers farther 'west' toward the Aegean?

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[152]M. Astour, "The Origin of the Terms 'Canaan,' 'Phoenician,' and 'Purple,'" JNES 24 (1965) 346-50.

[153]M. Astour (1965) 346-50.

[154]M. Mazar, "Canaan and the Canaanites," BASOR 102 (1946) 7-12.

[155]W. Albright (1942) 25.

[156]M. Astour (1965) 348 and n. 22.

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Palatial Administration

The earliest evidence for the administration of the first palace economies on Crete is Minoan seals and sealings.[157] The records suggest that textile production was one of the first craft activities administered by the Minoan palaces. Exactly when these seals take on an administrative or organizational function is a debated point. Pini maintains that Minoan seals always had a sphragistic, organizational function,[158] while Weingarten believes Minoan seals did not take on an administrative purpose before the palaces, not until MM II.[159]

Seal impressions on loom weights and spindle whorls found at Old Palaces suggest some degree of administration and organization of the textile industry.[160] Examples

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[157]For two volumes devoted to sealing systems and the ancient economy, see T. Palaima (ed.) Aegean Seals, Sealings and Administration, Aegaeum 5 (1990) and P. Ferioli, E. Fiandra, G. Fissore and M. Frangipane (eds.) Archives Before Writing, Proceedings of the International Colloquium, Oriolo Romano, 23-25 October 1991, (1994), and review by J. Younger, "Seals and Sealing Practices: The Ancient Near East and the Bronze Age Aegean," AJA 100 (1996) 161-165. Also highly informative is the "Session on Administration and Bureaucracy," in the Function of the Minoan 'Villa' (1997) 215-217.

[158]I. Pini, "The Hieroglyphic Deposit and the Temple Repositories at Knossos," Aegean Seals, Sealings, and Administration, Aegaeum 5 (ed.) T. Palaima (1990) 33-60. See also K. Branigan, "The Earliest Minoan Scripts - the Prepalatial Background," Kadmos 8 (1969).

[159]J. Weingarten, "Three Upheavals in Minoan Sealing Administration: Evidence for Radical Change," Aegean Seals, Sealings and Administration, Aegaeum 5 (ed.) T. Palaima (1990) 105-120. See also M. Vlasaki and E. Hallager, "Evidence for Seal Use in Pre-Palatial Western Crete," CMS Beiheft 5 (1995) 251-270.

[160]See Poursat's response to Pini in Aegean Seals, Sealings and Administration, Aegaeum 5 (ed.) T. Palaima (1990) 55 and M. Vlasaki and E. Hallager (1995) 253-254. See also J. Aruz, "Seal Imagery and Sealing Practices in the Early Aegean World," in P. Ferioli, E. Fiandra, G. Fissore and M. Frangipane (eds.) Archives Before Writing (1994) 211-235, especially 215-218. In addition to seal-impressed loom weights, two nodules found in the "Room of the Olive Press" at Knossos dating to the early MM period show impressions of cloth on the rear. The date for these seals is disputed between Weingarten and Pini. The proximity of this room to the Loom Weight Basement may be significant for locating early palatial textile activity at Knossos. See M. Panagiotaki, "Sealings from the Olive Press Room, Knossos: New Information from the Unpublished Notes of Sir Arthur Evans," BSA 88 (1993) 29-48.

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include: stamped disc-weights from Palaikastro,[161] another of MM II date at Monastiraki,[162] stamped cuboid weights from Mallia[163] and Khamezi,[164] and a stamped spindle whorl from Khamaleuri.[165]

Disc weights with seal impressions or graffiti continue at Amnisos,[166] Zakro[167] and elsewhere on Late Minoan Crete. Seventeen stamped pyramidal weights were found in LM I B destruction at Palaikastro, yet the shape does not fit any of the well-known loom weight types.[168] Further investigation of these stamped weights is warranted. One wonders if the same stamp design recurs on other loom weights, in other media, or at other sites. Do these stamps signify owner, product, or weight?

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[161]The stamped disc loom weight was found at EM II Palaikastro, House D 32, HM 4814; E. Eccles BSA 40 (1939) 49 no. 41, fig. 27. See also P. Warren, "The Primary Dating Evidence for EM seals," Kadmos 9 (1970) 31. Warren dates the seal and loom weight to the EM II period, *contra* R. Dawkins, "Excavations at Palaikastro II," BSA 10 (1903/4) 202.

[162]The loom weight from Monastiraki was inscribed in Linear A. Nine hundred clay sealings from this provincial Protopalatial site also show early organizational features; L. V. Watrous (1994) 742-4.

[163]The cuboid weight at Mallia was found in a questionable EM/MM I context. J.-Cl. Poursat, "Sceaux et Imprints de Sceaux," in Fouilles Exécutées à Mallia. Le Quartier Mu II. ÉtCrét XXVI (eds.) B. Detournay, J.-Cl. Poursat and F. Vandenabeele (1980) 204-205, no. 295, fig. 275.

[164]For the MM I A cuboid loom weight from Khamezi, see Eccles (1939) 48. Loom weights are also reported from the ten rooms at this site. See also, C. Davaras, "The Oval House at Chamaizi Reconsidered," AAA 5 (1972) 283-288. I am grateful to Mrs. Ione M. Shear for discussing her thoughts on stamped loom weights with me.

[165]Finally, for the stamped spindle whorl from Khamalevri, also of MM I A date, see M. Vlasaki and E. Hallager (1995) 253, 259.

[166]M. Guarducci Inscriptiones Creticae. Opera et Consilio Friderici Halbherr Collectae I (1935) 2 no. 3.

[167]D.G. Hogarth, "Excavations at Zakro, Crete," BSA 7 (1900-01) 127 fig. 38. These are the 'pierced clay objects from pits'.

[168]J. Weingarten, "Some Stamped Weights from Eastern Crete," Abstracts of the 1996 Cretological Conference (1996) 156.

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A new source of evidence informing us about textile production may be found in Minoan prism seal stones.[169] These seals, most of which come from MM I B - MM II contexts, *i.e.* Old Palace period, show what I believe are three to five loom weights attached to bars at the bottom of a warp-weighted loom (see fig. 2.4 a-d and Table 1). They have the potential for indicating the administration of textile production in the earliest palaces. In total, there are over twenty-five different seal stones in the CMS volumes with this motif. Many examples are chance finds, or were purchased by early collectors and their original provenance is unknown. Seven of the one hundred seals found in the Seal Engraver's Workshop of Quartier Mu, Mallia, show these suspended weights.[170] It is possible that these seals were used in the administration of a textile industry. Other seals from this area show potters and other crafts-people.[171] These seals have been cited as evidence for the early administration, or organization, of craft activities at Mallia, so textile production would not be an unlikely choice for a seal image.

The round objects on the seals suspended from poles are very similar to the disc shaped weights known from many excavations on Crete and in the Aegean (fig. 2.4 a). Wear marks on discoid loom weights from Akrotiri and elsewhere show that the one and two holed weights were attached by straps to something, rather than having the warp

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[169]For a discussion of the three-sided prism seal stone, see V. Kenna, Cretan Seals (1960) 20-23; and P. Yule, Early Cretan Seals: A Study of Chronology. Marburger Studien zur Vor- und Frühgeschichte 4 (1980) 66-67. Kenna does not believe three sided prisms were administrative seals but that they were originally used as amulets.

[170]A. Xenaki-Sakellariou, Les Cachets Minoens de La Collection Giamalakis, ÉtCrétoises, 10 (1958) Pl. xvii 61b, 62b, 63d, Pl. xviii 78a, 83c, 84c, Pl. xix 98b; J.-Cl. Poursat (1980) 157-234; J.-Cl. Poursat, "Hieroglyphic Documents and Sealings from Mallia, Quartier Mu: a Functional Analysis," Aegaeum 5 (1990) 25-33 (with discussion); P. Yule, (1980) 212-213; J. Younger, "Aegean Seals and other Minoan-Mycenaean Art Forms," CMS Beiheft 5 (1995) 336-339 and J. Younger, "Review of P. Yule, *Early Cretan Seals*" Göttingische Gelehrte Anzeigen 240 3/4 (1988) 201-203.

[171]See, for example, V. Kenna (1960) 93, no. 39 B which shows, "A potter with two-handled jar. He is dressed in a skin or cloak...The rayed device behind the jar has been identified by A. Evans, PM i (1921) 124, as the oven."

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threads directly attached to the weights.[172] The grooves found on the upper edges of the loom weights were most likely from poles to which the weights were attached, just as we see depicted on the seals (*cf.* fig. 2.5 b and c-d).[173] This would give equal tension to all the warp threads and could account for the variability in their measured weights, reducing the need for large numbers. Many of the seal images show tassels or fringes running upward, perpendicular to the bar and the weights. These may be the warp threads of the loom.

The other sides of these three-sided seals, when cut, often depict some combination of a human figure (or two), occasionally with a long tool that may be a sword beater used for beating the weft threads up the warp.[174] Also carved on the other sides of the prism seals are horned animals (perhaps indicating wool?), insects, bucrania, birds, and spirals. The spider motif may even indicate an early association of the spider and weaving known in the Arachne myth.[175] Any further interpretation of these representations would be mere speculation.

Evans first suggested that the images on these seals showed vessels attached to poles for transport.[176] Publications of similar seals have since followed Evans' identification. Yule, for example, classifies these seals along with other vessel motifs on

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[172]See fig. 2.4b, adapted from I. Tzachili (1991) 382, fig. 3.

[173]This rod could function as both a shed and a spacing bar as illustrated in M. Hoffmann (1964) fig. 2, d and k.

[174]For an illustration of the sword beater used with looms, see M. Hoffmann (1964) 279-283, figs. 5, 14, 11.2-6. One example of a sword beater from modern Greece is on display at the ethnographic museum in Nauplion.

[175]This idea was suggested to me by N. Marinatos.

[176]A. Evans, Scripta Minoa. The Written Documents of Minoan Crete with Special Reference to the Archives of Knossos. Vol. I. The Hieroglyphic and Primitive Linear Classes (1909) 131-32 figs. 69a, 70b, 71a.

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seals, as do Younger and the CMS descriptions.[177] There are other prism seals that show vessels with lips, handles and rims clearly delineated by the glyptic artist. The loom weight seals seem to be showing something completely different from suspended globular vessels. I believe that they are the one or two-holed Minoan disc loom weight attached to the bottom of a warp-weighted loom, indicating weaving.

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[177]P. Yule (1980) 166-167. Younger calls them, 'Vertical Supports with Globular Attachments'. See J. Younger (1995) 336 and table 1 for CMS references. As I understand it, the editors of the CMS volumes follow the excavators' descriptions of the seals and are not accountable for the interpretations.

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Table 1 - Minoan Seal Stones

	CMS	Inv. No.	Proven.	Description
1	II.2 107	Iraklion 1777	Mallia Workshop a	a) 5 lw; yellow-green
2	II.2 124	Iraklion 1795	Mallia Workshop a	a) 4+ l.w. (frag.), b) 2 spiders, c) 2 handled jug; br-yellow
3	II.2 125	Iraklion 1797	Mallia Workshop a	a) crouching animal, b) 3 l.w., c) 3 goat heads; green w black veins
4	II.2 151	Iraklion 1823	Mallia Workshop d	a) 2 water birds, b) 4 l.w., c) four legged animal; green
5	II.2 214	Iraklion 333	Artsa - chance find	a) standing fig. w/ 3 l.w., b) whirl c) horned animal and birds
6	II.2 224	Iraklion 80	Gonies - chance find	a) standing fig. w/ 4 l.w., b) framed spider, c)horned animal
7	II.2 225	Iraklion 1458	Krassi - chance find	a) 4 l.w. on frame, b) insect, c) 2 standing figs., deep olive green
8	II.2 297	Iraklion 1475	unprovenanced	a) line and puncture design, b) 3+ l.w. (frag.), c) ? ; olive green
9	II.2 302	Iraklion 114	unprovenanced	a) fig. with tool(s), b) fig. with 3 l.w., c) 2 water birds; light olive
10	II.2 306	Iraklion 1298	unprovenanced	a) two copulating animals, b) animal c) fig. w/ 5 l.w. & tool
11	IV 10	Iraklion 1255	Mallia	a) fig. w/ 2 vases, b) goat, c) 4 l.w. and 2 jugs; brown
12	IV 11	Iraklion 184	Mallia	a) fig. w/ 3 l.w., b) scorpion, c) bird and fish; grey and brown
13	VII 17	BM1934.11-20.3	unprovenanced	a) 2 arrows, 2 branches, b) fig. w/ 5 l.w., c) dog; black
14	VII 206	Cambridge, F. 78.1901	Sitanos	a) Cretan dog framed, b) 6 l.w., c) 2 two-handled jars and sheep head;
15	VII 212	Cambridge, F. 73.1901	Anavlochos	a) bucranium, b) 3 l.w., c) toothed quadrilateral; green grey
16	VII 216	Cambridge, F. 72.1901	Epano Zakro	a) fish, b) Cretan dog, c) 3 l.w. (attached); grey and black
17	VIII 13	SLG. Dawkins Paris, CdM. N	unprovenanced	a) 3 l.w. on toothed rod, b) bucranium, c) ant; brown, cream
18	IX 13	3437 Zürich,	unprovenanced	a) goat, b) 5 l.w. w/ tool, c) spirals; yellow
19	X 272	Bollmann Berlin St. M.	unprovenanced	a) bucranium, b) 4 l.w. w/ line above; yellow buff
20	XI 7	FG 62 Wien Mus. IX	unprovenanced	a) fig. with 4 l.w., b) three fish, c) 2 figs., 2 insects; yellow-green
21	XI 298	1979 NY, Met. M.	unprovenanced	a) 2 figs. w/ 9 l.w., b) 2 animals (?), c) 4 figs. in line; light brown
22	XII 29	46.111.3 NY, Met. M.	unprovenanced	a) 2 joined birds, b) 5 bird heads, c) 3 l.w. with heads (?)
23	XII 43	26.31.90 NY, Met. M.	unprovenanced	a) fish (or tool?) and 3 l.w. below, b) 2 joined Hs, c) fig.
24	XII 47	26.31.139 NY, Met. M.	unprovenanced	a) 4 l.w. with tools, b) hound, c) S-spiral with flower bud
25	XII 48	26.31.140 NY, Met. M.	unprovenanced	a) 4 goat-heads, b) 5 l.w. w/ fish (or tool?), c) hound; dark
26	XII 64	26.31.134	unprovenanced	a) bucranium w/ leaves (frames?), b) hound, c) 4 l.w. w/ toothed tool

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Linear A

Along with the first palace on Crete the Linear A script developed. Among other uses, Linear A recorded economic transactions throughout the Old and New Palace periods on Crete and in othe parts of the Aegean .[178] Examples of the Minoan script used for administrative purposes are found at eleven different palaces or villa sites on the island, from Khania in the west to Zakros in the East. Inscriptions on non-administrative objects also occur at seventeen sites on the island and at three places outside of Crete. Unfortunately, scant mention is made of textiles in Linear A.

Killen tentatively suggests that the differences in the organization of Mycenaean textile production at Knossos compared to Pylos may be attributable to a well-organized Minoan system already in place before the Mycenaean administration began.[179] Killen himself acknowledges that this hypothesis of an earlier Minoan textile industry is tenuous because of the insufficient textual evidence from Minoan Crete. While acknowledging that Minoan and Mycenaean Greek are completely different languages, at least three Linear A records used to document textile administration in the Minoan period foreshadow tendencies of the later Mycenaean record-keeping system (fig. 2.5 a-c).[180]

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[178]For a good review of Linear A on Crete, see T. Palaima,"Palatial Control of Economic Activity in Minoan and Mycenaean Crete," in Function of the Minoan Palaces (1987) 301-305 and J. Bennet, "Knossos in Context: Perspectives on the Linear B Administration of LM II-III Crete," AJA 94 (1990) 193-211.

[179]J. Killen, "The Textile Industries at Pylos and Knossos," in Pylos Comes Alive (eds.) C. Shelmerdine and T. Palaima (1984) 60-61. "One possibility is...that the greater decentralization at Knossos reflects the fact that the incoming mainland Greeks found centrally-controlled textile industries already established at a number of the Minoan centers, and decided to keep these in being, merely subordinating them to overall central direction". See also T. Palaima (1990) 83-99 for a discussion of the origins of administrative systems on Crete in the Bronze Age. A discussion of Mycenaean textile production is presented in Chapter Three.

[180]As R. Palmer comments in The Function of the 'Minoan Villa' (1997) 217, the Linear A administration of textile production among other crafts seems to be more regionalized compared to Linear B. This is the same difference noted by Killen for Pylos and Knossos.

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The first document is a sealing from Hagia Triada with the Linear A sign identified as LOOM (fig. 2.5 a).[181] This ideogram is a schematic representation of a warp-weighted loom with suspended weights attached to warp threads. The representation probably derives from earlier Minoan seal stones also showing the bottom of a warp-weighted loom (*cf.* fig. 2.4 a-d). Large quantities of loom weights were found at Hagia Triada, in the 'servants' quarters' of Villa A, along with 45 clay sealings (in room 27), indicating that textiles were being produced and stored in Villa A.[182] This sealing is perhaps one record of the administration of such craft activity.

Next, a recently published sherd from Akrotiri contains interesting signs possibly related to textile administration (fig. 2.5 b).[183] The incised sherd is local Theran pottery, slightly curved, and reused as an inscribed document. The very sketchy inscription was probably made sometime after the pot was no longer complete, a true ostrakon. The horizontal stroke indicates units for tens and is a late characteristic in Linear A that also occurs in Linear B suggesting a date of LM I A for the sherd. Michalaidou suggests a Linear B reading for these Minoan signs and their units: 40 units of MA, 5 PU, 7 TE, 9 ZO, and 4 TA.[184] The MA, for example, could be an abbreviated form of MA+RE, the

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[181]The earliest interpretation of the Linear A sign representing a loom is found in A. Evans (1921) 253. For the ideogram, see W. Brice, (ed.) Inscriptions in the Minoan Linear Script of Class A (1961) Table I: L70, 75, 75' and Pl. xiv, IV.7, and E. Barber (1991) 92, fig. 3.12. For administrative issues, see J. Weingarten (1987) 1-43, especially 25-9 and T. Palaima (1994) 307-330, especially 317-318.

[182]L. Banti, F. Halbherr and E. Stefani (1980) 37-61; L.V.Watrous, "Ayia Triada: A New Perspective on the Minoan Villa," AJA 88 (1984) 123-134. For a variety of views on the Minoan 'villa', see essays in R. Hägg (ed.) The Function of the Minoan 'Villa' (1997). For Hagia Triada, see especially the articles by V. La Rosa, R. Koehl, and P. Rehak.

[183]Fig. 2.5 b, adapted from A. Michailidou, "'Ostrakon' with Linear A Script from Akrotiri (Thera). A Non-Bureaucratic Activity?" Minos 27-28 (1992-93) 7-24.

[184]A Linear B reading of these signs seems legitimate since only 16 of the 89 Linear B syllabograms have no Linear A predecessor; C. Shelmerdine (1997) 559. See also P. Rehak and J. Younger (1998) 133-134, 159.

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Linear A monogram sign Lc 46 meaning 'wool'.[185] Evans first connected this ideogram with wool when he suggested that MA depicted a pair of shears attached to the RE syllabogram.[186] More than likely, the word is related to Greek *mallos*, a somewhat rare word without an accepted etymology, meaning wool.[187] The other signs possibly indicate commodities such as the *te-pa*, *pu-ka-ta-ri-ja*, and *zo-ta* that also occur in Linear B.[188]

Finally, another very recent discovery in Linear A is of particular interest for issues related to the presence of Minoans overseas (fig. 2.5 c).[189] A Minoan graffito found at the Middle Bronze Age site of Tel Haror, in the western Negev, Israel, shows Linear A signs inscribed before firing.[190] Here also we find a Linear A sign taken to mean TELA + TE, a familiar type of Mycenaean cloth. The other two ideograms probably indicate figs and a bull's head rhyton.[191] Further research into the use of Linear A will better inform us about

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[185]A. Evans (1952) 68. See also M. Ventris and J. Chadwick, Documents in Mycenaean Greek (1973) 52, 314; J. Hooker, Linear B: An Introduction (1980) 39-40. This subject is addressed more fully in Chapter three. Other examples of such monograms in Linear B are ME+RI, *meli*, for honey, or TU+RO_2, *turoi*, for cheeses. These words are still used in modern Greek today for the same agricultural products.

[186]One problem with this interpretation is the lack of shears in the Bronze Age. Most wool was probably plucked or combed from sheep by the Mycenaeans. The flexibility of iron allowed for shears to be developed in the subsequent Iron Age. R. Forbes (1964) 7; E. Barber (1991) 20-30. Plucking consequently results in lower yields of wool, the need for a special plucking tool, and a different quality of wool. See J. Melena, "On the Linear B Ideogrammatic Syllabogram ZE," in Studies in Mycenaean and Classical Greek Presented to John Chadwick. Minos 20-22 (1987) 403-404. See Chapter Three.

[187]E. Barber (1991) 260 cites Hesychius' reference to a variant of the word, *mallukes*, meaning *trikhes*, 'hairs'.

[188]See Chapter Three.

[189]Fig. 2.5 c, adapted from, J.-P. Olivier, "The Graffito" in "Minoan Graffito from Tel Haror (Negev, Israel)," E. Oren, Cretan Studies 5 (1996) 99-109, 91-118.

[190]This indicates that the sherd is not a reused ostrakon like the Theran example but purposefully inscribed.

[191]The bull-head sign occurs in Linear B and probably represents a bull-head rhyton, from its Linear B contexts and from representations on Theban tombs of the 18th dynasty. The significance of this Minoan find abroad was addressed by M. Finkelberg and others at the conference, "The Aegean and the Orient in the Second Millennium B.C." held at the University of Cincinnati, 18-20 April, 1997. See M. Finkelberg, "Bronze Age Writing: Contacts between East and West," The Aegean and the Orient in the Second Millennium B.C.: Proceedings of the 50th Anniversary Symposium. Aegaeum 18 (eds.) E. Cline and D.

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the early administration of craft activities like textile production. Applying our knowledge of Mycenaean administrative features may be helpful in clarifying the earlier Minoan material. It should be stated, however, that the distribution of Linear A finds on Crete and in the Aegean show that it reflects a different administrative system than Mycenaean. It is rare for Linear A to refer to large quantities, especially luxury or military goods. There is also a lack of standardization in the Linear A script and few sign groups that suggest toponyms, unlike Linear B.[192]

Middle Minoan Knossos

One of the largest single deposits of Minoan discoid loom weights was found by Sir Arthur Evans at Knossos in 1902, below the East Wing of the palace.[193] The stratigraphic evidence, however, for the date of this deposit is problematic. Evans dated it to MM II (Old Palace period) but others are now suggesting an MM III date instead (New Palace).[194] The weights have one or two holes in their upper halves with grooves along the tops. Evans' publication of the weights in reference to the Loom Weight Basement, which was named for this equipment, amounts to the following:

> That this chamber belonged to the women's apartments may be inferred by the large stores of loom-weights that it had contained. Over four hundred of these had been precipitated below from its floor-level and were found above the relics belonging to the basement proper. These are pear-shaped in outline and flatter than the oval Late

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Harris-Cline, forthcoming. An abstract of the paper is found on-line at http://ucaswww.mcm.uc.edu/classics/program.html.

[192]J. Bennet (1990) 193-200.

[193]A. Evans, "The Palace of Knossos," BSA 8 (1901-02) 24; Palace of Minos vol. I (1921) 253; J. Carington Smith (1975) 279-281.

[194]P. Rehak and J. Younger (1998) 144, refer to architectural 'models from the Loom Weight Basement (traditionally MM II, but probably MM III).' Unfortunately no citation for this revision in date is given. On this area of the palace, see, however, F. Hentschel, The Basis for the Standard Chronology of the Late Bronze Age at Knossos, Ph. D. Dissertation, Yale University (1982) 192-195.

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Minoan type. It may be mentioned that looms with pendant weights supply a recurring sign of the Linear Script A.[195]

In May 1997, I was permitted to weigh, draw and catalogue the same 45 loom weights from the Loom Weight Basement also analyzed by Carington Smith.[196] Most of the loom weights were complete and for those that were not, the percentage preserved was calculated in estimating the complete weight. The 45 weights, representing perhaps 10% of the total found by Evans, were striking in their uniformity, especially in comparison to other deposits of weights elsewhere in Crete. All were flat, fired, discoid weights with a single suspension hole. They measured between 9 and 10 cm. in height and about 7.5 to 8.5 cm. in width. The pronounced, flattened top ranged from 2.5 to 4.5 cm in length. Three of the weights showed incisions of a faint but deliberate cross or X. Oddly this same cross mark appears on the other two types of Minoan loom weight discussed below.

The weights ranged from 127 to 205 grams (table 2). Following the method used in weighing the Phrygian textile tools, these loom weights were grouped by ten gram units. 24% were between 127 and 140 grams, 35% of them weighed between 141-150 grams, 20% between 151-160 grams. The remaining 26% weighed 161 to 205 grams. The total deposit of 400-plus loom weights within the Old Palace at Knossos indicates 10-15 working looms if we assume each loom used about 20-30 weights for plain weave.[197] The uniformity of the weights with single suspension holes, similar fabric and shape, and the concentrated numbers of them (400), suggests that these weights were part of a regulated textile industry administered by the Old Palace at Knossos. The cross mark incised on

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[195]A. Evans (1921) 253. A deposit of similar disc weights, numbering 450, was found in the West House at Akrotiri, dating to the eruption of Thera, probably around 1628 BC. See I. Tzachili (1991) 380-389.

[196]Carington Smith determined at least three different sets based on weight and size; J. Carington Smith (1975) 281. I thank the British School at Athens and the Keeper of the Stratigraphical Museum for allowing me to study these and other weights from British excavations on Crete.

[197]E. Barber (1991) 104. If the weave was patterned, twill, or for an extra-wide textile, more loom weights would be used on one loom.

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some of the weights may also have been used to indicate sets of weights. This deposit is good evidence for retainer workshops with full-time weavers working for the Minoan state, centered at the Palace of Knossos.

Table 2. Knossos Loom Weight Basement - disc type:

Size	Count	%	Range
small	10	24%	127-140 grams
medium	16	35%	141-150 grams
large	9	20%	151-158 grams
very large	9	20%	167-205 grams

The New Palace Period and Spherical Loom Weights

The Old Palaces are destroyed sometime in the MM II period. The succeeding phase of rebuilding marks the New Palace period on Crete, lasting from MM III - LM I B (*ca.* 1750 B.C. - 1500 B.C.). There does not seem to be significant change in material culture from the Old to New Palace period. Towards the end of the Neopalatial period, palaces and other sites show evidence for a decline in trade and agricultural productivity, perhaps partly attributable to the eruption of Akrotiri and a series of earthquakes in LM I A.[198] Late in the LM I B period major destruction levels at Chania, Phaistos, Hagia Triada, Mochlos, Gournia, Petras, Zakros, and Tylissos mark the end of the Neopalatial period.[199] These destructions may or may not be contemporary but they do seem deliberate

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[198]Significant deposits of LM IA pottery do not continue into LM IB at Knossos and Phaestos. New excavations at Galatas, Petras, and Kommos show evidence for destruction and abandonment of buildings during LM IA; P. Rehak and J. Younger (1998) 101.

[199]I was able to view the collection of textile equipment from H. Boyd Hawes' excavations at Gournia, housed in the University Museum, University of Pennsylvania, in a preliminary fashion. There are nine single hole disc weights, eight double holes, two cuboid weights with four holes, 19 slightly flattened,

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and more severe than the natural destructions which mark the Old and New Palace periods. The destructions probably occurred early in the 15th century B.C., just before the reign of Tuthmosis III (1479-1425 B.C.). According to Rehak and Younger, 'These destructions appear selective and carefully planned to destroy as many administrative centers as possible. Settlements, and specific buildings that housed Linear A administrations, are particularly badly hit.'[200] Interestingly, Knossos is left relatively intact, as are a few other key sites, like Kommos.

Spherical or melon-shaped Minoan loom weights seem to be exclusive to the island of Crete and occur most often in the Neopalatial period.[201] None have been found at Akrotiri, Kythera, Kea or Rhodes, areas with a great deal of Minoan contact. Most of the Cretan examples seem to be concentrated in the north and eastern parts of the island, especially around the area of Knossos. The weights seem to be confined to the Neopalatial period. Examples are found at Mallia,[202] Vathypetro,[203] Archanes - Phourni Building

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single holed globular clay objects. Most interesting were the four well-formed rectangular weights with a very consistent, u-shaped groove directly above the single suspension hole. These may not date to the Minoan period. I thank D. Romano, Keeper of the Classical section, for his help in locating this material.

[200]P. Rehak and J. Younger (1998) 148.

[201]M. Popham *et al.* Minoan Unexplored Mansion at Knossos. BSA Suppl. 17 (1984) 249 states that these weights appear from MM III-LM III. He also believes that they 'are the direct descendants of the plain spherical weights known to exist since EM times'.

[202]J. Deshayes and A. Dessene Mallia: Exploration des maisons et quartiers d'habitation (1948-1954). ÉtCrét XI (1959) pl. xxii 2.3; O. Pelon, Mallia. Exploration des maisons et quartiers d'habitation (1963-1966). ÉtCrét XVI (1970) Maison III, Quartier E has a seal impressed spherical weight, pl. xxvi.3.

[203]S. Marinatos Πρακτικά (1952) 594, fig. 7. See the recent analysis of this 'villa' with religious associations by J. Driessen and J. Sakellarakis (1997) 63-77.

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4,[204] the North Building of the Stratigraphical Museum excavations at Knossos,[205] the Houses by the Acropolis at Knossos,[206]and the Unexplored Mansion.[207] The context of these finds, only on Crete, within special workshop facilities, suggests that, as with the Loom Weight Basement deposit, these weights are part of retainer workshops of full-time weavers employed by the ruling elites at Knossos.

These weights are fairly large, approximately the size of an orange (rather than a melon).[208] They often have grooves running parallel to the suspension hole across the entire surface of the weight. The number of grooves varies but most often they quarter the sphere. These can be fairly broad grooves, 0.5 cm in width and depth, or they can be very faint and worn, perhaps indicating thread wear if the weights were suspended with the grooves. Both the grooves and the interior of the central suspension hole were sometimes painted with the same dark red-brown iron-based paint used on fine wares, perhaps in order to prevent snagging when the weights were threaded.[209] In addition, while these weights appear fairly spherical, close analysis shows that some have slightly flattened sides or 'resting surfaces', as if they were placed on a flat surface while the clay was still soft. The suspension hole is often at a 45 degree angle to this flat surface, perhaps indicating that

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[204]J. Sakellarakis and E. Sapouna-Sakellaraki (1991) 87, fig. 62; J. Sakellarakis, 'Ανασκαφή Αρχανών 1974,' Πρακτικά (1974) 209, pl. 153; and Ε-Ε. Δεληγιάννη. Στοιχεία Βιοτεχνικών εγκαταστασεών απο το Κτήριο 4 του Νεκροτάφειου στο Φουρνί Αρχανών.' 187-196. I have been granted permission to study these weights by the excavator, Professor J. Sakellarakis, to whom I am very grateful. Unfortunately the official permission did not arrive in time for inclusion here. There are 46 loom weights reported from Rooms 1 and 2 of the East Wing of this Building 4.

[205]P. Warren, "Knossos: Stratigraphical Museum Excavations, 1978-1980. Part 1," Archaeological Reports (1980-81) 73-93, especially 85, fig. 42-3.

[206]H. Catling, et al. "Knossos 1975: Middle Minoan III and Late Minoan I Houses by the Acropolis," BSA 74 (1979) 1-78, especially 61, fig. 43-44.

[207]M. Popham, *et al.* Minoan Unexplored Mansion at Knossos. BSA Suppl. 17 (1984) 247-249.

[208]M. Popham (1984) 247 compares them to pomegranates.

[209]M. Popham (1984) 249.

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the malleable clay ball was resting when the suspension hole was pierced through with a stick or finger at a 45 degree angle.[210] Wear marks found at both ends of the suspension hole indicate that the weights were suspended with the hole on the horizontal plane, parallel to the ground.[211]

Minoan Unexplored Mansion

Spherical loom weights from the Unexplored Mansion are probably the best-published of the known examples (table 3). The MUM (Minoan Unexplored Mansion) was initially constructed in LM IA after completion of the Little Palace immediately to the east (see plan in fig. 2.6). It was occupied and damaged in LM II and then reused until LM III B2.[212] The context for the loom weights seems to be the end of the Neopalatial period, probably during LM I B.[213] According to Hitchcock and Preziosi's recent analysis, the Unexplored Mansion should be identified as the 'annex' to the 'villa' that is the Little Palace.[214] This villa-annex building system is also found at Tylissos (Villa A and 'Annex'

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[210]I noticed this pattern when I photographed some of the spherical loom weights. All the weights lined up easily on their resting surface without rolling away and the suspension holes faced me at a uniform angle.

[211]Future research comparing the Knossos deposits of spherical loom weights to those at outlying sites such as Vathypetro and Archanes, or at another palace like Mallia, might show distnctive palatial features of textile equipment in form, weight, and context.

[212]M. Popham (1984) 2-3.

[213]LM II is generally considered a short-lived ceramic phase. The naturalism of LM IA and LM IIIA:1 is very difficult to tell apart, according to E. Vermeule, and the lack of both Marine Style (LM IB) and LM II at the Palace make it difficult to alot them the 100 year separation Evans gives them in his chronology. Vermeule suggests that the two phases, LM IA and LM IIIA:1 are successive and in part overlapping. This would also fit in with the pottery sequence found elsewhere on Crete. Likewise, Doro Levi questioned the significance of the entire LM II style because of his finds at Gortyna of a mixed deposit of LM IA and LM III pottery. He suggests that the entire history of Minoan pottery should be rewritten. According to E. Vermuele, "There is not a single stratified vase in the Palace Style (LM II) inside the bounds of the great palace at Knossos from which it took its name...The last ceramic material in the fallen palace is Late Minoan III A:1." See E. Vermeule, "The Fall of Knossos and the Palace Style," AJA 67 (1963) 195-199.

[214]L. Hitchcock and D. Preziosi, "The Knossos Unexplored Mansion and the 'Villa-Annex Complex,'" in The Function of the Minoan 'Villa' (1997) 51-62.

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B) and Hagia Triada (Villa A and 'Annex' B). The more regular, rectangular annexes are seen as functional complements to the more sophisticated villas. The remains of buttresses found on the eastern side of the Unexplored Mansion probably once supported a bridge connecting the two structures, as at Tylissos.[215] According to the most recent analysis, buildings like the Unexplored Mansion were most probably workshop and storage areas rather than any kind of elite residence, dining area or ritual area.[216]

This new interpretation of the Unexplored Mansion fits very well the evidence of textile equipment found within. I was able to examine 147 of the 150 weights reported by Popham in the Stratigraphical Museum at Knossos. Most of these (87) came from room N 31, identified as a storage cupboard under a set of stairs in the southwest corner of the building.[217] Other deposits were found in the four-pillared hall, H (22 examples), in corridor L (8), or under the other set of stairs, J/K (1). Unprovenanced weights numbered 29 within the Unexplored Mansion, including two discoid types and one probable spindle whorl.

Of the 147 objects I measured, the weights or restored weights ranged from 64 to 769 grams. The lightest object, weighing 64 grams, was shaped like a biconical spindle whorl. Most of the spheres were made of fired beige-red clay. Small pieces of white, gray and black stones were used as grog. Some of the loom weights showed evidence of

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[215]L. Hitchcock and D. Preziosi (1997) 51.

[216]M. Popham (1984); J. Graham, "The Banquet Room of the Little Palace," AJA 79 (1975) 141-145. There are, however, elite furnishings known from the Unexplored Mansion such as a floral fresco. See the recent analaysis by A. Chapin, "A Re-examination of the Floral Fresco from the Unexplored Mansion at Knossos," BSA 92 (1997) 1-24.

[217]These are catalogued at the Stratigraphical Museum in two boxes labelled "MUM 14/7/72, SMS 467-9, SMT 2334, Room N, locus 21 below lot 19, SF 16." I do not know why there were three missing from Popham's original count of 90.

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burning.[218] All had a single suspension hole, ranging from 0.5 to 2 cm. in diameter, some examples showing string wear-marks.

From the 87 loom weights that were found under the stairway N, 58% (50) of the weights had traces of grooves, sometimes painted, along the exterior surface. Some of the interiors of the suspension holes were also painted. Six of the weights showed the same, deliberate cross marking scratched on the surface of the loom weight as was found in the discoid examples from the Palace of Minos in the Loomweight Basement. Of the 90 weights measured by Popham, he found that they clustered into three or four broad groups based on their weight. From my own measurements, I calculated three basic groups, small, medium, and large: 11% (10) weighed between 86 and 130 grams (small). The majority of weights, 55% (48), were between 150 and 250 grams (medium). 21% (19) were between 270 and 350 grams (large). The remaining ten weights varied greatly, from 370 to 710 grams. These outliers may have added tension to the selvedges.

Table 3. Minoan Unexplored Mansion Loom weights - spherical type:

Size	Count	%	Range
small	10	11%	86-130 grams
medium	48	55%	150-250 grams
large	19	21%	270-350 grams
outliers	10	11%	370-710 grams

[218]M. Popham (1984) 247.

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Unfortunately there did not seem to be a clear correlation between the grooved and ungrooved weights, or those marked with the x by the various sizes. One interpretation I would suggest, however, is that this assemblage of loom weights was used for several different looms rather than Popham's interpretation which assumes that all 90 weights found in Room N were intended for a single loom.[219] Perhaps one light-weight loom with weights of the first cluster, five looms of the medium size weights, and two of the large set, assuming that each loom used about ten weights. Alternatively, the weight groups could represent one loom each but produced textiles of different width. The medium weight group (of 48 weights) would make a fairly wide textile.

Twenty-two spherical weights were found in the so-called Pillar Hall H of the MUM. As Hitchcock and Preziosi demonstrate, the identification by Popham of this area as a Pillar Crypt, a domestic shrine and/or living area, seems unfounded.[220] The abundance of metalworking remains, cast trimmings, bellows nozzles, crucibles, and scrap metal, clearly indicates that this area was used for manufacturing metal. The 22 loom weights found in the same area give a general impression of an industrial quarter. The weights averaged 280 grams, giving a total of 61 kilograms of textile making equipment in this hall. These weights cluster into two groups, light and heavy. Thirteen weighed between 150 and 300 grams, eight were between 400 and 500 grams. Either they were used on the same loom, with the heavier weights providing tension along the sides to keep the selvedges straight, or they were used on two separate, although smaller, looms.

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[219]Popham's analysis works 'on the assumption that the Room N group is a more or less complete set intended for one loom.' MUM (1984) 248.

[220]L. Hitchcock and D. Preziosi (1997) 53-54; M. Popham (1984) 21-26.

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North Building

Comparing the weights from the Unexplored Mansion to the nearby Cult Room Basement of the North Building excavated by P. Warren, one finds a similar distribution of spherical weights (table 4).[221] Measurements of the 79 weights, kindly provided to me by Professor Warren, range from 100 to 670 grams. 17% (14) weighed between 100 and 150 grams, within the small range. The majority, as at the Unexplored Mansion, (64% or 51 weights) were medium size, between 150 and 250 grams. Five were considered large, and the rest were outliers (*i.e.* three weights were in the 420 range, one at 440, one at 460, another at 470, and the heaviest three weighing 500, 510, and 670 grams).

Table 4. Stratigraphical Museum Site Excavations, North Building Loom weights - spherical type:

Size	Count	%	Range
small	14	18%	100-150 grams
medium	51	64%	150-250 grams
large	5	6%	270-310 grams
outliers	9	11%	420-670 grams

[221]The melons are photographed although the weights were not published in P. Warren (1980-81) 85, fig. 42 with scale, and 43 *in situ*. This excavation is taking place near the Stratigraphical Museum site at Knossos.

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Buildings Near the Acropolis

Another set of loom weights was not available to me for study but is well published by H. Catling, *et al.*[222] In what is referred to as the main deposit, Deposit F, of the LM I A building there were substantial finds of spherical and cylindrical terra-cotta loom weights that had originally been stored in a clay tub.[223] Other finds include LM I A vases and stone vessels. At the edge of the group of loom weights was a small bronze implement, a rod square in section with flattened ends, like a spatula.[224] Perhaps this was a sword beater for pushing up the warp threads?[225] Also found with the loom weight group were 6 conical cups (v.244-249) all of which had a hole knocked in their bases, doubtless so that they too could be used as loom weights.[226] The range of weights for these spherical loom weights is well within what we find in other examples in the area of Knossos, 200-750 grams.

South House

Brief mention should be made of eleven loom weights from the South House entrusted to me for publication by P. Mountjoy, who is studying the pottery from this house in the southwest corner of the Palace of Minos. All are of similar red-brown clay and show pronounced resting surfaces created when the clay was still moist. Three are grooved and one is impressed with three fairly large cross markings, the familiar X as on loom weights mentioned above. Three X-marks are found on this one weight, however,

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[222]H. Catling, *et al.*"Knossos 1975: MM III and LM I Houses by the Acropolis," BSA 74 (1979) 1-80, especially 2, 15, 61, 78.

[223]Catalogued as T.3-T.55. The tub is V. 229.

[224]Catalogued as B 1.

[225]See M Hoffmann (1964) 279-282 and p. 60 above for a discussion of sword beaters.

[226]H. Catling (1979) 61.

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not scratched-on but firmly impressed into the once soft clay. The distribution of weights here was between medium and large sized weights. One set of six ranged from 137 grams to 214, while the remaining five weighed between 252 and 336. These weights seem well in-line with other comparable examples found at Knossos.

Unprovenanced Weights

I was able to study one other collection of spherical loom weights at the Stratigraphical Museum (table 5). There was, however, no context provided other than that they were found somewhere at Knossos. Based on the examples from the Unexplored Mansion and the North Building we can be reasonably certain that these 47 spherical loom weights date to sometime in the New Palace period . A similar distribution in weight to the medium and large size loom weights from the MUM and the North House was found. Only two examples weighed less than 150 grams. Of the others, 38% weighed between 150 and 260 grams (medium) and 25% were between 260 and 350 grams (large). The remaining weights were outliers, 23% between 340 and 600 grams. While we cannot be certain that these weaving tools came from a single deposit, the distribution of their measured weights is thoroughly in line with what we would expect for this type of implement.

Table 5. Unprovenanced Loom weights from Knossos - spherical type:

Size	Count	%	Range
small	2	4%	119-130 grams
medium	18	38%	150-260 grams
large	12	25%	260-350 grams
extra large	11	23%	340-600 grams

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Organized Textile Production at Non-Palatial Sites

Many Minoan palaces were built in rather densely populated areas, as evidenced by buildings such as the Unexplored Mansion at Knossos or the Houses associated with the palace at Zakros.[227] Excavations on Crete have uncovered non-palatial settlements that seem to have had mixed economies dependent upon agriculture, craft production and sea-faring like the palaces, suggesting also that there should be textile production located there as well. The LM IA-B 'villa' in eastern Crete at Prophetes Elias, Praissos, provides evidence for organized textile production not in a palatial context, but still at a regional center.[228] Twenty loom weights were found in the fill above rooms M, N, Ξ, O, and Π suggesting the 'women's quarters' of the Villa to Platon.[229] A potter's wheel, grape press (dye vat?) and millstones were also nearby, adding to the impression that this was an industrial quarter.

Near to Platon's 'villa' at Praissos, certain features uncovered in the 'palatial building' of Petras, Siteia, fit the criteria for a Minoan palace: storage magazines, ashlar masonry, a central court, masons' marks, and evidence for administration.[230] A true archive in Minoan Hieroglyphic has recently been found in MM IIB levels suggesting a long history of organized society at Petras.[231] This is the first hieroglyphic archive on Crete excavated in primary context. Four-sided clay bars, medallions, crescents, noduli,

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[227]For the houses at Zakros, see D. Hogarth (1900-1901) 121-149; N. Platon (1985) 24-33, 247-249.

[228]See L. Platon, "The Minoan 'Villa' in Eastern Crete," Function of the Minoan 'Villa' (1997) 187-202.

[229]N. Platon, Πρακτικά (1960) 298; L. Platon (1997) 198-200.

[230]Features summarized most recently in M. Tsipopoulou and A. Papacostopoulou, "'Villa' and Villages in the Hinterland of Petras, Siteia," Function of the Minoan 'Villa' (1997) 211-14.

[231]M. Tsipopoulou, "The Hieroglyphic Archive from the Palatial Building at Petras," AJA 101 (1998) 391.

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and chest sealings, some in an unfinished state, suggest that the archive was in use at the time this building was destroyed in MM IIB. Various types of textile equipment have been found at Petras but they have unfortunately not yet been analyzed.[232] The remains will most probably provide a great deal of useful information on the organization of cloth production at a Proto- and Neopalatial center in east Crete.

Neopalatial House II at Petras also shows evidence for textile production yet it is located away from the palace building.[233] Textile equipment from this house might perhaps be comparable to the loom weights found at Praissos. In three areas of House II there were dye vats and water channels (areas A, B, and N), a large number of tripod vessels sitatuated on hearths that the excavator associates with dyeing, and discoid loom weights. The ideogram for cloth appears on two conical cups and on some of the loom weights.[234] Similar to Praissos, these installations and equipment suggest that there was a nucleated textile industry aggregated within the community of Petras. Production here might be contrasted with that from the Palatial building at Petras.

Other settlements in Eastern Crete, such as Gournia, Mochlos, Pseira, Palaikastro, and Kommos, to the south, are large towns located near the sea that experienced destructions and rebuilding during the Old and New Palace periods.[235] Some of these sites exhibit palatial features, such as Gournia with its open court, west facade, storage facilities

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[232]Permission to study this material has just recently been approved by the Greek authorities. I am very grateful to Dr. Tsipopoulou for her permission to study this material in the Siteia Museum.

[233]See M. Tsipopoulou and A. Papacostopoulou (1997) 203-214.

[234]M. Tsipopoulou and A. Papacostopoulou (1997) 208-209; M. Tsipopoulou, "Κεραμεικά Σημεία από την ανασκαφή Πετρά Σητείας (1989-1990)," Proceedings of the 7th Cretological Conference, A2 (1995) 931-971.

[235]The eastern sites are particularly interesting in the Late Minoan period because they seem to lie outside of the reach of the dominating palace center at Knossos. See J. Bennet, "The Wild Country East of Dikte: The Problem of East Crete in the LM III Period," Studies in Mycenaean and Classical Greek Presented to John Chadwick. Minos 20-22 (1987) 77-88; A. Kanta, The Late Minoan III Period in Crete: A Survey of Sites, Pottery, and their Distribution SIMA 58 (1980) 125-198.

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and cult areas, and Kommos' Building T consisting of four winged stoas built around a large central court.[236] Textile equipment from two of these sites, Palaikastro and Kommos, were studied in the spring of 1997 for information regarding organized cloth production outside a major palace center like Knossos.

Excavations at Palaikastro provide a good opportunity to examine textile production in an urban settlement in eastern Crete. Late Minoan Palaikastro is particularly interesting because of its leading position as a center of East Crete.[237] Unfortunately no complete publication of the textile equipment from Palaikastro has yet appeared. From references in the site publications, evidence for cloth production occurs in the earliest levels of the site. Stamped disc-weights were found dating to the EM II period at Palaikastro.[238] The presence of spinning bowls and murex deposits is noted as well.[239] Dating to the palace period are examples of over one hundred rectangular, 'block'-shaped weights from Palaikastro, including some that are stamped.[240] Stamped disc loom weights of MM III - LM I date are also known.[241] In the LM II period, seventy-one of the one hundred weights from Palaikastro were from one deposit in House E 36. Fifteen of these weights

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[236]For the Gournia 'palace', see J. Soles, "Towards a Reconstruction of the Palace at Gournia," Temple University Aegean Symposium 4 (1979) 11-17; "The Gournia Palace," AJA 95 (1991) 17-78. For Building T at Kommos, see J. Shaw and M. Shaw, "Excavations at Kommos (Crete) during 1986-1992," Hesperia 62 (1993) 161-188; J. Shaw and M. Shaw, (eds.) Kommos I: The Kommos Region and the Houses of the Minoan Town part 1 (1995), part 2 (1996); J. Shaw, A. Van de Moortel, P. Day, and V. Kilikoglou (1997) 323 - 332.

[237]See J. Bennet (1987) 80-88; A. Kanta (1980) 125-198.

[238]The stamped disc loom weight was found at EM II Palaikastro House D 32, HM 4814. E. Eccles (1939) 49 no. 41, fig. 27. See also P. Warren (1970) 31.

[239]R. Bosanquet (1904) 317-329 and "Dikte and the Temples of Dictaean Zeus," BSA 40 (1939-40) 60-77; R. Bosanquet and R. Dawkins (1923) pl. 6c.

[240] R.W. Hutchinson (1939-40) 47-49; E. Eccles (1939) 48-49, fig. 42 no. 45 HM 4797; CMS V Suppl. 1A, nos. 61.

[241]E. Eccles (1939) 48-49, fig. 41 no. 44 HM 4815; CMS V Suppl. 1A, nos. 64. Both the disc and the cuboid weights are in the Hagios Nicolaos Museum, room 3.

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had seal impressions or graffiti on them suggesting organization, if not administration, of production.[242] Many of the seals show goats or horned sheep, possibly suggesting wool production.

At the Stratigraphical Museum at Knossos I was able to study some examples of loom weights from the British excavations at Palaikastro. Unfortunately I was not able to determine the exact find spots or the relative dates for the equipment. Presumably the examples I looked at were found during the 1962-63 seasons, before proper storage facilities were built at the site. The material provided a good sample of many different types of Aegean textile equipment from a site other than Knossos.

At the Stratigraphical Museum only one spherical loom weight with the typical four quarter grooves from Palaikastro was found. Also one well-made disc of the Knossos Loom Weight Basement type was present. Several 'spools' or 'bobbins', of fired clay or stone, were examined. The scant examples from Palaikastro do not help to shed light on these puzzling bobbins.

Fifteen of the typically Minoan one and two-holed discoid weights, with grooved tops, were also studied, averaging 125 grams.[243] These examples are comparable in form to Minoan loom weights known at other Aegean sites, including Mallia, Kythera, and Akrotiri.[244] Most informative for a survey of Minoan textile equipment was the opportunity to study examples of the cuboid weights from Palaikastro. As was mentioned before, this type of weight seems to occur exclusively on eastern Crete, occurring only as

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[242]E. Eccles (1939) 47.

[243]Catalogue numbers PK 62/63, SMT 811-825 (missing is number 815).

[244]For example, see O. Pelon, "Maison d' Hagia Varvara et architecture domestique à Mallia," BCH 90 (1966) 579, fig. 28; N.Coldstream and G. Huxley, Kythera (1973) pl 60; I. Tzachili (1991), (1997) 183-193.

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far west as Mallia.[245] Carington Smith suggests that these weights may be related to the much earlier Middle Neolithic cuboid weights found at Knossos. The tentative theory suggests the first Minoans displaced the original Neolithic settlers, pushing them into eastern Crete, where they retained one of their cultural traits, the cuboid loom weight.[246] The area of east Crete is known for its long-lived traditions. The Eteocretans are historically associated with the area of Praissos in East Crete and if anyone preserved old-ways on Crete, it would have been them.

Twelve examples of the four-holed clay weights from Palaikastro were available to me in the Stratigraphical Museum.[247] One of them, interestingly, had a cross or X inscribed on one side. This was the lightest weight, measuring only 67 grams. The holes were all fairly uniform in their placement in the four corners. The clay was similar pinkish-orange color but generally with less grit than the disc weights. The range of weights was from 67 to 284 grams. The weights seem fairly well-distributed over this range without any obvious evidence for clustering into sizes.[248] There was evidence of wear marks on the string holes so it is probable that these weights were suspended, possibly for making textiles. The idea that these blocks were used in tablet weaving would fit with the four-holed corners but the blocky nature and uneven size of the weights makes this seem unlikely.[249] However, the similar fabric and the cross marking of these block weights, along with the measured weights falling within the range of other known loom weights, possibly suggests a weaving association for these weights.

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[245]J. Carington Smith (1975) 295.

[246]J. Carington Smith (1975) 296. This is discussed above with Neolithic Knossos, pp. 29-30.

[247]Catalogue numbers PK 62/63, SMT 826-837.

[248]The recorded weights were 67, 83, 104, 115, 115, 131, 140, 142, 153, 186, 204, and 284 grams.

[249]J. Carington Smith (1975) 293.

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The tablet weaving method is ideal for making narrow decorative bands with fairly simple equipment. Barber describes the method where 'each warp thread is fed through a hole in one corner of a card or tablet that has at least two (but possibly several more) perforated corners' (see figure 2.7 a).[250] Although there is no evidence for tablet weaving at Palaikastro, there is perhaps some evidence for it at other Minoan sites. Square bone or ivory plaques with holes in all four corners are published along with other loom weights from Protopalatial Phaistos by D. Levi (fig. 2.7 b).[251] These objects came from storage pits so it is difficult to determine whether the plaques were used in a cloth production context, but the form and size seem like they could be easily handled in a deck and used for decorative tablet-band weaving.[252] Other examples of such ivory plaques are on display in Room 3 of the Agios Nicolaos museum. The find spot is not given but they are labeled 'furniture plaques', in the same case as the famous gold spatula/pin inscribed with Linear A. Ivory would provide a smooth, tangle-free surface through which warp threads could be fed for making a tablet weave.

Kommos undoubtedly had strong connections to the nearby palatial structures at Phaistos and Hagia Triada, only 7 km. away.[253] Kommos also shows evidence for contacts with the Near East, beginning in the Middle Bronze Age and continuing through the Iron Age.[254] There is also a sizable amount of textile equipment from all Bronze Age

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[250]E. Barber (1991) 118. Figure 2.7 a is adapted from Barber's illustration, figure 3.33.

[251]D. Levi, Festos e la civilita minoica (1976). Textile equipment is found on pls. 240, 241 and pl. 243, o, q, s and reproduced here as figure 2.7 b.

[252]See E. Barber (1991) 118-122 and fig. 3.33. If these objects are related to tablet weaving, these Minoan tablets are the earliest known examples in Europe.

[253]J. Shaw, *et al.* A Great Minoan Triangle in South Central Crete: Kommos, Hagia Triada, Phaistos Scripta Mediterranea 6 (1985).

[254]See S. Morris (1992) 151, 154-158 for Phoenician elements at Kommos.

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levels of the site published in great detail by M. Dabney.[255] Reference has already been made to the evidence for purple dyeing at MM I/II B Kommos.[256] Examples of loom weights and spindle whorls that were not published by Dabney or were of particular stratigraphic significance were analyzed in June 1997 with the help of Professor J. Shaw at Kommos.[257] Deposits from Building T were of great interest as this monumental structure is the most 'palatial' feature at Kommos, with four wings surrounding a large central court, evidence for storage facilities, craft production, and perhaps administration.[258] Of the available data, three assemblages are examined for how they reflect economic organization at Kommos. These date to 1) the Protopalatial period, 2) the Neopalatial period, 3) the 'Final Palatial' period.[259]

1) The earliest loom weights that I studied at Kommos dated to the MM II period and were from the fill of the Protopalatial building AA. These 10 weights were first uncovered in 1993 in trenches 80B, 86D, and 88B. They are all very fragmentary but seem to be of similar type, two holed discoids. Estimated restorations of their weights shows them also to be fairly consistent, around 140 grams each.

Excavations during 1995 revealed 16 loom weights in the Eastern part of Room E in Building T, trench 97 E. Since Building T was built in early LM IA and the ceramic evidence gives a date of MM II B - early LM I for these loom weights, it is probable that they should be associated with the leveling operations for the construction of Building T. If these weights are associated with the earlier Protopalatial Building AA underneath

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[255]See M. Dabney, "Ceramic Loomweights and Spindle Whorls," in Kommos I: part 2 (eds.) J. Shaw and M. Shaw (1996) 244-259.

[256]See above and D. Ruscillo (1997).

[257]I am very grateful to the Shaws for their permission to include the weights here.

[258]J. Shaw, *et al.* (1997) 323.

[259]See P. Rehak and J. Younger (1998) 149 -166 for the phase 'Final Palatial'.

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Neopalatial T, we can see a long history of centralized textile production at Kommos. The equipment includes two double-perforated half discoid weights that may be net-weights. The remaining fourteen weights are typical single and double pierced discoid weights weighing an average of 116 grams.

2) According to Dabney, the majority of loom weights at Kommos were found in mixed deposits, described as fill or dump. A substantial number of loom weights from Room 29 of Building T, however, were found in primary contexts and provided the best evidence for textile production at Neopalatial Kommos during the LM IA period. The weights were fairly uniform; discoid types with one or two holes, grooves along their flattened tops, unpainted, weighing an average of about 70 grams.[260] The lightness of these weights suggests the weaving of finer textiles, perhaps for export.

3) Some other discoid loom weights were found on the Hilltop House X dating to LM III A1. The weights clustered into two groups, one around 90 grams, the other at 170 grams, plus a few outliers in the 260 gram range.[261] Dabney makes the ingenious suggestion that since the first two cluster weight-groups equal in total the average weight of the outliers (90 + 170 = 260), weaving three different types of fabrics would have been possible by using the lighter weights alone or in combination.[262]

While textile production is found in houses at Kommos, such as House X, centralized production does not continue in the LM III buildings overlying Building T (Buildings N and P).[263] The LM III period is a time when many other sites on Crete are severely destroyed, marking the end of the New Palace period. Again, Kommos and

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[260]M. Dabney (1995) 245. One outlier weighed 160 grams.

[261]M. Dabney, "Ceramic Loomweights and Spindle Whorls," in Kommos I (eds.) J. Shaw and M. Shaw (1995) 246-47.

[262]M. Dabney (1995) 248.

[263]M. Dabney (1995) 247.

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Knossos are unusual sites on Crete in that they do not have the same severe LM I B destruction levels that are found at administrative centers elsewhere. If this period marks the arrival of mainland Mycenaeans to the island, they may have centered their operations at Knossos and preserved the port town of Kommos for trade. While some textile production in LM III is located in domestic contexts at Kommos, the loom weight evidence suggests that large-scale textile production is no longer located at the center. This evidence fits very well with what would expect for a Mycenaean system of cloth production, in comparison to the Greek mainland, examined in Chapter three. There is very little archaeological evidence for weaving or spinning in the LM/LH III periods at palatial centers.

Final and Post-Palatial Crete

It is generally accepted that the New Palace period is over by LM III A1. Some of the destructions may be related to social unrest, climatic changes, or shifting political and economic fortunes, all serving to weaken the indigenous Minoan power structure for the arrival of Mycenaeans from the mainland.[264] While this is probably the most intriguing period of inter-regional exchange and influence within Crete and beyond, the LM III A-B transition is perhaps the most complicated phase of Minoan history.[265]

After the Neopalatial period, new features, such as chamber tombs, tholos tombs, warrior burials, and mainland Greek pottery forms, appear on the island of Crete. This evidence has suggested to some that Mycenaeans either caused the LM I B destructions or supported a ruling elite strata of Minoans at Knossos who made for an island-wide take-

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[264]For a current review of the chronological issues related to Minoan Crete, see P. Rehak and J. Younger (1998) 91-73, especially 98-102.

[265]For recent research helping to clarify the LM III period on Crete, see E. Hallager, The Mycenaean Palace at Knossos (1977); A. Kanta (1980); W.-D. Niemeier, "Mycenaean Knossos and the Age of Linear B," SMEA 23 (1982) 219-287; J. Bennet, "The Structure of the Linear B Administration at Knossos," AJA 89 (1985) 231-249; (1987); and J. Dreissen An Early Destruction in the Mycenaean Palace at Knossos (1990).

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over.[266] Whatever the cause, after LM I B we can say that the Minoan Palace period is over. Knossos continues to function as an administrative center, evidenced most clearly by the Mycenaean Linear B tablets, described in Chapter three. These tablets provide the best evidence for textile production on Crete in the Late Bronze Age. Artistic representations of Minoan garments have been studied by Barber and others to understand the exchange of ideas and textiles throughout the Mediterranean region.[267]

In terms of loom weights and spindle whorls the evidence in LM III A2 - B Crete is scanty. The Late Minoan site of Kephala-Chondrou in eastern Crete provides some archaeological evidence for textile production on Crete after the New Palace period. N. Platon reports that many clay loom weights were found in the few houses scattered along the Kephala saddle.[268] Dating to LM III A2, this settlement provides useful information on the material remains of cloth production that is otherwise lacking in the archaeological record of LM III Crete. It would be interesting to see if these weights are representative of household weaving that continues into the archaic and classical periods, or if they indicate any organized production under the dominant Mycenaeans. As we will see in the next chapter, frustrating archaeological evidence is not unexpected for the Late Bronze Age Aegean.

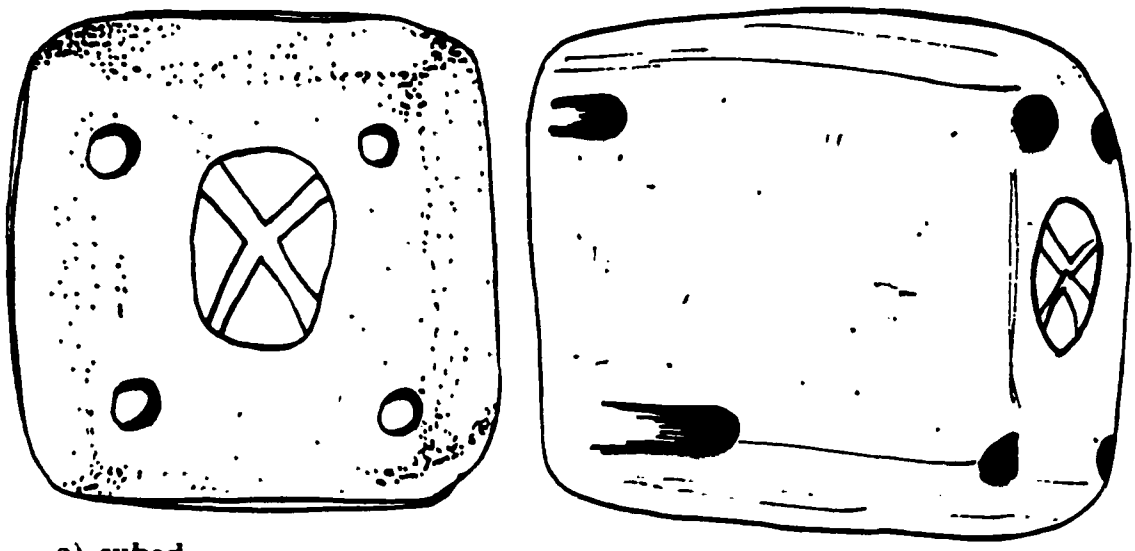
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[266]These arguments are well-summarised by P. Rehak and J. Younger (1998) 148-49. See M. Popham, "Mycenaean - Minoan Relations between 1450 and 1400 B.C." BICS 23 (1976) 119-121 and H. Catling, Some Problems in Aegean Prehistory c. 1450-1380 B.C. (1989).

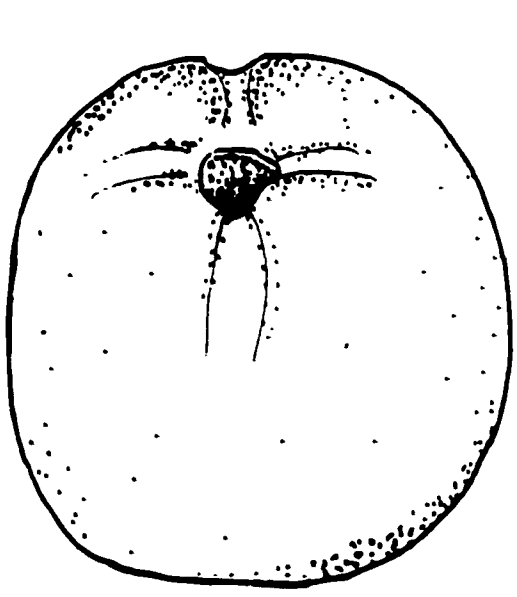
[267]E. Barber (1991) 330-351; (1994) 101-116; P. Rehak (1996) 35-51; L. Morgan, The Miniature Wall Paintings of Thera: A Study in Aegean Culture and Iconography (1988), especially 93-98.

[268]"Chondros," Aerial Atlas of Ancient Crete (eds.) J. Myers, E. Myers and G. Cadogan (1992) 82; N. Platon. "Ανασκαφή Χονδρου Βιάννου." Πρακτικά (1957) 143. Unfortunately no photographs or counts of the weights are included in these reports.

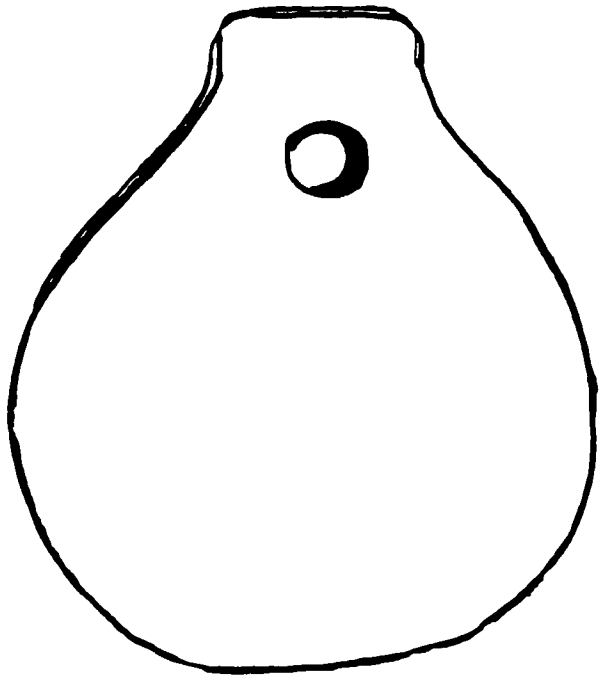
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a) cubed



b) spherical or melon-shaped



c) discoid

Figure 2.1 a-c Minoan loom weights

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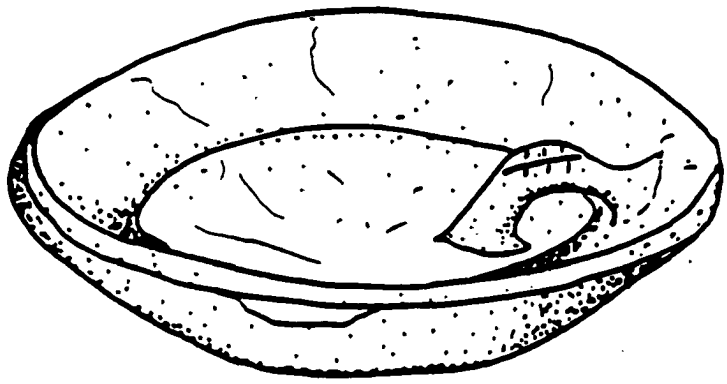


Figure 2.2 Minoan Spinning Bowl

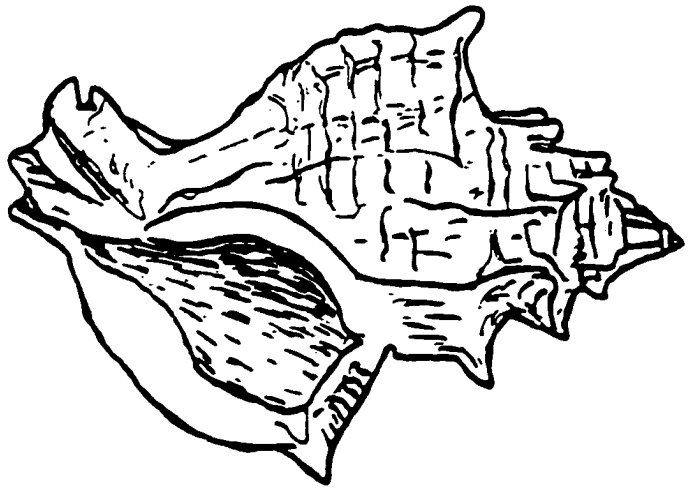
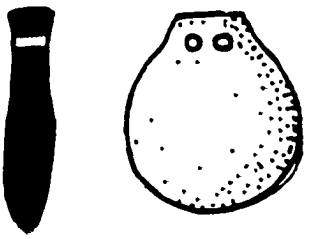
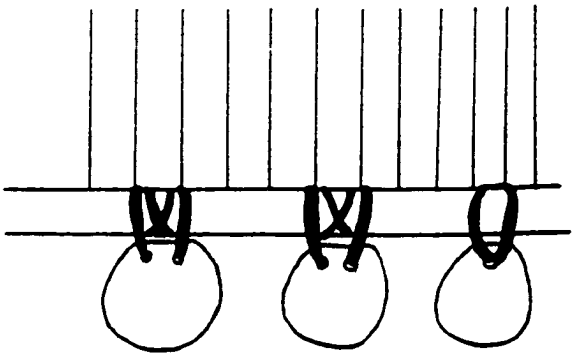


Figure 2.3 Murex brandaris

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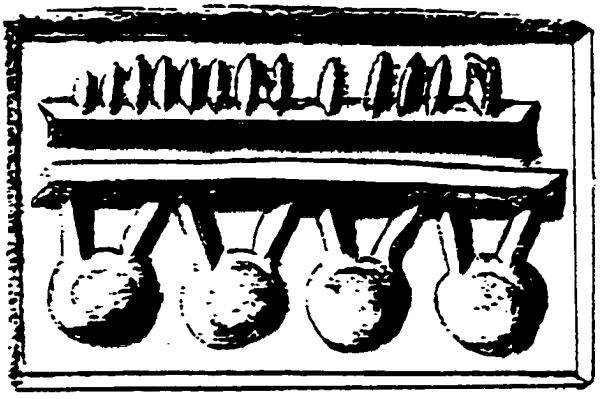
a) Minoan loom weight



b) hypothetical loom



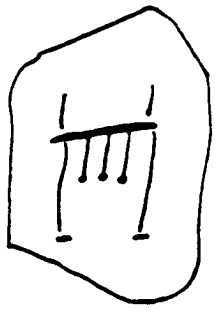
c) CMS IX no. 13 (Paris)



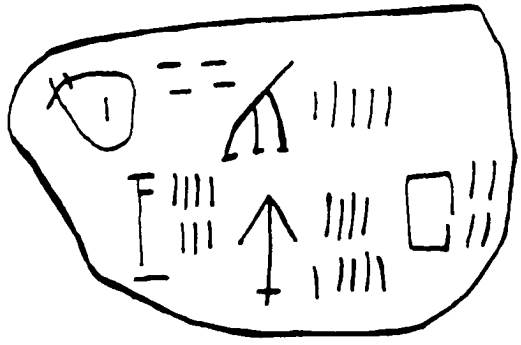
d) CMS II 2 no. 151 (Iraklion)

Figure 2.4 a-d - Attached loom weights and Minoan seals

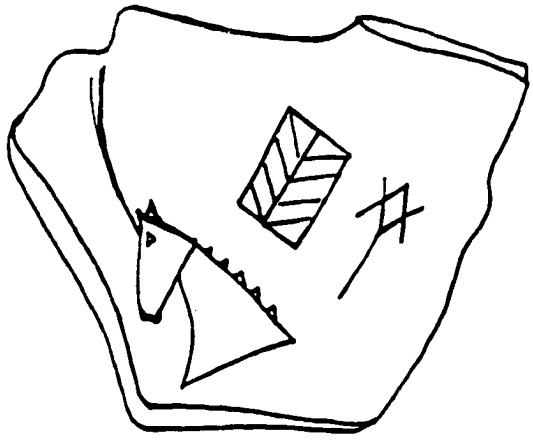
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a) Hagia Triada sealing



b) Akrotiri inscription



c) Tel Haror

Fig. 2.5 a-c Linear A Inscriptions dealing with Textiles

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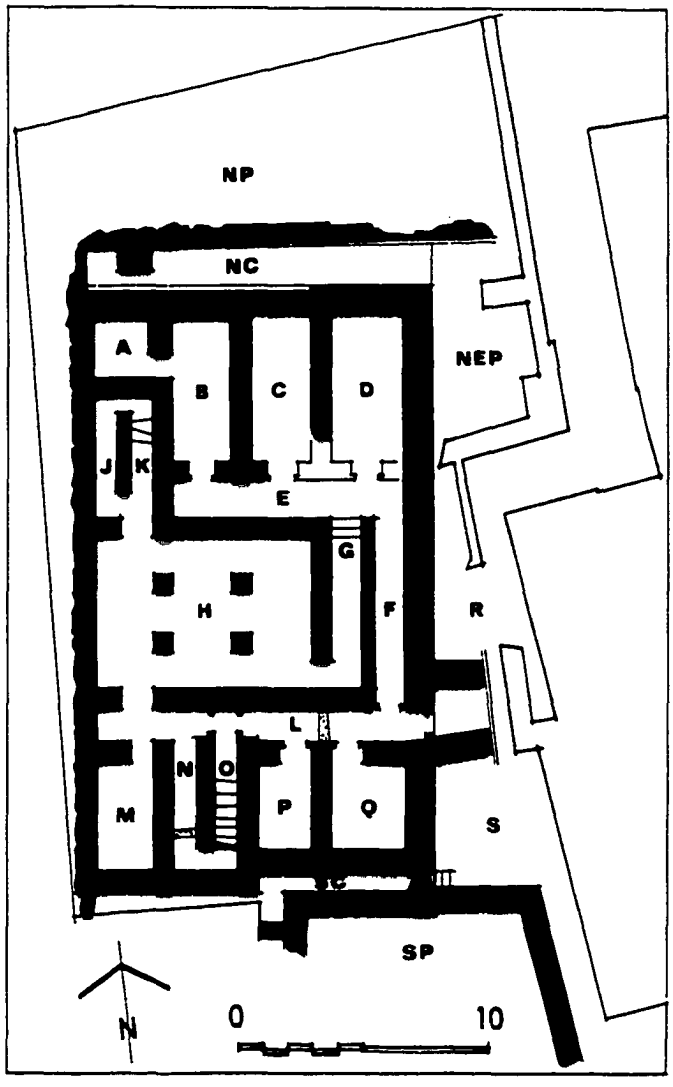


Fig. 2.6 Plan of the Minoan Unexplored Mansion

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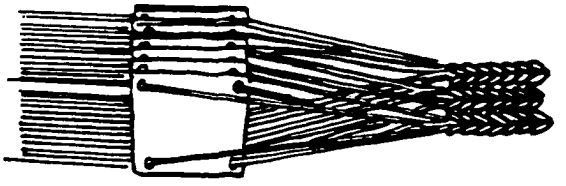


Fig. 2.7 a Tablet weaving illustration

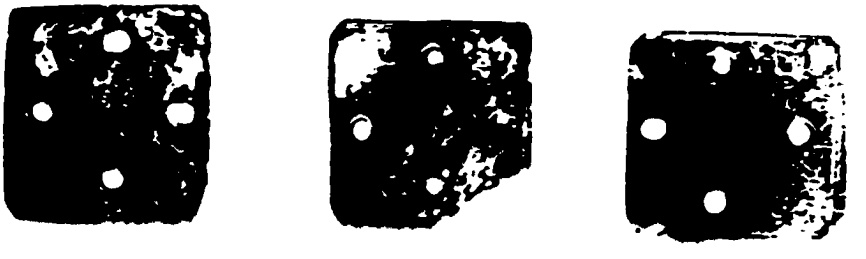


Fig. 2.7 b Possible tablets for weaving from Phaestos

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Chapter Three - Textile Production in the Mycenaean Economy

Evidence for textile production in the Late Bronze Age Aegean derives from four types of sources: artifactual remains of textile tools, artistic representations of different kinds of garments, later Greek literary references to cloth and palace society, and administrative records in Linear B. This chapter examines all of these sources but focuses most attention on Mycenaean Greek documents from major palace centers. In contrast to Minoan and Phrygian evidence discussed in other chapters, archaeological evidence for Mycenaean cloth manufacture is much less plentiful. Representations of Mycenaean cloth in Aegean art often illustrate textiles in elite or ritual settings and provide information on various types of garments. The Homeric tradition preserves, however faintly, remnants of Bronze Age economic and social organizations. Yet for an understanding of the role of textiles in Mycenaean palace economies, our best source of evidence is documents in Mycenaean Greek.

These texts were meant to be temporary records written in a kind of scribal shorthand for transactions of concern to the palace center. This chapter presents a synthesis of Linear B scholarship on textile administration and addresses the issue of how cloth manufacture was organized in the Late Bronze Age. The first part of this chapter presents some general remarks about Linear B documents related to textile production, including the topics of Mycenaean 'collectors', wool evaluation, and various types of cloth known from the texts. The second part focuses more specifically on the detailed phases of textile manufacture as recorded by scribes from Knossos and Pylos. Attention is given to the archives from Knossos and Pylos because they have the largest number of tablets of any of the Mycenaean

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citadels and a great many of these documents relate to textile production.[1] The final section concludes with an analysis of some other sources of evidence for textile production, including archaeological and textual data from Thebes and Mycenae.

Part 1 - Mycenaean Economy and Cloth

Wealth Finance

The economic system illuminated by the Linear B tablets demonstrates that the Mycenaeans operated within a wealth finance system. Wealth finance involves the production of goods which have some standard of value and are used by the state possibly as some form of payment to craft specialists, administrators, or other elites.[2] Specialists and second-order administrators are able to draw from the central places their own rations and are granted access to provincial land, labor, and tribute.[3] These goods may be converted into staples or exchanged for other wealth goods. Stein and Blackman suggest four criteria for distinguishing prestige goods described by administrative texts from Mesopotamia.[4] I use these criteria to suggest a wealth finance model for the Mycenaean economy:

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[1]At least 3,373 documents in Linear B are reported from Knossos, 1,112 from Pylos. In comparison, Thebes has only 400 tablets, many of them recently discovered. T. Palaima, "Preliminary Comparative Textual Evidence for Palatial Control of Economic Activity in Minoan and Mycenaean Crete," The Function of the Minoan Palaces (eds.) R. Hägg and N. Marinatos (1987) 303; V. Aravantinos, "Old and New Evidence for the Palatial Society of Mycenaean Thebes: An Outline," Politeia (1995) 613-622.

[2]T. D'Altroy and T. Earle, "Staple Finance, Wealth Finance and Storage in the Inka Political Economy," Current Anthropology 26 (1985)188. For a Mesopotamian example of this kind of finance, see Th. Jacobsen, "On the Textile Industry of Ur under Ibbi-Sin," Toward the Image of Tammuz and Other Essays on Mesopotamian History and Culture (ed.) W. Moran (1970) 216-230. Wool from the bookkeeper in charge of the copper paid the positive balance on his account in wool rather than in silver.

[3] For a fully integrated discussion of the staple and wealth finance model as it applies to the Inca, see T. Levine, (ed.) Inka Storage Systems (1992).

[4]G. Stein and M. Blackman, "Specialized Craft Production in Mesopotamia," Research in Economic Anthropology 14 (ed.) B. Isaac (1993) 50.

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1) the presence of appropriate craft specialists in the texts as dependents of a 'great institution' (*e.g.* the palace),

2) textual evidence for state, or palatial, procurement of the necessary raw materials,

3) a high price/value equivalent for the raw materials or finished products,

4) records indicating that these particular items were presented as gifts to high-status individuals of known social rank.

This chapter demonstrates that the textiles described in the Linear B tablets fulfill these criteria for prestige goods. In addition there is evidence to suggest that textiles functioned as a kind of currency in Mycenaean exchange systems where certain goods were valued in terms of another. It is demonstrated that cloth plays a particularly prominant role in these transactions.[5]

Documents of Mycenaean Textile Production

From the Linear B tablets we are able to reconstruct separate, localized, redistributive economies centered at major palaces of the Aegean, at Knossos, Pylos, Mycenae, Thebes, Tiryns, and possibly Chania.[6] The often fragmentary clay tablets incised with Linear B signs were accidentally fired in the final conflagrations that ended the Late Bronze Age palace system. These records preserve descriptions of raw materials and the phases of production

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[5]T. D'Altroy and T. Earle (1985) 187-197. See also Primitive, Archaic, and Modern Economies: Essays of Karl Polanyi (ed.) G. Dalton (1968) 321; E. Brumfiel and T. Earle (1987) 1-9. For this model specifically applied to the Aegean sphere, see J. Bennet, "'Outside in the Distance': Problems in Understanding the Economic Geography of Mycenaean Palatial Territories," Texts, Tablets and Scribes. Minos Suppl. 10 (eds.) J.-P. Olivier and T. Palaima (1988) 33; P. Halstead, "The Mycenaean Palatial Economy: Making the Most of the Gaps in the Evidence," Proceedings of the Cambridge Philological Society (1992) 59-61; G. Walberg (1995) 157-161 and discussion following.

[6]For an introduction to the decipherment and classification of the Linear B tablets see J. Hooker, Linear B: An Introduction (1980) 19-47.

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for both wool and linen cloth, among other Mycenaean industries. Less explicit in the tablets are references to the final use and consumption of textiles in Mycenaean society.[7]

Linear B documents record four basic types of transactions: taxation, agricultural production, personnel maintenance, and craft production.[8] In all of these transactions cloth plays a substantial role. For example, one commodity *146, prevalent in the Pylos Ma taxation records, is most likely a type of cloth produced in outlying areas of the kingdom and received along with other taxation commodities.[9] As an agricultural good, flax and sheep are extensively recorded by scribes at Pylos, Knossos, Mycenae, and Thebes. Personnel records describe a variety of laborers, from the general textile worker to the specialist, *e.g.* shepherds, spinners, weavers, fullers and decorators, all under the administration of the palace.[10] Finally, in terms of craft production, Linear B records describe various kinds of cloth produced by specialists at Mycenaean centers mostly under the *tarasija* system whereby workers receive raw materials from the palace with the expectation that targets of production

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[7]For use and consumption of textiles we can look at artistic representations of cloth and costume in Mycenaean art. E. Barber (1991) is a valuable source for representations of Aegean cloth.

[8]P. Halstead (1992) 59-61.

[9]The ideogram at Pylos looks like the syllable *pte* with an inserted *we*. There are, however, discrepancies in the handwriting of Knossos scribes for these two signs and the ideogram *146. Other known products of taxation include ox-hides and possibly honey or sheep/goats. See Y. Duhoux, "Idéogrammes textiles du linéaire B: *146, *160, *165 et *166," Minos 15 (1976) 116-132; J. Chadwick, "Pylos tablet Un 1322," in Mycenaean Studies (ed.) E. Bennett (1964) 19-26; C. Shelmerdine, "The Pylos MA Tablets Reconsidered," AJA 77 (1973) 261-275; M. Ventris and J. Chadwick, Documents in Mycenaean Greek (1973) 289-90; J. Killen, "The Textile Industries at Pylos and Knossos," in T. Palaima and C. Shelmerdine (eds.) Pylos Comes Alive (1984) 61-62; *Idem*, "The Linear B tablets and the Mycenaean Economy," Linear B: A 1984 Survey (eds.) A. Morpurgo Davies and Y. Duhoux (1985) 241-305; H. Morris, An Economic Model of the Late Mycenaean Kingdom of Pylos Ph. D. dissertation, University of Minnesota (1986) 99. L. Palmer, The Interpretation of Mycenaean Greek Texts (1963) 300-301 discounts the textile identification for *146.

[10]See P. Halstead (1992) 61; S. Hiller, "Dependent Personnel in Mycenaean Documents," Society and Economy in the Eastern Mediterranean (c. 1500-1000 b.c.). OLA (eds.) M. Heltzer and E. Lipinski (1988) 53-68.

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will be met. These aspects of cloth production in the Mycenaean economy are elaborated upon below.

Most of the tablets relevant to the textile industry have been deciphered and published at least in preliminary form.[11] The textual data are, however, limited in what they can tell us about the Late Bronze Age. The documents are chance discoveries from one year's administration at a single Mycenaean palace. Many are fragmentary, broken, or incomplete and it is difficult to say how many tablets were lost since the final days of the Bronze Age. These documents were never meant to be permanent records detailing the intricacies of a textile industry or any other craft. Through the careful and deliberate work of Mycenaean philologists often varying interpretations of these documents are possible. Certain terms and words remain problematic, such as the so-called 'collectors' and the *tarasija* system, both integrally linked to the textile industry. In addition, the date for the majority of tablets found at Knossos has not been established with certainty. The generally accepted date for most of

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[11]The most recent tablets uncovered from Thebes are awaiting final publication. For a brief survey of the material, see V. Aravantinos (1995) 613-622; V. Aravantinos, L. Godart, and A. Sacconi, "Sui nuovi testi del palazzo di Cadmo a Tebe: Note preliminari," RendLinc 6 (1995) 809-845. Preliminary notes suggest that some of these documents refer to textile production and other activities often associated with female labor. For textile production in the Mycenaean tablets, see articles by J. Killen over the last thirty-five years: "The Wool Ideogram in Linear B Texts," Hermathena 96 (1962) 38-72; "Some Adjuncts to the SHEEP Ideogram on Knossos Tablets," Eranos 61 (1963) 69-93; "The Wool Industry of Crete in the Late Bronze Age," BSA 59 (1964) 1-15; "The Knossos Lc (cloth) Tablets," BICS 13 (1966) 105-109; "Minoan Woolgathering: a reply I," Kadmos 7 (1968) 105-109; "Minoan Woolgathering: a reply II," Kadmos 8 (1969) 23-38; "A Problem in the Knossos Lc(1) (cloth) Tablets," Hermathena 118 (1974) 82-90; "The Knossos Ld(1) Tablets," Colloquium Mycenaeum (ed.) E. Risch and H. Muhlestein (1979) 151-81; (1984) 49-63; (1985) 241-305; "Epigraphy and Interpretation in Knossos WOMAN and CLOTH Records," in Texts, Tablets, and Scribes (1988) 167-183; "Some Further Thoughts on 'Collectors,'" in Politeia (1995) 213-226. Other Linear B studies relevant to textile production include: J. Chadwick, "The Women of Pylos," in Texts, Tablets, and Scribes (1988) 43-96; Y. Duhoux (1976) 116-132; P. Halstead, "Lost sheep? On the Linear B Evidence for Breeding Flocks at Mycenaean Knossos and Pylos," Minos 25-26 (1991) 343-366; H. Landenius-Enegren, "A Prosopographical Study of Scribal Hand 103, Methods, Aims and Problems," in Politeia (1995) 115-123; M. Lejeune,"Les documents pyliens des series Ns, Ng, Nn," Études Myceniénnes (1956) 137-165; J. Melena, Studies on Some Mycenaean Inscriptions from Knossos Dealing with Textiles Minos Suppl. 5 (1975); J.-P. Olivier, "L'industrie textile dans la Grèce mycénienne," SMEA 4 (1967) 120f. A very useful chart summarizing each phase of production as recorded by Linear B scribes can be found in J. Cherry, "Beazley in the Bronze Age? Reflections on attribution studies in Aegean prehistory," EIKΩN. Aegaeum 8 (eds.) R. Laffineur and J. Crowley (1992) 137 table.

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the Linear B tablets is LM III A2.[12] For Pylos, the destruction date is set at the end of LH III B2/LH III C early.[13] The discussion here, however, interprets the texts as broadly representative of administration during the Mycenaean period from approximately 1400 - 1150 B.C.

The nature of the Mycenaean economy not directly related to the palace center and therefore not recorded in the Linear B tablets has also been the subject of recent debate.[14] T. Palaima and others agree that there were economic areas and geographical regions beyond the scope of the palace bureaucracy.[15] J. Bennet has made explicit the extent of Knossos' administration and influence in his regional studies of archaeological and textual evidence from Late Bronze Age Crete.[16] Cottage industries and second-order markets for exchange may have existed outside of the realm of the palace. Examples of activities not recorded in the palace archives include pottery production and the collection of certain food groups

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[12]That Linear B script was in use as early as LM II is shown by J. Driessen, An Early Destruction in the Mycenaean Palace at Knossos: A New Interpretation of the Excavation Field-Notes of the South-East Area of the West Wing (1990). The scribal tradition and administrative system of the palaces is developed enough that we can safely assume a few centuries of use. For chronological issues, see L. Palmer, The Penultimate Palace of Knossos (1969); *Idem*, "Knossos: Some Instructive Recent Errors," Minos 15 (1974) 34-67; *Idem*, A New Guide to the Palace of Knossos 1969; *contra* M. Popham, The Last Days of the Palace at Knossos: LMIIIB Vases (1964); *Idem*, The Destruction of the Palace at Knossos: LMIIIA Vases (1970); L. Palmer and J. Boardman On the Knossos Tablets (1963). More recently, see C. Shelmerdine, "Historical and Economic Considerations in Interpreting Mycenaean Texts," Mykenaïka. BCH Suppl. 25 (1992) 569-590.

[13]P. Mountjoy, "The Destruction of the Palace at Pylos Reconsidered," BSA 92 (1997) 109-137.

[14]J. Bennet and H. Morris have each attempted studies combining textual and archaeological data with geographic modeling to reconstruct economic processes not explicitly recorded by the Linear B scribes. See J. Bennet (1988) 19-42; H. Morris (1986). See also the discussion in Politeia (1995) 171-72.

[15]Palaima states, "Textual and archaeological evidence suggests that sites within the separate regions of Crete had some degree of economic autonomy in the LB period." T. Palaima (1987) 302. See also *Idem*, "Inscribed Stirrup Jars and Regionalism in Linear B Crete," SMEA 25 (1984) 189-203; H. Haskell, "From Palace to Town Administration: the Evidence of the Coarse-Ware Stirrup-Jars," in Minoan Society (1983) 121-128; A Kanta, The Late Minoan III period in Crete (SIMA 58) (1980) 318-321.

[16]J. Bennet, "Knossos and LM III Crete: A Post-Palatial Palace?" in Function of the Minoan Palaces (1987) 307-311.

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known from archaeological sources to be an important part of Mycenaean life.[17] Industries or products which the palace labor force was self-sufficient in, *i.e.* not in need of raw materials from outside the immediate palace vicinity, may not have been recorded in the archives.[18] The question of inter-regional trade beyond the Aegean has also been a topic of recent discussion.[19]

Pylos and Knossos

The two most informative Mycenaean archives were found at Knossos and Pylos. In terms of textile administration, the tablets from Pylos show a greater interest by the palace scribes in flax and linen production compared to Knossos.[20] Most likely this can be attributed to the geographic predisposition of the Pylian kingdom to linen production: well-watered loamy soil in a moderate climate well-suited for flax cultivation even into modern times.[21] At Knossos, the tablets show a heavy bias toward sheep and wool production. The economic benefits of sheep rearing, milk, wool, and meat, are particularly well-suited to the terrain of Crete in both ancient and modern times.

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[17]There are references to individuals who are thought to be potters (*ke-ra-me-we/wo* and *ke-ra-me-ja*) at Pylos, Mycenae, and Knossos but nothing in the texts refers to palatial administration of this craft. Most of the plant species that have been found in paleobotanical studies are not recorded in the archives at all, particularly pulses. Similarly, the faunal remains of animal husbandry do not correspond to the livestock records of the Linear B tablets. See P. Halstead (1992) 57-86 and J. Bennet (1988) 19-41, especially 26.

[18]P. Halstead (1992) 64.

[19] See T. Palaima, "Maritime Matters in the Linear B Tablets," Thalassa. Aegaeum 7 (eds.) R. Laffineur and L. Basch (1991) 276-278; J. Killen (1985) 268-69.

[20]The linen industry is discribed below. See also A. Robkin, "The Agricultural Year, the Commodity SA and the Linen Industry of Mycenaean Pylos," AJA 83 (1979) 469-474; C. Shelmerdine, "Nichoria in Context: a Major Town in the Pylos Kingdom," AJA 85 (1981) 320-25; J. Killen (1984); J. Carington Smith, "Textiles at Nichoria," Excavations at Nichoria in Southwestern Greece Vol. II: The Bronze Age Occupation (eds.) W. McDonald and N. Wilkie (1992) 674-711.

[21]J. Chadwick The Mycenaean World (1976) 153.

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The tablets from Knossos and Pylos show that these centers not only emphasized different fibers, linen and wool, but also that the kingdoms themselves were organized differently.[22] The archives from Pylos reflect a highly centralized economy with contributions of raw materials to the palace center coming from sixteen smaller place-names within the kingdom. The Pylian state itself was divided into two regional areas, the Hither and the Further provinces.[23] Based on the tablets from Pylos, more work groups and more workers were concentrated at the two major centers of these provinces, Pylos and Leuktron, than in any of the outlying place-names.

Documents from Knossos, however, suggest a decentralized economic system with many secondary centers spread throughout the central and western parts of the island, giving and receiving assessments of raw materials and finished goods.[24] Most likely, the differences visible at Knossos compared to Pylos should be attributed to the Mycenaeans adopting an already organized Minoan system of administration on Crete. The decentralized economy of Mycenaean Crete is also reflected in cloth production and is attributed to the Mycenaeans taking over an older Minoan textile industry.[25]

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[22]Recent work on regional variation at Mycenaean centers includes, J.-P. Olivier, "Administrations at Knossos and Pylos: What Differences?" in PCA (1984) 11-18; J. Killen (1984) 11-18, 49-64; J. Hooker, "Minoan and Mycenaean Administration: a Comparison of the Knossos and Pylos Archives," in Function of the Minoan Palaces (1987) 313-316; G. Kopcke, "The Argolid in 1400 - What Happened?" in Politeia (1995) 89-94; and C. Shelmerdine, "A Comparative Look at Mycenaean Administration(s)," Floreant Studia Mycenaea. Proceedings of the 10th Mycenaean Colloquium, Salzburg, May 1995, (eds.) S. Hiller, S. Jalkotzy, and O. Panagl (forthcoming) (*non vidi*).

[23]See J. Chadwick, "The Two Provinces of Pylos," Minos 7 (1963) 125-141; "The Geography of the Further Province of Pylos," AJA 77 (1973) 276-278.

[24]J. Hooker (1987) 313-16; T. Palaima (1987) 301-306; J. Bennet, "The Structure of the Linear B Administration at Knossos," AJA 89 (1985) 231-249.

[25]Evidence for the organization of textile production on Minoan Crete discussed in the previous chapter is understood from textile equipment and Linear A records found in many parts of the island. Shelmerdine notes that these differences between Pylos and Knossos might also be explained by different economic conditions prevailing at the time when the archives at each palace were burnt. The Pylos tablets may reflect a greater state of emergency in LH III B2 than those at Knossos, which Shelmerdine dates to the

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The Mycenaean textile system is elaborated below by synthesizing administrative documents of Mycenaean Greek. Linen and wool were only one part of a wide-ranging Mycenaean industrial economy. This investigation of one product, textiles, should nevertheless reflect general aspects of economic organization and production in the Bronze Age Aegean. Stages of production recorded by the palace scribes provide a fairly secure outline of the textile industry during the Late Bronze Age Aegean.

'Collectors'

From the detailed Linear B records it is evident that Mycenaean textile production was subject to the central authority of the palace at Knossos. Some tablets, however, have personal names most often written in the genitive that seem to imply some raw materials, laborers, and finished textiles themselves, 'belonged' to certain individuals other than the palace. Records and totals for production appear against either a place-name or a personal name. These named figures are known in Linear B scholarship as 'collectors' or 'owners' and occur in tablets from Pylos and Knossos, on sealings from Thebes, and on inscribed stirrup-jars.[26] The interpretation of these individuals as 'collectors' is based on the Mycenaean word *a-ko-ra*, related to later Greek ἀγορά, in the sense of 'collection' or *a-ke-re*, Greek ἀγείρει, the third person singular verb, 'he collects'.[27] This vocabulary item occurs most frequently at Pylos on livestock records. For example:

LM/H III A2. C. Shelmerdine, "Architectural Change and Economic Decline at Pylos," Studies in Mycenaean and Classical Greek Presented to John Chadwick, Minos 20-22 (1987) 567-8.

[26]Translations of these named figures in the tablets have ranged from "collectors" (Documents (1956) 200-201), 'owners' (L. Palmer, "Review of Documents (1956)" Gnomon 29 (1957) 569-570 and The Interpretation of Mycenaean Greek Texts (1963) 164-171), 'bénéficiaires' and 'collecteurs' (J.-P. Olivier, "La série Dn de Cnossos," SMEA 2 (1967) 84 and "KN: Da-Dg," Texts, Tablets and Scribes (1988) 252-257), and 'overseers' (J. Bennet (1985) 240). For a thorough review of the subject, see J. Bennet, "'Collectors' or 'owners'? An Examination of their Possible Functions within the Palatial Economy of LM III Crete," in Mykenaïka (1992) 65-101 and J. Killen (1994) 213-221.

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PY Cc 660

.1 a-ke-o , a-ke-re

me-ta-pa , pa-ro ka-ra-su-no CAPm 30

On this tablet, one Alkeos *collects* at Metapa from Karasuno 30 goats (male).[28] The function of these figures in the Mycenaean economy has been the focus of study for scholars since the script was first deciphered.[29] The 'collectors' operate in various contexts of the Mycenaean economy, not just in textile production, but also in bronze-working and perfume manufacture. While the terms *a-ke-re* or *a-ko-ra* are not fully attested at Knossos, the system of named individuals associated with large flocks of sheep on Crete has been compared with the collector tablets known from Pylos.[30] At Thebes the terms *a-ko-ra* and *a-ko-ra-jo* occur among the Wu-series sealings.[31]

Killen summarizes three main points to consider when trying to understand the role of collectors:[32]

1) The tablets that record their activities are found in palace archives, indicating that they were under the supervision of and probably controlled by the Mycenaean palace.

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[27]J. Bennet (1992) 68; Documents (1973) 200; T. Palaima in discussion of J. Killen's paper, Politeia (1994) 226.

[28]Documents (1973) 200-201. There are only four known collectors named on the Pylos tablets, *A-pi-me-de*, *A-ke-o*, *A-ko-so-ta*, and *We-da-ne-u*.

[29]At the Ninth Colloquium of Mycenaean studies an entire session was devoted to this enigmatic group. See papers in Mykenaïka (1992) by L. Godart, J. Bennet, P. Carlier, and J. Driessen.

[30]J. Bennet notes that a shearing record from Knossos, Dk(1) 969, may preserve the term *a-ko-ra*[, although the tablet is broken to the right of the *-ra*; J. Bennet (1992) 68.

[31]Also on these sealings is the term *qe-te-o* discussed below. See V. Aravantinos, "The Mycenaean Inscribed Sealings from Thebes: Problems of Content and Function," Aegaeum 5 (1990) 149-174; C. Piteros, J.-P. Olivier, and J. Melena, "Les inscriptions en linéaire B des nodules de Thèbes (1982) : la fouille, les documents, les possibilités d'interprétation," BCH 114 (1990) 103-184.

[32]J. Killen (1994) 213.

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2) The flocks and work groups that they 'own' or 'collect' are administered in the same way that all the other flocks and work groups are recorded in the tablets. The only exception is that they have separate totals for collectors and non-collectors.

3) References to 'ownership' by named collectors are sometimes found in parallel with references to 'ownership' by divinities. For example, at Knossos the goddess Potnia is recorded as an 'owner' [of sheep] (*po-ti-ni-ja-we-jo*).

The occurrence of similar names in different palaces, such as the collectors named *a-ka-i-jo* and *ko-ma-we(-ta)* in texts from both Thebes and Knossos, might suggest that these figures were divinities.[33] Killen suggests that a more likely explanation is that these names were from a limited stock of possibly the same dynasty and compares it to the many English monarchs named Edward or Henry for example. Killen defines these 'collectors' as 'prominent members of the ruling élite (members of the royal family, high palace officials and the like) who have been assigned part of the productive capacity of their kingdoms for their own benefit (that share, however, still being managed on their behalf by the central authorities).'[34] In support of this hypothesis, Killen cites comparative evidence from Ur III where the queen and officials of both the state and the temple are listed as owners of sheep and wool next to the king.[35] It is very likely that similarly high ranking individuals in the Greek world might be listed in Linear B tablets by their names in the genitive as collectors and at other times are referred to by their rank, title, or position in the Mycenaean hierarchy, *e.g. da-mo-ko-ro*, *e-qe-ta*, or even possibly, the *ra-wa-ke-ta*.[36]

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[33] J. Killen (1979) 177.

[34] J. Killen (1994) 213.

[35] J. Killen (1994) 213 -14; H. Waetzoldt, Untersuchungen zur neusumerischen Textilindustrie (1972) 107-111.

[36] This idea is described in the discussion in Politeia (1994) 226. On the *e-qe-ta*, 'Followers', see S. Deger-Jalkotzy, E-QE-TA. Zur Rolle des Gefolgschaftswesens in der Sozialstruktur mykenischer Reiche. (1978).

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Sheep and Other Animals

Beginning with sheep management, the records found at Knossos show a high degree of interest by the palace in all phases of wool production. Ideograms for animals, such as horse, pig, and deer, were identified by their naturalistic representations before the Linear B script itself was deciphered. The signs for sheep, goat and cattle, however, proved more difficult. Sign *23 is identified as OX based on its occurrence on a tablet at Pylos with the word *qo-o* heading the list of animals; this word is almost certainly equivalent to Greek ὁ βοῦς, 'ox, cattle'.[37]

Caprids are identified by two general signs in Linear B, one of which is much more common than the other and frequently associated with the sign WOOL and textiles. This is taken to refer to sheep. While goat-hair is not impossible it is much more likely that the main animal exploited in the Knossos economy was the sheep. The ideograms also distinguish between the sexes of the animals. This is of interest to the study of wool production because female sheep (ewes) generally produce better wool than intact males (rams). The highest quality wool, however, is made from neutered males (wethers). All together there are at least eight different signs that refer to sheep and goats in the Linear B tablets: two generic signs that do not differentiate the sex of the sheep and goat, male and female sheep or goat signs, and two other signs for certain types of sheep, one ligatured with TA and the other using the *we* sign (*75). With an educated guess, Evans correctly distinguished the marking for sex

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The *ra-wa-ke-ta* or *lawagetas*, 'leader of the people,' has been identified as the second highest ranking Mycenaean official after the *wanax*. The two titles, *wanax* and *lawagetas*, also appear in Phrygian over the Midas City monument, discussed in Chapter four. See A. Leukart, Die frühgriechischen Nomina auf *-tâs* und *-âs* (1994) 68-70, for the interpretation of *lawagetas* and *eqeta*. The suggested identification of the *lawagetas* as a 'collector' is made by M. Lindgren, The People of Pylos: The Use of Personal Designations and their Interpretation II (1973) 134.

[37]The ox is qualified by the term *we-ka-ta*, taken to mean 'working', discussed below. Cf. *bous ergates* in Archilochus, 39, etc. Documents (1953) 195.

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among the animal signs: two horizontal bars indicating males and a bifurcated stem identifying a female. Along with RAMS and EWES these forms also occur on signs for MAN and WOMAN, OX and BULL, and he-goat and she-goat. Kober, however, was the first to demonstrate conclusively the sex differences when she read totals for sheep with grammatical inflections, *to-so*, 'so many', for RAMS and *to-sa* for EWES even before the script was deciphered.[38]

Wool Evaluation

The average yield of wool per sheep, based on ancient sources, ranged from 500 to 1,000 grams.[39] From the Knossos tablets, Killen demonstrated that in the Dk series the ratio of sheep to wool was four sheep to one wool unit. This wool-unit, Linear B sign *145 = LANA, is a unique measurement used only for quantifying wool.[40] The sign appears to be a monogram of two syllabograms clustered together into one sign, MA + RE, probably inherited from a Minoan Linear A sign (Lc 46).[41] A. Evans first connected this sign with wool when he suggested that it depicted a pair of shears attached to a syllabogram *RE*. One problem with this interpretation is the lack of shears in the Bronze Age. Most wool was probably plucked or combed from sheep by the Mycenaeans. The springiness of

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[38]A. Kober, "'Total' in Minoan (Linear Class B)," Archiv Orientalni 17 (1949) 386-98 (*non vidi*).

[39]See M. Ryder Sheep and Man (1983).

[40]J. Killen (1962) 38-72; K. Petruso, "Wool-Evaluation at Knossos and Nuzi," Kadmos 25 (1986) 26.

[41]A. Evans Scripta Minoa II (1952) 68. See also Documents (1973) 52; J. Hooker (1980) 39-40; J. Melena, "On the Linear B Ideogrammatic Syllabogram *ZE*," Studies in Mycenaean and Classical Greek Presented to John Chadwick. Minos 20-22 (1987) 397. Other examples of such monograms include *ME+RI*, μέλι, for honey, or *TU+RO2*, τυρί, for cheese. These words are still used in modern Greek today for the same agricultural products.

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iron allowed for shears to develop in the subsequent Iron Age.[42] More than likely, the sign *145 indicates the word related to Greek ὁ μαλλός. a somewhat rare word without an accepted etymology, meaning 'wool'.[43] The more common word for wool in Mycenaean is*we-we-e-a*, related to Homeric Greek τό εἶρος. The unit of weight indicated by the wool sign is equivalent to three double-minas or one-tenth of a talent, approximately three kilograms. Measuring commodities such as metals, spices, and grain in terms of this unit of weight would be possible but from the known Linear B tablets this sign was used only in connection with the textile industry.

For other goods, Petruso has shown that the Linear B scribes at Knossos mainly used four denominations of measurements, ranging from the talent = L (28,900 gr.), double mina = M (964 gr.)[44], half mina = N (241 gr.) and the 1/24 mina = P (20.1 gr.).[45] Based on this measurement system, the weight for the wool unit was 2,892 grams, or three times the double mina (964 X 3). Petruso has shown that at Knossos, as in the metrological system at Nuzi, 'the mathematical relationship between the mina and the talent is sexagesimal'... and 'that the corresponding units are approximately of the same mass.'[46] Petruso then states that this same feature of a unique measurement unit for wool is also found at Nuzi. In Nuzi documents the special unit is a fairly uncommon word, known as the *nariu*.

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[42]R. Forbes Studies in Ancient Technology IV (1956) 7; E. Barber (1991) 20-30. Plucking consequently results in lower yields of wool, the need for a special tool for plucking, and a different quality of wool. See J. Melena (1987) 403-404.

[43]E. Barber (1991) 260 cites Hesychius' reference to a variant of the word, *mallukes*, meaning *trikhes*, 'hairs'.

[44]K. Petruso (1986) 27 has this number, 964, incorrectly published as 946. On p. 30 the total for three double minas should read 2,892 grams (964 X 3), not 2,982.

[45]These letters (L, M, N, P) are arbitrary units used to designate specific Linear B signs used for measurement. See J. Hooker (1980) 41, 44. For the value of the talent, see C. Zaccagnini, "Aspects of Copper Trade in the Eastern Mediterranean during the Late Bronze Age," Traffici micenei nel mediterraneo: problemi storici e documentazione archeologica (eds.) M. Marazzi, S. Tusa, and L. Vagnetti (1986) 413-424.

[46]K. Petruso (1986) 28-29.

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This unit weighed just over three kilograms, equivalent to the wool unit of the Knossos tablets.[47] While it would be very interesting to speculate that this common element in wool recording shows some connection between the Minoans and the Hurrians, this common unit most probably derives from the average per-animal yield of wool. Dividing 2,892 grams by four sheep gives an average yield of 723 grams per animal. This amount fits well within weight ranges calculated from sheep flocks in Medieval Britain[48] and from modern Crete.[49] As Petruso concludes, 'It is evident that, in L.B.A. Crete, the bureaucracy had occasioned a shift away from the natural unit of reckoning wool (the yield of the individual animal) to the yield of four animals.'[50]

Types of Mycenaean cloth

Types of Mycenaean cloth can be listed in three ways: 1) by ligatured cloth ideograms, *i.e.* cloth signs with abbreviated Mycenaean words superimposed within them, 2) by other rather obscure signs that also resemble various types of cloth, or 3) by a full Mycenaean word either from the abbreviation in the above mentioned ligatures or as a non-ligatured type of cloth. These three types of cloth references are described below.

(1) The Linear B ideogram for CLOTH (Latin TELA = *159) occurs frequently in the L-series records with various numbers of fringes or tassels along the bottom.[51] This sign

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[47]See also the comparison of the *nariu* to the Mycenaean LANA in J. Melena (1987) 398–401.

[48]J. Drew, "Manorial Accounts of St. Swithun's Priory, Winchester," English Historical Review 62 (1947) 20–41 (*non vidi*, *apud* J. Killen (1984).

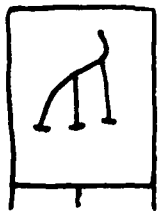
[49]K. Ktistopoulos, cited in D. Young, "Some Puzzles about Minoan Woolgathering," Kadmos 4 (1965) 113.

[50]K. Petruso (1986) 31.

[51]See Y. Duhoux (1974); J. Melena (1975); I. Tzachili (1997) 46–49. The number of tassels are indicated by a supra-script number after the TELA sign, *e.g.* TELA^2 would be the cloth ideogram with two central tassels hanging down.

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probably represents a warp-weighted, vertical loom or possibly a type of garment from such a loom. It is highly telling of an organized textile industry of long duration that this sign for cloth appears in all three Minoan scripts: Hieroglyphic Minoan, Linear A, and Linear B. TELA occurs in Linear B on its own and with five different ligatures.[52] Each of these variations indicates different types of textiles by an acrophonic abbreviation within the cloth ideogram:[53]



• TELA + PU is the most common of the ligatured ideograms, and is identified as Mycenaean *pu-ka-ta-ri-ja*, related to the Greek adjective πυκνός/πυκινός, 'thick, folded', perhaps meaning cloth that is of double-thickness or tightly woven. Large numbers of this cloth are listed, such as L 5561 which lists 980 pieces.[54] It occurs at Knossos, Pylos and Mycenae. On Mycenae tablet MY X 508 from the House of the Shields, *pu-ka-ta-ri-ja* cloth is being transported *te-qa-de*, 'to (Boeotian ?) Thebes'.[55] If this cloth is doubly-thick it may be a kind of felt, a fabric from matted wool. In describing various types of felt, Barber cites Swedish peasants who made caps from 'knitted felt...they knit a cap fully twice the size needed, and then felt it and shrink it down to half the size and *double the thickness.*'[56] This may compare to the

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[52]See also J. Melena (1975) 107-112.

[53]For a similar problem in discerning meaning of textiles from ideographic representations in the Near East, see K. Szarzynska, "Records of Garments and Cloth in Archaic Uruk/Warka," Altorientalische Forschungen 15 (1988) 220-230.

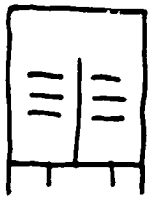
[54]L. Palmer (1963) 296.

[55]This maybe the only indication of extra-regional exchange or gift-exchange between palaces in the Mycenaean tablets. See analysis by T. Palaima (1991) 276-277. *PU* is also discussed by L. Palmer (1963) 295-296.

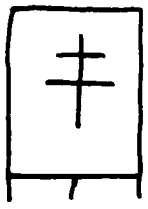
[56]E. Barber (1991) 216, n. 1, (emphasis is my own). For the various types of felt, see E. Barber (1991) 215-222.

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Homeric cloth ἡ χλαῖνα. In Il. 10.133-134, Nestor '...pinned about him a great vermilion mantel / sweeping in a double fold, with a thick fleece of wool upon it.'[57]



•TELA + TE is often paired with TELA + PA and is probably an abbreviation of *te-pa* cloth.[58] The ratio of wool delivered for the production of TE is one unit of cloth to seven units of wool, nearly four times as heavy as the *pawea* cloth described below. This cloth was probably a heavy duty textile like a shepherd's cloak or a rug. It is possible that this heavy cloth was also made of felt or felted after weaving. In Homer, ὁ τάπης occurs most often as thick purple bedding or it is used as a covering for furniture.



•TELA + PA stands most probably for the plural, *pa-we-a/a*$_2$, *pharwea*, identified with Homeric τό φᾶρος with the various meanings of 'a sail', 'a web' woven by Penelope, or 'a heavy cloak'. In Linear B PA refers to *pawea koura*, a common type of cloth in the tablets. Units of wool used to make this garment are in the ratio of five wool units for every three units of finished cloth (*i.e.* 1.66 units of wool for one PA cloth). Barber suspects that this was a heavy woolen cloth with three closed edges that could be used as a cloak, a blanket or a woman's peplos.[59] There are also references to *pawea* (KN L 594, *e.g.*) where it is linen instead of wool. If τό φᾶρος in Homer can also refer to a sail, it would have to be made of linen or some other bast fiber rather than wool.

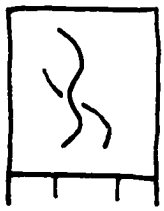
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[57]R. Lattimore (translation) Iliad of Homer (1961) 221.

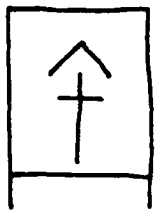
[58]J. Melena (1975) 108, fig. 11.

[59]E. Barber (1991) 358 n. 1.

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• TELA + KU is difficult to explain. No term beginning with KU has been found on the textile tablets and this is one of the least common textile ligatures. The KU sign might refer to *ku-do-ni-ja* cloth (*i.e.* cloth from or for 'Chania')[60] or possibly *ku-pi-ri-jo* ('Cyprus', 'Byblos').[61]



•TELA + ZO occurs rarely in Linear B. One suggested meaning sees a connection with Homeric ὁ ζωστήρ (Il. 4.132 and the verb ζώννυμι Il.14.181), 'a type of belt', 'to gird oneself'.[62] Melena, however, believes that since ZO also appears in Linear A it 'clearly indicates that both the cloth and the ideogram are of Minoan heritage. Any possible relationship of *zo-ta* with the Homeric *zoster* or *zostra* (Od. 6.38) must then be discarded.'[63]

(2) Signs *146, *160, *162, *164, *165, *166 and possibly *167 also depict textiles.[64]



*146 is a rectangle with a triangular or v-shaped mark on the upper side. It often has the *we* sign superimposed on it and can be fringed.



*160 occurs at Pylos in the La tablets and is very similar to *146 in shape.



*162 is shaped like a tunic with shoulder holes.

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[60]J. Melena (1975) 108.

[61]See T. Palaima (1991) 280-281.

[62]For men, ὁ ζωστήρ is modeled by Menelaus, as he is protected by Athena who deflects most of the iron-tipped spears hurled upon him, like a mother brushing away flies from a sleeping child. For women, it appears in the Iliad on Hera who wears one with one-hundred tassles in an attempt to seduce and distract Zeus. E. Barber, Women's Work: the first 20,000 years (1994) 42-70, examines belts of this kind all over Europe and compares the description of the 'string skirt' in Homer to the earliest figural representations of female figurines of the Paleolithic (20,000 years B.C.) through modern folk-costumes still worn in eastern Europe, all relating to displays of marriageability by young women through dress.

[63]J. Melena (1975) 111-112; Melena views Minoan as a non-Indo European language.

[64]See Y. Duhoux (1976) 116-132.

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*164 is a rectangle, sometimes fringed, with half-circles on all four sides (*cf.* KN L 520). The wool units show that this is a fairly heavy cloth of about six kilograms of wool per garment.



*166 is shaped like an ox-hide ingot and may be related to a Mycenaean kilt.[65]

The syllabogram *we*, which in some contexts indicates a product of the perfume industry, distinguishes *167 from *166.[66] The sign is superimposed over the ingot shape and may actually be some kind of (sheep?) hide and not a textile.

(3) The final method Mycenaean scribes use to list textiles is to write out the whole word Three types of cloth are most common in the target assessment records from Knossos, *pa-we-a, tu-na-no*, and *pe-ko-to* cloth. The abbreviation TELA + PA discussed above refers to this same *pawea koura*.[67] These terms are discussed below with the Lc tablets.

In addition to the various references to cloth described above, we also have color and status terms particularly associated with Mycenaean textiles. The word *po-pu*, related to Greek ἡ πορφύρα. 'purple', occurs on four tablets from Knossos.[68] Palaima suggests that since there are no convincing Greek or Near Eastern etymologies for ἡ πορφύρα. it is

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[65]J.-P. Olivier (1976) 117, 123-132.

[66]C. Shelmerdine, The Perfume Industry of Mycenaean Pylos (1985); *Eadem*, "Shining and Fragrant Cloth in Homer," The Ages of Homer (eds.) J. Carter and S. Morris (1995) 103-104; *Eadem*, "Workshops and Record Keeping in the Mycenaean World," TEXNH (1997) 390.

[67]J. Killen (1964) 9; (1966) 105-109; (1974) 82-90; (1979) 157, n. 13; J. Chadwick (1988) 82.

[68]Kn Lc(1) 561, L(7) 474 + fr., X 976 + 8263, and L 758. Palaima suggests that the absence of purple references on tablets at other sites may indicate that the Cretan textile industry had a monopoly on purple-cloth. For purple in Linear B, see T. Palaima, "Potter and Fuller: The Royal Craftsmen," TEXNH (1997) 407-412; *Idem* (1991) 289-291. This history of purple dyeing in the Aegean was explored fully in Chapter two. By the sixth century B.C. in Greece, purple dye was worth its weight in silver, according to Athenaeus 12.526. In the Diocletian price edict of 301 A.D., the price of Murex purple was equivalent to that of gold.

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possible that the word itself derives from Minoan.[69] Units of cloth are described on three of the four tablets attesting *po-pu*. One unit of TELA[1] + TE, two units of TELA[2] + PU, and 21 units of TELA[3] + PU are recorded. The fourth tablet refers to professional 'purple' women of *da-*83-ja* making so much of something *'wa-na-ka-te-ro'*, 'royal'.[70] There is no ideogram on this last tablet to indicate that cloth is being made or dyed but this seems likely.[71] While it is probable that this word *po-pu* refers to murex dyeing, the marine source is not indicated by the tablets and we know that many other dye sources yielded a similar color.

Large quantities of cloth and wool in the Lc target records (*e.g.* Lc 525) are also described as being *wanakteros* or in its abbreviated form, *wa*. There is, however, no suggestion that they are purple in color although this is possible.[72] One other indication that the Mycenaeans were using purple dye is found in the Ma taxation records. The last commodity listed in the Ma tablets, sign *13 (Shelmerdine's commodity F), is identified as ME, possibly related to Greek τό μέλι. 'honey'. In this context, however, the interpretation of honey is doubtful to Shelmerdine. She offers two alternative readings of the sign: Greek τό μῆλον, 'sheep/goats', or μελικῆρα. found in Aristotle as referring to the 'the dark spawn of murex' (HA 546.b.19).[73] It is possible that this refers to purple-dyed textiles. The high value of purple-dyed cloth would make this commodity unusual among the other staple

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[69]T. Palaima (1991) 289.

[70]J. Killen suggests that this place name, *da-*83-ja*, and *e-ra-ja* on tablet Lc(1) 561, also referring to purple, may be located in the area of Tylissos. See J. Killen, "Piety Begins at Home," Tractata Mycenaea (eds.) P. Ilievski and L. Crepajac (1987) 165-170.

[71]T. Palaima (1991) 291.

[72]Palaima also cites two tablets from Pylos that may be connected with the *wanax*. PY La 622 refers to a certain amount of a *wa-na-ka-to*, thought to be some type of textile. PY Na 334.a refers to land owned by the *wanax* that is free from the flax assessment. See T. Palaima (1997) 408-9.

[73]C. Shelmerdine (1973) 261-278.

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goods assessed in the Ma tablets so perhaps it is the raw dye, or snails? The commodity *146 also listed in the Ma tablets is almost certainly a cloth although probably not of high quality (*supra*). Even if the sign ME does refer to honey, this might also have been used in the purple-dye industry, since experiments have shown that sugar added to the stewing murex snail extract facilitates the oxidation of the dye.[74]

Whether the word *po-pu-ra* in the tablets actually refers to purple obtained from the murex snails so famous in antiquity is open to debate, but there is no doubt that imitations of this color made from cheaper plant dyes were exploited by the Mycenaean dyers.[75] We can be relatively certain that dyer's madder and alkanet were used for making red, blue, and purple shades. This imitation suggests that the purple color itself made textiles more valuable regardless of its source.[76] Linear B *po-ni-ki-jo* is another word that suggests some purple color. It is understood by Ventris and Chadwick as describing a spice [77] but is interpreted as dyer's madder by Astour.[78] Dyer's madder (*Rubia tinctorum*) is a plant for red or imitation purple-dye, local to the Levant and Egypt. To Foster, *po-ni-ki-jo* refers to alkanet (*Alkanna tinctoria*), another kind of root used for red-purple dye.[79]

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[74]Classical loom weights in the Athenian agora stamped with the words *glyk* or *mel* may refer to certain sugars added to purple dye mixtures. Alternatively, the stamped *mel* could refer to Aristotle's μελικηρα, certainly indicating murex dye.

[75]For a discussion of *po-pu-ra*, see M. Astour, "The Origin of the Terms 'Canaan,' 'Phoenician,' and 'Purple'," JNES 24 (1965) 349-50.

[76]See recipes of purple substitutes in O. Lagercrantz, Papyrus Graecus Holmiensis. Rezepte für Silber, Steine und Purpur (1913) (*non vidi*).

[77]Documents (1973) 222. Tablet Ga 418+ records men of Surimo and three kilograms of Phoenician spice (?), and 60 l. of coriander seed at Knossos.

[78]M. Astour (1965) 348-9. Melena first believed this word referred to olive oil and other sorts of oil in the Mycenaean tablets but now he identifies it as date palm.

[79]E. Foster, "*Po-ni-ki-jo* in the Knossos Tablets Reconsidered," Minos 16 (1977) 52-66.

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The use of roots, alkanet and madder, as dyes calls to mind the Linear B word *wi-ri-za*, Greek ἡ ῥίζα, 'root'. Still used in modern Greek for root, the diminutive, ῥιζάρι, refers to dyers' madder that grows wild in Greece, especially on Crete.[80] This word in the Mycenaean texts is often associated with wool and on one tablet associated with both wool and perfume (PY Un 249). Based on modern sheep terminology, the term *wi-ri-za* has been interpreted as the thicker, base-part of the wool fiber, the 'root' rich in lanolin.[81] Shelmerdine and others suggest that wool listed among raw materials used in the perfume industry was used for straining perfumed oil (*e.g.* PY Un 267, 249). Along with wool, *wi-ri-za* also appears on Un 249, transcribed and translated by Shelmerdine as follows:[82]

PY Un 249

.1a po-ti-ni-ja-we-jo

pi-ra-jo , a-re-pa-zo[-o] ku-pa-ro$_2$ AROM 2 T 5

.2 wi-ri-za LANA 2 [ ] *157 10

.3 vacat [ ] KAPO T 6

'Philaios the perfume-boiler of Potnia (the main deity of the Mycenaeans):
cyperus 240 l.
root wool 6 kg. *157 10 units
fruits 58 l.'

Rather than interpreting the root to refer to the oily base of the wool fiber, I suggest that it is related to the dyes from plant roots, dyer's madder or alkanet. Its association with an

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[80]E. Barber (1991) 232. See n. 10 for the etymology of alkanet, related to the Arabic word for henna.

[81]J. Killen (1962) 42, 70-71; Documents (1973) 224; L. Beck and C. Beck, "*Wi-ri-za* wool on Linear B Tablets of Perfume Ingredients," AJA 82 (1978) 213-215; C. Shelmerdine (1985) 18-20; Melena (1988) 406. The word also occurs at Knossos on Od 2026, Od 8202, always in connection with wool, but not with reference to the perfume industry.

[82]C. Shelmerdine (1985) 20. The underlined syllables refer to probable readings of the text, usually indicated with a dot underneath the syllable in Linear B scholarship.

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unguent boiler would be appropriate for dyes and we know that the Mycenaeans colored some of their perfume red because they are described as *po-ni-ki-jo*.[83] The wool modified by *wi-ri-za* may have been red and given color to the perfume, or the other way around, the dyed perfume added a red color to the wool.[84]

The Linear B term *tu-ru-pte-ri-ja* may also refer to dyeing textiles if we accept its identification with Greek ἡ στυπτηρία. 'alum,' used as a mordant to fix dyes.[85] The word occurs on PY An 35, Un 443, TI X 6, and possibly at Knossos on X 986. An 35 and Un 443 record the commodity, *tu-ru-pte-ri-ja* given 'in exchange for' wool, grains, figs and other commodities. That this good is being 'bought' is suggested by the Mycenaean word *o-no*, connected with the Classical Greek verb ὀνίνημι, 'to derive advantage'. Alum is a generic term for a naturally occurring material composed of sulphates of aluminium and potassium, with iron, manganese or magnesium sulphate, according to Forbes.[86] Found only in certain parts of the Mediterranean, such as Cyprus and Syria, alum was used in antiquity as a fire-retardant, a medicine, and was one of the best mordants for fixing certain dyes to textiles. The occurrence of alum in Linear B with the word *ku-pi-ri-jo* suggests that it may by an imported commodity from Cyprus, possibly in connection with the textile industry.[87] This is one more indication from the tablets that there are wealth goods being bought or exchanged, valued in terms of one another.

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[83]C. Shelmerdine (1985) 23.

[84]The connection between oil and cloth is discussed below.

[85]C. Shelmerdine (1985) 136-137; L. Baumbach, "Mycenaean *tu-ru-pte-ri-ja* and Herodotus II.180," Tractata Mycenae (eds.) P. Ilievski and L. Crepaja (1987) 49-54; E. Barber (1991) 276.

[86]R. Forbes, Studies in Ancient Technology IV (1964) 134-5.

[87]See C. Shelmerdine (1985) 137; E. Barber (1991) 238.

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Part II.A - Textile Production at Knossos

Linear B tablets from Knossos describe flock management (tablet sets Da-Dg, Dn), sheep shearing (Dk/Dl), and wool allocations (Od) recorded by the palace scribes for outlying areas of the kingdom.[88] Wool as a raw material was distributed to textile workers with the expectation that set production targets would be met with finished textiles delivered back to the palace. This system of distribution and requisition is known as the *ta-ra-si-ja* system.[89] The word *tarasija* occurs in later Greek as ἡ ταλασία where it is only associated with textile work, especially with wool working.[90] In Mycenaean times the word is attested in bronze smiths' obligations at Pylos, to materials used in making wheels, and in textile industries of Pylos, Knossos, and Mycenae.

At Knossos the Lc(1) and Lc(2) sets record the completion of production targets for the textile workers. Once these targets were met, the laborers, predominantly women, would receive food rations listed in another set of tablets, the Ak series.[91] By cross-checking sheep-shearing documents, target accounts, and workers' rations with various place-names and ethnic descriptions, a better understanding of the textile industry on Crete is possible.

Concluding the manufacturing stages were the delivery, receipt, and decoration of finished textiles listed in the Ld, Le, and Od series. Other smaller groups of the D series, *e.g.* Dv, Dw, and Dx, are too fragmentary to be assigned to more specific classes but do refer in some way to the wool industry. Some sheep are also recorded in the C series although these

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[88]J. Bennet (1988) 27; P. Halstead (1991) 343-366.

[89]For a recent discussion of the *tarasija* system, see M-L. Noch, "Acquisition and Distribution: *ta-ra-si-ja* in the Mycenaean Textile Industry," Proceedings of the Trade and Production in Premonetary Greece Conference VI in Athens, December 1996 (1997) *ms*. I am very grateful to M.-L. Noch (nee Gregersen) for sending me her manuscript before publication.

[90]E. Barber (1991) 265.

[91]See. M.-L. Gregersen, "Pylian Craftsmen: Payment in Kind/Rations or Land?," TEXNH (eds.) R. Laffineur and P. Betancourt (1997a) 397-405.

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tablets are also fragmentary. The decorating stage of the textiles was fairly complicated and of obvious importance to the ruling elites. Small allotments of wool were made to various specialists who edged and finished cloth.

D-Series

Of all the Linear B tablets from Knossos the largest group is the D-series, recording an estimated 100,000 sheep under the administration of Knossos. Sixty thousand of these are estimated to have been used in the wool industry. References to these sheep and their wool occur on over 1,100 separate tablets.[92] Most were found in the upper east-west corridor at Knossos in the East Wing of the Palace, formerly known as the Domestic Quarter. While the number of sheep is very large, according to modern pasturalists these sheep could have been accommodated within one-fourth to one-third of the total land area of Crete.[93] Most likely these sheep were located in the central and western parts of the island since we have no indication that Knossos controlled eastern Crete in LM III.[94]

Halstead and others have investigated the evidence for restocking the flocks under palatial control at Pylos and Knossos.[95] While the evidence may be a result of poor preservation of certain tablets, there is some suggestion for the role of the non-palatial sector supplying the palace with sheep. Using analogies with English medieval sources, Killen has argued that at five to six years of age sheep under palace control at Knossos would have been

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[92]J. Hooker (1980) 88; Killen (1964) 5, n. 23 states there are over 800 tablets; J.-P. Olivier, "La série Dn de Cnossos," SMEA 2 (1967) 89; P. Halstead (1991) 343.

[93]P. Halstead, "From Determinism to Uncertainty: Social Storage and the Rise of the Minoan Palaces," Economic Archaeology (eds.) A. Sheridan and G. Bailey (1981) 204.

[94]J. Bennet, "The Wild Country East of Dikte: The Problem of East Crete in the LM III period," Studies in Mycenaean and Classical Greek Presented to John Chadwick. Minos 20-22 (1987) 77-88.

[95]P. Halstead (1991) 343-366; (1992) 68.

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replaced.[96] This is at a rate of 17-20% of the total each year, amounting to 10-12,000 new sheep. As Killen notes, of tablets that refer to the age of sheep a remarkable 19% of them indicate 'yearling' perfectly in line with replacement estimates of 17-20%.[97] If 10-12,000 new sheep were added each year, that would mean the same number died or were removed by some means from the palace flocks. The tablets at Knossos that list missing (*o.*) or old (*pa*) sheep, however, do not record numbers anywhere near 10,000. At Pylos the tablets do record numbers of old sheep replaced by young ones suggesting that this discrepancy is attributable to the time of year the tablets (and palaces) were destroyed, early autumn for Knossos and late spring for Pylos.[98]

In the records that indicate the sex of the sheep, the ratio of males to females is eight males for every one female in the palace flocks.[99] Assuming a 1:1 sex ratio at birth and an optimistic 100% lambing rate, Halstead estimates that 18-22,000 female sheep (breeding ewes) were needed to complete the sheep population listed in the Da-Dg tablets. This is much greater than the number of ewes recorded in the tablets. To Halstead, this suggests that there was a substantial number of sheep contributed from outside the palace sector to replenish their flocks.[100] The Mycenaean Cretans were also keeping records of large amounts of wethers and a much smaller percentage of female ewes, some for breeding purposes, others not.

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[96]J. Killen (1963) 87.

[97]J. Killen (1963) 88.

[98]L. Godart (1977) 39-40. This time of year also accords well with what we know of the flax harvesting. See below.

[99]J.-P. Olivier (1967) 81-2.

[100]P. Halstead (1992) 68.

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Da-Dg Sets

Ideograms of sheep in the Da - Dg subsets list large numbers of wool-producing animals pasturing in central Crete.[101] Even before Linear B was deciphered, different interpretations of these large lists of sheep were put forward. Sundwall suggested that they were part of sacrificial hecatombs for a divinity.[102] Others suggested sheep as tokens of exchange, *i.e.* a form of currency similar to cattle used in Homeric society. Ventris and Chadwick believed that the records were tribute by Minoan subjects to the Cretan overlord.[103] Killen's arguments against the Da-Dg records being tribute lists are most convincing, based primarily on the RAM ideogram.[104] If the ideogram of the tablets reflected actual RAMS, *i.e.* uncastrated male sheep, it is unlikely that they would have been run together with EWES, also listed but in the minority.[105] In sheep rearing, the normal ratio of ewes to mature rams is not that reflected in the Mycenaean tablets. Usually one ram can service up to fifty or sixty ewes for breeding purposes.[106]

Killen suggests that the thousands of rams listed by ideogram are actually wethers, or castrated males. Citing a parallel with the BULL ideogram where the intactness of the animal is not indicated, Killen believes bulls identified by the adjective *we-ka-ta*, ἐργατικός, meaning 'working', were castrated. A whole bull would be much more difficult to work as would a complete sheep compared to castrated animals. There are also Akkadian records

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[101]For a thorough analysis of these documents, see J.-P. Olivier (1988) 219-267.

[102]J. Sundwall, "Altkretische Urkundenstudien," Acta Acad. Abo., Humaniora, 10.2 (1936) 1-45 (*non vidi*).

[103]Documents (1956) 198.

[104]J. Killen (1964) 1-4.

[105]J. Killen (1964) 2.

[106]J. Killen (1964) 9. See also J. H. Thomas, Sheep 3 (1957) 57. Indications are that in medieval England a much higher number of breeding ewes was serviced by a single male. K. J. Allison, "Flock Management in the Sixteenth and Seventeenth Centuries" in Economic History Review xi.1 (1958) 104.

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from Mari that refer to both castrated and entire male sheep by a single term, UDU.NITÁ.[107] This suggests that the Mycenaeans could have meant both castrated and whole animals with the same sign. The advantages to keeping wethers are many. Compared to ewes, wethers are much more resistant to diseases and they yield substantially greater amounts of wool.[108]

Ventris and Chadwick believed that the Da-Dg lists were records of tribute assuming that sheep owners were simply assessed a contribution of so many sheep to the palace, regardless of the animals' sex. To a shepherd, rams would be the least valuable member of their flocks and their first choice for tribute. Ventris and Chadwick state that if these records were only a census of flocks, it is unlikely that they would have round numbers and a record of missing sheep.[109] But if the Da-Dg tablets refer to predominantly male wethers and female sheep, it is still possible that these sheep were grouped in round numbers as tribute to the central palace.

Using modern sheep populations of approximately 530,000 on Crete, Killen has shown that extracting 100,000 sheep as tribute each year would be neither possible nor practical.[110] That the palace would welcome so many rams as tribute is equally unlikely. Rams do not contribute milk and their flesh is inferior to ewes or castrated males (wethers), as is their wool. In addition, correspondences between references on the Da-Dg tablets and other D-series texts, such as the Dk-Dl records, suggest that these sheep were not tribute but were part of an organized textile industry. The D-series tablets are most likely a kind of

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[107]J. Killen (1964) 4; J. Bottéro, Archives royales de Mari VII (1957) 248; and B. Landsberger, "Studien zu den Urkenden aus der Zeit des Ninurta-tukul-Assur," AfO 10 (1935) 155.

[108]In recent times, however, emphasis has been placed on meat, especially lamb, and therefore wether flocks are rarely kept, and male lambs most often slaughtered.

[109]See also M. Finley, "The Mycenaean Tablets and Economic History," Economic History Review x.1 (1957) 130.

[110]J. Killen (1964) 4.

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census or inventory of the palace's holdings and not a record of tribute owed to the palace.[111]

Tablets of the subset Da-Dg list only sheep, the majority of which are by one scribe (hand 117) and show sheep distributed widely over central Crete.[112] The Da set numbers rams from specific toponyms while the Db tablets list rams and ewes. Only rams are listed in the Dc tablets by either the male sheep ideogram alone or with an adjunct syllable that qualifies the sheep in some way.[113] Finally, the Dd, De, and Df sets record rams and ewes also either plain or with adjuncts. I include a few examples:

KN Da 1221

ai-ta-ro-we / RAMS 200[

pa$_3$-ko-we [

KN Dd 1171

RAMS 20 EWES 72

po-ro / pa-i-to pa. RAMS 8

The tablets are all very similar and each one records a separate entry beginning with what is presumed to be a shepherd's name in tall characters. The tablet then usually divides horizontally into two lines. On 31% of the tablets the top line names a 'collector' in the nominative or genitive.[114] A round hundreds number, or a series adding up to a round total

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[111]Documents (1956) 197-200; J. Killen (1964) 9.

[112]See J. P. Olivier (1988) 219-267. The records have been most recently examined by P. Halstead showing the detailed care and thorough understanding of sheep breeding the Mycenaean administrators had; P. Halstead (1991) 343-366.

[113]See J. Melena (1975) 190ff.

[114]For a thorough review of the subject, see J. Bennet (1992) 65-101.

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of RAMS and/or EWES, is listed by ideograms, with the great majority (*ca.* 87%) being RAMS.[115] The lower line gives a place name and information regarding missing, old, or young sheep, most often RAMS again.

The description of some rams being 'missing' or 'owed' is not uncommon in the Linear B corpus. This absence is indicated by *o.* before the commodity and the total number missing. Most likely this was an abbreviation for *o-pe-ro* [*-me-no?*], related to Greek, ὄφελλον, 'deficit'.[116] Killen states that while this term may have already had its later meaning of 'something owed' on some of the tablets, it may, in the context of the sheep records, have possibly two different meanings:

1) It could refer to something that is not there but should be, as in an inventory.

2) It may refer to something lacking which needs to be replaced, such as sheep totals under the palace administration.[117] This could mean that for the flock to be at full strength, the number of '*o.*'-sheep had to be supplied by the palace.

Dk - Dl Sets

Each Da-Dg tablet recorded large numbers of wool producing sheep. The target and the actual wool yield from the Da-Dg sheep were recorded in the Dk and Dl tablets.[118] Killen estimates approximately 30 to 50 tons of wool were produced by these sheep annually for the palace workshops.

The Dk records have been divided into two sets based on the two scribes who wrote them. Dk(1) was written by h. 120 while Dk(2) was by h. 119. It is generally believed that

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[115]P. Halstead (1991) 344.

[116]J. Chadwick (1988) 46.

[117]J. Killen (1964) 8.

[118]J. Killen (1984) 50.

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these tablets are shearing records showing both the target and actual yield of wool from the palace flocks. One peculiar feature of these tablets is that deficits in wool are almost always noted, but never a deficit in rams.

There are correspondences between the Dk and the Da-Dg records. These include toponyms and a single 'collector' name also found in the Da-Dg series. Some of the same flocks listed in the Da-Dg series also reappear in the Dk tablets, such as Dk(2) 1071 corresponding to the flock on Dc 5228.[119] The flocks with female sheep are set the same wool targets as flocks exclusively made up of wethers.[120] Male sheep dominate in the Dk records so it is likely that these refer to wool flocks only, and not to breeding flocks.

In the Dl series the female flocks are assessed a reduced wool target (one unit of wool per five ewes plus five lambs) indicating that they were used for breeding purposes. At least 1939 ewes are listed, with the majority in subset Dl(1), by hand 118. Dl(1) totals a minimum of 1503 ewes with reduced wool production targets and lambs at a rate of one lamb per ewe. Both the Dk and Dl sets record sheep and wool in fixed ratios; the Dl texts have ten sheep per one unit of wool, the Dk texts have four sheep per unit of wool.[121]

The Lc Tablets

The L-tablets refer to the receipt and delivery of cloth from outlying villages. In association with terms for cloth, words such as *a-pu-do-si*, from Greek ἡ ἀπόδοσις, 'restitution, delivery', and *o-pe-ro*, ὄφελλον, 'deficit' occur in this series. The raw material

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[119]J. Killen (1984) 49.

[120]P. Halstead (1991) 345.

[121]J. Killen (1964) 4. The absolute values of symbols of weight in the Linear B records is fairly secure. One unit of wool weighs roughly 3 kg, or 6.4 lbs., thus one ram is expected to yield 1.6 lbs. The equivalent standard unit of wool in Medieval England and second millennium Alalakh is roughly the same amount. See K. Petruso (1986) 26-37.

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listed in the Dk and Dl records (wool) finds corresponding targets of cloth production recorded in the Lc series.[122] These tablets refer to various types of cloth expected from various work groups and provide the most detailed information about the amounts of wool used in Mycenaean textiles and their decoration. Most of the Lc texts generally begin with a feminine ethnic or occupational term written majuscule (in large characters), *e.g. se-to-i-ja* (*cf.* Lc 525), *da-wi-ja* (*cf.* Lc 526), *i.e.* 'the women workers at Siteia' or 'at *da-wo*'.[123] Variations of this pattern do exist. For example, some tablets might have a place name written large followed by a small occupational description, such as *ko-no-so te-pe-ja*, 'the (female) *te-pa* makers at Knossos' (*cf.* L 641). This pattern recurs on tablets by the same scribe in the personnel records of the Ak tablets. Following the patterns established by the Ak set, Killen reasonably infers that some of the Lc variants, *i.e.* tablets listing individuals in the genitive such as Lc 551 *e-me-si-jo-jo*, are the 'owners' or 'collectors' of the textile personnel.[124]

After the opening signs, the Lc tablets most often divide into two lines. The upper line will read either 'plain' TELA, TELA + TE (as on Lc (1) 526) or both. 'Plain' TELA is always preceded by the phrase *pa-we-a ko-u-ra*, describing a generic kind of textile. The TELA ideogram is then followed by a wool entry. TELA + TE is listed once or twice and also followed by a wool entry. The second TELA + TE entry, when it occurs, always has an extra stroke in the characteristic 'fringe' at the bottom of the sign and the first ideogram is always preceded by the qualifications *pe-ko-to* (abbreviated to *pe.* on 551).[125] This type of cloth, *pe-ko-to*, is most probably related to the Mycenaean professional force of *pe-ki-ti-ra*$_2$,

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[122]See J. Killen (1966) 105-109; (1974) 85-86; (1988) 167-184; M.-L. Noch (1997).

[123]For a more thorough review of the Lc tablets see J. Killen (1966) 105-109.

[124]J. Killen (1965) 105. On the subject 'owners' see above.

[125]J.Killen (1965) 106.

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women who combed wool off of sheep for spinning.[126] Barber further suggests that this was a luxury cloth of soft, fine, kempless wool. The lower line of the Lc tablets, when inscribed, begins with 'plain' TELA qualified as *tu-na-no* (see 526) followed by a wool entry.

The Lc tablets fall into two groups: Lc(1) by hand 103, and Lc(2) with rectos by hand 113 and versos by hand 115.[127] Killen has suggested that these two sets deal with different geographic areas of Crete: the Lc(1) tablets refer to central Crete, including laborers at Knossos, Phaistos, and Tylissos; the Lc(2) tablets record groups in the far west, at Kydonia (Chania) and *si-ra-ro*.[128] The Lc(1) records list two main types of cloth, TELA + TE, and the 'plain' cloth ideogram, without fringes, described as *pa-we-a ko-u-ra*. Both subsets record the targets, or production quotas, established for various work groups of women and children textile workers also listed in the personnel tablets, Ak and Ap, discussed below. Often these two different series (the Lc targets and the A personnel tablets) refer to different aspects of the economy (craft production and personnel maintenance) but are written by the same scribe for the each work group and relate to the same target of production.[129] For example:

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[126]E. Barber (1991) 262, 283; J. Chadwick (1988) 82.

[127]J. Killen (1966) 105-109. Scribe KN 115 has recently been the subject of intensive study because of the newly discovered tablets at Chania which were, at first, attributed by J.-P. Olivier to the same scribe at Knossos, 115; J.-P. Olivier, "KN 115=KH 115" BCH 117 (1993) 19-33. This would have had great significance in the chronological debates surrounding the final destruction of Knossos. The attribution has, however, been retracted for various reasons. See T. Palaima, "Ten reasons why KN 115≠ KH 115," Minos 27-28 (1993) 261-281.

[128]J. Killen (1976) 121; (1988)167.

[129]See C. Shelmerdine (1992) 569-590.

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KN Lc(1) 540 + 8075 (103)

.1 'pa-we-a' ko-u-ra TELA[1] 3[

.2 da-te-we-ja / [

.1 3[: arrangement suggests 9.

KN Ak (1) 612 (103)

.1 *TA* 1 *'DA* 1' MUL 9

.2 ko-wa ,/ me-zo 1 ko-wa / me-u-jo 1

.3 da-te-we-ja ko-wo / me-zo 1 [[ko-wo me-]]

Scribe 103 writes on Lc(1) 540 + 8075 a target record for probably nine units of *'pa-we-a ko-u-ra'* cloth for the workers of *da-te-we-ja*. Hand 103 also writes tablet Ak(1) 612 recording nine adult women and three children workers, described as *da-te-we-ja*.[130] Killen suggests that these workers were expected to produce the same nine units of cloth referred to in Lc(1) 540 + 8075. The corresponding numbers of nine units of cloth in Lc(1) and the same number of female workers in Ak(1), both at *da-te-we-ja*, indicates that these tablets refer to the same workers. This implies the ratio of one unit of cloth per Ak worker, most probably a dependent female weaver. Further, the Lc(1) tablets record approximately 1000 cloths, suggesting that the total Ak work force was about the same, 1000 workers.[131]

M.-L. Noch states that this theory of Killen's that estimates the targets and workforce may be correct but it cannot be confirmed by the other tablets.[132] The poor preservation of the Lc and Ak tablets precludes proof as does the discrepancy between targets for

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[130]For the meaning of DA and TA, see J. Killen, "TA and DA," Concilium Eirene (eds.) P. Oliva and A. Frolikova (1982) 121-126 and J. Chadwick (1988).

[131]J. Killen (1984) 52 note 10.

[132]M.-L. Noch (1997). I am extremely grateful to M.-L. Noch for sharing so generously with me her research on textile labor in Linear B.

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professional groups and targets for ethnics. *Pa-we-a* is mainly produced by professionals and TELA + TE is made most often by personnel identified soley by ethnicity. Noch states that records for TELA + TE show no correspondence with the number of women workers in Ak. In her investigation of the target assessments, Noch has identified a pattern where the targets fall into a general interval between 150-200 units of wool for the ethnic groups and are adjusted slightly based on the labor force. A similar target pattern emerges with professional groups although with a smaller number of target units, *i.e.* 15-28 units of wool. Noch suggests that this may be explained if we take the *tarasija* targets as only one part of the professional groups' duties.[133]

The Lc(2) 'stint' records by h. 115 take a standard form.[134] The recto sides are always by h. 113 and record *pa-we-a* TELA and wool units from western Crete. The term *to-u-ka* occurs on some of these tablets. This word has been analyzed by G. Björck who suggests it corresponds to the Homeric verb τεύξειν, 'to make, render'.[135] The participial form, *te-tu-ko-wo-a*, is also found on a WHEEL record at Pylos, Sa 682, and, as with the textile records, probably refers to objects that were 'decorated' or 'finished'. Since all the cloth on the rectos of the Lc tablets by h. 113 are identified as 'plain' TELA and the wool listed there is presumably for the manufacture of those cloths, the wool on the verso that is described as *te-u-ka*, by h. 115, is most likely that wool used to decorate or finish the cloth.[136] Killen also suggests that there is a fairly constant ratio in the wool used for

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[133]M.-L. Noch (1997).

[134]See J. Killen (1979) 163 for recto by 113.

[135]The word appears five times on the reverse of the Lc tablets. G. Björck, "Pour le vocabulaire des tablettes 'à bannieres' de Knossos," Eranos 52 (1954) 275.

[136]J. Killen (1979) 164.

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decorating (on the verso, by h. 115) and the number of units of cloth (on the recto, by h. 113), *i.e.* a ratio of 1:4.

The best source of information about the most common types of Mycenaean cloth are found in the two totaling tablets of the Lc(1) series, Lc(1) 535 and Lc(1) 536.[137] One tablet totals the amounts of cloth for 'collectors' (535), the other for 'non-collectors' (536):

KN Lc(1) 535 + 538 F10[138] (103)

.1 ta-ra-si-ja pa-we-a [

.2 ke-ri-mi-ja tu-na-no [

.3 to-sa / pe-ko-to [

.3 trace at right consistent with TELA[.

KN Lc(1) 536 [+] 7383 + 7731 F10 (103)

.1] 'ta-ra-si[-ja']pa-we-a[ ]TELA^{x} 200

.2] 'vest. ['] tu-na-no[ ]TELA^{1} 48 a-ro-zo 'ki-to' TELA1 1

.3 to-]sa / pe[-ko-to TELA^{1}+TE]18 TELA$^{?}$+TE 267

Since Lc(1) 536 lists a total of cloth less than one of the single collector targets (Lc(1) 7392), it is likely that this is a total of non-collector cloth.[139] Lc(1) 535 + 538 should therefore be the fragmentary remains of the collector totals. Of the three types of cloth common to both

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[137]See the analysis of these tablets in M.-L. Noch (1997).

[138]This is the findspot as established by J.-P. Olivier.

[139]J. Bennet (1992) 71; J.-P. Olivier (1967) 91; J. Killen (1974) 85-86. M.-L. Noch has also noted that the amounts of the first four types of cloth listed (200 + 48 + 1 + 18) add up to the total of the final TELA + TE cloth, 267. What this means exactly is difficult to say. It is possible that it is just a coincidence.

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tablets, *pa-we-a*, *tu-na-no*, and *pe-ko-to*, Killen has determined how much wool was necessary for each type of cloth:[140]

pa-we-a: LANA	=	3:5
tu-na-no: LANA	=	1:3
pe-ko-to: LANA	=	1:10

TELA + TE, as was mentioned above, is made up of seven units of wool for every one cloth unit.[141] This gives us a total amount of wool from the cloth listed in Lc(1) 536 for the non-collector targets of 2,529 units.[142]

Restoring the total targets for the collector tablets of the Lc series is more difficult. Referring to a related target record of *pa-we-a ko-u-ra* cloth Killen suggests that Ld(1) 587 records a total of 453 cloth units.[143] Building on this theory M.-L. Noch has recently used the individual quotas of collectors for the other types of cloth (*tu-na-no*, *pe-ko-to*, and TELA + TE) to get an idea of the total collector targets listed in Lc(1) 535.[144] A restored total collector target of 915 units of wool is proposed.

Adding the collector target total to the total of wool units from the non-collector targets (Lc(1) 536) [*i.e.* 2,529 units] we get a total of 3,444 units of wool calculated in the *a-ra-si-ja* tablets. The percentages for collectors and non-collectors of this total for wool units corresponds to the estimated percentages of collectors vs. non-collectors in the sheep tablets

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[140]J. Killen (1964) 9; (1966) 105-109; (1974) 82-90; (1979) 157, n. 13.

[141]E. Barber (personal communication) has pointed out that if one unit of wool weighs approximately three kilograms, a seven-unit TE cloth would weigh about 46 lbs. - a very heavy textile. One *pe-ko-to* cloth would weigh over 65 lbs.

[142]This sum is calculated by adding the following: PA, if 5/3 = x/200 then x = 333; TU 48 x 3 = 144; PE 18 x 10 = 180; TE 267 x 7 = 1869, plus one chiton of 3 units of wool equals 2529 total wool units.

[143]See below the section on the Ld tablets for this tablet.

[144]M.-L. Noch (1997) *ms.* 3. She uses totals from Lc(1) 55, Lc(1) 7398, and Lc(1) 532.

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listed in the Da-Dg series, *i.e.* about 30% collectors and 70% non-collectors.[145] The total of 915 wool units for collectors is 26.6% of the 3,444 units of wool; 2,529 is 73.4% for collectors, showing that the restored targets for the collectors are not too far off in comparison to the sheep totals.

Based on the *ta-ra-si-ja* target tablets, Noch has suggested that certain types of cloth are produced more often by different types of work groups, collectors and non-collectors. Collectors have been discussed but the non-collectors deserve some mention. Non-collector work groups are identified by either their profession or some form of the name of their village. The latter type is referred to as an ethnic. Of the major types of cloth, Noch states that *pa-we-a* is mostly made by collectors or professional groups; *pe-ko-to* by collectors and non-collectors listed by ethnic names; *tu-na-no* is made by all groups, collectors and the non-collector professionals and ethnics. One type of cloth *pa-we-a ko-u-ra *161* is made only by non-collector professionals.[146]

Ld Tablets

The Ld delivery records are divided into two sub-sets, a major and a minor one.[147] The major group totals cloth in fairly large numbers, including Ld(1) 587, which most probably records the totals from all Ld tablets. The minor set has smaller numbers and uses terms not included in the major set for cloth, such as *e-ta-wo-ne(-we)* and *o-nu-ka*. The store tablets number 14, with only seven of them preserving cloth figures. From these there is an

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[145]This line of reasoning is M.-L. Noch's. She follows estimates of sheep totals on the Da-Dg tablets by Olivier.

[146]M.-L. Noch (1997) 4.

[147]J. Killen (1979) 151-181.

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average of 27 cloth units on each tablet, giving a projected total of 378 units for the set of 14 tablets.[148]

The Ld(1) tablets are all by hand 116 and refer to either the delivery of cloth with its source indicated or to the storage of cloth, without a source given. Most probably these two types, the delivery tablets and storage records, refer to the same cloth but in different phases of administration. The tablets refer to cloth of the same varieties and in the same relative quantities, with the most common type of cloth being that described as *re-u-ko-nu-ka*.[149] Other descriptions include *po-ki-ro-nu-ka* or *ko-ro-ta*$_2$. Interestingly, these epithets are not found in the Lc(1) work group records.[150] From these documents it can be shown that many of the delivery tablets did not come directly from the weaver work groups but, more probably, they came from decorators or finishers who in turn sent on the completed textiles to the palace.

In the Linear B tablets from Knossos *o-nu-ke* is written by two scribes, h. 103 and 115. On tablets by h. 115 we find *o-nu-ka* followed by LANA, in parallel with the term *ko-ro-to*.[151] If we accept that Linear B *o-nu-ka* corresponds to Greek ὁ ὄνυξ, 'nail, claw', it may refer to either an item of decoration or to a type of wool, since it is followed by LANA. This explanation also follows for *ko-ro-to*, probably in the nominative as well. If these decorative terms are compared to *po-ki-ro-nu-ka*, *re-u-ko-nu-ka*, and to *ko-ro-ta*$_2$, which are followed by the TELA ideogram, it seems likely that these refer to cloth decorated with

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[148]J. Killen (1979) 151-153.

[149]See J. Killen and J.-P. Olivier, "155 raccords de garments dans les tablettes de Cnossos," BCH 92 (1968) 119.

[150]See below.

[151]J. Killen (1979) 162-3.

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woollen *o-nu-ka* or *ko-ro-to.*[152] The term *po-ki-ro-nu-ka* is related to Greek ποικίλος + ὄνυξ, 'multicolored edgings,' while *re-u-no-nu-ka* probably signifies white colored edging.[153] Killen suggests that this wool was used to make ὄνυξ-decorations, wool not as yet woven into bands but likely to be used by the finishers, as an elaboration of the cloth.

One might wonder why some cloths would be described as simply *o-nu-ka, i.e.* with decorative edging and the others with a more specific *po-ki-ro-nu-ka*, or *re-u-ko-nu-ka.* Killen notes that the pattern in all the Ld(1) delivery tablets is first a cloth entry described by an ethnic or trade-name and then the specific *po-ki-ro-nu-ka* or *re-u-ko-nu-ka.* Following this entry is the formula *pa-ro e-ta-wo-ne(-we)'* and then the *o-nu-ka* description.[154] Killen suggests that *e-ta-wo-ne-u* was a finisher of cloth and that the cloth *'pa-ro e-ta-wo-ne(-we)'* refers to cloth *'chez* Etawoneus' workshop', not completely finished with any specific type of decoration, *i.e.* either the variegated or white-banded embellishments left undistinguished, but bound to have some form of ὄνυχες.[155]

Other types of cloth indicated in the Ld records include *e-qe-si-ja* and *ke-se-nu-wi-ja pa-we-a* TELA. That these terms are written parallel to one another on Ld receipt tablets suggests that their meanings are related.[156] *E-qe-si-ja* derives from *e-qe-ta,* probably from

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[152]Documents (1973) 556 suggest that *ko-ro-ta*$_2$ is connected with the Greek κροσσοί, 'tassels' or 'fringes'.

[153]See E. Barber (1991) 324-327.

[154]J. Melena (1975) 84 also notes the correlation between *o-nu-ka* and *e-ta-wo-ne (-we).*

[155]For a detailed critique of Melena's suggestion that the terms *e-ta-wo-ne* differs from *e-ta-wo-ne-we*, the former being a textile material and the latter a specialized craftperson, see J. Killen (1979) 159-60 n. 17. Killen suggests a scribal error in Xe 5540 for *e-ta-wo-ne*, which should read *e-ta-wi-ne-we.* C. Shelmerdine has also noted that this decorator, Etawoneus, on tablet KN Xe 7711, receives textiles back from a perfumer who has treated the cloth with scented oil and then returned it to the decorator. See C. Shelmerdine (1995) 103.

[156]Compare, for example, KN Ld 571 and Ld 573.

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Greek ἕπεἰσθαι, 'to follow'.[157] These individuals are often referred to as 'the Followers' and seem to be noblemen who acted as liaison officers between the military contingent and the Mycenaean palace.[158] They have fairly high status in the Mycenaean hierarchy, below the *wanax* and *lawagetas*, and probably acted as leaders on military expeditions as indicated by the so-called '*oka*' tablets, *e.g.* PY An 654. This tablet lists groups of 50 and 60 unnamed men under certain *e-qe-ta*. Cloth described as *e-qe-si-ja* often varies in color and decoration but is sometimes described as *a-ro$_2$-a*, 'of better quality', further suggesting a high status for the *e-qe-ta*.

The term *ke-se-nu-wi-ja* is used to modify 35 units of *re-u-ko-nu-ka pa-we-a* cloth on KN Ld 573 and also describes cloth on Ld 649. This term, *ke-se-nu-wi-ja*, is thought to relate to Greek ὁ ξένος, 'guest-friend', or ἡ ξεχνία, 'hospitality', perhaps meaning cloth that is meant for 'foreigners', 'guest-gifts', or 'for export'.[159] Driessen, however, has suggested that the context of the word, in parallel construction with *e-qe-si-ja* and the military associations of this term, implies that *ke-se-nu-wi-ja* is also related to warfare. Citing the later Greek usages of *xenoi* in Homer (Il. 24.202 and Od. 14.102), Herodotus (I.77), Thucydides, Aristophanes (Ploutos 173) and Xenophon (Anabasis 1.10; 1.2.1), Driessen argues that *ke-se-nu-wi-ja* cloth in Linear B refers to mercenary soldiers.[160] Cloth described as *ke-se-nu-wi-ja* is always the same type, *pa-we-a* cloth with red patterns and white fringes, unlike the varied *e-qe-si-ja* cloth. This quite possibly indicates payment in cloth or a

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[157]See S. Deger-Jalkotzy (1978).

[158]J. Driessen, "Mercenaries at Mycenaean Knossos," BSA 79 (1984) 49-56.

[159]Documents (1973) 318.

[160]J. Driessen (1984) 55-56. Driessen also cites the Knossos fresco, the "Captain of the Blacks" as iconographic evidence for the use of foreign military service at Knossos. See A. Evans, The Palace of Minos ii (1921-1936) 755-57, pl. 13.

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standardized mercenary uniform for the hired soldiers. The word is also found on oil tablets (PY Fr 1231, 1255) and implies that hired foreign warriors were supplied with garments and oil by the Mycenaean palaces during the very troubled times of the Late Bronze Age. The association of cloth with mercenary soldiers further supports the wealth finance model, where certain types of cloth have such a standardized value that they can be used as payment for service.

As was mentioned above, the storage and delivery tablets only mention one type of cloth, 'plain' TELA. The store tablets also record cloth qualified as *pa-we-a*, referring to the same *pa-we-a ko-u-ra* of the Lc(1) records. Killen and Melena have both stressed that *pa-we-a* does not simply describe cloth generally, but that it is a specific variety of textile.[161] Only two ideograms are associated with *pa-we-a*, either 'plain' TELA, *i.e.* cloth without fringes or ideograms, or TELA + PA, (*pa* stands most probably for *pa-we-a*). This is confirmed by a tablet from Pylos showing TELA + PA preceded by *ko-u-ra*.

Le Tablets

Another possible corresponding set of tablets to the Lc(1) records is found in the Le series, also by hand 103. These tablets record the receipt of cloth most probably from the same work groups listed in the Lc(1) tablets.[162] An example of a corresponding Le tablet is:

KN Le 641 + frr. F14 103

.1 o-a-po-te , de-ka-sa-to , a-re-i-jo , o-u-qe-po[

.2 pa-i-ti-ja , '*pe*' TELA + TE 2 mi TELA[1] + TE 14 da-wi-ja , pe TELA[x] + TE 1[

.3 do-ti-ja mi TELA + TE 6 qa-mi-ja TELA[1] + TE 1[

.4 ko-no-so , / te-pe-ja 'mi' TELA + TE 3 tu-ni-ja TELA[1] + TE 1 [

.5 .6 vacat [ ] vacat [

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[161]J. Killen (1979) 153 and J. Melena (1975) 43, 106.

[162]J. Killen (1966) 105; (1974) 82.

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This tablet is classified in the receipt series based on *de-ka-sa-to*, the Homeric verb ἐδέξατο 'received'.[163] Who the subject of the verb is remains unclear, either *a-po-te*, if this is a personal name as seems likely from tablet Od(1) 562,[164] or *a-re-i-jo*. The varieties of qualified and unqualified cloth (TELA + TE) from *pa-i-ti-ja*, *da-wi-ja*, *qa-mi-ja*, and *tu-ni-ja* (women workers of these places), plus the female *te-pe* makers at Knossos, compares well with records on other Lc(1) tablets. KN Lc(1) 546, 526, 543, 547, and 549 refer to similar place names as those on Le 641 and, when the variety of TELA + TE is noted, it also corresponds to the types on Le 641. For example, Lc(1) 543 refers to plain TELA + TE from *qa-mi-ja*, just as we see plain TELA + TE from *qa-mi-ja* on Le 641. Similarly, the *da-wi-ja* on Lc(1) 526 send TELA + TE of the *pe-ko-to* type, just as the 'pe' indicates for the cloth from the same workers at *da-wo* listed on Le 641.[165] The Le records are interesting because they only record one type of cloth, TELA + TE, and they do not mention *pa-we-a ko-u-ra* so common on the other cloth tablets.[166] This is in contrast to the Ld(1) tablets, the storage and delivery records, which show no TELA + TE and only the 'plain' TELA, most certainly the same basic type as the *pa-we-a ko-u-ra* on the Lc(1) records.[167]

Since the Le delivery records only list TELA + TE, it may be that these cloths went directly to the palace without requiring a separate finishing. On Le 641 we see the receipt of TELA + TE from six different sources. Five of them are feminine ethnics, presumably made

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[163]J. Killen (1966) 107.

[164]See the text of Od(1) 562 below. On Od(1) 562, *a-po-te* precedes the verb *pe-re* in line three, corresponding with the other subjects of the tablet. See J. Killen (1966) 106. Palmer suggests that *a-po-te* is a place name; L. Palmer (1963) 299.

[165]J. Killen (1979) 152-53.

[166]See J. Killen (1979) 153 n. 8 for the list of full references to the TELA + TE in the series.

[167]See below for a discussion of the Ld(1) series.

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by the women of these places, e.g. *pa-i-ti-ja*, *da-wi-ja*, and *tu-ni-ja*. Four of these five ethnics recur on Lc records by the same scribe and further suggest that this tablet ought to be classified amongst those. The sixth entry on Le 641 has TELA + TE described as *ko-no-so / te-pe-ja*, that is cloth from the Knossos *tepe*-making women.[168] If we take TELA + TE as an abbreviation for *tepe* cloth, we recall that this was also a type of cloth recorded by Linear A scribes on Crete. The word itself suggests, possibly, Greek ὁ τάπης 'tapestry', found also in the later Phrygian word *tepas*.[169]

In contrast to the TELA + TE cloth which went directly from weaver to the palace, Killen suggests that the *pa-we-a ko-u-ra* cloth of the Lc(1) tablets were sent to finishers where they received various elaborations. These finishers, in turn, delivered the decorated *pa-we-a ko-u-ra* to the palace as recorded in the Ld(1) tablets with the specific designations, such as *re-u-ko-nu-ka*, *po-ki-ro-nu-ka*, etc. These words are clearly related to the word *o-nu-ke/-ka*. While Palmer disagrees with Ventris and Chadwick who interpret this word as 'border' or 'fringe',[170] this explanation of some kind of edge seems more likely based on the uses of the word and on the actual practice of weaving.[171] As Barber notes, the only two objects in the Linear B corpus with the epithet *po-ki-ro-nu-ka, poikilonukh*, are textiles and chariot wheels.[172] Representations of chariots on frescoes, such as those at Tiryns, show

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[168]J. Chadwick (1988) 83.

[169]See Fig. 2.3, p. 89 and also J.-P. Olivier in E. Oren, "Minoan Graffito from Tel Haror (Negev, Israel)", Cretan Studies 5 (1996) 91-118. A form of the TELA + TE logogram has been found on a Minoan sherd along with two other signs that refer to figs and an ox-head (rhyton?). Olivier is uncertain whether this sign is Minoan hieroglyphic or Linear A but it is clearly related to the later Linear B sign found on the textile tablets from Knossos. Olivier, however, does not associate the word *te-pa* with the later word for tapestry.

[170]L. Palmer (1963) 293.

[171]See J. Killen (1979) 157 n. 14.

[172]E. Barber (1991) 327-28, n. 8 and fig. 15.11.

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the rims of the wheels with alternating black and blue bars around the edge.[173] A Scandinavian heading band for a warp-weighted loom, as illustrated by M. Hoffmann, shows an alternating bar-pattern quite similar to that on the chariot frescoes and also on garments depicted on other Aegean frescoes.[174]

The type of cloth found in the Lc(1) target records, *pa-we-a ko-u-ra*, is also found in the delivery tablets of the Ld(1) type, but not at all in the Le tablets. What is the connection between these two types of records, the work group targets of Lc(1), and the receipt records Ld(1) and Le? That Ld(1) records were found in Magazine XV along with Le tablets, by the same scribe, hand 103, does suggest a connection between the two types of storage tablets. In addition, the figures listed as targets of production in the Lc(1) records should correspond to the records of the received cloth. We will examine the totals presently.

From individual collectors, the total number of *pa-we-a ko-u-ra* is at least 356 units of cloth listed in the Lc(1) tablets. In the non-collector work-group tablets there are 200 units of *pa-we-a ko-u-ra* cloth. This gives a total of at least 556 units for both collectors and non-collector work groups. From tablet Ld(1) 587 we see that the total number of received *pa-we-a ko-u-ra* cloth is 453. This is 103 short of the corresponding Lc(1) total. What to make of this discrepancy? Killen suggests that the non-collector work group totals were not considered in the major records of the Ld(1) series.[175] The most likely correspondence, then, would be between the 356 + units of *pa-we-a ko-u-ra* cloth listed on the major Lc(1)

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[173]Recall also the discussion above with the Lc tablets which mention the participle of τεύχο found in a description of chariot wheels, similarly 'finished' with decoration.

[174]M. Hoffmann, The Warp Weighted Loom (1974) 66, fig. 26. And see E. Barber (1991) 325 for fresco representations. The word ὄνυξ also occurs in Classical Greek building inscriptions for ornamental bands painted or carved on the Erechtheion.

[175]J. Killen (1979) 155.

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tablets from individual collectors with the total in Ld(1) 587 of 453 cloth units. It seems that the Ld tablets only deal with 'collector' work groups.[176]

Two of the ten minor Ld(1) records refer to what are thought to be non-collector work group, Ld(1) 5955.1 beginning with the ethnic term, *da-*22-ti*[, and Ld(1) 656 *pe-ki-ti*, probably restored as the occupation of *pe-ki-ti-ra*$_2$ (or *-ri-ja*), combers of wool.[177] Of the ten, only five have totals preserved, and some only in part. These totals do suggest an average of about 15 units of cloth per delivery tablet, for a total of 150, very close to the 200 *pa-we-a ko-u-ra* listed in the Lc(1) totals for the non-collectors.

In summary, the situation seems to be that two varieties of cloth are targeted for production by the Lc(1) work groups (TELA + TE and *pa-we-a ko-u-ra*) but they are treated differently at the receipt stage. The Le tablets record the delivery of TELA + TE. Ld(1) tablets record *pa-we-a ko-u-ra* cloth, but with epithets such as *re-u-ko-nu-ka, po-ki-ro-nu-ka,* or *ko-ro-ta*$_2$. As was mentioned above, these terms do not appear in the Lc(1) records. It is probable that the Lc targets merely enumerated the *pa-we-a ko-u-ra* expected by the palace, without further description. But upon delivery to the palace store room, this cloth had gone through further processing, such as decorating, which gave it these various qualities. The fullers or decorators may have worked on this cloth prior to final delivery. This system is paralleled in the Neosumerian textile industry. Waetzoldt suggests that fullers were also responsible for adding decorations of wool to woven garments, such as borders or embroidered designs, since they are known to have received small allotments of wool.[178]

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[176]J. Killen cites tablet Ld(1) 598 which begins with a reference to the known collector workgroup of *wi-jo-qo-ta* also on the SHEEP series, Dq.

[177]The attribution of these tablets to hand 116, the scribe of the Ld(1) series, is 'virtually certain', according to Killen (1979) 151 n. 3, although not established. Barber suggests that these workers were not only the combers of already plucked wool, but that they also probably collected the wool by combing it directly off the backs of the sheep; E. Barber (1991) 262.

[178]H. Waetzoldt (1972) 168; J. Killen (1979) 157.

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The Ak and Ap Series[179]

The Ak and Ap tablets have already been referred to briefly. These tablets provide detailed information about work groups made up of women and children in the two regions of Crete, the center and western parts, as delineated in the Lc records. Based on the tablet descriptions and parallels with the L series it is certain that some of this was going to male workers.

The Ak tablets achieved two purposes for the palace administration: they recorded the productive potential of the separate work groups at the time it was calculating the targets and provided the basic information for allocating the monthly rations.[180] The workers listed in the Ak tablets were probably fully dependent on the palace and fed year round.[181] Before 1984, however, no such personnel records had been found that corresponded to the Lc(2) work groups in the far west of Crete.[182] Following an analogy with Pylos ration tablets, however, Palmer suggested that the missing personnel tablets might be located in the region where the laborers worked, *i.e.* in an archive somewhere in western Crete.[183] Finds of new fragments of tablets from the Herakleion museum in 1984 allowed for new joins with known tablets that have changed our understanding of western Crete personnel management. One tablet, Ak 7827 has been joined to record 50 women, probably in the 'ownership' of some

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[179]See J. Killen, "Two Notes on the Knossos Ak Tablets," Acta Mycenaea (ed.) M.S. Ruipérez (1972) 425-440.

[180]J. Killen (1988) 168-9.

[181]J. Killen (1984) 52.

[182]J. Killen (1988) 168.

[183]See L. Palmer (1963) 113. Palmer suggests that rations for the workers of the Further Province at Pylos would have been recorded at *re-u-ko-to-ro*. No Pylian tablets, however, have been found outside of Ano Englianos, the palace at Pylos. This phenomenon is discussed by J. Killen (1988) 169-70.

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individual, whose name is in the genitive, *i.e.*]*jo-jo*.[184] This tablet, Killen also suggests, not only includes features such as the owners which figure so prominently in the Lc(2) tablets, but also is likely to have been written by the same scribe, hand 113.[185] Two other newly joined tablets are suggested by Killen as possible records of work groups in the far west whose production is recorded in the Lc(2) group.[186]

Women of the Ap tablets at Knossos are listed by profession though these references are fewer than comparable terms at Pylos. One tablet will serve as an example.

KN Ap 694

.1]-ja ko-u-re-ja MUL [

.2] ka-ra-we MUL [

.3] a-ze-ti-ri-ja MUL [

Occupational terms listed in these and other tablets provide information on the successive manufacturing stages of Mycenaean textile production and the work force. In tablet Ap 694, the first line probably refers to a certain number (now lost) of women who made cloth. The second line is taken by Ventris and Chadwick to refer to 'old' women.[187] The third group of women are probably decorators.

Linear B terms for 'wool gatherers' (*pe-ki-ti-ra*$_2$) and 'flax workers' (*ri-ne-ja*) give evidence for the processing of raw materials into textiles.[188] The *a-ra-ka-te-ja* probably have

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[184]J. Killen (1988) 171.

[185]It should be noted, however, that this is based on the writing of one sign only (*jo*) since scribe 113 never wrote the ideogram MUL (woman).

[186]Ad 7830 + FR. reading]ja MUL 23 [, and X (= Ak?) 8330 reading] 43[(trace at left possibly MUL).

[187]Documents (1973) 165.

[188]The *pekitira*$_2$ are generally accepted as preparers of wool, perhaps as pluckers, carders and combers, related to Greek πεκεῖν, 'to comb'; M Lindgren (1973) 114-115; E. Barber (1991) 262, 283. The *ri-ne-ja*

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something to do with the Greek word, ἡ ἠλακάτη, 'distaff', or more likely, 'spindle', and may refer to *alakateiai*, 'spinning' women.[189] They receive allotments of wool and flax and presumably spin it into usable thread.[190] Other workers, like the *a-ze-ti-ri-ja*, (*cf* KN Ap 694), may have been weavers of heading bands. The weft threads of these starting bands would be used as the warp threads on a warp-weighted loom. As discussed above, some of the Ld tablets refer to finished or decorated cloth, *i.e.* cloth with 'ὄνυξ'. This term is taken to mean 'nail-' or 'claw-' patterns in alternating colors either woven in the heading band on warp-weighted looms or added as a specific type of decoration.[191] Two tablets, Ln 1568 and M(1) 683 include the term *a-ze-ti-ri-ja* which may be an alternative form of *a-ke-ti-ri-ja* / *a-je-ti-ra*$_2$.[192] Ventris and Chadwick first suggested an interpretation of this term as ἡ ἀκέστρια, 'seamstress'.[193] Lejeune, however, followed an alternative suggestion of Ventris and Chadwick and interpreted the term in relation to later Greek ἀσκεῖν, 'to work with skill'. From this interpretation, Lejeune suggests that the women listed as such were apprentices of the wool industry.[194] Killen, however, favors the verb ἀσκεῖν but interprets it as 'to decorate or dress' particularly since the *a-ke-ti-ri-ja* are closely associated with *o-nu-*

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are certainly flax workers since Linear B *ri-no* corresponds to Greek λίνον, 'flax'. See J. Chadwick (1988) 83; M. Lindgren (1973) 138.

[189]See Documents (1973) 533; E. Barber (1991) 264.

[190]This term also occurs at Thebes (Of 34.2). It is clearly associated with textiles in many instances and not only with yarn. J. Chadwick (1988) 79; M. Lindgren (1973) 23-24.

[191]E. Barber (1991) 272 n. 10 and 283.

[192]See J. Chadwick (1988) 78; A Morpurgo-Davies, "Terminology of Power and Terminology of Work in Greek and Linear B," Colloquium Mycenaeum (eds.) E. Risch and H. Muhlestein (1979) 91, n. 15, 99, n. 44. M. Lindgren (1973) 17-18 gives occurrences of this word at Pylos, Knossos, and Mycenae. For a summary of the linguistic difficulties in associating these two terms, see E. Barber (1991) 272-3 n.10.

[193]Documents (1956) 387.

[194]M. Lejeune, Mémoires de philologie mycénienne II (1971) 105

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ke.[195] Using both the linguistic evidence and a knowledge of weaving, Barber has shown that in the Mycenaean textile industry *a-ze-ti-ri-ja* and *a-ke-ti-ri-ja* are probably women who made both decorative edgings and also constructed warps by weaving heading bands.[196] Both of these processes would produce the typically Mycenaean 'claw' or 'nail' pattern. The heading band makers would be at the beginning stages of weaving while the decorators would probably be closer to the end.

Both male and female weaving specialists are indicated by references to certain laborers on the tablets, *i-te-ja-(o)* and one *i-te-we*, related to Greek ἱστος, 'a standing loom'.[197] After the cloth was woven, it could then be decorated or finished in some way. Occupations describing these activities include the *o-nu-ke-ja*, makers of decorative borders, the *ka-na-pe-u*, corresponding to Greek οἱ κναφῆς, '(male) fullers', and the *ra-pi-ti-ra$_2$/ra-pte(-re)*, related to Greek ῥαπτεῖν, 'to sew', workers who would stitch the textiles.[198] One interesting occupational term that recurrs in two Ak tablets, and on other tablets associated with textile workers, is *ne-ki-ri-de*, analysed recently by J. Killen.[199] The parallel position of this word to other professional textile names suggests that this was a specialized worker. Killen demonstrates that these women were probably makers and

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[195]Killen 1979:165-1667. He also notes that the term *a-ke-te-re* found in the Pylos bronze tablet, Jn 832.1, is probably the masculine equivalent of *a-ke-ti-ri-ja*. These workers do not receive allocations of bronze as Lejeune notes (Mémoires II 180 and n. 49) and this may explain why they did not require more bronze if their task was decorating. One can imagine well-known Mycenaean examples of engraving, repoussé or polishing.

[196]E. Barber (1991) 272-3 n.10. The evidence seems to indicate that the two terms (*a-ze-ti-ri-ja* and *a-ke-ti-ri-ja*) are probably not variations of the same word.

[197]M. Lindgren (1973) 58.

[198]Three fullers are known by name in the tablets. One is described as *wanakatero*, 'royal'. See T. Palaima (1997) 407-409; M. Lindgren (1973) 71-72; E. Barber (1991) 283. Ventris and Chadwick suggests that the male sewers were associated more with leather working rather than cloth. Documents (1973) 578.

[199]J. Killen, "Two Mycenaean Words," in A. Etter (ed.) *o-o-pe-ro-si*: Festschrift für Ernst Risch zum 75. Geburtstag (1986) 279-284.

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decorators of funerary shrouds. The word *ne-ki-ri-de* is related to Greek ὁ νεκρός, 'corpse'.

Od Tablets

Ventris and Chadwick suggest two possible explanations for the presence of a wool entry after the cloth on the Lc tablets: either these tablets show the receipt of cloth and wool separately from outlying areas, or they 'record the weight of wool used in the cloths themselves'.[200] As was mentioned above, ratios of various types of cloth to wool units are adhered to fairly strictly on the tablets of the Lc series, suggesting that the latter interpretation is correct. Killen has demonstrated that these ratios more than likely refer to the wool that was used for these cloths.[201] One case, Od 562, from the series closely related to the Lc series, shows that the three separate wool allotments to certain individuals are 91, 42, and 69:

KN Od 562

.1	]o-pi , no-nu-we ,	'a-ti-pa-mo' pe-re	LANA 91
.2	]si-da-jo , pe-re	'po-ro-to'	LANA 42
.3	a-po-te , pe-re		LANA 69

In the first two instances, these are multiples of seven. The last entry is one short of 70. The preposition *o-pi* probably means 'in the charge of'[202] followed by the personal name *no-nu-we* in the dative. *A-ti-pa-mo* is probably the subject of the verb *pe-re* (Greek φέρει, 'he brings') in line 1. This suggests a pattern for the other two lines: individuals 'bringing'

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[200]Documents (1973) 315-6.

[201]J. Killen (1966) 106.

[202]See A. Morpurgo-Davies, "Mycenaean and Greek Prepositions: *o-pi*, *e-pi* etc." Res Mycenaeae (eds.) A. Heubeck and G. Neumann (1983) 293; J. Killen, "The Knossos *o-pi* tablets," Atti e memorie del 1.o congresso internazionale di Micenologia (1968) 636-643.

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certain amounts of wool. As was mentioned above, the Lc tablets recording TELA + TE have a 1:7 ratio of cloth to wool units and are written by the same scribe of Od 562, hand 103. It seems highly likely that the Od tablets and the Lc records refer to the same wool in different phases of production; in Od the wool is raw and unwoven. In the Lc tablets it is a completed cloth of the TELA + TE variety.

A group of tablets from Knossos, also by hand 103, records units of wool allocated to workers in the textile industry yet the category of tablets is fairly non-distinct and rather miscellaneous. Amounts of wool are recorded in very small quantities which suggests that this wool was not only sent to weaving workshops but also to finishers or decorators.[203] New joins of tablet KN Od 765, for example, provide interesting information on craft specialists:

KN Od 765 + 7320 + FRR.[204]

.1		pa-i-to	[
.2	]-ka-ra /	a-me-a LANA M	[

What is of note here is that the activity here is textile finishing and the named individual receiving the small amount of wool is a man. M is a fractional accounting unit. *A-me-a* recurs on Da 1189 as a shepherd and we know that male workers are often associated with finishing.[205] Other Od tablets (7312, 7326, 7779 + FRR., and 8628 + FR.) list similar quantities of wool and we can presume that they were going to male workers.

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[203]Killen (1988) 172 discusses new joins in the Od series based on the 1984 finds in the Herakleion museum. See J. Killen, Kadmos 24 (1985) 149ff for wool recorded in precise quantities on Od 487, Od(1) 667, and Od(1) 696.

[204]Capital FRR. indicates new joins of Knossos tablets in the Iraklion Museum in 1984. See J. Killen (1988) 173.

[205]J. Killen (1979) 167.

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Pp Tablets

While no explicit reference to wool or textiles is made in these tablets, Melena has noted a connection to the wool industry.[206] The tablets of the group identified as Pp were found so close together that the modern inventory numbers should reflect the prehistoric filing system.[207] By looking at the excavation photographs, however, one can see that the order of the tablets as found is not reflected in the inventory numbers.[208] In addition, one can never be certain which was the beginning 'header' tablet which provides important clues to the subject of the rest of the tablets. The scribe is identified as 119, responsible also for the first set of the Dk series that deals with wool from the collector *da-mi-ni-jo* at the town of *ku-ta-to*.[209] The Pp tablets were found in the West Wing at Knossos, in Magazine VIII, and probably originated in the textile and wool archives on the floor above.[210] Of the seven tablets in the Pp set, six are one line long with the ideogram *168 + SE preceded by a toponym and sometimes a personal name. One of the men's names here is *mu-ka-ra* who may be the same person who delivers a quantity of wool recorded in tablet Od 666.[211] The seventh, Pp 498, has a central line division for two lines and does not include the SE adjunct with *168 but does include two place-names. Melena suggests that this compression of two place-names on one tablet indicates that this was the last tablet of the series and that the scribe

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[206]J. Melena (1975) 118-134.

[207]J. Chadwick, "The Classification of the Knossos Tablets," Acta Mycenaea I (1972) 26; J. Melena (1975) 119.

[208]See J. Melena (1975) 120 n.1. He states that this error was already indicated in the editions previous to the fourth edition of The Knossos Tablets.

[209]J.-P. Olivier, Les scribes de Cnossos (1967) 64.

[210]J. Melena (1975) 119.

[211]J. Melena (1975) 132.

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made two entries on one tablet because he was running out of records. Melena also gives a revised order of the place-names based on the location of the tablets when found.[212] The significance of the order of place-names is also problematic. Palmer makes the suggestion for 'scribal routes', itineraries reflected in the order place-names were recorded in the tablets.[213] This idea is critiqued by Melena and Chadwick among others as being very tenuous. Melena does identify the place-names listed in the Pp tablets as being in the northern and Messara plains of Crete where important flocks of sheep were known to graze.[214]

The ideogram *168 and the ligature *168 + SE should also be explained since Melena connects these tablets with the textile industry. The sign *168 was first identified as possibly reflecting adzes based on its resemblance to the wood-cutting tool. The sign itself is a four sided rectangle of fairly wide width in contrast to its height. Often one end seems to taper toward the left giving an unequal appearance to the quadrilateral. At both ends, or rarely just the left end, there is an internal vertical stroke possibly resembling blades. The interpretation by Melena suggests that the sign indicates a basket used for the delivery or storage of wool about to be spun. This object occurs in later Greek as καλαθίσκος, 'wool-basket', the diminutive form of ὁ κάλαθος. which may occur in Linear B as *ka-ra-to* (Ge 603.1a, 605.1).[215] This revised view of the sign and ideogram better fits with our understanding of the extensive wool industry on Bronze Age Crete. Melena further suggests that the 'Pp

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[212]See the map on J. Melena (1975) 127.

[213]L. Palmer, "Mycenaean Inscribed Vases II: The Mainland Finds," Kadmos 11 (1972) 33.

[214]J. Melena (1975) 132.

[215]J. Melena (1975) 134; E. Barber (1991) 262, n. 3. The etymology of κάλαθος is probably related to Greek κλωθεῖν, 'to spin'.

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tablets record deliveries of baskets to be used as part of the equipment for wool plucking.'[216]

Linen at Knossos

Evidence for linen at Knossos is found on the Nc and L(1) tablets.[217] The flax or linen ideogram is generally assumed to be SA, often found on tablets at Pylos and discussed below. At Knossos the ideogram SA is measured by the textile unit M and is found on three fragmentary tablets classified as the Nc series. Does SA at Knossos have the same meaning as at Pylos, where it is not measured, counted, or weighed? Does it mean linen or flax as we assume? The sign is not a syllabic abbreviation for flax and appears to have nothing to do with the actual Greek word for the item, similar to the use of sign NI to indicate 'figs'.[218] The contexts of the ideogram at Knossos and Pylos are comparable and may indicate similar meanings. On tablets where it occurs, there seem to be references to military matters. SA as linen or flax could have been used as a trophy or payment for soldiers, just as we saw with the *ke-se-nu-wi-ja* cloth, or perhaps it was made into woven linen for tents and ropes used by the squadrons. Finally, Homer refers to a λινοθώρηξ in Il. 2.529, 830, where linen is clearly used as part of lesser Ajax's armor.

The six L(1) tablets by scribe 103 all refer to *ri-ta pa-we-a, linon pharwos*.[219] One tablet reads:

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[216]J. Melena (1988) 406.

[217]J. Killen, "The Knossos Nc Tablets," Studia Mycenaea (1968) 33-38; (1988) 167-184.

[218]This is to say that the syllable NI is unrelated to the later Greek word for fig, τό σῦκον. Documents (1973) 295.

[219]See J. Killen (1988) 174-183.

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KN L(1) 594 (103)

.1] ri-ta , pa-we-a

.2]da-te-we-ja TELA[1] 1 TUN + KI 1

This presumably records a suit of clothes: a linen tunic and a cloak worn over it. Sign *162 depicts a rectangular garment or corslet with two shoulder-loops and appears often on the Sc tablets with lists of chariots. The Mycenaean word, *ri-ta*, corresponds to the adjectival, Homeric, accusative, plural λῖτα, 'linen cloth', probably a light garment worn underneath a heavier tunic or corslet. This tablet and others in the series probably refer to the receipt of finish linen cloth, just as we saw on the Le tablets, also by scribe 103. The L(1) tablets are probably not targets of production since no raw materials, such as flax or SA , are alloted to workers.[220] The work group *da-te-we-ja*, however, is familiar from the Lc(1) target records, also by scribe 103. On Lc(1) 540 + 8075 (discusses above) nine units of *pa-we-a ko-u-ra* cloth are made by the *da-te-we-ja*. In addition, the rations for these workers are listed in the Ak tablets (*infra*). Killen speculates that *pa-we-a* cloth might be made by the same work group but out of different materials, wool (*ko-u-ra*) on the Lc tablets and linen (*ri-ta*) on the L(1) records. The various uses of τό φᾶρος in Homer (cloak, web, sail) would need to be made of different materials (wool for a cloak, linen for a sail).

One well-known tablet from Knossos deals with linen and bronze together.[221]

KN L 693 (103)

.1 ri-no , / re-po-to 'qe-te-o' ki-to AES M1 [

.2 sa-pa P 2 Q 1 e-pi-ki-to-ni-ja AES M1[

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[220]J. Killen (1988) 176.

[221]Documents (1973) 320-21, tablet 222.

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Often used to modify chitons, *ri-no re-po-to* almost certainly refers to fine linen. Compare Il. 9.661, λίνοιο λεπτὸν ἄωτον, 'a fine linen bedspread'.[222] Most interpretations agree that tablet KN L 693 is a list of fine linen with the verbal adjective *qe-te-o*, meaning 'to be paid', followed by the Semitic loan word, *ki-to* for chiton (the word itself is related to linen), and then an amount of bronze.[223] This term, *qe-te-o*, is fairly important in terms of the economic organization of Mycenaean kingdoms for it probably suggests equivalencies of value, fitting in with a wealth finance model outlined in Chapter one. There are 18 attestations of some form of *qe-te-o* in Linear B.[224] Six of them deal with textiles: L(5) 513, L(5) 5092, L(5) 7380+ 7500, L(5) 7834, L(5) 8441 and L 693.[225] All of the L(5) tablets follow the same formula. The first line contains *qe-te-o* and an amount of cloth. The second line begins most probably with a personal name and then more quantities of cloth. For example:

KN L(5) 7380 + 7500

.1] qe-te-o TELA[1] 2[

.2]-ra/ TELA[x] 4 po TELA[x][

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[222]That the chiton is of linen is not surprising since the Semitic root actually means flax and linen. C. Shelmerdine (1995) 102-103.

[223]See M. Lejeune, "Sur quelques termes du vocabulaire économique mycénien," Memoires II (1971) 301-306; J. Hooker (1980) 65-66. E. Hamp, however, does not accept *qe-te-o* as meaning 'to be paid' and discounts the term as an Indo-European verbal adjective of necessity since 'there is no digamma to be restored'. He instead closely associates the word with the modifier *lepton*, restoring it to read *qe-te-(j)o, kweiteion*, an adjective meaning 'costly' fine linen. E. Hamp, "Kn L 693 *qe-te-o* and *mantis*," Minos 19 (1985) 53.

[224]See W. Hutton,"The Meaning of *qe-te-o* in Linear B," Minos 25-26 (1990-1991) 105-131. Many new attestations have come from the recently published sealings from Thebes.

[225]J. Killen restores *qe-te-o* on tablet M 683 but this is questioned by Hutton.

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The modifier *po* before the last cloth sign suggests possibly that this was an abbreviation for *po-ki-ro-nu-ka* cloth, decorated with multi-color edges.[226]

Written in small signs and elevated above the main line, *qe-te-o* is clearly set-off by the scribe (103) in L 693. As Hutton has shown, this scribe often tends to elevate certain words to modify specifically the following word, just as modern writers might use a parenthesis mark.[227] In this case, then, *qe-te-o* would only modify the chiton. On the second line, *sa-pa* is interpreted as a garment but this is uncertain.[228] Units of weight measurements P and Q show that whatever *sa-pa* was, it was fairly light.[229] The meaning of *e-pi-ki-to-ni-ja* is fairly clear, an over-coat of some kind. Fragmented at the end of this line is also an amount of bronze. The bronze may refer to attachments on garments, *i.e.* a kind of armor (*cf.* λινοθώρηξ mentioned by Homer in Il. 2.529, 830 or ὁ χιτῶν χάλκεος in Il. 13.439); it may indicate the weight of the garment in bronze units; or it may be the value of the garments in terms of bronze.[230] All of these are possible interpretations. The weight values given by Ventris and Chadwick are, 1 kg. of bronze for the tunic, 45 g. for the *sa-pa*, and 1 kg. for the epichiton.[231] As a comparison to this tablet listing fine linen, Ventris and Chadwick cite Il. xviii, 595, where young men on the Shield of Achilles are described wearing finely woven tunics 'touched with olive oil'.[232] An exchange of fine linen is also

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[226]Hutton has suggested that the fragmentary nature of the other tablets allows for the possibility that *po* might be restored at the end of each tablet; W. Hutton (1990-1991) 108.

[227]W. Hutton (1990-1991) 110.

[228]The word also appears on tablet Oe 108 at Mycenae.

[229]P equals 1/24 mina, approximately 20 grams, and Q is 1/6 of that.

[230]I. Προμπονᾶς, Ἀνθολογία Μυκηναϊκῶν Κειμένων (1983) 125; J. Hooker (1980) 98-99.

[231]Documents (1973) 320-321.

[232]Documents (1973) 320. Textiles treated with oil will be discussed in reference to the textile industry at Pylos.

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found at Pylos, on PY Un 1322.5 for example, where the price of this cloth is fixed not in terms of bronze, as at Knossos, but in terms of wheat.[233]

PY Un 1322.5

1. we-a_2-no[ri]-no , re-po-to *146 GRA 5

The weight of 5 units of wheat equates to about 480 liters. The bronze amounts in L 693 is at least one double mina (M), approximately 964 grams. The association of one amount of a certain commodity for another amount of textiles suggests strongly that there is an equivalence of value for certain goods. The one item that is constant in such transactions is textiles, giving the impression that cloth functioned nearly as a kind of currency in the wealth finance of the Mycenaean Bronze Age.

Part II.B - Textile Production at Mycenaean Pylos

As we have seen, woollen and linen textiles played a major role in the Mycenaean economy of Knossos. Looking at the texts found at the palace at Pylos we also see references to textiles and their manufacture administered by the palace authorities. Two major points distinguish textile production at Pylos from Knossos:[234]

1) In comparison to Mycenaean Knossos, the palace economy at Pylos shows a greater degree of regionalized specialization among its various provinces and towns. The Linear B records found at the Palace of Nestor reflect centralized control of craft activities at two major centers, Pylos and Leuktron, monitoring activities dispersed throughout the kingdom.[235] As

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[233]This part of tablet Un 1322 is discussed by C. Shelmerdine (1995) 102-103. See also, J. Chadwick, "Pylos Tablet Un 1322," Mycenaean Studies (ed.) E. Bennett (1966) 23-24.

[234]See P. Halstead (1991) 343-366; J. Killen (1984) 49-63.

[235]See C. Shelmerdine, "Industrial Activity at Pylos," Tractata Mycenaea (eds.) P. Ilievski and L. Crepajac (1987) 333-342.

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Killen summarizes, Pylos had 28 separate work groups with nearly half containing over 20 women in each group. Leuktron has six or seven work groups, one with 37 women. The other place names in the tablets usually only have one work group with less than a dozen adult workers.[236]

2) Linen production had a much larger role in the palace textile industry at Pylos than at Knossos. The various phases of linen cloth production in centers of differing size are fairly well-attested in the Pylian records although much less numerous than at Knossos. The textual and archaeological evidence for Pylian cloth, both wool and linen, is summarized and discussed below.

Pylian Wool

Wool is not as prevalent in the tablets from Pylos compared to Knossos possibly because at the time the palace was destroyed sheep shearing had not yet taken place that year.[237] There are, however, records of sheep. Corresponding to the D tablets at Knossos, the Cn class at Pylos record primarily male sheep, in the same order of magnitude as the sheep at Knossos and also in round numbers.[238] The Cn set often begins with a place name in the dative, a personal name and then livestock information. This sequence and variations on it are described by J. Bennet.[239] One important difference between sheep tablets at Pylos

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[236]See J. Killen (1984) 53-60; J. Chadwick (1988) 43-95.

[237]J. Chadwick (1976) 130.

[238]J. Killen (1984) 51. See also L. Godart, "The Grouping of Place-Names in the Cn Tablets," BICS 17 (1970) 159-161; "Valeur des idéogrammes OVISm, OVISf, CAPm, SUSm, SUSf, BOSm et BOSf dans les tablettes de Cnossos," Minos 13 (1973) 113-129; P. Ilievski, "A Re-Examination of the PY Cn Tablets," Atti e Memorie del 1o Congresso Internazionale di Micenologia, = Incunabula Graeca XXV (1968) 616-632.

[239]J. Bennet (1992) 67-68. Also recall the example PY Cc 660 discussed under the 'collector' section.

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compared to Knossos is that only at Knossos are the sheep connected with amounts of wool and consequently may not reflect the same sort of wool industry as on Crete.[240]

At Pylos there are no *ta-ra-si-ja* target attestations for wool as there are at Knossos but there is evidence for the administration receiving cloth produced according to the *ta-ra-si-ja* system.[241] As in KN Le 642 which also records TELA+TE and *ta-ra-si-ja*, Chadwick, Duhoux, and Killen have restored the Pylos text La 1393 as:[242]

PY La 1393 SW area 13 Scribe S622-H13 (?)[243]

.1 (o-)de-ka-]sa-to , a-ri-wo , ta-ra-[si-ja

.2]TELA+TE 40 [

One individual, Aleios, receives ? according to the *tarasija* system and is assessed forty units of TELA + TE cloth. As was mentioned before, TELA + TE is most likely a woollen garment using seven units of wool for each unit of cloth, giving a total of 280 wool units for Aleios.

Pylian Linen

The commodity SA is recorded on the Na, Ng and Nn records and is identified as *ri-no*, either 'linen' or 'flax', on Nn 228.1.[244] The Na-series from Pylos consists of a place

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[240]J. Hooker (1987) 313.

[241]M-L. Noch (1997) 5-7.

[242]Y. Duhoux Aspects du vocabulaire économique mycénien (1976) 70 and Killen (1984) 51.

[243]T. Palaima, The Scribes of Pylos IG 87 (1988).

[244]The term λίνον in classical Greek has many referents: linen thread, linen cloth, linen garments, and the raw material, flax, itself. Robkin suggests that SA is unlikely to refer to linseed since other types of seeds are designated *pe-mo*, τό σπέρμα, 'seed,' by the Linear B scribes; A. Robkin (1979) 469.

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name followed by the ideogram SA and a number.[245] One problem in identifying SA, however, is that in the Pylos documents there is no indication of whether the commodity was weighed, counted, or measured.[246] Chadwick has suggested, based on modern and ancient land-use productivity, that one unit of SA indicates one Mycenaean talent of un-spun flax fiber, after initial processing, equal to 29 kg.[247] The modern region of Messenia, which probably covers an area roughly equivalent to the extent of the ancient Pylian kingdom, is well-suited to the cultivation of flax, the raw material for linen. In the 1950's this region produced over half of the linen in Greece.[248]

The N series from Pylos includes references to flax/linen collection.[249] This is not at all surprising based on the soil, climate and landscape around Pylos.[250] The N series are a part one type of collection record at Pylos. Collected goods are recorded on 221 tablets and comprise 18% of the total tablets from Pylos. The types of good collected are ones which the palace was not self-sufficient in producing itself.[251] Textiles and textile crops from over 100 different towns and villages compose 78% of the collection records while goods such as food and livestock comprise only 16%. The Na tablets record SA collection and are written in a variety of formats which suggests to A. Robkin that the ultimate distribution of the

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[245]A. Robkin (1979) 469-474; M. Lejeune, "Les documents pyliéns des series Na, Ng, Nn," Études Mycéniennes, 137-165.

[246]In contrast, at Knossos, SA is weighed according to KN Nc 4479.

[247]J. Chadwick (1976) 153; Documents (1973) 54.

[248]A. Robkin (1979) 471 n.29; W. McDonald and G. Rapp, Jr. (eds.) Minnesota Messenia Expedition (1972) 240.

[249]M. Lejeune (1956) 137-165.

[250]W.McDonald and G. Rapp, Jr. (eds.) (1972) 240.

[251]H. Morris (1986) 91-99.

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commodity was still in process at the time the palace was destroyed, sometime in early spring.[252] One tablet serves as an example:

PY Na 396

1. e-ko-si

2. wo-no-qe-wa, ko-ro-ku-ra-i-jo SA 30

'At Wonoqewa, Korokuraijo have 30 units of flax.' Ventris and Chadwick suggest that individuals or officials, such as (the?) Korokuraijo, administered palace land throughout the kingdom and were assessed certain amounts of flax as payment in exchange for use of the land.[253] Eight of the Na tablets refer to payment for use of land, but Morris suggests that perhaps all of the Na tablets were both flax collection and land-lease records.[254] Robkin estimates the amount of flax being produced at Pylos based on modern productivity records from Messenia. Based on the collection targets in Ng 319 from the Hither province a yield of nearly 50,000 kg. of fiber is estimated.[255]

A unique series of tablets from Pylos, the Ma series, record a 'regulated taxation system of six set commodities indicated by Linear B ideograms'.[256] Presumably these goods were agricultural commodities that each subdivision of the kingdom's two provinces was capable of producing as payment or as a tax to the center. Only a few of these signs are understood. Suggestions include beeswax, oxhides, honey.[257] *146 is probably unfinished, linen cloth most likely manufactured throughout the Pylian kingdom and sent to

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[252]A. Robkin (1979) 472. The tablets show a great many erasures, alterations and even palimpsests.

[253]Documents (1973) 469.

[254]H. Morris (1986) 91-99.

[255]A. Robkin (1979) 473.

[256]C. Shelmerdine (1973) 261-75.

[257]H. Morris (1986) 91-99.

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the palace for finishing by specialists. H. Morris speculates that this was probably a simple, home-made fabric, possibly required for religious or military purposes.[258]

Some Mycenaean documents refer to laborers receiving rations from the palace. One group is of particular interest because of the connections between women workers and cloth production. The Aa, Ab and Ad tablets from Pylos provide a great deal of information concerning women and their children, many of whom were employed by a centrally controlled textile industry.[259] Ethnic descriptions are attached to these women. These place names indicate an origin for the women workers in the east along the Anatolian coast. Women of Knidos (*ki-ni-di-ja*), Lemnos (*ra-mi-ni-ja*), Miletos (*mi-ra-ti-ja*) and possibly somewhere near Halikarnassos (*ze-pu$_2$-ra$_2$*, 'Zephyra') are mentioned in reference to textile workers at Pylos.[260] As Chadwick notes, any one of these references might be only interesting but the coincidence of all these names from the same general region is highly suggestive of contact and exchange with Asia Minor in the Bronze Age.[261] Ventris and Chadwick relate the word *ra-wi-ja-ja* occurring in three related Pylian tablets, Aa 807, Ab 586, Ad 686, to the Homeric adjective λητ̈ας, 'captured'.[262] Although the women with Anatolian names are never explicitly described as *ra-wi-ja-ja*, Ventris and Chadwick suggest that they may have been prisoners of war won from raids conducted by Mycenaean warriors

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[258]H. Morris (1986) 99 refers to the Mn tablets where all but one of the fourteen tablets records a collection of *146.

[259]Documents (1973) 410, 417.

[260]Also possible is a reference to women of Xios (*ki-si-wi-ja*). For the most recent analysis of these terms, including linguistic difficulties, see J. Chadwick (1988) 78-93.

[261]J. Chadwick (1988) 91. For archaeological evidence, see the recent work at Miletos by W.-D. Niemeier, "The Mycenaean Potter's Quarter at Miletos," TEXNH (1997) 347-352.

[262]Documents (1956) 156, 162.

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along the Ionian coast.[263] Alternatively, these women may have been rescued Mycenaean women abducted in the past by Anatolian pirates and taken back by Greeks. Or they were Anatolian refugees skilled in weaving and attracted to Pylos because of its extensive linen industry.[264] Xenophon's reference in the Memorabilia (117.2-8) to women refugees being put to work in Athens making clothes and baking bread during the Peloponnesian war is an interesting parallel.[265] One final interpretation of these foreign names is that they were the names of ancestors of Pylian Mycenaeans who came originally from Anatolia. A few tablets refer to the occupations of these women with foreign names. Certain ethnics are associated exclusively with particular craft activities suggesting that these women specialized in different aspects of textile production. It may be that their original homeland gave a designation to a kind of trade or craft-guild. For example, the Milesians are spinners and Tinwasians weavers (Ad 684), the Aswian and Zephyrian women are listed as flax-workers (Ad 326, Ad 664), and the Chians are makers of *o-nu-ka* (Ab 194, Ad 675).[266]

Archaeological evidence for textile production in the kingdom of Pylos is slight, as is the case with other Mycenaean centers. Significant numbers of loom weights and spindle whorls were found at the site of Nichoria, identified as the Pylian town of *Ti-mi-to-a-ke*.[267] This Mycenaean settlement was assigned one of the largest amounts of flax in the Pylos

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[263]J.-C. Billigmeier and J. Turner, "The Socio-Economic Roles of Women in Mycenaean Greece: a Brief Survey from the Evidence of the Linear B Tablets," Reflections of Women in Antiquity (ed.) H. Foley (1981) 4. They believe that this connection is tenuous. See, however, J. Driessen (1984) 50 where the term is taken to refer to captive women at Pylos.

[264]This is the suggestion by J.-C. Billigmeier and J. Turner (1981) 4.

[265]See also E. Barber, "The Peplos of Athena," Goddess and Polis: The Panathenaic Festival in Ancient Athens (ed.) J. Neils (1992) 103-118.

[266]J.-C. Billigmeier and J. Turner (1981) 5.

[267]J. Carington Smith, "Spinning and Weaving Equipment," Excavations at Nichoria in Southwestern Greece II: The Bronze Age Occupation (eds.) W. McDonald and N. Wilkie (1992) 674-711. See also, C. Shelmerdine (1981) 319-325 where Nichoria is identified with the Pylian town of *ti-mi-to-a-ke-e*.

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tablets, suggesting a specialization in linen production. Archaeological data included 153 spindle whorls from houses containing mixed ceramic deposits dating to the Middle and Late Bronze Age (fig. 3.1).[268] These whorls were generally of two types, large and normal, with the majority bi-conical in shape. The chronological span of spindle whorl types suggests that no new types of whorls were introduced into Messenia since the Middle Bronze Age. At the Middle Bronze Age site of Malthi, 350 spindle whorls of similar shape to Nichoria examples were excavated. Many of them were found in one room suggesting centralized spinning or at least communal storage of spinning equipment.[269] Nothing that was clearly a loom weight, however, was found.

Carington Smith compares the Messenia whorls to those found at Troy. She sees a similarity in the new forms appearing in Greece in the Middle Bronze Age to those of Troy VI.[270] There is, however, no correspondence between the new type of whorl of the MH period to an appearance of comparably new loom weights in Greece. Carington Smith suggests that different weaving technologies (*i.e.* non-warp weighted looms) were employed during the MH period. Twelve Mycenaean loom weights were found at Nichoria, the majority probably from a set of LH II-LH III A1 date in Area VI (fig. 3.2).[271] One loom weight and a possible spinning bowl were of Minoan type. These data provide a unique

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[268]J. Carington Smith (1992) 676.

[269]N. Valmin, The Swedish Messenia Expedition (1938) 29, fig. 8, 31-41, 50, 87-93, 104, 137-147, 158-160, 335-6; "Malthi-epilog," Op Ath (1958) 29-46.

[270]H. Goldman reports small, biconical, incised 'buttons' in MH levels at Eutresis that are absent in the Early and Late Helladic strata. She states that they are as 'distinctive of the MH period as Minyan pottery', H. Goldman, Excavations at Eutresis in Boeotia (1931) 198. See E. Barber (1991) 303-310 for a thorough tracking of certain spindle whorl types.

[271] J. Carington Smith (1992) 688-89.

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view of a non-palatial, subordinate center from the Mycenaean period, comparable perhaps to the non-palatial Minoan sites of Crete discussed in Chapter two.[272]

In areas surrounding the Palace of Nestor at Pylos four loom weights similar in form to those at Nichoria, although better formed, were found in the aqueduct in the southeast corner of the palace.[273] Five loom weights with five holes each were found in late Mycenaean levels at Malthi, comparable in shape to the Nichoria examples.[274]

Part III - Additional Sources of Mycenaean Textile Production

Pylos and Knossos are not the only Mycenaean centers that preserve evidence for centralized textile production. While the Linear B records for cloth are fewer at Mycenae and Thebes, the documents that we do have suggest that these centers also relied on a well-organized textile industry. In addition, archaeological evidence in the form of textile tools and painted representations of cloth inform us about Mycenaean textile production.

Thebes

At Thebes, on the corner of Epaminondas and Metaxas Streets, the remains of a two-room structure have been associated with textile production and distribution (fig. 3.3). The excavator states that the remains of dye vats can be discerned in the form of pits cut into the floor level.[275] The area in front of these pits shows a small channel with three post holes along the sides. Spyropoulos suggested that this was a drying rack for newly washed or

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[272]For a thorough discussion of the Pylian kingdom, see H. Morris (1986) and J. Carothers, The Pylian Kingdom: A Case Study of an Early State Ph. D. Dissertation, UCLA (1992).

[273]C. Blegen, *et al.* The Palace of Nestor vol. 1 (1966) 301, fig. 305.1, pl. lv; vol. 3 (1973) 7-16, 336, fig. 105.1-3.

[274] N. Valmin (1958) 38, 43, pl. II4.

[275]T. Spyropoulos and J. Chadwick The Thebes Tablets II. Minos Suppl. 4 (1975).

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dyed textiles although Shelmerdine has shown that this is unlikely.[276] In what was identified as a bathroom, however, there were two holes cut in the floor, in the center of the room, which might indicate post-holes of a warp-weighted or two-beam frame loom. The raised platform in front could have functioned as a bench for the weavers working at the loom. The bath feature may have held spun wool or linen ready for weaving.[277]

The 16 tablets found in the Room 1 are classified in the Of series and deal with distributions of wool.[278] The disbursements of wool (LANA) range between 1 to 30 kgs. for a total of 69 kg. Shelmerdine notes that this is not a very substantial amount of wool. It was probably stored in this room, some for perhaps a long time since the qualifiers meaning 'new' and 'last year' are included on Of 34, for example. The wool would also be sent out to outlying areas away from Thebes, such as *a-ma-ru-to-de*, Euboean (?) Amarynthos, mentioned on Of 25. Finally, the new sealings found at Thebes, already mentioned in reference to collectors, also show a strong interest by the central administration in livestock, especially sheep and goats.

Mycenae

As was mentioned before, the archaeological evidence for textile production at Mycenaean centers is slight. The best known Mycenaean center is no exception. Excavations of Grave Circles A and B at Mycenae yielded blades and a wooden dish that

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[276]That this area was not a palace archive or workshop has been demonstrated by C. Shelmerdine (1997) 387-389. She suggests instead that this building was a clearing house for wool collected and sent on to workers in outlying areas.

[277]Shelmerdine, however, reconstructs these post holes as the supports for shelves containing the 16 inscribed Mycenaean tablets found in this room listing quantities and distributions of wool.

[278]C. Shelmerdine (1997) 387; See the transcriptions of the Of series also in J. Melena and J.-P. Olivier Tithemy: The Tablets and Nodules in Linear B from Tiryns, Thebes and Mycenae. Minos Suppl. 12 (1991) 35-37.

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were covered in fine linen, and some whorls.[279] Nineteen loom weights were reported by Wace from the Guard's House at Mycenae but none of these can be located in storage either in Athens, Nauplion, or at Mycenae.[280] These weights date to the beginning of the LH III A period.[281] Made of clay, stone, bone and sometimes steatite, these small weights usually three to five centimeters in diameter are often confused with or interpreted as beads, buttons, dress-weights, or spindle whorls.[282] The actual function of these objects is difficult to determine. Iakovides' interpretation of dress-weights, based on representations of weighted tassles in Mycenaean frescoes, is convincing for the very light stones with small diameters. Objects two cm. or less in diameter and/or weighing less than ten grams are generally considered suspect as spindle whorls, according to J. Carington Smith.[283] The textual evidence is slighly more informative.

The West House group at Mycenae provides some Linear B evidence for the administration of a textile industry (fig. 3.4). These four houses were excavated in the 1950's by the British School at Athens. Their contents have been studied most recently by

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[279]See E. Barber (1991) 174, n. 12 for textile fragments from the Aegean and Cyprus. S. Marinatos, "Kleidung, Haar- und Barttracht," Archaeologia Homerica 1 A'-B' (1967) 16, no. 69, identifies the cloth as *spartium junceum*. For Grave Circle A: H. Schliemann, Mycenae (1880) 155, 283; G. Karo Die Schactgräber von Mykenai (1930) 71, 142, 145, 251; A. Snodgrass, Arms and Armour of the Greeks (1967) 16; A. Åkerström, "Mycenaean Problems," Op. Ath. 12 (1978) 52. Grave Circle B: I. Papdemetrios, "Ἀνασκαφή εν Μυκήναις," Πρακτικά (1951) 203; G. Mylonas, Ὁ Ταφικός Κύκλος Β' των Μυκηνών (1973) 22.

[280]Personal communication with E. French, February 1997.

[281]A. Wace, "Excavations at Mycenae, VII," BSA 25 (1928) 38-61.

[282]Iakovides presents his own opinion based on weights found in the tombs at Perati and theorized that they were sewn onto the bottom parts of women's dresses; S. Iakovides, "On the Use of Mycenaean Buttons," BSA 67 (1972) 113-115. See also E. Barber (1991) 51-52. I have been told recently that the stone spindle whorls from Mycenae will be published by S. Diamant.

[283]J. Carington Smith (1992) 674.

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I. Tournavitou.[284] As at Thebes, the 29 Oe tablets from the West houses at Mycenae record disbursements of wool to different outlying areas. What Shelmerdine has shown in her discussion of these tablets is that the amounts of wool being disbursed are much higher than originally thought.[285] On one tablet alone, presumably a total tablet, Oe 111 records at least 300 wool units, amounting to over 900 kilograms. In addition, one tablet recording disbursements of oil, Fo 101, lists amounts of oil sent to textile workers.[286] Shelmerdine's analysis of the long tradition associating textiles and oil has already been discussed. The theory finds good support in the documents at Mycenae.

Finally, one tablet from Mycenae provides slight evidence for the extra-regional exchange of textiles between palaces.

MY X 508

.1] te-qa-de , ta-[

] ze-ta , / pu-ka-ta-ri-ja ma-ri-ne[

This tablet appears to describe a transaction of *pu-ka-ta-ri-ja* textiles from Mycenae to Boeotian (?) Thebes. Palaima questions why such a relatively common type of cloth would be shipped from Mycenae to an area like Thebes which, as we have seen, was capable of producing its own woollen textiles.[287] Possible explanations include the technical advantages of Argolid wool and weavers or that this was some sort of gift-exchange between elites or a dedication to a divinity.

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[284]I. Tournavitou, The 'Ivory Houses' at Mycenae (1995).

[285]C. Shelmerdine (1997) 390. See the discussion of wool-houses in Mesopotamia in Chapter five.

[286]J. Killen, "Some Puzzles in a Mycenae Personnel Record," ZAnt 31 (1981) 37-45.

[287]T. Palaima (1997) 276-277.

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Mycenaean Art and Artifacts

The study of Aegean costume as depicted on wall-paintings and figurines is significant for issues related to cultural influence, self-presentation, and modes of production. The Minoan costumes of elite members of society, most especially the women, are generally seen as more elaborate and elegant. The flounced skirt of the snake goddesses and other costumes worn by women in Minoan art imply a great deal of effort by individual seamstresses and weavers.[288] This does not seem to be the work of mass-produced textiles in an industrial setting, as we infer from both the Mycenaean texts and Mycenaean representations. Elite Minoan costumes were probably individually made with a great skill. This may have been the work of ladies of leisure or their servants. In contrast, representations of Mycenaean garments are often show plain, white cloth elaborated only with decorative edgings or bands. Elite Mycenaean women continue to use some styles inherited or borrowed from the Minoans, such as the open bodice and full skirt. The textiles themselves, however, seem simpler and less individualistic. For the men, the costumes of both the Minoans and the Mycenaeans remained fairly simple. Minoan men wore a fairly skimpy loin-cloth or breech-cloth which can most often be seen in bull-jumping frescoes and on male figurines.[289] The Mycenaean men wore a kilt both in religious and military contexts. Some wall-paintings from Egypt actually show the underpainting of a Minoan breech-cloth replaced by a Mycenaean warrior's kilt.

In other works of art, a few examples of spindles made of precious materials, like gold and ivory, are preserved in the archaeological record, most often from burials. These include a gold spindle from Grave Circle A at Mycenae, five ivory ones from Perati, a bone

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[288]E. Barber (1994) 101-126.

[289]See E. Barber, "Late Bronze Age Kilts and the Reconstruction of Aegean Textile Connections," AJA 97 (1993) 350 (absract); P. Rehak, "Aegean Breechcloths, Kilts, and the Keftiu Paintings," AJA 100 (1996) 35-51.

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one at Asine, all comparable to examples found at Megiddo, Hama and Cyprus.[290] None have been reported from Crete. It may be that most of the spindles there were made of wood or that they did not exist at all. Bronze needles are also known from some Late Helladic sites, including Prosymna, Dendra, Malthi, the Menelaion, Nichoria, and a child's grave in the Athenian Agora.[291]

Homer

The epic poetry of Homer records a great variety of textiles and demonstrates that the later Greeks used cloth in ways similar to the Mycenaeans. As Shelmerdine states, 'the Homeric epics are not a reliable guide to the culture and habits of Mycenaean Greeks, but it is sometimes possible to work the other way, and to show from Late Bronze Age data that an object or practice in the Iliad or the Odyssey does accurately reflect that period.'[292] While Homer uses at least a dozen different words to refer to cloth, they fall into two basic types, plain and fancy.[293] These roughly parallel the two types described above as depicted by Minoan and Mycenaean artists and fit the basic outline of textiles known from the Linear B tablets. There is the elaborate, fine textile work of the elites, particularly the weaving of Homeric women used to display status (Il. 3.125-28; 16.220-224; 22.440-44;

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[290]See G. Karo, Die Schachtgräber von Mykenai (1930-1933) 57, pl. xvii 93, 95, 106; S. Iakovides, Perati II (1969) 350 fig. 155; O. Frödin and A. Persson, Asine. Results of the Swedish Excavations 1922-1930, (1938) fig. 252; P. Guy, Megiddo Tombs (1938) 170, fig. 175.6; P.J. Riis, Hama. Fouilles et recherches 1931-1938. II 3: Les Cimitières à crémation (1948) 173 fig. 217; F. Schaeffer, Enkomi-Alasia (1952) 181, pl. xlii; E. and J. Stewart, Vounous 1937-1938 (1950) 99, no. 6.

[291]S. Immerwahr, The Athenian Agora XIII: The Neolithic and Bronze Ages (1971) 106, 196, pl. 42.

[292]C. Shelmerdine (1995) 99.

[293]See M. Schoenhammer, "Women and Textiles in Homer," APA Abstracts (1993) 16. C. Shelmerdine (1995) analyzes cloth in Homer but focuses on specific types described as shining and fragrant. E. Barber (1991) Chapter 16, (1992), and (1994) 117-126 also discusses textiles in Homer and in later Greek literature. Finally, for a recent analysis of textiles in Classical Greek, see M. Stieber, "Textiles in Euripides' Ion," AJA 102 (1998) 392-393.

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Od. 3.349-351; 7.95-97). The other type is the plainer, every-day textiles made by slaves and captives (Il. 1.29-31; 6.456; Od. 7.103-106). The varieties of cloth are perhaps best summarized in tabular form:

Table 1. Homeric words for cloth

Cloth word	Translations	Iliad	Odyssey
ἡ ἑσθής	Elite clothing, often mentioned with gold and bronze given to highly honored people, e.g. to Odysseus from the Phaiakians, to Telemachus from Menelaus, to Eumaios from Odysseus.	24.94 (τό ἕσθος - the veil of Thetis)	1.165, 2.339, 5.38, 6.74, 83, 192, 7.6, 8.440, 13.136, 14.510, 15.207, 16.231, 23.290, 341, 24.67
ἡ ὀθόνη	Cloth, usually fine linen. Always plural.	3.141, 18.595	7.107
ἡ καλύπτρη	Veil, shroud	22.406	
ἡ χλαῖνα	A plain, thick, fleecy mantle worn over the chiton. A personal garment used as a blanket when sleeping. Simpler than the φᾶρος.	2.183, 262, 10.133, 16.224, 22.493, 24.163, 230, 646	3.349, 351, 4.50, 115, 299, 5.229, 7.338, 11.189, 14.460, 520, 17.86, 19.225, 318, 337, 20.4, 95, 143, 21.118, 23.180
ὁ ἑανός	Most often a robe worn by a goddess.	3.385, 419, 14.178, 16.9, 21.507	
ὁ πέπλος	Skillfully woven robe (e.g. by Athena, Sidonian women). Purple robe wrapping the body of Hector	5.194, 315, 338, 734, 6.90, 271, 289, 302, 8.385, 24.229, 796	6.38, 7.96, 15.105, 124, 18.292

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ὁ τάπης	Most often, purple bedding or coverings for furniture.	9.200, 10.156, 16.224, 24.230, 645	4.124, 298, 7.337, 10.12, 20.150, 24.276
ὁ χιτών	A man's undergarment.	2.42, 262, 416, 3.57, 5.113, 736, 9.490, 10.21, 13.439 (of bronze?)	1.437, 439, 3.467, 4.50, 6.214, 8.441, 13.434
τό εἷμα	A general term for a man's clothing (usually a cloak and tunic).	2.261, 3.392, 5.905, 16.670, 680, 18.517, 538, 22.154, 510, 23.67, 24.162	2.3, 4.750, 5.167, 6.26, 7.234, 8.249, 10.542, 13.218, 14.132, 154, 214, 320, 341, 396, 501, 516, 15.338, 368, 16.79, 17.550, 21.339, 22.487
τό φᾶρος	Most often a mantle or cloak worn by a man of distinction. Also, a sail cloth in Od. 2.58 and the web Penelope weaves.	2.43, 8.221, 18.353, 24.231, 580, 588	2.58, 97, 3.467, 5.230, 258, 6.214, 8.84, 392, 13.108, 15.61, 19.138, 142, 24.132, 147
τό ῥῆγος	A woollen cloth laid down as bedding or thrown over the back of a chair.	24.644	3.349, 351, 4.297, 6.38, 7.336, 10.352, 11.189, 13.73, 118, 19.318, 337, 20.141, 23.180.
τό σπεῖρον	A cloth, sheet, shroud, or sail.		2.102, 4.245, 5. 318 (sail), 6.179, 269 (sail), 19.147, 24.137

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The Homeric epics also refer to women who are 'keybearers' and control access to the palace storerooms (Od. 8.438-441; 15.99-110) that contain textiles and metals used as guest-gifts (Od. 8.387-393; 24.271-279). Women of similar function are listed in the Linear B tablets as *ka-ra-wi-po-ro* (PY Ae 110, Eb 338, Ed 317, Ep 704, Jn 829, Un 6, Vn 48) *klawiphoroi*, 'keybearers'. They are often mentioned with cultic associations. One of the women is a named Keybearer, *Karpatia*, at the sanctuary of Pylos, *Pa-ki-ja-na* (Ed 714, Eb 338). From the Homeric references and the parallels in the Linear B tablets, we see that some craftswomen and female administrators had fairly prominent roles in early Greek society and many of them were connected to the textile industry.

Conclusion

This chapter began with four criteria for determining prestige goods in early state economies.[294] The first criterion requires the presence of craft specialists who are dependent upon the palace center. This is demonstrated by personnel records listing textile specialists of various crafts who receive rations from the palace centers. The Linear B tablets also record disbursements of wool and flax acquired by the state from outlying areas for these same dependent workers to process into cloth. This fulfills the second criterion of workers receiving raw materials from the central administration. The third criterion, a high price/value equivalent for the raw materials or finished products, is evident in Mycenaean tablets that refer to the exchange of one good for another, most often involving textiles. The Mycenaean word, *qe-te-o*, has been discussed and its meaning, 'to be paid' was shown to refer to cloth and other commodities. Exchanges in bronze, alum and aromatic oil are among the goods evaluated and administered by the palaces. The fourth criterion, items presented as gifts to high-status individuals of known social rank, has also been

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[294]G. Stein and J. Blackman (1993) 50.

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demonstrated by various types cloth. *Ke-se-nu-wi-ja* cloth listed in the Ld tablets probably refers to cloth that was fit for οἱ ξένοι, *i.e.* foreign, mercenary soldiers employed by the palaces just before the great conflagrations that ended the Late Aegean Bronze Age. This cloth may have been payment for services rendered, or part of equipment supplied by the Mycenaeans for their fighting force. Other types of cloth are fit for the '*wanax*' or are meant for the 'followers'. Also fulfilling this last criterion are the Aegean tribute bearers depicted on Egyptian wall-paintings, bringing tribute, including textiles, to foreign elites.

The high value of Bronze Age cloth and its use as a medium of exchange finds parallels in the Homeric epics. In the very emotional conclusion to the Iliad, (Il. 24. 228-232) Priam puts together a ransom of his most valuable possession to offer to Achilles for the return of his son Hector's body. Priam takes from 'his clothes-chest ...twelve robes surpassingly lovely and twelve mantles to be worn single, as many blankets, as many great white cloaks, and also the same number of tunics." He also includes ten talents of gold and other objects of metal. A comparable amount of goods was supposedly given as guest-gifts to Odysseus, according to the lie he tells his father Laertes in Od. 24.274-279. Along with the metal goods (although only seven talents of gold here) and the dozen mantles, cloaks and tunics, Odysseus includes the four highly skilled female textile workers as part of the gift, perhaps reminiscent of the skilled textile women from Asia Minor listed in the tablets. Telemachos (Od. 2.339) takes metals and textiles from Ithaka for his trip and he receives the same from Menelaos and Helen as gifts of honor (Od. 15.207). Similarly, Odysseus receives cloth, gold, and bronze from Alkinoös and the Phaiakians (Od. 8.440, 13.136, 16.231) before his departure homeward.

From the evidence presented above, textiles played a vital role in the Mycenaean palace economies. In this chapter I have presented most of the textual data relevant to textile production from the Linear B records. I have not included every textual reference related to cloth but have instead isolated key issues and phases of cloth production. Since

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the archaeological evidence is so meager from the Late Bronze Age Aegean we rely on the fragmentary records of the Mycenaean scribes. From this information we see that many different types of wool and linen cloth were manufactured under palace supervision and most probably used as prestige goods and as payment for service within a wealth finance system. These varieties are paralleled in later Homeric Greek. Evidence exists also for common textiles that were assessed as a kind of tax by surrounding communities, especially in the Pylian kingdom (commodity *146). This cloth was probably a plain textile of bast fibers, perhaps linen. Large amounts of plain linen cloth coming into the center might have been used as sails for palace-sponsored overseas trade. The Linear B tablets do not provide much information on ship construction but the possibility that the Mycenaeans relied on the subordinate population to manufacture sails seems likely.[295] The best way to ensure an adequate supply is to require cloth as a tax.

This exploration of textile production provides a synthesis of views regarding the complex and multifaceted subject of Mycenaean cloth industries and tries to emphasize the archaeological and economic aspects. With the end of the Bronze Age we see a change in social organization in Greece. Palaces and controlling elites no longer administer organized textile production. In Iron Age Anatolia, however, we see a holdover of the Bronze Age palatial economy. Although the documentary evidence for textile production is not as rich in Anatolia as it is in the Aegean, the archaeological record regarding organized textile production at the palace of King Midas at Gordion in the Phrygian highlands provides almost an ideal opportunity to study cloth production.

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[295]F. Hocker and T. Palaima, "Late Bronze Age Aegean Ships and the Pylos Tablets Vn 46 and Vn 879," Minos 25-26 (1990-91) 297-317 and T. Palaima (1991) discuss the relevant evidence for maritime activity in the Linear B tablets.

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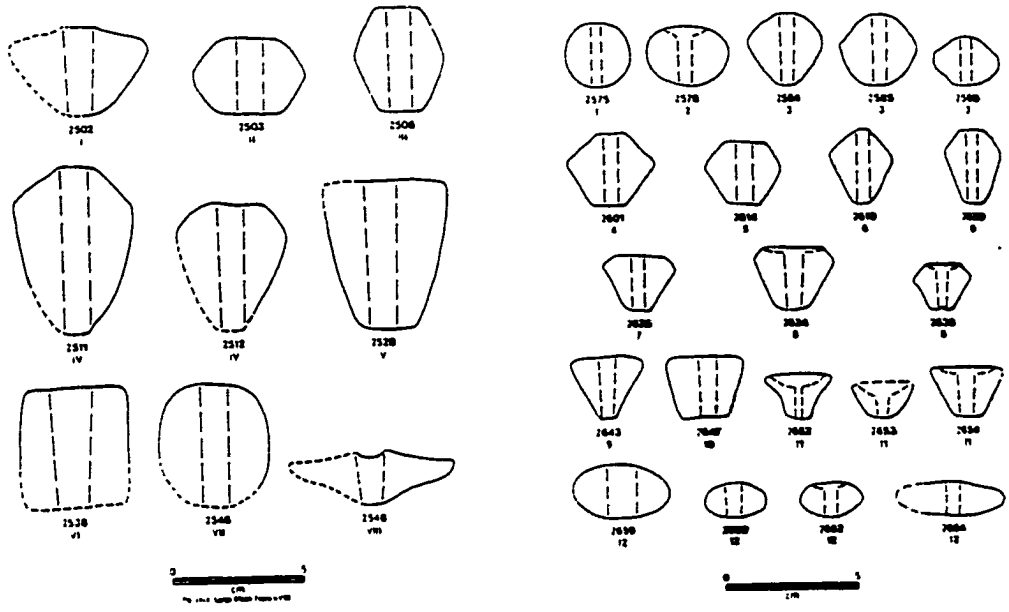


Fig. 3.1 Mycenaean spindle whorls from Nichoria



Fig. 3.2 Mycenaean loom weights from Nichoria

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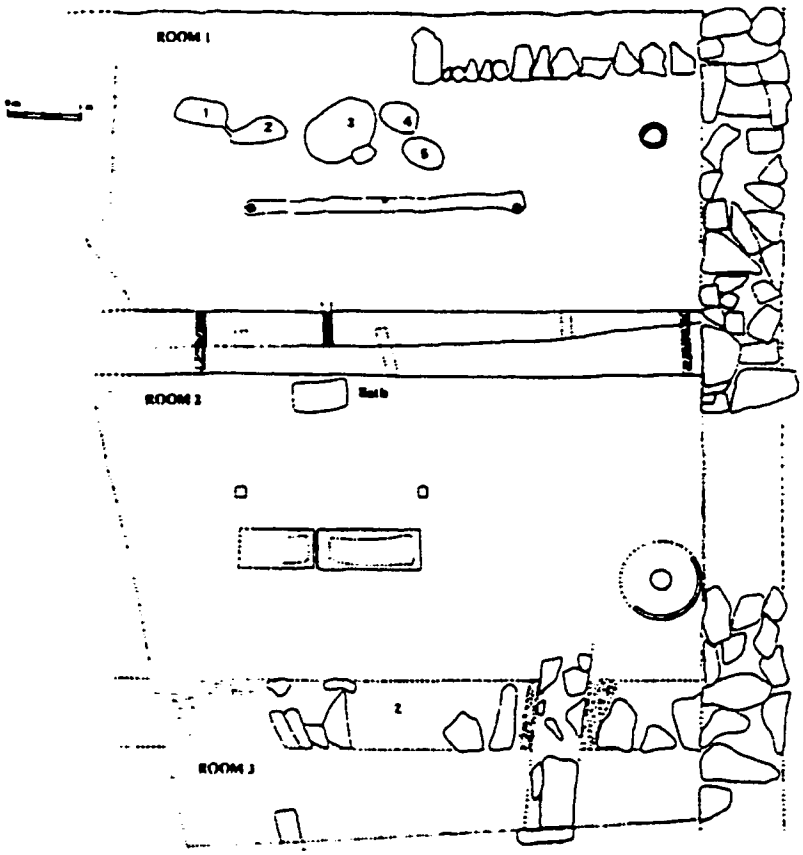


Fig. 3.3 Thebes Wool-House

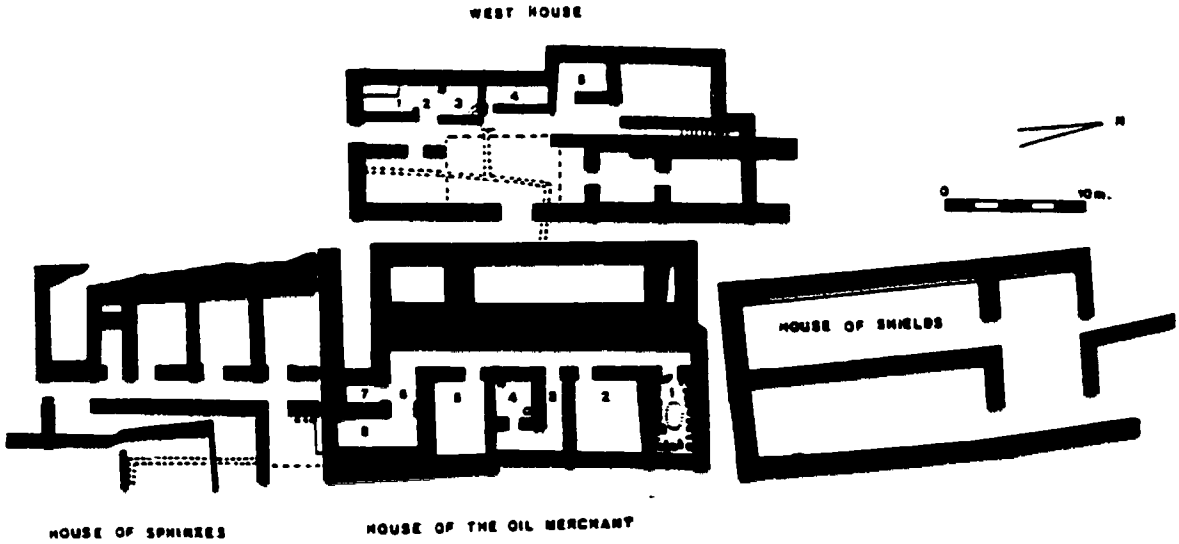


Fig. 3.4 West House Group at Mycenae

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Chapter Four - Textile Production at Gordion and the Phrygian Economy

The site of Gordion, located near the modern Turkish village of Yassıhöyük about 100 kilometers southwest of Ankara, provides almost an ideal opportunity to study the organization of a well-developed textile industry and how this craft functioned in the political economy of first millennium Anatolia. Gordion lies in the Sakarya river plain, about 688 m. above sea level.[1] The main settlement is located on a 'flat mound', giving the Turkish name Yassıhöyük to the nearby village. This area of major excavations is conventionally known as the Citadel Mound or City Mound (fig. 4.1). The höyük measures approximately 10-12 ha. and rises 24 m. above the plain. A lower town in the flood plain was of comparable size and was enclosed by a mud-brick fortification wall, remnants of which still survive. A 40-60 ha. Outer Town was located on the opposite bank of the Sakarya River although little is known about the surrounding settlement.

This chapter examines the organization and technology of the textile industry at Gordion to characterize modes of craft production that will help explain the wealth base of a ruler like the Phrygian King Midas and his early Iron Age kingdom. The discussion is divided into three parts. Part one provides the Bronze Age background to the site of Gordion, concentrating particular attention to the textile equipment found there and how these tools compare to other sites in Anatolia. In the second section the cultural context of Early Phrygian Gordion and the sources of evidence for textile production are presented.

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[1]The archaeological site corresponds to descriptions by ancient authors that say Gordion is on the Sangarios (Sakarya) river, equidistant from the Pontic and Cilician Seas (Hist. Alex. III.1.11).

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The third part describes the craft-production quarters at Gordion and the tools of textile manufacture to document the organization of Phrygian cloth production in Iron Age Anatolia. The archaeological evidence from Gordion provides the most complete picture of organized textile production at a palace center. As in the Minoan and Mycenaean kingdoms, the evidence from Gordion suggests that textiles functioned as both a medium of exchange and a valued wealth finance good in the Phrygian kingdom.

Ideally, one would like to compare the organization of textile production at a center such as the Phrygian capital with textile workshops in secondary sites or in household contexts. The extent of Phrygian culture and economic influence is only hinted at by the presence of Phrygian pottery at other Anatolian sites. As Sams states, Gordion differs from other sites in the Halys region, 'where Iron Age pottery appears suddenly and already in a fully developed stage...Since no other Phrygian center of the period is so extensively known through excavation, Gordion has the distinction of being the type-site of Phrygia.'[2] For this reason the data relevant to textile production will be limited to Gordion.

History of Excavations

The site was first identified as Gordion in 1893 by the German classicist Alfred Körte.[3] In 1900 Körte returned to Gordion with his brother Gustav, for a single excavation season. The German brothers dug five tumuli and a trial trench on the south side of the City Mound and promptly published the results in 1904.[4] For nearly fifty years after that the site lay relatively dormant.

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[2]G. K. Sams, The Gordion Excavations, 1950-1973: Final Reports vol. IV, Early Phrygian Pottery (1994) xxxi.

[3]Körte was exploring parts of Anatolia for a German railroad company when he came upon Gordion. A. Körte, "Kleinasiatische Studien II. Gordion und der Zug des Manlius gegen die Galater," AthMitt 22 (1897) 1-51.

[4]A. Körte and G. Körte, Gordion: Ergebnisse der Ausgrabung im Jahre 1900 JdI Ergänzungsheft 5 (1904).

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In 1948 the director of the University Museum, University of Pennsylvania, F. Rainey, and J. Daniel, Curator of the Museum, planned a reconnaissance mission to Turkey looking for an excavation project for the University Museum to administer. Rodney S. Young, then of the Athenian Agora excavations, accompanied Daniel and Rainey to Gordion and to other possible excavations in Turkey.[5] Daniel's unfortunate death in 1949 did not deter the University Museum's commitment to excavating in Turkey. The decision was made to begin excavations at Gordion on April 1, 1950. Under the direction of Rodney Young, with G. Roger Edwards as assistant field director, and with the assistance of Machteld Mellink of Bryn Mawr College, major excavation seasons continued alternately on odd numbered years until 1973, ending with the sudden death of Rodney Young in 1974. From 1974 to 1988 Keith DeVries was Director of the Gordion Project. Although no substantial excavations were carried out during this period, the project focused on the analysis and publication of the excavations directed by Young.

In 1988 direction of the Gordion project was entrusted to G. Kenneth Sams of the University of North Carolina at Chapel Hill. His research has focused on site conservation and the continued publication of Young's excavations. In co-operation, Mary Voigt of the College of William and Mary, renewed excavations at Gordion, concentrating on chronological and stratigraphical problems associated with the Citadel Mound and the surrounding settlement. The most recent excavations have been 'aimed at documenting changes in the form and function of the city during a series of dramatic political changes, as Gordion and the Phrygians were dominated by Lydians, Persians, Macedonians, Galatians, and Romans.'[6] With this new phase of research, scholarly focus at Gordion has

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[5]Other sites considered include Pamphylia-Pisidia and Sardis. The Harvard-Cornell project at Sardis began in 1958.

[6]M. Voigt, "New Views on an Ancient City: Excavations at Gordion, 1993-1995," AJA 99 (1996) 362. Part of Voigt's research method focuses on domestic architecture and past living habits that are traceable in the archaeological record, including the use and construction of storage pits, ovens, hearths and mud-brick.

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expanded in its multi-disciplinary scope, to include systematic paleobotanical and faunal collection, surface survey, geomorphology, geochemical characterization of ceramics, and new investigations of archaeological conservation methods.[7]

Part I - Bronze Age Gordion

Roughly contemporary with the destruction levels that ended the Mycenaean palatial system was the collapse of the Hittite empire.[8] In the first decade of the 12th century B.C., we have the last documents of the state archives from Hattusas, the Hittite capital. After that time, the citadel was then raided and destroyed by an unrecorded enemy. As early as the mid-13th century B.C. Egyptian sources refer to the 'Peoples of the Sea' as a belligerent collection of marauders of different ethnic groups causing trouble along the eastern Mediterranean.[9] The Egyptian pharaoh Ramses III, in the early twelfth century B.C., states, 'Not a single (country) remained standing before them; beginning with Hatti,

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[7]This is not to say that the Gordion project under Young's direction lacked an interest in paleobotany, faunal studies, or conservation. Studies by G. Hillman and S. Payne have been helpful for reconstructing Phrygian subsistence. The pot mending and storage system developed by E. Kohler conserved a great many artifacts that would have otherwise been lost. Since 1988, paleobotanical research has been directed by N. Miller; "Recent Research at Gordion," AJA 99 (1995) 92-93. Reconnaissance of regional settlement patterns has been undertaken by W. Sumner, K. Dickey, A. Goldman, and L. Keilhoffer; K. Dickey, "The Gordion Regional Survey: Preliminary Report for 1987-1992," AJA 97 (1993) 304; R. Henrickson has been analyzing ceramic production and technology throughout the Bronze and Iron Ages at Gordion; "Continuity and Discontinuity in the Ceramic Tradition of Gordion during the Iron Age," Anatolian Iron Ages 3: The Proceedings of the Third Anatolian Iron Ages Colloquium (eds.) A. Çilingiroğlu and D. French (1994) 95-129; R. Henrickson and M. Blackman, "Large-scale Production of Pottery at Gordion: Comparison of the Late Bronze and Early Phrygian Industries," Paléorient 22.1 (1996) 67-87. B. Marsh has investigated the ancient course of the Sakarya River and the alluviation on the valley floor around the site. Faunal analysis is presented by M. Zeder and S. Arter, "Changing Patterns of Animal Utilization at Ancient Gordion," Paléorient 20.2 (1994) 105-118.

[8]The date for the actual end of the Hittite empire is difficult to specify depending on how one defines 'empire'. I am mainly referring to the destruction of the Hittite central authority at Boğazköy.

[9]For a recent analysis of the Sea Peoples, see S. Sherratt, "'Sea Peoples' and the Economic Structure of the Late Second Millennium in the Eastern Mediterranean," Mediterranean Peoples in Transition: Thirteenth to early Tenth Centuries BCE (eds.) S. Gitin, A. Mazar, and E. Stern (1998) 292-313.

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Kizzuwadna (Qede), Carchemish, Arzawa, and Cyprus (Alasia) were destroyed.'[10]

The earliest named 'Sea People' is the *Srdn.* The *Srdn* first served as mercenaries for the Egyptians in their war with the Libyans. Other Sea peoples include the *'kws* (Achaeans?), *dyny* (Danaians?), *rk* (Lycians?), *Skrs* (Sicilians?)[11], *prst* (Philistines) and the *trs* (Etruscans or Thracians?).[12] These people destroyed the Hittite empire as well as many other capitals and were finally stopped by Ramses III. They seem to have established themselves along the shores of the Mediterranean. M. Dothan has suggested that new settlers, the Sherdana, brought purple dyeing to the Levant, beginning the intensified purple-dye production for which the region became so well-known in the Iron Age.[13]

Most of our information about the Bronze Age settlement at Gordion comes from ceramic evidence and from several Hittite burials.[14] The Bronze Age levels of the Citadel Mound were excavated in three trenches during different excavation seasons: 1) The North Central Trench (NCT), begun in 1950, dug through the back of the cella of what became known as Megaron 12; 2) Trench PN-3/3a, started in 1961, in the front of the anta of Megaron 12; and 3) a deep sounding begun in 1965 below Megaron 10. From 1951 to 1953 a Hittite cemetery was excavated under Tumuli H and I, near the location of the present Gordion Museum. These burials included inhumations, cist graves and pithos

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[10]Quoted from I. Diakonoff, The Prehistory of the Armenian People (1984), translated by L. Jennings. See also J. Breasted, Ancient Records of Egypt (1927) 4; sec. 39-44, cf. secs. 77, 81, 403.

[11]See M. Dietrich and O. Loretz, "Das 'Seefahrende Volk' von Sikila," Ugarit-Forschungen 10 (1978):53-56.

[12]I. Diakonoff (1984) 154-55.

[13]M. Dothan, "The Significance of Some Artisans' Workshops along the Canaanite Coast," in Society and Economy in the Eastern Mediterranean (c. 1500-1000 B.C.) (eds.) M. Heltzer and E. Lipinski (1988) 295-303.

[14]A. Gunter, Gordion Excavation Final Reports III: The Bronze Age (1991). See also, M. Mellink, A Hittite Cemetery at Gordion (1956); and I. Diakonov, Phrygian (1985).

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burials, which were thoroughly published by Mellink in 1956. The burials date from the Old Assyrian Colony period to the Old Hittite period and show an unbroken development in ceramic and burial customs for the second millennium.

By the early second millennium, Gordion adopts the wheelmade, red-slipped and burnished pottery that is characterized as 'Hittite'.[15] Gunter states that the pottery types at Bronze Age Gordion show a conservative orientation, lacking the sophisticated and exotic imports from the east that are found at Boğazköy.[16] Elsewhere in northwestern central Anatolia, at Polatlı and Asarcık Höyük for example, the pottery is also fairly regional. As Gunter suggests, this region probably did not engage directly in the international commerce of the Old Assyrian Colonies. Similarly, the 'destructions that brought an end to the Old Assyrian Colony period settlements at Boğazköy and Kültepe find no counterpart at Gordion'.[17]

During the Hittite Old Kingdom, Gordion may be associated with Arzawa which included the cities Sallapa and Parduwata, recorded in Hittite documents as being along the Sehiriya river, possibly the Sangarius.[18] Hittite documents found at Gordion also suggest some degree of contact if not administration by a Hittite center. These include two handles, each with a seal impression;[19] a sealed clay bulla with hieroglyphic signs in Old Hittite

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[15]See M. Mellink (1956) 51-55. Mellink believes that the term Hittite is appropriate for the Middle and Late Bronze Age in Anatolia because of the continuity in pottery styles. She also notes, however, that similar pottery styles should not imply or suggest linguistic or political associations.

[16]A. Gunter (1991) 104.

[17]A. Gunter (1991) 104.

[18]J. Garstang and O. Gurney, Hittite Geography 77-78; G. del Monte and J. Tischler, Die Orts- und Gewässernamen der hethitischen Texte. (1978) 333, 544-45; J. Mellaart Festschrift Bittel, 345-348; A. Gunter (1991) 104.

[19]For the seal impression found below Megaron 10, see M. Mellink (1956) 41, pl. 23e; H. Güterbock, "Seals and Sealing in Hittite Lands," From Athens to Gordion (ed.) K. DeVries (1980) 51; A. Gunter (1991) 72. For the other jar handle seal, see R. Young, "The Gordion Campaign of 1965," AJA 70 (1966) 277, pl. 74, fig. 24.

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from the rubble foundation of Megaron 12;[20] and a four-sided Hittite bead seal from Northeast side over the Phrygian City Wall.[21]

Spindle whorls and loom weights from below the Early Phrygian destruction stratum show that textile production had a long history at Gordion even before the arrival of the Phrygians. Two spindle whorls from the Early Bronze Age are catalogued by Gunter although many more labeled 'Early Bronze Age' were found in storage at the Gordion museum depot.[22] Dating to the Middle Bronze Age, from the sounding below Megaron 10, one spindle whorl, catalogue no. 278, and one loom weight, no. 334, were found. From below Megaron 12, spindle whorls nos. 471 and 492 also attest Bronze Age textile production.

From the Hittite cemetery, equipment associated with textile production includes a wooden comb, though it may also be a hair comb, comparable to those found at the Middle Bronze Age cemetery at Jericho.[23] A total of nine spindle whorls were also found in the cemetery excavations. Whorls 1 through 4b seem to have profiles typical for the 18th- to 15th-century, as noted by Mellink. Parallels are found at Boğazköy, Alaca, Polatlı, and Kusura.[24] Six whorls are catalogued and described by Mellink:[25]

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[20]R. Young (1966) 277, pl. 74, fig. 25; H. Güterbock (1980) 55; A. Gunter (1991) 85, 104, pl. 29.

[21]A. Gunter (1991) 85, pl. 29. Güterbock (1980).

[22]A. Gunter (1991) pl. 16 and 17, catalogue nos. 79, 80. See also R.Young (1966) 277-8.

[23]M. Mellink (1956) 42-3, pl. 23f, g catalogue no. BI 181 (from burial H 41). For combs at Jericho, see Illustrated London News (October 3, 1953) 520, 522 fig. 11. Some of these combs were found in situ on the skulls of skeletons.

[24]Boğazköy (WVDOG 60, 25), Alaca (Ausgraben (1936) Pl. 52; Rapport (1937-1939) Pl. 90-93), Polatlı (Anatolian Studies I (1951), 62, fig. 15: 11, 13) and Kusura (Archaeologia 86 (1936), 32, fig. 13: 25, 26, 29; cf. Archaeologia 87 (1937), 254).

[25]M. Mellink (1956) 43.

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1) MC 58 (burial H 18)
H. 0.021, D. 0.039. Abraded on surface. Terracotta. Conical whorl beveled around bottom and depressed at base for piercing. Buff clay.

2) MC 61 (burial H 24)
H. 0.017, D. 0.032. Mended, partly broken away. Terracotta. Biconical whorl truncated at both ends, where deep conical depressions lead to narrow central piercing. Coarse gray clay, smooth on outside.

3) MC 84 (burial H 37)
H. 0.015, D. 0.033. Terracotta. Plain flattened conical whorl, with concave depression in base. Buff clay.

4) MC 95a, b (burial H 13)
a) H. 0.021, D. 0.034. Undecorated, truncated biconical. Clay buff-grayish.
b) H. 0.017, D. 0.034. Terracotta. Similar but lower, with incised design on base consisting of three concentric circles around hole, and four groups of concentric semicircles or triangles, alternating. Fine incisions.

5) Not catalogued (burial H 30)
H. 0.014, D. 0.025-0.026. Limestone. Irregular, undecorated. Biconical whorl with flattened top and base. Pierced with a hole of 0.005 diameter.

One of the most intriguing aspects of Anatolian textile tools is the banana or crescent shaped 'loom weight'. These objects were suspended weights of some kind but it is not certain that they were used for textile production. Some examples in Anatolia have been found in work shop areas with other tools of textile production. The average weight for the crescents are also within the known range of loom weights. The crescent form, however, is difficult to reconstruct on a vertical warp-weighted loom. While it is probable that there existed weaving technologies that we are not familiar with, it may be that these weights were not used for weaving. The weights are generally called 'Hittite' loom weights although in Anatolia they first occur in the Early Bronze Age, before the Hittites.

Five crescent weights were found in various contexts at Gordion (fig. 4.2). They are similar to other examples known as 'Hittite loom-weights' by the most excavators

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working in Anatolia. Their actual function is discussed below. None has been reported from the Aegean.[26]

1) MC 102A
Notebook 33, p.94. From Clay Cut building 1, excavated in 1955. 0.4 cm string hole diameter, 2.75 cm thick, 12.5 cm. long. 166 grams.
Bronze Age crescent shaped loom-weight, complete, fired and hand-made. Stamped with crescents.

2)MC 102b
Notebook 33, p.94. From Clay Cut building 1. 0.6 cm string hole diameter, 3.5 cm thick, 11 cm long, 154 grams.
Bronze Age crescent shaped loom-weight. Fairly straight, and less curved compared to other similar weights. Orange-brown color fabric.

3) MC130
Notebook 67, p. 39. From trench MWR clay. 0.9 string hole diameter, 3.5 cm thick, incomplete length, 133 grams.
Bronze Age fragment of crescent shaped loom weight

4) MC 200
Notebook 112, p.4. From the Citadel Mound. 0.6 cm string hole diameter, 3.4 cm thick, 12 cm. long, 147 grams (incomplete).
Bronze Age crescent shaped loom weight, broken end, slightly warped. String holes appear to be on different axes than would be expected if the weight was suspended.

5) MC 312
No notebook reference, found in bag 7e5, from the City Mound- Building N. 0.9 string hole diameter, 4 cm thick, incomplete length. 132 grams, broken weight.
Bronze Age fragment of a crescent shaped loom weight, only one end preserved.

G. Vogelsang-Eastwood and J. Weingarten have recently studied other assemblages of these crescent shaped weights and conclude that they were not used for textile production and should not be called loom weights.[27] Weingarten, studying the seal impressions made on 175 crescents from Karahöyük, cites Alp's tentative interpretation that the weights were 'receipts for goods, with the seal impression made by the recipient of

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[26]See A. Gunter (1991) 84-85, fig. 526-529. See also, F. Fischer, Die hethitische Keramik von Boğazköy. WVDOG 75 (1963) 76 and n. 483.

[27]G. Vogelsang-Eastwood, "Crescent Loomweights?" Oriens Antiquus 29 (1990) 99-113 and J. Weingarten, "The Sealing Structure of Karahöyük and Some Administrative Links with Phaistos on Crete," Oriens Antiquus 29 (1990) 63-95.

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the goods'.[28] This explanation, however, does not account for the unstamped weights, unless they are taken to be 'blanks'.

Seven other sites in Anatolia have reported these same weights, all from Middle Bronze Age contexts. These are: Karahöyük with 310 weights, of which 175 are stamped with seal impressions and 104 marked in some way with slashes, stars, dots, circles and geometric patterns.[29] Demircihüyük yielded 74 weights, plus two miniature fragments, 22 with seal impressions or some kind of marking.[30] Beycesultan had 31 weights and Kusura had 30.[31] Tarsus had only had twelve crescent weights,[32] as did Alişar.[33] Finally, Troy yielded only one example.[34]

Vogelsang-Eastwood bases her argument that these crescents were not used as loom weights on three points: the number and position of the crescent holes, the various sizes of the crescents, and the distribution within the various sites. While I also question whether these weights can be associated with weaving activities, I do not find Vogelsang-Eastwood's argument completely persuasive. She states that 'all man-made (loom) weights have one hole'. From looking at Minoan evidence, weights for warp-weighted looms often have more than one suspension hole, sometimes up to four. Some of the Minoan weights were probably attached to the bottom of a pole or rod, which would have been attached to the bottom of the warp threads. The crescent shaped loom weights may not have been

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[28]S. Alp Karahöyük (1974) 75.

[29]S. Alp, Zylinder und Stempelsiegel aus Karahöyük bei Konya (1968).

[30]B. Kull, Demircihüyük vol. 5 (1996).

[31]W. Lamb, "Excavations at Kusura near Afyon Karahisar I-II" Archaeologia 86-87 (1937-1938) 256.

[32]H. Goldman, Excavations at Gözlü Kule, Tarsus (1956) 11-12, fig. 441.

[33]E. Schmidt, The Alishar Huyuk Season of 1928 and 1929. OIP XIX.Vol. IV. (1932) 122.

[34]C. Blegen, Troy VIIa, VIIb, and VIII, vol. IV (1958) 152.

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directly attached to the warp threads as well, but attached to some sort of bar. This could account for their variability in the size of the weights as well. As for the distribution of the crescents within sites, the examples cited by Vogelsang-Eastwood are fairly concentrated in certain areas of each site. However, as is the case in the Aegean, find spots of loom weights do not necessarily mean that weaving took place in that same area. Storage of loom weights is a very likely possibility, where the weights could be retrieved for a loom set up in a well-lit area, perhaps in a court yard or on the roof.

Part II - Early Phrygian Gordion

The Phrygians most likely migrated into central Anatolia from southeastern Europe sometime after the collapse of Late Bronze Age culture, based on a variety of evidence.[35] Linguistically, Phrygian is a new branch of the Indo-European family in the east, standing outside the Anatolian language groups.[36] It is related to Armenian and Greek and shows affinities to Thracian and Illyrian. Rock-cut inscriptions at Midas City in the Phrygian highlands and wax dedications from the Midas Mound at Gordion show that the Phrygians adapted the Graeco-Phoenician alphabet to their language. One of the red-painted rock-cut inscriptions at Midas City shows that the Phrygians retained the use of Mycenaean titles, *wanax* (*vanaktei*) and *lawagetas* (*lavagtaei*) in reference to Midas.[37] These inscribed titles

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[35]For arguments against this theory of migration,on the grounds that it is based on the faulty historical method of Herodotus and Xanthus of Lydia, see R. Drews, "Myths of Midas and Phrygian Migration from Europe," Klio 75 (1993) 9-26. Drews does not adequately explain the archaeological evidence in my opinion, specifically the intrusive Knobbed Ware and Early Iron Age Handmade wares in the Aegean and Anatolia. Most scholars today agree that, "...as Greek tradition claims, [the Phrygians] were newcomers from southeastern Europe during or after the turmoil of the early twelfth century B.C." M. Mellink, "The Native Kingdoms of Anatolia," CAH III.2 (1991) 621.

[36]J. Mallory, In Search of the Indo-Europeans (1982) 30.

[37]See G. Huxley, "Titles of Midas," Greek, Roman, and Byzantine Studies 2.2 (1959) 85-99; M. Lejeune, "A propos de la titulaire de Midas," Athenaeum 47 (1969) 179-192; A. Leukart, Die frühgriechischen Nomina auf -tas und -as (1994) 68-70. The titles and other Old Phrygian lexical correspondences with Greek are summarized by F. Woudhuizen, "Old Phrygian: Some Texts and Relations," Journal of Indo-

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at Midas City, in the dative, are taken to refer to the time when Midas held both of these titles. Based on this attestation, Puhvel has remarked that it is possible that in Mycenaean society, one person could have two titles, *wanax* and *lawagetas*, perhaps as both priest-king and 'leader of the people'.[38] A recent find from Daskyleion also shows that one of these titles (*wana*[*x*]) continued in use by the Phrygians even after the Old Phrygian period.[39] We are, however, severely limited by the scanty examples of the Phrygian language.

Phrygian is also related to Armenian but does not seem related to the local Hittite and Luwian language groups of central Anatolia.[40] This may be the origins of Herodotus' notion that the Phrygians sent colonizers to settle Armenia. Evidence for Phrygian spoken at Gordion at the time of the destruction level comes from wax tablets inscribed with Phrygian names found in the largest tumulus, the so-called Midas Mound, and from inscriptions found in the City Mound.[41] A Phrygian inscription far from Gordion, known

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European Studies 21.1-2 (1993) 22-23. Woudhuizen also suggests that, based on the public uses of the Phrygian alphabet and the more secure, historical dates for the earliest inscriptions, "...the odds are clearly in favour of the chronological precedence of Phrygian over Greek literacy," 4-5. While this suggested revision in the history of the alphabet seems a bit radical, stress on the use of the earliest alphabetic inscriptions does seem reasonable.

[38]J. Puhvel, "Helladic Kingship and the Gods," Minoica Festschrift for J. Sundwall (ed.) E. Grumach (1958) 327-333.

[39]An Attic Red-figure sherd was found at Daskyleion inscribed with the title FANA (*wana*); M. Mellink, "Archaeology in Anatolia," AJA 97 (1993) 121.

[40]See I. Diakonoff (1984); J. Mallory, In Search of the Indo-Europeans (1982) 30-33; M. Brixhe, "Du Paleo- au Neo-Phrygien," Academie des Inscriptions & Belles-Lettres Comptes Rendus (1993) 323-344; M. Brixhe, "Le Phrygien," in F. Bader (ed.) Langues Indo-Europeenes (1994) 165-178; M. Brixhe and A. Panayotou, "Le Thrace," in F. Bader (ed.) Langues Indo-Europeenes (1994) 179-203; K. DeVries, "Phrygian Gordion Before Midas," Anatolian Iron Ages I 1987:7; G. K. Sams, "Midas of Gordion and the Anatolian Kingdom of Phrygia," Civilizations of the Ancient Near East vol. 2 (eds.) J. Sasson, *et al.*(1995) 1147. See also J. Friedrich in *RE* 20, 1 (1941) 879-880 and 882-885 (*non vidi*).

[41]For Midas City, see E. Haspels, The Highlands of Phrygia. Sites and Monuments I-II (1971). For Old Phrygian texts, see R. Young, "Old Phrygian Inscriptions from Gordion: Toward a History of the Phrygian Alphabet," Hesperia 38 (1969) 270-275; C. Brixhe and M. Lejeune, Corpus des inscriptions paléo-phrygiennes (1984) 98-99. The geographical extent of Phrygia, as indicated by Phrygian inscriptions, is

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as the Black Stone of Tyana, also indicates that in the eighth century King Midas actually spoke Phrygian and used it in diplomatic discourse.[42] Mellink notes that this alphabetic, Phrygian inscription in Neo-Hittite, Luwian territory is a political statement that Midas is an ally of King Warpalawas of Tyana. But the use of the alphabet and Phrygian language stresses Midas' central Anatolian and European roots.

At Gordion the handmade and burnished ware pottery found in post-Bronze Age levels is probably related to similar wares found in the Aegean and suggests the arrival of newcomers in both regions.[43] Tracing this pottery and its derivatives leads one to Troy, Thrace, the Balkans, and the Greek mainland and ceramic studies by French, Rutter, Henricksen, and Sams among many.[44] Recent finds of this ware east of the Halys may better inform us of this population group but it is generally believed that they originated in Europe.[45] Drews argues that Homer's stated alliance between the Phrygians and Trojans demonstrates a Bronze Age presence of Phrygian people in Anatolia.[46] This argument, however, needs to account for the historicity of the Trojan War and the associated Homeric problems of the Trojans themselves.[47] The Trojan people Homer refers to were probably

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traced by J. Friedrich, "Phrygische Sprache," in Kleinasiatische Sprachdenkmäler (1932) 123-140; and O. Haas, "Die phrygischen sprachdenkmäler," Ling. Balk. 10 (1966) 1-260 and 19 (1976) 3.49-82, 4.53-71.

[42]J. Garstang and J. Myers, "The Black Stone of Tyana, Midas beyond the Halys," LAAA 1 (1907) 10-16; M. Mellink, "Midas in Tyana," Florilegium Anatolicum (Mélanges E. Laroche) (1979) 249-257; *Eadem*, (1991) 625.

[43]G.K. Sams (1994) 7; *Idem*, "The Early Phrygian Period at Gordion: Towards a Cultural Identity," Source 7 (1988) 13-14; R. Henrickson, "Politics, Economics, and Ceramic Continuity at Gordion in the Late Second and First Millennia B.C." Social and Cultural Contexts of New Ceramic Technologies, Ceramics and Civilization VI (ed.) W. Kingery (1993) 111-122; F. De Graaf, "Midas, Wanax, Lawagetas," Thracians and Mycenaeans (eds.) J. Best and N. DeVries (1989) 153-155.

[44]See references in G. Sams (1994).

[45]M. Mellink reports that hand-made 'barbaric' ware has been reported at Kaman-Kalehöyük; M. Mellink, "Archaeology in Anatolia," AJA 96 (1992) 130; AJA 97 (1993) 114.

[46]For a thorough critique of the ancient sources, see R. Drews (1993) 9-26.

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not indigenous to Anatolia themselves but derived from southeastern European stock instead, just as the Phrygians were. Herodotus and Strabo both believed that the Phrygians originated in Europe and migrated to central Anatolia.[48]

Architecturally, the organization of the Phrygian citadel does not resemble oriental palaces with multiple interior courtyards.[49] But rather the free-standing megaron structures with similar orientation find better parallels in buildings at Troy or in the Aegean during the Bronze Age.

Finally, the textile remains from Gordion also support a southern European origin of the Phrygians. Pieces of fabric from the tumuli around Gordion have been identified as hemp, the textile uses of which Herodotus associates with the Thracians.[50] While these sources of evidence require further analysis, the archaeological remains at Gordion shows that by the end of the eighth century Phrygian speakers had built the largest known regional capital in central Anatolia.

From one early excavation season in 1900 by the Körte brothers and continuing excavations since 1950 by the University Museum, the capital of the Iron Age Phrygians is

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[47]If we accept Troy VIIb2 as Homer's Troy, and this is not at all certain, the dominant population at the time was using Knobbed Ware pottery which is thought to have orginated in Thrace, or in some other area bordering Greece, such as Illyria. See recent articles by S. Hood and W. Burkert in The Ages of Homer: a Tribute to Emily Townsend Vermeule (eds.) S. Morris and J. Carter (1995). Burkert discounts the historicity of a Trojan War. For Thracian orgins of the Trojans, see G. Kirk, The Iliad: A Commentary II (1990) 45. The finds related to textile production at Troy during this transitional period do not show signs of cultural change; K. Balfanz, "Bronzezeitliche Spinnwirtel aus Troia," Studia Troica 5 (1995) 137.

[48]Herodotus 7.73, and Strabo 7.3.2.

[49]M. Mellink (1991) 629.

[50]L. Bellinger, "Textiles from Gordion," The Bulletin of the Needle and Bobbin Club 46.1 (1962) 8, 13, 32. This identification was not verified, however, by R. Ellis in his more recent analysis of the textiles; R. Ellis, "The Textile Remains," in Gordion Excavation Reports I: Three Great Early Tumuli (P. MM. W) (1981) Appendix V. The passage in Herodotus (4.74) refers to a description of the Scythians who used hemp for its narcotic qualities. Barber infers from this reference and other data a north and eastern origin of hemp as both a textile and narcotic in the early Iron Age into the Mediterranean region; E. Barber, Prehistoric Textiles (1991) 18 and n. 4. The Greek word *kannabis* is most probably of northeastern origin. See E. Barber (1991) 36-38 for much more details.

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being uncovered. The present investigation of textile production focuses on workshop debris from the Early Phrygian destruction level, circa 700 B.C.E., in which the Citadel Mound was destroyed in a great conflagration.[51] After this destruction, the citadel was subsequently rebuilt with few changes to the Early Phrygian plan. This rebuilding lay upon a thick layer of clay over the destroyed remains. The date of this reconstruction has been the subject of renewed investigations.[52] M. Voigt has noted that some traditionally ascribed Middle Phrygian constructions may have begun before the great Early Phrygian destruction.[53]

R. Young had first attributed the rebuilding to the Persian conquests in the 540s but then dated it to the earlier Lydian period of the late seventh or early sixth century, with a gap of fifty to one-hundred years between the Early and Middle Phrygian periods.[54] Renewed excavations by M. Voigt have obtained clear stratigraphic evidence indicating 'that very little time elapsed between the Early Phrygian destruction and a leveling operation that constituted the first stage of reconstruction.'[55] K. DeVries has recently identified Corinthian Late Geometric and Early Protocorinthian pottery in parts of the rebuilt settlement suggesting a fairly early date for the rebuilding, in the late eighth to early seventh century B.C.[56] The Early Phrygian destruction was attributed by R. Young to the raiding

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[51]According to the revised stratigraphic sequence from Gordion, this is Phase 6A; M. Voigt, "Excavations at Gordion 1988-89: The Yassıhöyük Stratigraphic Sequence," Early Iron Ages III (eds.) A. Çilingiroğlu and D. French (1994) 265-293.

[52]M. Voigt (1994) 274-275; *Eadem*, "Gordion: Changing Views of a First-Millennium City," AJA 102 (1998) 381.

[53]For example, M. Voigt suggests that the terrace for Building C (Middle Phrygian/YHSS 5) was begun before the destruction of Megaron 2 (Early Phrygian/YHSS 6A).

[54]R. Young, *s.v.* "Gordion," in R. Stillwell (ed.), Princeton Encyclopedia of Classical Sites (1976) 360; K. DeVries, "The Gordion Excavation Seasons of 1969-1973 and subsequent research," AJA 94 (1990) 391.

[55]M. Voigt (1994) 274-275.

[56]K. DeVries, "The Assyrian Destruction of Gordion?" AJA 102 (1998) 397.

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Cimmerians, whose invasion, according to ancient sources, caused King Midas to commit suicide.[57] K. DeVries has recently suggested that the Early Phrygian destruction might more appropriately be attributed to an Assyrian sack of Midas' capital in 710 or 709 B.C. based on finds of Corinthian Late Geometric and Early Protocorinthian pottery at Gordion and a close reading of the textual sources.[58] The subsequent sealing of the destruction debris by a thick layer of clay and rebuilding of the citadel has preserved a great deal of archaeological evidence from which we can reconstruct aspects of the early Phrygian economy.

Literary Evidence

Although no epigraphic testimony exists which identifies this site with historical Gordion, the location and size of the citadel correspond to ancient accounts. These references include:

Table 1. Literary references to Gordion

Ancient Source	Information about Gordion
Herodotus	-I.14.35 Midas as the first barbarian to dedicate at Delphi, a wooden throne; -V.49 The Milesian Aristagoras is quoted in a speech to the Spartans, that the Phrygians had more flocks and more fertile land than any other people in the world; -VII.73 European origin of Phrygians/Brygians from Thrace or Macedonia.

[57]The Christian chronographer Eusebios dates the destruction to 696/5 B.C. Eusebios, Chron. (Armenian version) canon, year 1321 after birth of Abraham. Julius Africanus dates it to 676 B.C.: Cramer, Anecd. Par. 2.264. See G. and A. Körte (1904) 20, 23-4; K. DeVries (1990) 388-9; G. K. Sams (1994) 1.

[58]K. DeVries (1998).

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Aristotle	-Fragments 611, 37 refers to Midas marrying Hermodice, the daughter of the King of Kyme; -*Politics* 1.3.16 (1257b16), Midas had the ability to turn what he touched into gold.
Xenophon	-Hell 1.4.1 Description of city.
Strabo	-VII.3.21, XII.5-8, XIV.5.29 records the Thracian origin of the Phrygians and presents a description of their capital.
Pollux	-Onomastikon IX.83 Midas' marriage to Demodice, daughter of 'Agamemnon' of Cyme.
Julius Africanus	-II.264 Date of Cimmerian attack in 676 B.C.
Eusebios	-Chronica 0.1.21.1 (Latin) 21.2 (Armenian) Cimmerian invasion at 696/5 B.C. and Midas' ascent to the throne in 738 B.C.
Arrian	-Anabasis II.3 mentions Gordion as the residence of Midas and the visit of Alexander.
Hellenica Oxyrhynchia	-21.6 describes the Spartan attack on Gordion
Polybios	-XXII.18.8D Description of city.
Quintus Curtius	-Hist Alex III.1-2 records the story of the knot and a description of the city.
Ovid	-*Metamorphoses* 11.85-145 recounts a detailed version of the Midas story.
Justin	-XI.7, 3-5 refers to the tradition of kingship at Gordion and the Gordion knot.
Plutarch	-Life of Alexander 18 mentions the visit of Alexander and the knot.
Pliny	-Historia Naturalis VII.74.195 suggests a Phrygian invention of embroidery.
Livy	-XXXVIII.12.1-38 recounts the campaign of Manlius Vulso

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Other classical references to the Phrygians allied with the Trojans, or Midas' marriage to an Aeolian princess of Cyme, all point to a close relationship and familiarity between Greeks and Phrygians.[59] There are also references to Phrygian garments and textiles. Pliny, for example, credits the invention of embroidery to the Phrygians though no traces of such work have been found among the Gordion textiles. This statement is false since examples of embroidery have been found in Egypt dating to the late second millennium B.C.[60] Phrygian textiles, however, did have an extraordinary reputation in antiquity. Strabo and others refer to the long tradition of skilled weavers and dyers in western Phrygia.[61]

Traces of Hittite inscriptions have been found in Bronze Age levels at Gordion suggesting some degree of centralized administration.[62] The name Midas is recorded as early as the Bronze Age in Hittite documents, although none at Gordion. One reference is to a disloyal Hittite vassal named Mita or Midas, of Pakhuwas, presumably in the area of

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[59]See K. DeVries, "Greeks and Phrygians in the Early Iron Age," From Athens to Gordion: The Papers of a Memorial Symposium for Rodney S. Young (ed.) K. DeVries (1980); O. Muscarella, "King Midas of Phrygia and the Greeks," Anatolia and the Ancient Near East: Studies in Honor of T. Özgüç (eds.) K. Emre, *et al.* (1989) 335-344.

[60]Pliny, Natural History VIII, 74.196. See further references in C. Greenewalt, jr. and L. Majewski, "Lydian Textiles," From Athens to Gordion (ed.) K. DeVries (1980) 133-4, n. 6. Wace believed that Pliny was not referring to embroidery at all but merely the decoration of cloth; A. Wace, "Weaving or Embroidery?" AJA 52 (1948) 52. The art of tapestry was probably a Syrian technology as it appears suddenly in Egypt after Thutmose III's sack of Megiddo. Levantine cloth and craftsmen (women?) were brought into Egypt at the same time new weaving technologies appear. By the mid-fourteenth century B.C. the Syrians were using embroidery to decorate tunics like Tutankhamon's. See E. Barber (1991) 158-162.

[61]Stabo XII, 6.16; XIII, 4.14; and see Pliny, Natural History XXI, 27.51.

[62]This evidence for complexity at Bronze Age Gordion is further supported by the ceramic analysis by R. Henrickson of the Hittite pottery. See R. Henrickson and M. Blackman (1996) 70. For textile production at Bronze Age Gordion, see above. The Hittite documents are published in: M. Mellink (1956) 41, pl. 23e; R. Young, "The Gordion Campaign of 1965," AJA 70 (1966) 277, pl. 74, fig. 24-25; H. Güterbock (1980) 51; A. Gunter (1991) 72, 85, 104, pl. 29.

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the Armenian highlands.[63] The text is very similar in style, language, and probably date, to the more famous Madduwatta letter which plays such an important role in the Trojan War-Ahhiyawa debate.[64] The two rebellious vassals (Maduwatta in the west and Mita in the far east) act in similar ways, breaking a treaty with the Hittites and attempting to conquer land of their own. In the Chicago Hittite dictionary the personal name Mita is not listed but the noun and adjective, *mit(t)a* is.[65] The word *mita* in Hittite most often refers to colors of wool, threads, or cloth, usually red, and sometimes to ivory, gold and food. It often occurs in ritual passages, sometimes describing wool that was both blue and red and then twisted together. The significance of this, if any, is unclear but it seems very interesting that Mita of Pakhuwas is also located in the area of Armenia, which, according to Herodotus was settled by the Phrygians (vii.73). Armenian and Phrygian are thought to be related languages and, according to Herodotus, the Phrygians and the Armenians were both famous in antiquity for their sheep and wool production.[66]

In other Near Eastern records of the Early Iron Age, references to an Anatolian tribe known as the Mushki are believed to refer to the Greek equivalent of the Phrygians. What the Phrygians called themselves is unknown but it has been recently suggested that *vrekun* in the Areyastis inscription at Midas City refers to the king 'of the Phrygians' or 'the

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[63]M. Mellink, "Mita, Mushki and Phrygians," Anadolu Araşturmaları 2 (1965) 317-325; O. Gurney, "Mita of Pahhuwa," Annals of Archaeology and Anthropology 28 (1948) 32-47. G. Sams cites a personal communication with O. Gurney that the reference to Mita is now dated to the 15th century, well before the Hittite collapse.

[64]For just some of the views on this issue, see: T. Bryce, "Ahhiyawans and Mycenaeans - an Anatolian Viewpoint," OJA 8 (1989) 297-310; D. Easton, "Hittite History and the Trojan War," The Trojan War: its Historicity and Context (eds.) L. Foxhall and J. Davies (1981) 23-45; H. Güterbock, "Troy in Hittite Texts? Wilusa, Ahhiyawa, and Hittite History," in M. Mellink (ed.) Troy and the Trojan War (1986); M. Mellink, "Postscript," in Troy and the Trojan War (1986) 93-101.

[65]H. Güterbock and H. Hoffner, (eds.) Chicago Hittite Dictionary 301-304.

[66]For more on prehistoric Armenia, see I. Diakonoff (1984).

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Phrygian'.[67] Originally the Mushki and Phrygians may have been two separate groups. Either the two peacefully united or the Phrygian element overtook the Mushki by force at some point in the Iron Age.[68] The most likely border between the two was probably the Halys river with Phrygia to the west and Mushki to the east. Recorded in the Assyrian annals of Tiglath-Pileser's first year (ca. 1115/1112-1074 B.C.) is the defeat of the Mushku with their five 'kings' and 20,000 troops, on the western shore of the Tigris River.[69] Another text states that this area was controlled by the Mushku for over 50 years.[70] Mellink states that 'The possibility of an affiliation of Kaska, the traditional enemies of the Hittites, and the Mushki is strong.'[71] Whether this tribe should be associated with the Phrygians at such an early date is open to debate.

Over two hundred years later, the Mushki still have a reputation for military strength even before the great King Midas' rise to power and appearance in historical records. The Mushka appear in Yariris' inscription at Carchemish from the first half of the eighth century.[72] They also appear along with the Urartians as enemies of the Assyrian general Shamshi-ilu on the gate lions at Til-Barsib.[73] The Assyrian references to the

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[67]A. Lubotsky, "The Old Phrygian Areyastis Inscription," Kadmos 27.1 (1988) 9-26 provides a linguistic study and interpretation of the Areyastis inscription. See also F. Woudhuizen, "Old Phrygian: Some Texts and Relations," JIES 21.1-2 (1993) 1-25 for the Areyastis inscription among others.

[68]M. Mellink suggests that the Mushki are a metonym for Phrygians, and that the Assyrian identification of Midas of Mushki indicates Phrygian Midas' control over that one tribe in northeastern Anatolia among many; M. Mellink (1991) 622.

[69]D.D. Luckenbill, Ancient Records of Assyria and Babylonia vol. 2 (1927) 4.

[70]A. Grayson, Assyrian Rulers of the Early First Millennium B.C. (1114-859 B.C.) Royal Inscriptions of Mesopotamia, Assyrian periods, vol. 2 (1991) 14, 33, 42, 53.

[71]M. Mellink (1991) 623. See also E. von Schuler, Die Kaškäer. Ein Beitrag zur Ethnographie des alten Kleinasiens (1965) 66-69.

[72]J. Hawkins, "The negatives in Hieroglyphic Luwian," Anatolian Studies 25 (1975) 152.

[73]F. Thureau-Dangin and M. Dunand, Til-Barsib (1936) 149.

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Mushki ruler 'Mita' are probably the Semitic variant of the Phrygian name 'Midas'.[74] Midas emerges from these texts as an ambitious king who, early in his reign, tried to oppose Assyrian influence in Asia Minor with the aid of alliances, diplomatic moves, espionage and incidental aggression.[75]

Textual sources tell us that Mita provided moral and economic support to disloyal Assyrian allies, including Pisiri of Carchemish in 717 B.C. and the King of Tabal.[76] The Assyrians, however, quickly establish a policy of installing their own governors in key cities of southeastern Anatolia to counterbalance the Phrygian influence. Losses by the Phrygians between 715 and 709 B.C. to the Assyrians proved too great for Midas and his kingdom and a peace agreement was soon brokered. Part of this agreement included an Assyrian envoy stationed at Midas' court, the return of Phrygian captives, and a Phrygian delegation sent with tribute to Sargon II (722-704 B.C.) in Babylon, sometime between 717 and 709 B.C., according to his annals.[77] Also in the Annals of Sargon, in 707 B.C. the trans-Caucasian *Gimurru* first appear in Urartu. These may be the Cimmerians, possibly *en route* to Gordion.[78]

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[74]The similarities between the words 'Mushki' and Greek 'Moschoi' and Biblical 'Meshech' suggest a connection although no linguistic identification of the Mushki is possible; M. Mellink (1991) 623.

[75]D.D. Luckenbill (1927) 8. And see H. Saggs, "The Nimrud Letters, 1952- part IV," Iraq XX (1958) 182ff; J. Postgate, "Assyrian texts and fragments," Iraq 35 (1973) 13-36.

[76]H. Saggs (1958) 204-207.

[77]D.D. Luckenbill (1927) 8ff; Nimrud letter 39 published by H. Saggs (1958) 182-212; J. Postgate, (1973) letter 21-34; M. Mellink (1991) 622-23; A. Grayson, "Assyria: Tiglath-Pileser III - Sargon II," CAH III.2 (1991) 91-92.

[78]See, however, the discussion regarding the destruction level at Gordion (*infra*); K. DeVries (1998).

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The Legends of Midas and Gordion

The name Gordion has been interpreted as the contracted form of Gordieion, *i.e.* the seat of Gordius, father of Midas.[79] It seems more likely, however, that the term Gordion was misinterpreted by Greeks as a patronymic and that it is related to an Indo-European word referring to a protected place or enclosure.[80] Referring to Herodotus' statement that he saw the throne dedicated at Delphi by Midas, Drews speculates that the Greek inscription might have read 'Midas of Gordion,' rather than 'Midas, son of Gordios'. Because of the great distance between Greece and Gordion, the place-name may have been mistaken for a family name. There are no clear historical references to Gordios or Gordias, only to his famous son and the later references to the knotted chariot yoke dedicated at Gordion.[81]

According to Justin, Gordius became king of Phrygia because of a Phrygian prophecy. To consult the oracle at Gordion about circling birds Gordius drove his wagon to the 'Temple of Zeus,' traces of which have never been found at Gordion. When the Phrygians, who were in the midst of a civil war, saw this, they recalled the prophecy that said the man who first drives his cart to the temple would be their king. According to Arrian it was Midas himself who drove the wagon. This wagon was then dedicated to the temple and the twine of cornel bark holding the yoke in place became the 'Gordion knot'. The undoer of this knot would rule all of Asia. In 333 B.C., Alexander the Great visited the then insignificant community of Gordion and is said to have cut the famous Gordion

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[79]G. K. Sams (1995) 1148.

[80]See R. Drews (1993) 21 who compares Gordion to Germanic *garten*, and Slavic *gorod/grad*. One might also include Hittite *gurda*: L. Roller, "Midas and the Gordion Knot," Classical Antiquity 3.2 (1984) 260.

[81]M. Mellink (1991) 624. The use of a mythical foundation hero for a titular figure follows Hittite patterns.

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knot.[82] Most of the ancient sources (Plutarch, Arrian, and Curtius) contend that Alexander came to Gordion only for strategic, military reasons and was then told of the legend of the knot. Justin claims that Alexander was driven to Gordion by the legend of the knot in his quest to rule all of Asia.

Distinguishing between Midas as an historical figure and legendary character can sometimes be difficult.[83] The most enduring legendary image of Midas is as a man of excessive wealth. Tyrtaios (fr. 12, 6) in the mid-seventh century already uses Midas as an example of wealth. Aristophanes, Plato and Clement all cite the Phrygian king as the archetypal 'wealthy' man. Another story, often shown on Greek vases, has Midas capturing the satyr Silenos and asking him the meaning of happiness and life.[84] Around 425 B.C., for the first time, Midas is shown interviewing Silenos, but here with huge ass-ears, a characteristic that stays with him throughout antiquity. No one can adequately explain why he is shown with ass-ears, perhaps to show his search for knowledge, his paranoia of spies, or his foolishness for preferring Marsyas' music to Apollo's. The qualities of excessive wealth and ass' ears are combined in the best known legend of Midas, that he wished, and received, the ability to turn everything he touched into gold. This gift leads to Midas' inability to eat until he bathes in the Pactolos River, washing away his gift/curse and leaving the river beds rich in gold.[85]

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[82]For Gordion in the Hellenistic period, see the summary in F. Winter, "Phrygian Gordion in the Hellenistic Period," Source: Notes in the History of Art VII.3-4 (1988) 60-71.

[83]See, for example, R. Drews (1993) 9-26; L. Roller "The Legend of Midas," Classical Antiquity 2.2 (1983) 299-313; L. Roller (1984) 256-271; J. Mallory, In Search of the Indo-Europeans (1982) 31.

[84]For a catalogue of depictions of Midas on Greek vases see M. Miller, "Midas as Great King," Antike Kunst 31 (1988) 79-89, pl. 18-19.

[85]This legend is first mentioned in Aristotle Politics 1.3.16 (=1257b16). The best known account of this story is found in Ovid, Metamorphoses 11.85-145.

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Sources of Evidence

Literary and historical references to Gordion, the Phrygians and King Midas, help us to understand the cultural context of the great textile industry preserved in the Early Phrygian destruction level. Only a few inscriptions in the Phrygian alphabet, however, have been found in the excavations at Gordion, mostly of personal names and simple dedications.[86] None of the texts identify the site as Gordion by name and no Phrygian administrative documents have been found. Since there is nothing comparable to the Linear B documents describing the resources and labor organization, we must rely on the archaeological indicators of craft activity, workshop spaces and craft residues, for a reconstruction of the textile industry. These finds occur in the Terrace Buildings (TB) and Clay Cut (CC) area of the site and will be described in detail. The masses of loom-weights, spindle whorls, and other tools found in the destruction level of Gordion provide the best archaeological evidence for attached, specialized production of textiles of any site in the Mediterranean region.

In addition to the textile tools found at Gordion, there are fragments of wool and linen cloth preserved in the many tumuli surrounding the site and from the destruction level of the citadel itself. The evidence from the great Midas Mound is especially helpful because of its approximate contemporaneity with the conflagration that destroyed the center. Comparable textile pieces were found in the southeast corner of Megaron 3.[87] This fragmentary material is more properly the field of textile conservators and material scientists. In this study of Phrygian economics and craft organization, however, a

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[86]See R. Young, "Old Phrygian Inscriptions from Gordion: Toward a History of the Phrygian Alaphabet," Hesperia 38 (1969) 270-275; and also C. Brixhe and M. Lejeune, Corpus des inscriptions paléo-phrygiennes (1984) 73-214.

[87]K. DeVries (1980) 35, fig. 6.

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preliminary summary of the excavated textile fragments as it relates to cloth production is presented, although a more detailed publication of the cloth evidence is necessary.

Spindle Whorls

Comparable in concentration to the spindle whorl finds from Early Bronze Age levels at Troy, the whorls from Early Phrygian Gordion number well over one thousand.[88] (fig. 4.3, charts 1-14). These fired clay weights were most likely attached to the end of a wooden spindle and used for drawing out and spinning together wool or linen fibers into usable thread. The whorls from Gordion range in weight from very small, almost bead-like, to so large that some may actually have functioned as light loom weights. The average weight of a spindle whorl from Gordion was twenty-four grams. The distribution of these whorls, as will be demonstrated, shows that the majority weighed less than twenty grams. The lighter weighted whorls were probably used for the spinning of finer thread. While ethnographic studies of spinners suggest that experienced spinners can spin with a broad range of whorl weights, a lighter weight whorl is preferred for finer thread.[89] Heavier whorls were more likely used to ply two or more threads together, suitable for weaving fairly heavy textiles.

None of the whorls from the Terrace or Clay Cut buildings showed signs of distinctive decoration. Most examples were asymmetrical, biconical whorls, with fairly smooth surfaces, and some had a mica slip. Even though many of the whorls were

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[88]The total estimated number of spindle whorls from Troy, between 10,000-20,000, comes from all Bronze Age levels. The Trojan examples differ also in that many of them are incised and decorated. The ones from Gordion are universally plain. See K. Balfanz (1995) 117-144.

[89]M. Parsons, "Spindle whorls from the Teotihuacan Valley, Mexico," in Miscellaneous Studies in Mexican Prehistory, by M. Spence, J. Parsons, and M. Hrones Parsons. Occasional Papers, Museum of Anthropology, University of Michigan, no. 45 (1972) 65; *Eadem*, "The Distribution of Late Postclassic Spindle Whorls in the Valley of Mexico," American Antiquity 40 (1975) 209-215; J. and M. Parsons, Maguey Utilization in Highland Central Mexico: an Archaeological Ethnography (1990) 314-315.

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severely damaged by fire, some revealed traces of vegetable temper in a fairly standardized fabric. After cataloguing these whorls from Gordion I distinguished ten different types based solely on the shape and profile of the whorls (fig.4.3). Type one was by far the most common, followed by types three and nine.

After examining the entire Early Phrygian assemblage of whorls, however, and looking at the distribution of shapes, I was not able to discern a pattern based on the types. Spindle whorl typologies, such as the 31 types proposed by Blegen for the Trojan whorls, seem geared toward distinguishing chronological and cultural affiliation with the whorl-shape types.[90] The whorls from the destruction level at Gordion are all of the same time period and culture, *i.e.* Early Phrygian. For this reason, and because of the large numbers of whorls, the main classification system used here is based on the weight of the whorls, in ten-gram weight intervals. This seems the most meaningful method for reconstructing the types of threads and textiles being produced at Gordion.

Spindle whorls provide information about ancient thread. Lighter whorls were most probably used for spinning wool while the heavier outlying whorls could have been used for plying or spinning flax fibers together. Thread can be spun in either of two directions. With fiber held vertically in front of the spinner, the whorl and spindle are spun in one of two directions. Textile specialists refer to Z-spun and S-spun, based on the direction of the spun thread following the direction of the central part of the letters, Z or S. According to Ellis, all the fibers from the Gordion tumuli seem to be spun in the Z-direction, or, once spun, two threads are twisted together but in the S-direction, with a fairly tight angle of twist of about 45 degrees.[91]

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[90]C. Blegen, *et al.* Troy I. General Introduction. The First and Second Settlements (1950) Table 128; reproduced in K. Balfanz (1995) Abb 4. A typology of seven whorl shapes was made for the 227 spindle whorls from prehistoric Aphrodisias. See M. Joukowsky (ed.) Prehistoric Aphrodisias (1986) 373-378. Another recent study of spindle whorls explores the issue of Uruk expansion at Hacinebi; K. Keith, "Late Chalcolithic Spindle Whorls at Hacinebi Tepe, Turkey," (1996) *unpublished ms.*

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Loom Weights

The loom weights from Gordion are often referred to as 'doughnuts' because of their round shape and large central hole (fig. 4.4, chart 15). Other shapes of loom weight occur at Gordion in nearly every occupation level but none in such large numbers as the Early Phrygian doughnuts.[92] Catalogued examples vary in quality of preservation. Probably all of the weights were originally sun-dried clay from the banks of the Sakarya river and only preserved accidentally in the fire that destroyed the Citadel in the late eighth century B.C. Straw was mixed in with the pebbley clay for consistency. Only a small percentage of the recorded loom weights from Gordion were available for study, as most had either been discarded or deteriorated in storage.[93] Oddly, the two most substantial deposits of loom weights that were preserved enough for study were the first and last Phrygian loom weights discovered at Gordion. One assemblage was found in the anteroom of Terrace Building 2, excavated in 1989, and the other in Clay Cut 1, found in 1952. These weights were hand-made and fairly coarse. Small pebbles and vegetable matter were added to the clay as a kind of temper. As with the spindle whorls, the various shapes of Gordion loom weights, ovoid, conical, and cylindrical, are probably not related to different weaving technologies. All shapes of weight were found in piles together and the weights ranged across the shape types. That is, a certain shape of loom weight did not weigh within a limited range. The average weight for the Gordion loom weights was around five-hundred grams each. The majority of weights ranged from four to seven

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[91]R. Ellis (1981) Appendix V, 296. This compares to the fairly loose single twist which is set at about 30 degrees. Bellinger, however, identified some threads as S-spun; L. Bellinger (1962) 4-33.

[92]Interestingly, recent excavations in the Galatian period settlement at Gordion have uncovered loom weights of sundried river clay similar in shape to the Early Phrygian examples.

[93]See R. Johnston, Pottery Practices during the 6th-8th Centuries B.C. at Gordion in Central Anatolia: An Analytical and Synthesizing Study Ph. D. Dissertation, The Pennsylvania State University, (1970) 221-236 for an analysis of the clay and some of the loom weights from the destruction level.

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hundred grams. These are fairly heavy weights for weaving and suggest that the cloth being made in the workshop units was fairly thick and heavy cloth, probably not fine or elaborate fabrics.

Unlike some loom weights from antiquity, the shape of the Gordion loom weights suggests that warp-threads were directly attached to the unfired clay weights, through the central hole. It is estimated that at least fifteen warp threads could be attached to each loom weight. The two examples from the Terrace Buildings of loom weights found in a row, as if the loom were still standing at the time of destruction, suggest that the looms used about twenty weights each, in two rows for two warp sheds. Various deposits of weights as recorded in the notebooks will be described below.

Ivory Implements

The ivory, shell, and bone implements from Gordion have been catalogued and studied by P. Sheftel, in her 1974 dissertation.[94] None of the spindles and whorls catalogued by Sheftel date to the Phrygian destruction. The majority of these come from Roman burials. Flat bone bobbins, however, do date to the Early Phrygian level, and some come from the Terrace and Clay Cut buildings. These flat pointed implements have long notched heads and were possibly used to pass the weft threads back and forth through the warp. Pointed implements and awls of bone were also found in the City Mound of Gordion. It is possible that these also played a role in textile manufacture, such as punching holes in cloth or leather. So-called bodkins are also known from Gordion, although none were found in the destruction level so they are not included here.[95] Description of the bone shuttles and punches from the destruction level at Gordion, based on Sheftel's catalogue, are incorporated into the discussion of each room in

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[94]P. Sheftel, The Ivory, Bone and Shell Objects from Gordion from the Campaigns of 1950 through 1973 Ph. D. dissertation, University of Pennsylvania (1974).

[95]See P. Sheftel (1974) 229-237.

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which they were found. One extremely interesting object of ivory was found in the recent excavations directed by M. Voigt (fig. 4.5 a and b). The charred remains of a toothed weaving implement, perhaps a reed or comb, was found with the remains of woven fabric on one side and unwoven threads on the others. This object will be described in the discussion of TB 2 A. This tool, while not as elaborately decorated, compares well with an ivory comb dating to the Middle Phrygian period, perhaps in the sixth century, with a griffin and sphinx carved on it.[96]

Iron Objects

Along with the clay finds of spindle whorls and loom weights and the bone and ivory tools, the destruction level at Gordion yielded a significant number of bronze and iron implements probably associated with textile production. The iron artifacts have been catalogued and studied by J. McClellan in her University of Pennsylvania dissertation of 1975.[97] The bronze objects, however, have not yet been studied.

McClellan states that iron needles are fairly common in the destruction level at Gordion. They differ in shape and size, suggesting various uses such as stitching fine cloth, working heavier materials like leather, and for different techniques such as over-casting. The eyes of the iron needles, though often obscured by rust, seem to have been made by 'drawing the needle to a point and then bending it down against the shank, to form a thin loop', leaving the eye in the shape of an elongated oval.[98] The self-soldering character of iron makes this the preferred eye technique. That many of the needles were found with spindle whorls and loom weights, some in the same vessel, suggests that these were kits for textile workers. The rolled hems and other stitches on cloth from the tumuli

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[96]Persian comb published by R. Young AJA 60 (1956) plate 86, fig. 23.

[97]J. McClellan, The Iron Objects from Gordion. A Typological and Functional Analysis Ph. D. dissertation, University of Pennsylvania (1975).

[98]J. McClellan (1975) 655.

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and City Mound do attest to the use of needles in Phrygian textiles. Although Pliny incorrectly asserted that the Phrygians invented the art of embroidery, we can safely assume that they did know of the technique.[99] In addition, plaques of bronze with holes along the edges, scales of an iron corselet, a cheek piece of a helmet, and fragments of leather shoes demonstrate that needles were used on different types of materials at Gordion.

Two iron needles, one a 'canvas' needle and the other slightly curved, were found at Midas City.[100] All iron needles from Boğazköy come from Phrygian levels and are comparable in form and manufacture to the Gordion examples.[101] From Alişar iron needles are both slit-eyed and looped.[102] At these sites, and in the Aegean and the Near East, bronze needles far outnumber iron examples.[103] Bronze needles were also found at Gordion but not cataloged by McClellan since her focus was only on objects of iron. She states that these needles were not associated with other tools for sewing or weaving but are similar in form to the iron examples.[104]

A substantial number of bronze and iron knives were also found in the Terrace and Clay Cut building (fig. 4.6). They often occur with spindle whorls, needles and ceramic vessels. The knives are either straight or have a slightly concave shape and an interior sharp edge. They range in length from about eight to twelve centimeters. The base of

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[99]None of the textile fragments from Gordion show embroidery, however.

[100]C. Haspels, Midas City III (1971) pl. 42d:2, 4.

[101]R. Boehmer, Die Kleinfunde von Boğazköy: Boğazköy-Hattuša VII, WVDOG 87 (1972) pl. 49; 1351-4, 1361-2.

[102]E. Schmidt The Alishar Hüyük, Season of 1927 OIP 7 (1930) 99, fig. 70; H. von der Osten, The Alişar Hüyük, Seasons of 1930-1932 OIP 30 (1937) 179, fig. 200; c756, and 182, fig. 203, e1930.

[103]Sites mentioned by McClellan include Hama, Megiddo, Tell Fara. From later dates at Hazor, Naucratis, and Persepolis.

[104]J. McClellan (1975) 656.

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these knives often have pegs for hafting them to wooden or possibly bone handles. The majority of them were found in the more northern work room units, TB 7, 8, and CC 3.[105]

Fabrics

While McClellan observes that the scraps of clothing preserved from Gordion show '...no evidence for shaping the garments by cutting and sewing,' the analyses of Bellinger, Ellis and Ballard have shown that the textiles from Gordion are fairly sophisticated.[106] Stitching at least is indicated by the hems of certain fabrics. McClellan also mentions the short tunic with sleeves shown in Phrygian art, including the 'doodle stones', showing figures wearing tailored clothes. From this evidence she suggests that cut and sewn garments might be included among Phrygian textiles.

Looking briefly at the actual pieces of cloth from the tumuli and citadel mound will inform us about different qualities of cloth. The first account of textiles from Gordion was by Louisa Bellinger in 1962.[107] Bellinger, then at the Textile Museum in Washington D.C., identified various types of fibers: linen, hemp, wool and mohair. Cotton and silk were not present. How these identifications were made is not clear though this range of fibers is not unexpected for Iron Age Anatolia. Bellinger does not give the total number of textile fragments she examined, only 'groups of specimens'. Also, no context for the textile finds is mentioned so we are uncertain whether Bellinger is referring to finds from the tumuli or from the Citadel Mound.[108] The sketches accompanying the dull

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[105]Referred to by K. DeVries (1981) 40 and supplemented with information kindly provided by E. Kohler in the Gordion archives.

[106]J. McClellan (1975) 658.

[107]L. Bellinger (1962) 4-33.

[108]K. DeVries points out that one textile identified by Bellinger as coming from a tumulus actually came from Megaron 3; L. Bellinger (1962) 15-16, pl. 10-16 and K. DeVries (1981) 35, n. 13.

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photographs do indicate that the textiles from Gordion were elaborately decorated and finely made.

The only other published study of Gordion textiles is an appendix by Richard Ellis to the first Gordion Studies volumes, The Three Great Early Tumuli.[109] Ellis states that this is a preliminary study because it focuses only on textiles from the three great tumuli, MM, P, and W, not from the Citadel Mound itself, and because the examination of the fibers was made only with a binocular microscope rather than a scanning electron microscope or with chemical analysis. Consequently, no complete publication of all the textile remains from Gordion has as yet been made. Ellis was unable to confirm the fiber identifications of Bellinger and prefers a more cautious identification of threads as either animal or vegetable. The thread-spin is examined which provides interesting data in relation to the thousand-plus spindle whorls found at Gordion.

Continuing the textile analysis today is Mary Ballard, Senior Textiles Conservator at the Conservation Analytical Laboratory of the Smithsonian Institution. She reports that the condition of the textiles varies greatly; some are well preserved, fairly pliable and strong. Others, however, are brittle, nearly dust which makes identification of their structure difficult if not impossible. Some samples are very brittle but have a fairly easily identifiable structure due to preservation in the proximity of copper alloys, turning them into pseudomorphs. A preservative on some of the fabrics has left small grains of a smooth white substance on the yarns.[110] Ballard reports that many of the fibers at Gordion appear to be wool or goat and are very fine, some with diameters falling into the range of

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[109]R. Ellis (1981) 294-310.

[110]R. Ellis (1981) 295.

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cashmere, mohair, and merino, at 19-20 microns.[111] In some examples, the epicuticle is abraded off so none of the characteristic scales of animal fibers are present. Sometimes crystalline material obscures the fiber structure or, in other cases, the fiber has collapsed.

Of the bast fibers, Ballard reports that they are probably flax, with diameters in the 10 micron range, comparable to Egyptian linen in fineness. She was unable to distinguish between hemp and linen under the microscope, like Bellinger and Ellis. Further evidence is necessary to determine whether or not these textiles were manufactured at Gordion but it does not seem unlikely. The cloth remains show a very discerning and highly developed textile industry probably located at Gordion.

Three different types of fabric structure are reported for textiles from Gordion: plain weave with variations, weft wrapping ('Soumak'), and felt.[112] The tumuli textiles are only plain weave and felt. No examples of the Soumak were found in the burials, only in the City Mound at Gordion. Ellis groups the fragments into nine different 'fabric' groups, based on fiber and technical similarities. These are labeled Fabric A through K, numbering eleven different fabrics. Each fabric is thoroughly described by fiber, color, spin, weave, and a description of the starting borders and hems. Also, find spot and probable function of the fabric is given.

Fabric A[113]

Fabric A was found in the great tumulus MM, at the head and foot of the burial bed and in the bronze cauldron, MM 13. The fragments from the bed may have been the sheets

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[111]M. Ballard, personal communication (October 31, 1996). She also reports some of her research for the excavations at Sardis where she analysed a fiber inside a bead. This thread was found to be goat hair with a diameter of modern cashmere, between 6.1 and 6.4 microns. The yarn was 0.05 mm wide, extremely fine.

[112]R. Ellis (1981) 296.

[113]R. Ellis (1981) 301-2, figs. 137-139.

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underneath heavier blankets. From the cauldron, the fabric was probably a bag containing unidentified food, presumably grains. This Z-spun thread, double plied in the S direction (hereafter referred to as Z2S), is yellow-brown to orange-brown, and was identified by Bellinger as mohair, but to Ellis it is a vegetable fiber. The thread diameter varies between 0.1 and 0.6 mm. The fabric is interesting because the edges show a separately woven band that is only found on two other fabrics from the tumuli.[114] A band is woven separately with long looped weft threads which, when placed on the warp-weighted loom, act as the warp threads on the final textile.

Fabric A is a balanced plain weave, *i.e.* with the same number of warp threads as weft per centimeter, over one, under one. On average, there are 16 to 24 threads per centimeter. Some of the warp threads are fairly crowded, resulting in a slightly striped appearance on the fabric. One error in this fabric is the failure of some warp thread to actually weave. This, according to Ellis, is due to a heddle of improper length so that at the change of sheds, one warp thread does not fully return but gets caught in the other shed.[115] Paired Z-spun thread of the same fiber as fabric A was sewn through rolled hems of two fragments of Fabric A.

Fabric B[116]

This vegetal fabric varies in color from yellowish-white to brown and had strongly plied (Z2S) thread. The thread diameter is about 0.2 to 0.3 mm, with about 18 threads per centimeter. The fabric is generally very similar to Fabric A but comes only from Tumulus P. Young refers to this cloth as 'cheesecloth' and found fragments of it on one of the pegs

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[114]These three fabrics, A, C, and F are the only ones with preserved starting border edges.

[115]E. Barber (1991) 198-199, fig. 7.8 reproduces and discusses this technical feature.

[116]R. Ellis (1981) 302.

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of the tomb chamber walls, suggesting that it may have covered the walls. The preserved bit was red with a simple white stripe and was probably a wall hanging.

Fabric C[117]

This is a single piece of plain weave fabric, light yellow-tan in color, presumably of linen, in the Z2S ply, with a thread diameter of 0.15-0.2mm. This is a fairly fine, thin textile, with 18 warp threads per centimeter and 22 weft threads. It presumably was part of the cloth bag that contained many of the bronze fibulae found in the Midas Mound burial chamber. The identification is not certain because there is no textile sample number in the Gordion record system but Ellis suggests this identification based on the description of Bellinger and from the excavation notebooks.

Fabric D[118]

This fabric is similar to fabric C in general description, except that examples are found in narrow strips or tapes. There are on average 40 warp threads set at 24-26 threads per centimeter, and a widely placed weft at 10 to 12. These strips are found in all three major tumuli and may have been used to tie up bundles of cloth and wrap bronze objects.

Fabric E[119]

This is a plain weave fabric, probably of brown wool, though it is very poorly preserved and matted together making analysis very difficult. Ellis states that most of the examples of this fabric are balanced with about 18-22 threads per centimeter. All of the

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[117]R. Ellis (1981) 302-303, pl. 100b.

[118]R. Ellis (1981) 303.

[119]R. Ellis (1981) 303.

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examples come from Tumulus W, two from inside a bronze bowl, one from the wall of the chamber, and two others found between two pieces of leather. Brown felt and tape of Fabric D were also found in the same bronze bowl. There may be similarities between Fabric E and F.

Fabric F[120]

This fabric is interesting among the Gordion textiles in that it uses two different yarns, one animal, the other vegetable. The two different fibers are found in various combinations in the same fabric. Some areas are made completely of the animal fibers, others of the vegetable one, and finally, the third with the vegetable fiber as the warp, and the animal as the weft. Ellis suggests that it is possible these fibers were used to make checks or stripes. The vegetable fiber is yellowish-white, some blackened by the conflagration, Z-spun, double S ply at 0.1 to 0.5 mm in diameter. The animal fiber is either a natural yellow-brown, or sometimes dyed red to purple. This is a single Z-spun fiber, moderately twisted 0.15 to 0.25 mm in diameter. All four samples of this fabric were found in Tumulus MM and had been in contact with bronze objects: a cauldron, the situlae, and a fibula.[121] The cloth was probably used to wrap the bronze objects, or, in the case of the fibula, it may have fastened a group of various textiles together.

Fabric G[122]

As fabric F was found only in Tumuli MM and W, fragments of fabric G with decorative bands were all found in Tumulus P. The textile is made of two yarns, A and B.

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[120]R. Ellis (1981) 304, pl. 100e.

[121]Small bronze cauldron (MM 7), the lion situla (MM 45), the ram situla (MM 46), and a double-pinned fibula (MM 187). Ellis (1981) 304.

[122]R. Ellis (1981) 305, figs. 145-8, pl. 101a-e.

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A is similar in color, yellowish-white, to yarn A of fabric F.[123] Likewise, yarn B is similar to F's, in that it is dark red to purple in color. Fabric G seems to have been predominantly red or purple with white patterns. The warp threads dominate in this fabric, using both yarns A and B, in warp-faced plain weave and sometimes with complementary warps. Ellis reports that the system is set at 50-60 threads per centimeter. Some of the patterns of Fabric G are meanders in white against red, white triangles at the edges. These patterned rectangular designs are depicted on the Ivriz and Bor reliefs showing King Warpalawas of Tyana, a contemporary of King Midas.[124] Selvages found on Fabric G suggests that some examples were attached to other fabrics, including a piece of Fabric B.

Fabric H[125]

This fabric was found in various contexts, Tumulus MM, P, and W. It seems to be an animal fiber, yellow or bluish gray in color, spun in fairly tight Z-singles. Not many examples are preserved but what can be said is that they were a balanced weave with 9-18 threads per centimeter.

Fabric I[126]

This fiber is identified by Bellinger as 'mohair'.[127] Eleven samples were found in Tumulus MM, all Z2S, moderately twisted, of very consistent size, 0.1-0.2 mm. They

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[123]R. Ellis (1981) 305.

[124]R. Boehmer, "Phrygische Prunkgewänder des 8. Jahrhunderts v. Chr. Herkunft und Export," Archäologischer Anzeiger 11.2 (1973) 149-172; S. Dalley, "Ancient Assyrian Textiles and the Origins of Carpet Design," Iran. Journal of Persian Studies 29 (1991) 126 -127.

[125]R. Ellis (1981) 308

[126]R. Ellis (1981) 308, pl. 101f.

[127]L. Bellinger (1962) 14.

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were all found at shoulder, elbow and hip level beside the body and were in contact with the copper-bronze fibulae which helped in their preservation. This fabric probably is the remains of a funerary cloak or blanket made of layers of cloth fastened together by pins.[128]

Fabric J[129]

Examples of this fabric are fairly small so little can be said. The fiber is presumably wool. Made of two yarns, both brown in color, they are woven in plain weave. Yarn A is Z2S, moderately twisted at 0.3-0.7 mm. Yarn B is probably the weft thread, spun Z single and lightly twisted, with a smaller diameter than A, at 0.2-0.3 mm. Two fragments of this fabric are known, one from Tumulus MM, found with leather near a fibula, and the other from Tumulus P.

Fabric K[130]

The final fabric from Gordion is non-woven. Instead it is matted together as felt. This fabric was found under the bodies in all three Great tumuli and found as padding for a stool and possibly as wall-hangings and tablecloths. The samples were very difficult to examine due to their fragmentary state. No decorative pattern can be discerned from the many different colored threads found on the surface of the thickly wadded masses.[131]

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[128]R. Ellis (1981) 308.

[129]R. Ellis (1981) 309.

[130]R. Ellis (1981) 309.

[131]R. Ellis (1981) 309.

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Phrygian Fibulae and Textiles

Among the four hundred items that accompanied the man buried in tumulus MM at Gordion were many fibulae, elaborate bronze 'safety-pins' of distinctive Phrygian design.[132] In total, over 677 examples of Blinkenberg's Asia Minor type xii fibulae were found at Gordion, 512 from the excavated tumuli.[133] Some were placed on the burial shroud, while others were contained in woven bags (Fabric C), perhaps as dedications.[134] Fibulae were also found in the Citadel Mound, including gold and silver pins from the Terrace Buildings described below.[135] Examples of Phrygian fibulae are also found outside of Gordion, dedicated at a number of Greek sanctuaries along with other Phrygian artifacts. These sites include Lindos, Samos, Olympia, the Argive Heraeum, and perhaps Perachora, Paros and Sparta.[136] Similarly, east Greek imported 'leech-type' fibulae were found in the destruction level at Gordion and in two of the eighth century tumuli (G and Y).[137] That Phrygian clothing was also dedicated along with the Phrygian pins in Greek sanctuaries is a distinct possibility.[138]

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[132]O. Muscarella, Phrygian Fibulae from Gordion (1967); See also C. Blinkenberg, Fibules grecques et orientales (1926) 204 ff. XII Types d'Asie Mineure; R. Boehmer (1973) 149-172.

[133]O. Muscarella (1967) 12.

[134]Illustrated in R. Young, *et al.* Gordion Excavation Reports, Vol. 1: Three Great Early Tumuli (P, MM, W) (1981). The large numbers of pins suggests that they were dedications since the representations of pins worn on clothing only show one or two used by one person at a time. An abundant number of pins may be an excessive display in metal wealth.

[135]O. Muscarella (1967) 16, plate III.12-13.

[136]Muscarella has suggested an interpretation of the Greek word *peronai* to mean fibulae and the etiological story of their dedication in Greek sanctuaries is to be found in Herodotus (5.87-88). After the war between Aegina and Athens, Athenian women killed the one Aeginaetan survivor with their pins and then dedicated them at Argos and Aegina; O. Muscarella, "Fibulae Representation in Sculpture," JNES (1967a) 82. However, large, straight Doric dress pins are dedicated in Greek sanctuaries and may more accurately fit Herodotus' description of long pins, rather than fibulae.

[137]O. Muscarella (1967) 62, 82.

[138]See H. Herrmann, "Hellas," in RLA IV (1975) 303, 310; F. Prayon, Phrygische Plastik (1987) 193, n. 844; O. Muscarella (1989) 339. That some Phrygian fibulae themselves were votive objects independent of

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Representations of Phrygian fibulae in sculpture show how the pins were fastened and worn.[139] Most often one fibula was worn on the torso, attaching two parts of one large outer garment. One Neo-Hittite relief of King Warpalawas of Tyana (fl. 738 B.C.), shows that Phrygian fibulae and cloth were also likely gifts between ruling elites (fig. 4.7). The king is carved on a rock relief standing before a fertility god at the source of an important spring in south central Anatolia.[140] The king's outer garment is secured by a Phrygian fibula similar to those found at Gordion and Ankara, identified by Muscarella, as type xii. 9.[141] Not only is the pin of Warpalawas of Phrygian origin, but the garment itself worn by the king shows typical Phrygian design and may have been an item of royal gift-exchange.[142]

From historical sources we know that Warpalawas, or Urballa, of Tyana, sent envoys to the Assyrian governor of Que along with messengers sent by King Midas.[143] His diplomatic moves also parallel those of other southeastern Anatolian kings. He is known to have sent tribute to Tiglath-Pileser III in the late eighth century B.C., as did Pisiri of Carchemesh. The recent explorations at the site of Göllüdağ may have located one of

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clothing is shown by the flimsy gold pins found at Ephesus which could never have fastened clothes; D. Hogarth, *et al.* Excavations at Ephesus (1908) 98, pl. V.6; 117, pl. XI.12; O. Muscarella (1967) 52, 56, n. 16; (1967a) 85, n. 19. The chronology of the fibulae find-spots in the Greek world has been reviewed by Boehmer. He shows that none can be dated with certainty to the 8th century and that they may have been heirlooms dedicated in the 7th; R. Boehmer (1973) 149-172.

[139]O. Muscarella (1967a) 82-85. Besides the examples discussed here, see the male figure on the Khorsabad frieze, plate II, fig. 1.

[140]Warpalawas was a competitor of Tiglath Pilesar III (745-727). For Hittite elements of the relief, see. E. Akurgal, The Art of the Hittites, (1949) 139-140; for Phrygian aspects, see E. Akurgal, Phrygische Kunst, (1955) 45-47.

[141]O. Muscarella (1967). See R. Boehmer (1973) 151-159 for a thorough discussion of the textile elements in this relief. The fibula has the characteristic arc form and double pin masked by a rectangular transverse plate, rather than the unusual single exposed pin. The plate is detachable and serves to shield the pins.

[142]R. Boehmer (1973) 156.

[143]M. Mellink (1991) 622-23; *Eadem* (1979) 249-257.

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Warpalawas' strong-holds.[144] Warpalawas' dedication of the Stele of Bor (Kemerhisar) again shows himself wearing a Phrygian garment and probable fibula.[145] According to Mellink this stele might be best viewed as the Tabal equivalent of the Phrygian dedication by Midas preserved on the Black Stone of Tyana.[146]

Another probable representation of Phrygian cloth is seen on an ivory figurine found in a wealthy tomb at Bayındır, in the Elmalı valley, along with Phrygian inscriptions and metal objects of Phrygian style.[147] The garment of the young boy standing at his mother's side shows the typical Phrygian lozenge patterns. Finds from these tombs in the Elmalı valley suggest Phrygian contact, if not the presence of Phrygians themselves, in southern Anatolia.

Part III - The Industrial Quarter of Gordion

Within the massive fortification walls of Gordion, excavations have revealed the characteristic Phrygian building form, called the megaron after similarly shaped buildings in the Aegean. These structures were composed of a single stepped entrance leading to an anteroom and a main hall behind, often with a hearth located near the center of the room. Defined by courtyard surfaces and internal enclosure walls, the excavated portions of the Phrygian citadel divide into three principal zones, the Palace area, the workshops, and possibly a storage area (fig. 4.8).[148] The eastern area of the excavations is thought to be

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[144]For a summary of the very impressive architectural finds and recent survey work in the area of Tyana, see W. Schirmer, *et al.* "Die Bauanlagen auf dem Göllüdağ in Kappadokien," Architectura. Zeitschrift für Geschichte der Baukunst 23.2 (1993) 121-131.

[145]The stele is broken but a Phrygian fibula is reconstructed based on a comparison with the Ivriz relief; O. Muscarella (1967a) 82; R. Boehmer (1973) 156.

[146]M. Mellink (1991) 625.

[147]See the cover illustration on the Antalya Museum catalogue.

[148]For a thorough discussion of the destruction level buildings and pottery, see G.K. Sams (1994) 2-7.

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the residential quarter of the elite because of the elaborate nature of the freestanding megaron buildings and the luxury goods found within. The monumental Phrygian gateway, located at the southeast, leads the visitor in a direct east-west line through the gate structure, formerly known as the Polychrome house, to a reception area before the first of two courtyards.[149] The axis of entry through the Phrygian gate is obtuse to the orientation of the main Phrygian buildings, causing the visitor to turn sharply to the right on approach to this first courtyard.

Just prior to the great destruction an interior enclosure wall of mud brick was built from the south side-wall of Megaron 1 to the south side-wall of Megaron 9, defining the outer courtyard with stone-slab paving.[150] This thin wall was probably a temporary makeshift enclosure while renovations to the fortifications were being made.[151] Four freestanding megarons, two on each side, face this area: Megarons 9 and 10 to the east and Megarons 1 and 2 along the west side.

Megaron 1, also known as the Burnt Phrygian Building or the Brick Building, was constructed of mud-brick and wooden timbers and found fairly well-preserved after the conflagration. Recent efforts have been undertaken to conserve the architecture of this building by covering the remains entirely and leaving a reverse image out of white gravel along the line of the ancient wall remains. The visitor will see that there was a monumental building here and understand its form while the building itself is well protected. The building was found relatively empty of goods. The storerooms between the backwall of

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[149]R. Young, Gordion yearly reports: AJA 59 (1955) 12-16; AJA 60 (1956) 257-161; AJA 61 (1957) 320; AJA 64 (1960) 233; AJA 66 (1962) 157-159.

[150]G.K. Sams (1994) 2.

[151]See K. DeVries (1990) 387-8 for a discussion and illustration of these renovations, including the drainage channel through the Polychrome House. DeVries suggests that there was a terracing operation extending the terrace fill of the TB and CC units when the destruction occurred. If completed 'it would have left the zone of the two megaron courtyards as a sunken crater'.

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Megaron 1 and the terrace wall did contain some interesting finds including a jar with 494 astragals in one pot, wheat and barley in storage, and the occasional loom weight and spindle whorl.[152]

Megaron 2, the Mosaic Building or West Phrygian House, was constructed of soft limestone blocks within vertical wooden timber supports. Two small rooms with wooden plank floors flank the threshold to the main room. On the exterior of the walls were rock-cut doodles and graffiti, both abstract and representational images including birds, humans and building facades.[153] These designs may be the casual doodles of people awaiting an official audience, but some have suggested that the hawk and lion graffiti are good evidence for a cultic significance of Megaron 2, dedicated perhaps to an Anatolian mother goddess such as Kybele.[154] The main room of Megaron 2 was decorated with the earliest known pebble mosaic from antiquity (fig. 4.9). Multi-colored river stones were laid in various patterns often found in other kinds of Phrygian art, possibly imitating Anatolian carpet designs.[155] Unfortunately, Megaron buildings 1 and 2 were mostly empty of their contents at the time of destruction so further interpretation of the buildings' functions is difficult.

Another more substantial dividing wall separates the exterior court from the second one toward the north. Approximately one-third of this dividing wall between Megarons 2 and 3 was demolished when the terrace fill for the TB and CC rooms was constructed. Through a propylon the visitor entered the most elite area of the citadel thus far

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[152]K. DeVries (1980) 37.

[153]R. Young, "Doodling at Gordion," Archaeology 22 (1969) 270-275; L. Roller, Gordion Special Studies, Vol. 1: Nonverbal Graffiti, Dipinti, and Stamps (1987); *Eadem*, "Early Phrygian Drawings from Gordion," AJA 100 (1996) 362.

[154]For an analysis of Phrygian religion, see L. Roller, "Phrygian Myth and Cult," Source: Notes in the History of Art VII.3-4 (1988) 43-50

[155]G. K. Sams (1995) 1150-51, fig. 2. R. Young, "Early Mosaics at Gordion," Expedition 7 (1965) 4-13.

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excavated.[156] Megaron buildings 3 and 4 faced a pebble court, and lay in the same orientation as Megarons 1 and 2, facing east. Megaron buildings 11 and 12 ran perpendicular across the courtyard facing north.

Megaron Building 3 is by far the most impressive of all the buildings at Gordion.[157] Its large size required two rows of interior support posts along the side and back walls, presumably supporting a balcony.[158] This building contained very wealthy items, such as ivory furniture, fine bronze and clay vessels, a deposit of gold pieces which may be pre-stamped coinage, and finely woven textiles.[159] Three sides of the exterior walls of Megaron 3 were buried to a depth of nearly two meters by the piling of terrace fill on the north, south and west.[160]

Megaron 4 was less impressive than Megaron 3 and of fairly simple architecture.[161] Measuring 22 by 12.3 meters, the building was entered from the east by a steep ramp covered with cobbled paving. It was constructed entirely of brick without the vertical and horizontal timber supports of the other megaron units. It may have been one of the last Early Phrygian buildings constructed before the great destruction since it was constructed on an extension of the terrace fill after the TB rooms and at a level higher than Megaron 3.

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[156]See R. Young (1964) pl. 89, fig. 23 for a sketch of this propylon.

[157]Yearly reports of excavations in this area by R. Young AJA (1958) 144-45; (1960) 237-40; (1962) 160-63.

[158]For a discussion of ancient balconies, see R. Dyson, Jr. "The Question of Balconies at Hasanlu," From Athens to Gordion (ed.) K. DeVries (1980) 149-157, especially the reconstruction of Megaron 3 in figure 2.

[159]See K. DeVries (1980) 34-37.

[160]R. Young (1962) 164.

[161]R. Young (1962) 163-64; (1964) 286-88; G.K. Sams (1994) 2, 16.

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A flight of stairs was also excavated at the northwest corner of the front of the building.[162] These steps led down to the terrace which supported Megaron 5 at a lower level.

The burned deposit was also very thick in Megaron 4, but, unlike the other Megaron units, the debris was highly disturbed, with pieces of joining pottery found in highly disparate places. Luxury finds were also reported from this building including a large gold figure-eight plate probably once attached to a wooden box or piece of furniture. An ivory appliqué of two horses running abreast and a mounted rider holding reins gives a good representation of Phrygian horse trappings.[163] In addition, pieces of unworked ivory and many chips and flakes from ivory carving were also found here.[164] There were also about twenty-five spindle whorls found in the main room of this building. Behind Megaron 4, a store room similar to that behind Megaron 1 contained some interesting items, including hazelnuts and approximately seventy-five doughnut-shaped loom weights.[165]

To the north of Megaron 3 and 4 buildings of an earlier phase of the citadel were excavated and found mostly emptied of their contents (fig. 4.10)[166] These remains date to the Early Iron Age, well before the destruction of the late 8th c. B.C. Named Megarons 6, 7 and 8, these buildings are believed to have housed textile and cooking facilities similar to the later Phrygian buildings built perpendicular to them on a large terrace. Megarons 7 and

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[162]These stairs and terrace walls are well illustrated in R. Young (1966) pl. 69, fig. 9.

[163]See E. Kohler, "Ivory horse-trappings from Gordion - 1961," AJA 66 (1962) 198.

[164]Young notes that these are the obvious remains of an ivory carver's workshop but does not believe that Megaron 4 could have housed such a craft quarter. He suggests that this debris is from the filling of the terrace and should be dated before the construction of Megaron 4. See R. Young (1964) 287.

[165]K. DeVries (1980) 37.

[166]G. K. Sams (1994) 15; K. DeVries (1987) 6-12; K. DeVries (1990) 374-77.

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8 contained an oven and other food preparation equipment as well as evidence for textile production. As DeVries points out, already in the early Iron Age, the strong differentiation in function between areas in the City Mound were distinguished.[167] Presumably a growing economy dictated the demolition of these buildings for an even larger workshop space while still retaining the basic megaron plan and similar functional features (fig. 4.11 a and b). This organized expansion on the built-up terrace behind the Megaron buildings demonstrates that the activities located in them, cooking and weaving, were of vital importance to the ruling elites.

Young briefly refers in the 1957 report to the South Phrygian House bordering on the workshop area, just north of the TB 1-3.[168] The two meter high terrace upon which this 'house' was built is also noted. The pottery found upside-down in stacks revealed fragments of wicker baskets at the bottom. The pottery recovered from this depot was in storage rather than in use at the time of destruction. This room marks a transition between two different areas of the citadel; the elite quarters of the Megaron buildings, and the industrial workshop area on the terrace behind.

For the expansion that replaced Megarons 6-8, an extensive terrace approximately one meter high was constructed behind the back walls of Megarons 1-4, perpendicular to the demolished Megaron buildings. This terrace left limited access to the elite quarter of the Megaron buildings in the central court. On it were two very long buildings subdivided into nearly identical megaron units all in a row with party walls 1.5 m. thick (fig. 4.11 a). Twelve individual units have been uncovered to varying degrees but all include a main room and anteroom, and all measure approximately 11.5 m. in width by 21 m. in interior length. The anterooms were about 7 m. deep and the main rooms measured 13 m. wide.

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[167]K. DeVries (1990) 376.

[168]R. Young (1957) 323-324, pl. 90, fig. 13.

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The buildings were of different colored sandstone and limestone, with courses of stone, wood, and mud brick. The entire length of one, called the Terrace Building, or TB, has been fully excavated to its length of 105 meters. Six workshop units and two storage rooms (TB 1-8) of nearly identical megaron plan subdivide it. All of the main rooms had two rows of four post holes with a central one between the innermost pair. These posts would have supported a second story wooden gallery which surrounded the room on three sides. Twenty meters to the southwest, across a broad street, was the parallel building named the Clay Cut (CC), presumably of the same length as the facing Terrace Building although only portions of four megaron units have been excavated.[169] The two buildings were planned and constructed together relatively late in the Early Phrygian period during one very ambitious building program sometime in the eighth century B.C.

Some of the walls of the elite buildings (Megarons 1-4) served as a retaining wall for the terrace fill upon which the workshop units stood.[170] Areas between the terrace fill and Megaron 1 and 2 were partitioned into storage areas, as was mentioned. The entire area seems to have been an industrial center for Phrygian textile and food production and will be examined in detail below. Further to the north of these workshops was the so-called PPB, or Persian-Phrygian Building.[171] This building was the only Early Phrygian building to be reused in the later Middle Phrygian period. What remains is a cluster of rooms without entrances so they are thought to be basement rooms.[172] PPB was not destroyed and continued in use after the devastating fire that raged over the other two zones, the public megaron buildings and the workshop units on the terrace. Further

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[169]K. DeVries (1990) 371-406.

[170]G. K. Sams (1994) 3.

[171]See K. DeVries (1990) 379, 394-5, figs. 10, 29.

[172]The inscribed belongings of one Phrygian individual, *Woine(s)*, were found in these excavations.

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research into these units will help us better understand the organization of space at Gordion.

Textile equipment, food processing features and large quantities of pottery were the main finds from the Terrace buildings. Features related to food processing include saddle querns (fig. 4.12) and kneading trays, along with the charred seeds and grains of some of the foods being prepared. The plant remains include wheat, barley, lentils, hazelnuts, and cornelian cherries.[173] The anterooms usually had one or two round dome ovens of brick and clay, a U-shaped construction of crude brick presumably for broiling, and a round stuccoed hearth. The only exceptions were the anterooms of TB 1 and 2, which also lacked grinding facilities in the main rooms. These rooms did, however, have evidence for textile production, including loom weights and spindle whorls. Additionally, one should note that the distinctive doughnut shaped loom weights and fired spindle whorls are not limited to the Terrace and Clay Cut buildings. Examples of these loom weights were found along the south wall of Megaron 11 and in the rear room of Megaron 4. Some spindle whorls were also found in the storage area of Megaron 1.

The excavation of each Terrace Building and Clay Cut unit is described below. References to the various excavation notebooks and preliminary reports are provided for others who might wish to continue the study of Phrygian craft production.[174] One initial difficulty was being unable to identify exactly which room was being referred to in the excavation notebooks covering several seasons of work. The Terrace Building and Clay Cut building were excavated between 1956 and 1989 by many different excavators and in

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[173]K. DeVries (1980) 36, 39 citing a report by G. Hillam, November 17, 1977.

[174]The preliminary reports can be found in the AJA (1955) 1-18; (1956) 259-266; (1957) 319-331; (1958) 139-154; (1959) 263-268; (1960) 227-243; (1962) 153-158; (1964) 279-292; (1966) 267-278; (1968) 231-242.. The notebooks are stored at the Gordion Archives, in the University Museum, the University of Pennsylvania. I am very grateful to Dr. E. Kohler, Gordion archivist, for her generous and knowledgeable help. Copies of most of the notebooks can also be found at the Gordion excavation house in Yassıhöyük.

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over a dozen different notebooks. Young at first thought these rooms were the kitchens or pantries of a large palace, comparable perhaps to palaces of the Aegean or the Near East. He did not realize that these were a series of identical rooms of two massive buildings until later. For this reason, the nomenclature for the Terrace Building rooms changed over time. Some of the notebooks also refer to trenches in an unsystematized manner rather than the architectural features within those trenches. One needs to be aware of the many aliases for the Early Phrygian structures at Gordion to understand their publication. Finally, compass directions are also problematic in the description of each room. Magnetically, the two buildings run askew of true north-south, slightly east of north and west of south. Some of the excavation notebooks for the Terrace Building refer to the back of the rooms as the northern end, with the anterooms to the south. With the Clay Cut rooms, the orientation is a mirror image of the Terrace Building, yet the first two rooms excavated, CC 1 and CC 2, are referred to as the North and South rooms, shifting the conventional orientation once again. While this seems very confusing, the basic layout of each room is very predictable and the reader can easily follow the excavators' references in consultation with a plan. In the discussion below, most references to north in the Terrace Building rooms will be in the direction of the back walls.

Terrace Building 1 and Anteroom[175]

Excavations along the northern and western sides of TB 1 began in 1961. The entire room was excavated in 1963 and is described thoroughly in NB 100 (fig. 4.13). The room is referred to as E2, the Phrygian Room. Digging burnt fill in the northwest corner, the excavators comment on the well-preserved, greenish wall-plaster on both the North and

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[175]R. Young (1962) 167; (1964) 285-6. Gordion Notebooks (NB): Mainroom NB100 (1961/2) 21ff., trenches TBT 6a, 7a, 8a. Anteroom NB 107 (1963), trenches TB-E2-S1 and S2.

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West walls.[176] Along the west wall, the excavators discovered a large pile of doughnut-shaped loom weights. A conservative estimate puts the total number at 300 weights in the burnt fill. Found in the SE quarter of the main room, in level 7, described as the top of the burnt fill down to and including the floor, some 95 more loom weights of varying sizes and shapes were found.[177] The floor was cleaned and photographed with pottery and loom weights *in situ*.[178] Five loom weight deposits in the northwest corner of the main room are labeled A through E and sketched (see fig. 4.13).[179] Heap A contains 'hundreds',[180] heaps B and C total up to 27, heap D contains 23, and heap E has 45. The excavator notes that the loom weights in these piles are of varying sizes and shapes, as are the other loom weights found in the burnt Phrygian fill of this room. It is difficult to reconstruct what this deposit looked like since the excavation notebook records shapeless masses of objects in certain parts of the room. The number of weights reported, however, 23, 27, and 45, may be the number of weights for single looms, or perhaps two looms for heap E.[181] The total number of loom weights found in the northern half of the room in TB, including those in piles on the floor, is recorded at 732, the vast majority of which were discovered near the West wall. This is the largest number of loom-weights from one room. In contrast, only three spindle whorls can be associated with this main room.

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[176]It should be noted that 'north' in all notebook references and in this discussion is toward the back of the workroom, corresponding to magnetic 'north-east'.

[177]The excavator totals 380 loom weights for the entire E. room, yet this is probably not the complete count. Some examples are measured and drawn in NB 100: 10 x 9 x 5 inches, 14 in. diam x 10 in. thick, 7 in. diam. x 4 in. thick, 12 in. diam x 4 in. thick. Also found here was pot D, of coarse gray ware, with a base preserved and walls rising to 17 cm above the floor before breakage. Inside 'D' was a quantity of burnt grains which look like wheat, of which a sample was collected.

[178]NB 100, p. 89.

[179] NB 100, p. 96.

[180]These are also referred to in NB 100, p. 19.

[181]I thank E. Barber for pointing out this possibility to me.

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The excavation of the Anteroom of Terrace Building 1, from the 1963 season, is less well-described than the main room. Trenches TB-E2 S1 and S2 are believed to be in the anteroom. This room has spindle whorls found along with painted sherds and 43 doughnut shaped loom weights.[182] Again this number of weights may indicate one or two looms. Young states 'the excavation yielded little more than ordinary interest.'[183] Only five spindle whorls from this anteroom excavation can be located. TB 1 A also had a low staircase in front of four to five steps leading to the central doorway from the street toward the south, similar to the one excavated in TB 5 A.

At the time excavation of TB 1 began, five of the other Terrace Building rooms were already known. The main aim of the 1962 season was to find an eastern face and corner for this series of rooms on the terrace. As Young states in the season report, 'The Terrace Building was thus shown to have been a single unit consisting of a series of rooms each opening in the same direction to an open area, presumably a street.'[184]

What is most striking about TB 1 is what features are missing in comparison to most of the other Terrace Building rooms. While loom weights are abundant in TB 1, there is a curious scarcity of spindle whorls reported from the anteroom of TB 1 and both the main rooms of both TB 1 and 2. Neither of these megaron units (1 and 2) had grinding facilities or cooking installations. These two buildings clearly had a different function from the other rooms. Young describes the main room of TB 1 as 'an elegant apartment rather than a workroom'.[185] Some finds in this room do suggest that it was possibly a storehouse for more valuable items. Among the recorded finds from the main room is one

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[182]NB 107, p. 75.

[183]R. Young (1964) 285.

[184]R. Young (1964) 285.

[185]R. Young (1962) 167.

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small rim fragment of a large bronze cauldron. The fragment is bent and the whole cauldron undoubtedly was badly crushed in the destruction of the building.[186] Four other bronze cauldrons were also found in this room, with bull-protome attachments.[187] In the anteroom a heavy bronze cauldron with ring handles of iron was also found. It is possible that this could have been a more elegant workroom, in which higher class weavers worked.

Terrace Building 2 and Anteroom[188]

Also known as the TB East Room, the northern corner of the main room of this building was first uncovered in 1957 during the excavation of TB 3, the so-called 'Kitchen'.[189] The trenches for this room were labeled TBT 4-5-6 and are sketched in NB 46.[190] About 100 loom weights were found on the eastern side of the room, trench W1S, and about 25 were found in the south-eastern quarter, W2S.[191] Approximately 375 more loom weights were found in other parts of this room, totaling over 500, according to Young's 1962 season report.[192] The weight in grams of the loom weights, unfortunately, is not recorded and the weights themselves could not be found in the pottery depots at the Gordion dig house. As in the main room of TB 1, very few spindle whorls are reported from this room. Only two whorls were found from the Phrygian destruction level in TB 2.

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[186]This is comparable to the crushed bronze cauldron found in TB 2A, in the 1989 excavations.

[187]G. R. Edwards, "Gordion: 1962," Expedition 5.3 (1963) 45, fig. 22.

[188]AJA 66 (1962) 165-67; NB 70 (1957) 131-32; NB 89 (1961) 166-196; trenches TBT 4-5-6; NB 100 (1961/2) 59-74, 91-94, plan in notebook flap.

[189]NB 46, p. 187-88.

[190]See NB 46, p. 65.

[191]NB 46, p. 75-76.

[192]AJA 66 (1962) 165. Inexplicably, Young states that "the conventional name loomweights usually applied to (the doughnuts) is probably to be rejected out of hand."

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A flat bone implement from TB 2 may also be related to textile production if it was used as a weft-bobbin. These tools were described in Chapter two with equipment from Neolithic Knossos. Weft thread would be wrapped around the body of the object and then gradually unwind as it was fed through the warp threads.[193]

The anteroom of TB 2 was excavated in 1989 in the renewed excavations directed by Mary Voigt. With careful stratigraphic analysis Voigt has reconstructed the depositional sequence after the destructive fire.[194] We can safely extrapolate from this reading a similar pattern of destruction for all of the other Terrace Building rooms because of the total and simultaneous collapse of the entire structure. Wall collapse of stone and brick rested upon fire-damaged artifacts on the floor immediately adjacent to the walls of TB 2 A. This primary collapse layer decreased in thickness toward the center of the room. Above this layer and resting on the plaster floor of the room center were the remains of the charred roof debris. The roof seems to have been made of wooden beams covered by reed matting and then mud plaster. These remains were fairly well preserved which suggests to Voigt that the roof debris was not exposed for very long after the initial fire-destruction, meaning a fairly quick rebuilding in the Middle Phrygian period. Eroded bricky material and wall slabs lay on top of the roof collapse. In the center of the room was a concentration of stone blocks lying horizontally, while along the southern wall, collapsed wall blocks lay vertically. Based on the report of the destruction level in TB 7 A it seems possible that these wall blocks were knocked down after the fire before the thick layer of clay covered the debris.[195] This would have given the rebuilders a more even surface on which to work.

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[193]P. Sheftel (1974) BI 427 Terrace Building 2. Pl. 0.063; GW 0.011; Th. 0.002. Broken across at middle of sharp end; burnt white. Flat bone implement, probably pointed at missing end. Long head created by 5 small notches on each edge below a rounded tip. Iron stain near one notch.

[194]M. Voigt (1994) 265-293; See Voigt's YHSS 6A on p. 272-73.

[195]See NB 157 and 156.

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Also from Voigt's analysis of construction debris comes the interesting conclusion that, in TB 2 A at least, the proportion of mud-brick to stone slabs was very small. This suggests that the walls of the Terrace Building rooms were primarily built of stone with timber supports, with only a few courses of mud-brick.[196] These courses of mud-brick may have been used for a clerestory to allow light and air, or perhaps to position the roof beams. One major question, however, still remains about the Terrace Building architecture; was the roof flat or gabled?[197] There still does not seem to be conclusive evidence for either interpretation.

Large amounts of wood were found in both the main room and the anteroom of TB 2 A. Beams were analyzed by P. Kuniholm to derive a date for construction based on dendrochronology. The latest tree rings, however, were dated to 908+/-37 B.C. As Voigt comments, this seems too early based on the assumed destruction date of *circa* 700 B.C.[198] The problem may be that the latest tree rings and bark had been trimmed away or destroyed and thus the tree-ring tests would be inaccurate. Kuniholm, however, does note that other examples of wood from Gordion show that the Phrygian builders often re-used wood, sometimes many centuries after it had been cut.[199]

The recent excavation and detailed record keeping of the finds from this room make it the best example for detailed analysis of textile equipment. A total of 87 typical doughnut-shaped, clay loom weights were excavated in 1989 and catalogued by the excavations under M. Voigt. The average weight for the loom weights from TB 2 A was

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[196]M. Voigt (1994) 272.

[197]For further discussion of wood at Gordion see R. Liebhart, Timber Roofing Spans in Greek and Near Eastern Monumental Architecture during the Early Iron Age Ph. D. dissertation, University of North Carolina, Chapel Hill, (1988) 102-131.

[198]M. Voigt (1994) 273.

[199]P. Kuniholm, "Dendrochronology and Radiocarbon Dates for Gordion and other Phrygian Sites," Source: Notes in the History of Art VII.3-4 (1988) 8.

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557 grams, with sixty percent in the range of 400-700 grams. The distribution of their weights shows that standardization or uniformity was not strictly adhered to although there was a general range.

One interesting aspect of the loom weights was the change in the recorded weight over a short period of time for the same loom weight. These unintentionally fired weights absorbed moisture while buried in the destruction level and sealed by the later clay rebuilding level. Once excavated they began to dry out and consequently lose weight. Many of the thousands of reported loom weights from the destruction level were discarded during Rodney Young's excavations because it was thought they would not survive in storage. From studying the recently excavated doughnuts from the 1989 season this was a correct assumption. Since excavation, the poorly fired weights simply dried out and disintegrated into small mounds of clay and dust. The absorption and evaporation of moisture affected the measured weights to varying degrees. Many of the weights recorded initially in 1989 decreased dramatically in weight by 1996. For example, one weight decreased by as much as 91 grams between 1989 and 1996.

Along with the loom weights in TB 2 A were found 77 spindle whorls. This find of spindle whorls in the anteroom is slightly puzzling because of the scarcity of spindle whorls in the other rooms of TB 1 and 2. Perhaps the absence of whorls in those rooms is attributable to incomplete record keeping by the excavators of TB 1 and the mainroom of TB 2. The excavation, however, of these rooms was carried out over a long period of time and by a variety of excavators, suggesting that the absence might not be too easily attributed to poor excavation records. The evidence suggests that spinning was absent in most of the first two units of the Terrace Building.

Of the whorls that we do have from TB 2 A, the largest percentage weighed between 10 and 20 grams, approximately 37 percent (chart 1). The majority of the whorls weighed less than twenty grams. This is less than the average weight of the whorls from

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all of the Terrace Building rooms. Some of the spindle whorls were found in groups or clusters. These groups were probably stored in ceramic vessels often found nearby along with metal needles and knives, presumably used in textile production. These whorls are particularly interesting because they may reflect the spinning kits of individual workers. Most of the measured weights of these groups showed a fairly broad range. The Phrygian spinners seem to have used a variety of weighted whorls probably for different qualities of thread, as is demonstrated by the variety of textile samples preserved at Gordion. One set of thirteen whorls ranged from three to thirty-eight grams.[200] All were of the same pink-orange fabric and weighed an average of 13 grams. Whorls of another set were found together in pot YH33668 and were made of the same fabric as the other whorls from this room.[201] The measured weights were fairly uniform, ranging only from ten to fourteen grams.

As was mentioned above, the first two Terrace Building rooms (TB 1 and 2) were thought to be storage houses, possibly even treasuries. Valuable caches of exotic material were found within, particularly in TB 2.[202] While TB 1 and 2 seem to have been used for storage of loom weights and wealth items of precious metals and ivory, the cluster of twenty to forty plus loom weights and other tools indicate that weaving occurred even in these rooms.

Among the textile equipment was a carefully made comb found in the anteroom of TB 2 (fig. 4.5 a and b). This heavily charred, long-toothed comb is certainly associated

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[200]Catalogued as YH 33698, small find number 592, from locus 100, lot 211.

[201]Catalogued as YH 33643, small find number 591, from locus 100, lot 211.

[202] As DeVries points out, however, the other nine units which were certainly workshop areas may also have functioned as depots. A number of iron axes and axe-adzes were found throughout the Terrace Building rooms similar to ones found in the Megaron units. See K. DeVries (1980) 39 who suggests that these adzes may have been on display in Megaron 3. Also, a non-Phrygian, imported leech-type fibula was found in the Terrace Building; O. Muscarella (1967) 62.

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with textile production since it has unwoven warp threads adhering to one side and woven textile on the other. Its use with textiles suggests that it would have had a fairly smooth surface and was probably made out of ivory or bone.[203] The comb had a plain top and sides with a a slightly curving row of teeth. It is very fragmentary but E. Barber has suggested to me that this tool is quite possibly the earliest known reed. This weaving implement would keep the warp threads separate and push the weft threads down all at once. This reed or comb may have been used on a warp-weighted loom or for band-weaving. The relatively small size of the object from Gordion, approximately 6 cm. wide, suggests that it would be used for making small, decorated bands. This reed or comb and the loom weights found within TB 2A are the clearest indicators for cloth production at Gordion at the time of destruction.

Terrace Building 3[204]

This workshop was the first of the Terrace Building rooms excavated and was known by various names, including the Kitchen, TB Central Room, TBT 3, and TB W.[205] Portions of the back wall and cross-wall between TB 3 and 4 were uncovered during the 1955 campaign.[206] Based on the utilitarian objects found here, including grinding stones, loom weights, spindle whorls, and great quantities of pottery, Young called this room 'the

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[203]The comb is catalogued as YH33701, small find 583, from locus 100, lot 211, 8/21/89. The conservator, J. Johnson, identified the heavily charred remains as ivory. Voigt was unable to see the linear structure typical of ivory and remarks that the breaks look like wood. Wood, however, would probably cause the wool fibers to catch making weaving difficult.

[204]CC 1-2 were discovered earlier, in 1952. The Anteroom of TB 3 has not been excavated. AJA 60 (1956) 262-63; AJA 62 (1958) 147; AJA 64 (1960) 240-2; NB 46 (1957) 147-158, 160-89, trench MS-Q2; NB 70 (1957) 110-111, 119-120, trench MW2; NB 81 (1959) 87-99.

[205]In the 1958 AJA report, Young refers to the buildings southwest of the West Phrygian House (Megaron 2, Mosaic Building) and the Burned Phrygian Building (Meg. 1) as the Terrace Building for the first time. The designations 'Kitchen' was no longer accurate as the uniformity of each room emerged.

[206]R. Young (1956) 262-3, pls. 93-94, figs. 44-51.

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Kitchen' although he notes that 'pantry' might have been just as appropriate. Four pairs of grinding stones were found neatly arranged with boat-shaped upper grinders still in situ along the west end of the north wall.[207]

At this point of the Gordion excavations (1955) Young noted that not one sherd of Greek pottery had been found in the burnt Phrygian level to serve as a guide to the destruction date. Similarities, however, between the Phrygian pottery and examples from the Körte brothers' Tumulus III suggest a contemporaneous date for the two areas, somewhere around 700 B.C. Young concludes the discussion of TB 3 by stating, 'It is tempting to see in the conflagration which destroyed the Phrygian buildings at Gordion the effects of the destructive raid of the Cimmerians which took place at the beginning of the seventh century; and for this the commonly accepted dating of the pottery of Tumulus III and the traditional date of the Cimmerian invasion are in agreement.'[208]

In the main room of TB 3 138 spindle whorls were found with an average weight of 21 grams (chart 2). A few outliers weighed upwards of fifty grams. Fifty-eight percent of these spindle whorls weighed less than twenty grams. One cluster found together contained eight whorls and three needles, contained together in a cloth sack. This sack was found inside a Phrygian pinch-pot.[209] These eight whorls were very light compared to other spindle whorls in the room; five were the same weight, five grams each; one was eight grams, the next ten, and the final whorl weighed twelve grams. The range of only seven grams suggests a fairly uniform weight and probably indicates that fine thread was spun in this room. Another group of whorls was found with eight needles which are

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[207]R. Young (1956) 262, pl. 93, fig. 44.

[208]Young cites the two dates for the Kimmerian invasion from the ancient sources, 696/5 and 676 B.C. If the deposit from the Terrace Buildings is contemporary with the pottery from Tumulus III, the earlier date of the invasion is preferable since it is 'unlikely that any of the larger tumuli were made immediately after the invasion," R. Young (1956) 263 and n. 24.

[209]Pot #99 August 1959.

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catalogued by J. McClellan.[210] In addition, two bronze knives with curved blades were found.

In the excavation report for that year, Young mentions a great many loom weights found in the main room of TB 3. A total number, however, is not forthcoming and none of the weights themselves have survived in the storage areas at Gordion for analysis. In comparison to the other main rooms with substantial deposits of loom weights from the Terrace and Clay Cut Buildings, an estimate of approximately 500 loom weights for TB 3 would not be unexpected. The anteroom has not yet been excavated.

Terrace Building 4 and Anteroom[211]

The main room of this Terrace building unit is also known as TB West Room, or TB W1. From Young's difficult handwriting in NB 67, we get some understanding of the excavation of this room. He writes, 'We finish digging the burned fill... (finding) more whorls, doughnuts and iron knives. To recount in detail would be tedious.'[212] A total of 226 loom weights and 19 spindle whorls were found in the main room. Seven grinding stands were at the back of the main room and the brick outline of a bin nearby that may have contained unground grain.

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[210]See J. McClellan (1975) catalogue number 517, Gordion ILS 590: GPL (greatest possible length) 0.075; GPL longest needle 0.072; W. 0.011; W. single needle 0.003. TB-3, with piece of cloth and 4 spindle whorls in a bronze bowl, by main door. A clump of 8 iron needles, fused together. They are straight, round in section, with eyes formed by drawing one end to a thin wire and bending it down to form a loop. Complete but swollen.

[211]Mainroom is also known as TB West Room, TB W1. AJA 62 (1958) 146-147; AJA 64 (1960), 240-243 (on TB 3-5); AJA 66 (1962) 165. NB 67 (1957) 155-179; trench MW-2; NB 70 (1957) continuation of NB 67, 75-83, sketch on p. 83, 91-92; NB 80 (1959) 25-6, 29-32, 41-45, trench WML1/2; NB 81 (1959) 28-32, 34-8, 42-63, 69-71, 80-85, trench TBT1.

[212]NB 67 (1957) 78. On p. 161 he notes that 'this room too shows kitchenish characteristics', in reference to TB3.

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Young mentions the clearing of the east wall and an assortment of small clay whorls found in a jug.[213] The whorls had evidently been contained in a cloth bag since a few scraps of cloth were found still inside the jug. A total of only 19 spindle whorls from this room have been found but, on average, they are much heavier than those from other rooms (chart 3). The average whorl from TB 4 weighed 35 grams, over ten grams heavier than the typical Terrace Building whorl. Eighty-four percent of TB 4's whorls weighed more than twenty grams. Given the central location of this work room along the row, and the heavier whorls found inside, it is proposed that the spinning done in this room was the plying together one larger thread from two finer threads, suitable for heavier weaving. Fairly heavy textiles are likely, given the heavy weights of the doughnut-shaped loom weights, weighing upwards of 600 grams. We know from the textile finds in the Phrygian buildings and tombs at Gordion that much of the thread was two-ply; that is, fiber spun into one strand of thread and then spun together with another thread to make a stronger, thicker yarn. These heavy spindle whorls from TB 4 may indicate such plying.

There are two further explanations for this kind of spinning located in TB 4. Barber has suggested to me that this task of plying already spun thread together is a fairly simple task and may be assigned to younger, inexperienced spinners as a kind of practice. The central location of this workroom, TB 4, would be a good place to monitor junior spinners. Another possibility is that this simpler task was assigned to workers of a slightly higher status, or possibly an older age, who could in turn monitor the workers on either side of them.

The quality of other goods found in TB 4 is higher than that of the other workrooms. A cloth and leather piece was found folded up with both faces decorated with rows of bronze studs. This material was found 3.5 m. from the south wall and 0.75 m.

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[213]Miscellaneous Clay objects, nos. 127-9.

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from the west wall, 13 cm. above the floor. The total length is approximately 0.5 m. by 0.30 m. It had either fallen from the wall or been purposely placed over some pots.[214] A fine necklace was also found in TB 4.[215] Six knives were also found in this room, three straight edged, and three curved.[216]Nine other iron objects were found in the main room of TB 4 along the west side.[217]

Some of the pottery from both the main room and the anteroom was of unusual quality. The total estimate for the number of vessels is more than 150. Most of the pottery from the anteroom was coarse but two pieces are singled out by Young as noteworthy. One is a double looped 'python cothon' and the other a 'sipping cup' of fine buff-polish decorated with incised zigzags. This cup also 'had an umbrella handle at the bottom, its turned-up end decorated by a modeled bird; a hole through the handle below the bird may have held a tassel, or a string to hang the cup by.'[218]

The anteroom of TB 4 was excavated as TB W1S along with TB 5 A.[219] No loom weights from the anteroom could be located in the depots but forty-two spindle whorls were found, also of a slightly heavier weight on average compared to whorls from other rooms (chart 4). TB 4 A and 5 A were the first two anterooms fully excavated by Young and contained the typical cooking features found in all the anterooms, except TB 1 A and TB 2 A discussed above. The recorded dimensions are 7.4 m. long and 11.4 m. wide. In

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[214]NB 81 (1959) 119.

[215]K. DeVries (1980) 39.

[216]Information courtesy of E. Kohler. Gordion numbers B256, 257, 265, 289, 291, and 555.

[217]NB 81 (1959) 51; These finds were put into four artifact groups and sketched in the notebook. In the northwest corner, group 3, find #42 contained 26 doughnuts, 8-13 cm in diameter of unbaked clay. Find #45 records 13 spindle whorls of small size, inventory number 7968 MC 204.

[218]R. Young, (1962) 165.

[219]AJA 66 (1962) 165; photograph of TB 4 A pl. 45, fig. 18; plan and section of features in fig. 19. See also NB 89 (1961) 57-76, trench W1S.

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the northeast corner was a large brick and clay oven , raised above floor level.[220] The inside of the oven was covered with plaster and had a sloping floor. A round hearth 1.3 m. in diameter lay in front of the oven at floor level. A second, larger oven was located in the northwest corner, again of brick and clay with a sloping stuccoed floor opening to the east side. The inward curving walls of these ovens suggest that they were originally domed. A horse-shoe shaped wall of bricks was found in between the two ovens. Young suggests that this one, and that in TB 5 A, could have functioned as either a grill or broiler for meats, or possibly for the production of embers and charcoal shoveled into or out of the ovens.[221]

Terrace Building 5 and Anteroom[222]

Tracing the excavation of this room from the notebooks is difficult because many different trenches were under the supervision of the excavator and recorded in the same notebook, NB 79. Excavation of the main work room took place in July and August 1959. A well-sketched plan with labeled index can be found in the back notebook flap.[223] The sudden destruction of the Terrace Building is indicated by wheat found on one of the grinding querns in the back of TB 5. A hoard of 20 doughnut-shaped loom weights was found in the shape of an inverted 6 toward the back center of the main room. Fairly large pieces of wood were found tumbled together along with these loom weights.[224] The close

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[220]R. Young reports this as the northwest corner; R. Young (1962) 165.

[221]See a sketch of this room in NB 89, p. 65.

[222]Mainroom also known as TB W2. R. Young (1960) 240-242; R. Young (1962) 164-65; NB 79 (1959). Anteroom (room W2S2, TB S2) NB 79 (1959). TBS2 excavated simultaneously with main room so notebook is confusing for looking at both separately; NB 87 (1961); NB 89 (1961) 77-79. NB 89 (1961) pp. 77-79 ("SE quadrant); trench W1S.

[223]NB 79 Index on p. 164-5.

[224]NB 79, p. 141 "July 15, 1959 TB-W2 today 2 large chunks of charcoal, quite well preserved; evidence for cracks made in wood when it was being seasoned, as well as the usual breaks from burning. Evidence for tooling marks. One piece, larger than the second, was lying above it. The larger piece was 0.10m.

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proximity of one to the other suggest that these wooden pieces are the remains of a loom that was standing between the central post, a convenient location for lighting while still allowing traffic flow through the room.[225]

From this room came 18 spindle whorls. The weight distribution shows that half of these weights weighed less than thirty grams, while the other half was between thirty and sixty-two grams (chart 5). The notebook records groups of three or four spindle whorls collected on separate days in 1959 and it is assumed that these were from the discrete depositional loci. That is, these separate groups of spindle whorls were 'spinning kits', possibly used by one worker before the destructive fire. A curious description of one group of these whorls states that they were found 'with lots of phytoliths'.[226] This was the lightest of the four groups found, weighing an average of 16 grams each, with an interesting distribution of two whorls weighing sixteen grams exactly, the smallest one at seven, and the largest at twenty-four. This would mean that the whorls were nearly in multiples of 8 gram units. Another cluster of whorls was unusual in that their average weight was around fifty grams. Like the whorls from TB 4, these from TB 5 weighed on average more than the other whorls in the Terrace Building. Based on the heavier whorls, it may be inferred that here too was located the plying of thread from two yarns.

The anteroom of TB 5, also known as room W2S2 and TB S2, had been partly opened in 1959. It was excavated simultaneously with the main room so the notebook references are confusing.[227] Work concluded in 1961 along with the excavation of the anteroom of TB 4. Reference is made to the round hearth, the two large domed ovens, the

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thick, the smaller, 0.05m thick. The larger piece of charcoal was 0.78m long and 0.26m wide. The smaller, 0.76m long, and 0.13m wide."

[225]See the excavation of TB 7 A for a similar loom position. NB 156 (1971) 114.

[226]NB 79 p. 184. This could be the remains of burnt grains which were found so often in the other rooms or it could be the remains of textile material.

[227]NB 79 (1959). R. Young (1962) 164-5, pl 45, fig. 17 and 18 (top).

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U-shaped construction of crude brick probably used for broiling located in the north-west corner. These cooking features were damaged when the west wall was robbed in later times and are not as well preserved as those in TB 4 A. Many metal objects were inventoried but then thrown, made of both bronze and iron. A typical kneading board of unbaked clay was also found.[228] For textile equipment we have the usual stray finds but also a few found in clusters. Twenty-six spindle whorls were found together.[229] From the notebook references and the label on the spindle whorl bag in the depot, it is assumed that these formed a single group of tools that were originally stored together, perhaps the possession of one Phrygian worker. The weight distribution shows fifty-three percent of these whorls weighing between twenty and thirty grams (chart 6). The rest are fairly evenly divided on either side of this range, up to forty-seven grams.

The records for loom weights from both the main room and the anteroom are not as specific as one would hope; only a few references are found, such as a group of three loom weights removed from the floor.[230] Excavations resumed in 1961 revealing more grinding stones and loom weights but it is difficult to estimate the number.[231] In J. McClellan's catalogue of iron implements from Gordion, she records an iron needle found near some loom weights and pottery in the main room.[232] Based on the majority of anterooms with substantial deposits of loom weights, an estimate between fifty and one-hundred loom weights in TB 5 A would not be extraordinary.

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[228]NB 79 (1959) 169 Inventory number 6703P2579.

[229]Found on May 13, 1961, NB 89, 78, find number 2.

[230]NB 89 p. 100 July 3, 1959.

[231]NB 87 (1961); NB 89 (1961) 77-79.

[232]See J. McClellan (1975) catalogue number 518, Gordion ILS 250 GPL 0.045; GW across eye 0.003. TB-5, against west wall, with pottery and loom weights. Straight thin needle, round in section, with end flattened and pierced with a long vertical eye. Broken." GPL refers to the greatest preserved length.

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Terrace Building 6[233]

This room was excavated in 1963 along with the anteroom of TB 1, and two of the Clay Cut rooms (fig. 4.14).[234] It contained 16 grinding querns, the largest number of any of the Terrace Building rooms.[235] There is, however, a striking absence of references to loom weights in the notebooks for this room. It is possible that, after excavating five other large workrooms, counting them in detail would have been tedious for the excavators, as Young remarked in 1957 in reference to TB 4. What can be discerned from the notebook sketches shows at least twenty-five loom weights but it is likely that the actual number was greater.

Forty-eight spindle whorls were studied from this workroom, falling into typical weight distributions for an Early Phrygian workshop (chart 7). Seventy-one percent weighed between ten and thirty grams. The anteroom of TB 6 was left unexcavated.

Terrace Building 7 and Anteroom [236]

This is possibly the best documented Terrace Building complex, perhaps because of its relatively late excavation in the series and because of the careful record keeping by the excavators, especially in the anteroom. Architectural details, including evidence for reeds

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[233]The anteroom of TB 5 is unexcavated. The main room is also known as TB-W3. R. Young (1964) 285; NB 106 (1963) 19-29, Trench M4S1; NB 109 (1963) 64-79.

[234]R. Young (1964) 285-6 and plate 86, figs. 16 and 17.

[235]I count only fourteen in situ but DeVries reports that there are up to 16 querns in this Terrace Building room. Two querns may have been found displaced.

[236]Also known as TB-W4. R. Young (1966) 269-270; K. DeVries provides a thorough description of the finds in these rooms in K. DeVries (1990) 384-385. A good illustration of the doorway between TB 7 and TB 7 A can be found in R. Young (1966) pl. 70, fig. 8. Main room: NB 120 (1965) 27-91. Anteroom NB 157 (1971) 115 to 186; continues in NB 156 (1971) 55 to end. NB 161 (1973) for architectural detail.

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and wooden planks in roof construction, can be found in the excavation notebooks but will not be discussed here unless they relate to textile production.

The excavation of TB 7 and 7 A is the clearest description of finds from an entire TB unit at Gordion. In the anteroom DeVries records four basic shapes for the loom weights: flattened, globular, pyramidal, and cylindrical.[237] These types are often found in piles together and it is difficult to believe that there was some special function for each shape weight. The weight and dimensions are roughly the same. They are grouped by the generalized nick-name description, 'doughnut'. They were all made from local, coarse river clay and only accidentally fired, to different degrees, during the great destruction. The spindle whorls for the most part are fired clay and are of the usual range in shape and weight. Clusters of spindle whorls varying in size suggest spinning kits for individual spinners.

In the main room of TB 7, 450 loom weights and 89 spindle whorls were found.[238] Many of these tools seem to be concentrated along the eastern wall. For a main workroom, TB 7 had a fairly heavy concentration of loom weights. A group of thirty of these loom weights was studied in 1965 by R. Johnston.[239] The spindle whorls covered a broad range of weights, with a majority weighing less than thirty grams (chart 8). Heavy deposits of pottery were found along all the walls of the room while the center was fairly clear.

Along the backwall were three stations of grinding stands. The main stand showed evidence for at least six people working. A mud-plastered bin was located on its western side, perhaps for holding ground grain. Nearby were piles of burned wheat and barley,

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[237]K. DeVries (1980) 39; NB 156 (1971) 92-93.

[238]At least 46 are noted in K. DeVries (1980) 39 and (1990) 385.

[239]See R. Johnston (1970) 221-236.

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some of which were contained in baskets. In the northeast corner, two additional stands were set up adjoining one another. Two more stations for grinding continued along the back wall, facing the same direction as the other six, while the adjoining segment held one grind-stone and faced the eastern side wall. A large pit in front of these stations was found filled with burned debris but may have functioned as a storage area for unground grain.[240]

The destruction level of the anteroom was excavated in 1971 by K. DeVries (fig. 4.15). Work within the anteroom revealed a great deal of equipment related to textile production, including loom weights, spindle whorls, needles, metal tools and stone implements. DeVries notes that, 'doughnuts and spindle whorls begin being found at the top most (level) of the burned stratum.'[241] At least 150 loom weights were found in this room. Roughly in the center of the anteroom was a double line of 21 doughnut-shaped weights about 1.59 m. in length. This length gives us a good idea of the width of the textiles woven on this loom. Again the general cluster of weights numbers around 20. From the position of the weights, it is possible that these were two rows of loom weights attached to the warp sheds of a standing loom at the time of destruction.[242] As T. Cummer suggests in notebook 156, the location in the room for this line of weights, perpendicular to the door leading to the main room, was an excellent location for a loom which would not obstruct traffic yet still get the maximum amount of light for weaving.[243]

While most of the loom weights from this room seem to be concentrated in the eastern half, a cluster of ten loom weights was found along the walls of the large domed

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[240]R. Young (1966) 270.

[241]NB 156, p. 78.

[242]NB 156, p. 113-115.

[243]K. DeVries (1980) 39; K. DeVries (1990) 385; NB 156, p. 114.

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oven in the southwest corner.[244] These weights vary in shape and size, from the cylindrical type to the flattened globular kinds. Unfortunately, the distribution of different size weights cannot be determined. A spindle whorl and two stone tools are also found in the cluster. A low bench measuring 0.37 m. high and 0.43 m. wide was also found nearby, connecting the oven to the interior side of the south (front) wall of the anteroom.[245] The placement of the bench near the loom weights might suggest a seat for weavers but the location against the wall indicates that this would be fairly cramped. More likely, the bench was used for kneading trays in association with baking.[246] The loom weights in close proximity to the oven might indicate that they were in the process of being partially dried or hardened for use on a loom. More doughnuts were found in this anteroom, particularly along the eastern side in the corners, including a cluster of eleven wedged between two pots and the wall.

At least 101 spindle whorls were found in the anteroom. These whorls weighed slightly less than the average for the Terrace Building rooms (chart 9). Nearly sixty percent of the whorls weighed less than twenty grams. About thirty-five percent weighed between twenty and forty grams. The whorls were mostly found scattered but a few were found clustered together, presumably inside vessels. Iron needles were also found in the clusters, suggesting that these were textile tool kits. Eight whorls and six iron needles were also apparently kept in a small vase.[247] In the southeastern corner of the anteroom, just to the right of the entrance, there was a large concentration of loom weights, needles, and spindle whorls. Thirteen spindle whorls were found in the southeast corner, next to a heap of

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[244]The floor in this part of the anteroom is remarkably empty of pottery. NB 56 (1971) 162-63; 169; 171.

[245]K. DeVries (1990) 384-5; NB 56, p. 168.

[246]Also found in front of the bench area were an iron rod and an iron hoe. These tools are probably to be associated with tending the fire in the ovens. See K. DeVries (1990) 385.

[247]K. DeVries (1990) 385; NB 156 87.

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loom weights. References to the iron needles from this room can be found in McClellan's 1975 dissertation.[248]

Other unusual tools that might be related to textile production include iron rings, measuring between 0.03-0.05 m. in height, by about 0.025 m. wide. How these tools would have functioned is unclear at this point but the predominance of textile equipment found along with them is a good starting point. Examples of similar iron rings are noted in the excavation of other Terrace Building rooms, such as TB 1.[249] Egyptian scenes of spinners show that two threads that are plied together are often pulled through small rings attached to the wall. It may be that these rings had a similar function.[250]

At least four dome-shaped stone tools were found in TB 7 A and might have functioned as a kind of polisher or beater for finishing heavy textiles. The excavation notebooks describe half-dome shaped tools with a flat underside.[251] A long, narrow polished stone object with suspension holes on either side was also found. It measured .092 and is listed as a possible pendant. Its shape, however, suggests to me that it could have been a weft-bobbin, with thread wrapped around it while being sent through the warp sheds. At least seven knives were found in TB 7, two straight edged and five curved.[252] One knife may have been fitted with an ivory handle, although recent reports by S. Payne

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[248]J. McClellan (1975) Catalogue number 519a-d, ILS 662a-d: GPLs a) 0.102; b) 0.049: c) 0.054; d) 0.065; Ths. 0.002-0.0025. TB-7, anteroom, with pottery and 3 spindle whorls. Straight needles, tapering to point. Eyes are long, oval slits pierced through slightly flattened end. a) complete but bent; b) broken; c and d) fused, with points broken. Catalogue number 520, ILS 452:GPL 0.076; D. at eye 0.004. TB-7, doorway, in a trefoil pot, with ivory handle BI 543. Thin straight shaft tapering to point at one end. Eye pierced through flattened end. Complete but swollen from rust.

[249]See NB 100 (1961) 31-32.

[250]See E. Barber (1991) figs. 2.6; 13.1.

[251]NB 156 (1971) 173. Other stone tools of this shape have been noted in this room. See pg. 184 and NB 156 112.

[252]Information courtesy of E. Kohler: ILS 440, 445, 442, 443, 444, 487, 668.

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may suggest differently.[253] Finally, two more bone tools were found that might also have been weft-bobbins. These tools have been catalogued by P. Sheftel in her dissertation.[254]

Terrace Building 8 and Anteroom[255]

The main room and anteroom of this final Terrace Building unit were excavated in 1967 (fig. 4.16). The total length of the entire Terrace Building came to 105 m. with the excavation of TB 8. This room was similar to all the others except that it was less well-preserved. A deep cellar, lined with stones, had destroyed much of the northeastern part of the main room. No grind-stones were found on the main grinding stand at the back but there were some grinders on a smaller subsidiary stand to the southeast. Young comments on a fortunately well-preserved segment of the wall that fell from TB 8. The remains suggest that the walls were approximately 2.8 meters high of stone and wood, and then capped with a mud-brick superstructure, perhaps at the level of the gallery.

A total of 92 spindle whorls were found in the main room, while the anteroom produced only 3. The majority of the whorl weights of those in the main room fell within the range of ten to twenty grams (chart 10). Some of the whorls were found along with

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[253]K. DeVries (1980) 39.

[254]P. Sheftel (1974):
1) BI 452, Terrace Building 7, near South wall, west of door. L. 0.135; GW 0.012; GTh. 0.003. Complete as mended; burned black. Smoothly polished flat bone implement rounded along all edges and tapering to sharp triangular point. Head end rounded off and notched 4 times on one edge and three times on opposite edge.
2) uncatalogued, Terrace Building 7, found with shuttle no. BI 452 above. Pl. 0.048; GW 0.016; GTh. 0.006. Broken across shaft near head. Fragment of flat bone implement with smoothly rounded edges, tapering evenly down to rounded end notched four times on each side. Probably pointed at opposite (missing) ends as No. 3. A lot of bone marrow is visible on one side.
3)BI 563 Terrace Building 7, anteroom floor. Pl. 0.062; PW. 0.01; Th. 0.003. Broken roughly in half; warped and roughened by fire. Thin flat bone awl, smoothed to round on edges and ending in good point.

[255]AJA 72 (1968) 238-239; NB 123 (1967) 77-79, 97, 101; Trenches M6E, M6F, NB 128 (1967) 177-197; NB 131 (1967) 129; NB 137 (1967) 3-17. Anteroom NB 135 (1967) 123-140, 150-170; NB 137 (1969/1971) [continuation of 1969 work begun in NB 144; *n.b.* excavator's N = 'W'] NB 144 (1969) 66.

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iron needles and knives.[256] Nine bronze knives, all curved, were recorded.[257] As for the loom weights, there is the typical problem of no exact count recorded in the notebooks. From the excavation sketches and references in the iron catalogue, it seems that the anteroom had the usual number of nearly one hundred loom weights. The number in the main room is more difficult to estimate because of the later disturbance. There is, however, no reason to think that this room was any different than the other Terrace Building rooms 3-7.

In the anteroom, there was the usual large dome oven, a U-shaped broiler, and a smaller oven or hearth above the floor, but in this room, the cooking features were placed in the southeastern corner rather than the typical northwestern one (fig. 4.16). Associated with the ovens and broiling stands was a bench with fragments of clay kneading trays, leaning against the wall. These trays were located along the interior wall of the anteroom, toward the door which leads into the main room. The location of the cooking facilities at this opposite side of the room should be attributed to the entrance and staircase which gave access to the entire Terrace Building complex along the northern side wall of TB 8.[258]

Clay Cut Building

Traces of four parallel workshop units were located approximately 20 meters across a broad street in front of the Terrace Building. That this building was also severely and

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[256]See McClellan (1975) Catalogue number 521, ILS 570: GPL bundle 0.065; W. 0.007; Ls. needles 0.065; L. eyes 0.002. TB-8, anteroom, near a group of 47 whorls and 21 loom weights. A bundle of needles which are straight, thin and round in section, tapering to a sharp point. Eyes made by drawing end to a point and folding it down against shank. Complete though burned.
Catalogue number 522, ILS 520: GPL 0.06; D. at eye 0.004. TB-8. Straight rod, round in section with a blunt, rounded point. Eye is a loop made by bending the rod's thinned end back against itself. Mended to complete; burned.

[257]Gordion B 522, 523, 524, 525, 528, 531, 532, 536, 581.

[258]See K. DeVries (1990) 382, fig. 11, 14, and 15.

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suddenly destroyed by fire suggests that the destruction of Gordion was a planned, deliberate attack. The burning of Megarons 1-4 and the Terrace Building behind might be explained as an accident if these were the only areas of the Citadel Mound burnt. The destruction of the CC units at such a distance from the Terrace Building indicates that it was a separate, contemporaneous, and intentionally set fire.[259]

This building is referred to as the Clay Cut, or CC building, and was of identical though reversed plan to the Terrace Building, with megaron units composed of an anteroom and main room behind. The layer of clay placed over the Early Phrygian destruction level by the Middle Phrygian rebuilders gave the conventional name to this excavation area in the Clay Cut.[260] These two large workshop buildings, the TB and CC, were presumably built at the same time because they rest on the same terrace fill and the side walls of each room are aligned to the mid-point of the workshop unit across the street. For example, the cross-wall between CC 1 and 2, if continued, would run through the center of TB 6, and interestingly, continue on through the center of Megaron 4 further behind.

Unlike the Terrace Building, the exact length of the entire CC building is not known. Neither end of the building has yet been excavated. Only traces of the furthest north-west building, CC 4, have been excavated. It is assumed that the CC units continue at least as long as the eight units of the Terrace Building. The conventional names, CC 1-4, will perhaps someday cause confusion since CC 1 is probably located near the middle of the series of rooms and, if excavations were to continue, would reveal a CC 5 room immediately adjacent to CC 1 toward the south-east.

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[259]See K. DeVries (1990) 388.

[260]The date for this layer of clay and the rebuilding of the citadel is a debated point. R. Young had interpreted a gap of a generation at least between the Early and Middle Phrygian periods. K. Sams, M. Voigt and K. DeVries have recently argued that it was fairly soon after the destruction.

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Clay Cut 1 and Anteroom[261]

After the initial discovery in 1952, excavations continued in CC 1 in 1956 and 1963. This cutting was continued to expose the north end of the building, *i.e.* the front wall facing the TB units. The room was a typical terrace building-type, with cooking facilities in the anteroom and post-holes around the walls in two aisles of the back room. The walls were preserved highest at the SW corner, where several masonry courses can be counted. Above the sixth course, (counting from the bottom) apparently ran a horizontal beam course like the one at floor level flush with the inner face of the walls. The main room and anteroom of CC 1 were only partially cleared, along the north west side. Not completely excavated was the side wall to the southeast. It is assumed that CC 1 adjoined other identical workshop units running toward the south, parallel to the Terrace Building rooms across the broad dividing street. One of the ovens in the anteroom is particularly noteworthy in that it still contained a layer of wheat, presumably being roasted at the time of destruction.[262]

The grind-stone bench in the backroom was very similar to the one in the North room, CC 2. It was built in exactly the same way with mud-brick sides and a curved brick edge. The notebook records that a section of the installation with millstones *in situ* can be seen in the south scarp of the trench. The grindstone had slipped into the platform edge but the original arrangement is clear. The outer, western face of the bench was plastered down to floor level with thick mud plaster.

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[261] AJA 60 (1956) 262-264; AJA 68 (1964) 285-286; NB 33 (1952, 1955) 72-76, 80, 94ff. NB 55 (1955) 40-44, 84-86, 94-98, 106-110, 124; CC 1 is referred to as the 'South room'.

[262] NB 91 (1963) 111, 115-119, 183-192; NB 111 (1963) A-C (pottery), 1-5, 163 p. 62 12-17-viii-63, The North Room.

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Found in the main room were 133 spindle whorls covering a broad range of weights, from five to 68 grams (chart 11). Mellink notes that 124 spindle whorls from CC 1 and 2 were gathered together but from the tags on these whorls in the depot, it seems that a majority of them came from the main room of CC 1.[263] Some needles were also found in this room and three bronze knives.[264] The distribution of whorl weights in the main room of CC 1 was much more typical of the other Terrace Building workshops than those in the anteroom of CC 1.

The anteroom was relatively free of pottery but contained: 1 large oven with grains of barley inside, two smaller ovens, and a round hearth. Only one large pot appears to rest *in situ* against the western wall near the larger of the two ovens. At least 57 spindle whorls are known to have come from this anteroom. What is striking about these whorls is the limited range in weights (chart 12). A large majority, 90%, of the whorls from this room weighed less than 20 grams while the other ten percent were under thirty grams. This suggests that light weight, relatively fine thread was being spun in this room on these very light weight spindle whorls.

Clay Cut 2[265]

This unit is often referred to as the North room, as opposed to the South Room of CC 1. Only the main room was excavated. It had the usual arrangement of two rows of four post holes with a central one between the innermost pair, presumably supporting a

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[263]NB 33, 97.

[264]McClellan (1975) Catalogue number 523, ILS 220: GPL 0.103; W. at eye 0.004. BPH, south room, with 5 spindle whorls and several loom weights. Long thin shaft, round in section with long, narrow eye, presumably made by drawing out end and bending it down to form a loop. Broken. This needle is bent and is similar in shape to the kind of needle used for over-casting, if the bend is intentional. The knives are numbers 195 and 209 (two examples).

[265]AJA 68 (1964) 285-286; NB 55 (1955) 4, 8-12, 16-23, 26-40, 62-84, 86, 88-92, 100-104, 112-122; NB111 (1963) 18-19, 60-64, 67-96 (finds).

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balcony. Along the walls ran another set of post holes echoing the interior ones. Two large grinding stands were found along the back wall, one of which was excavated by Machteld Mellink in 1956. The north inner side wall had fallen out considerably so that from the top the bench seemed to resemble a long narrow boat. The pottery deposit was fairly heavy in this room. Broken jars near the wall contain burnt wheat and barley, samples of which were saved. A fragmentary red-polished jar with rim revealed a piece of fairly thick cloth inside with thick thread all bundled up. In the corner many loom weights were found in fallen rows and piles.

A total of five querns rest in place on the bench, three of these having hand grinders; east of the westernmost is the hollow for a sixth which was not present. Lying on the floor of the room scattered about are 8 complete querns (four upside down, four rightside up) and one end of a 9th; 3 saddle querns, all right side up (scraping end down) but one resting against the side of a rectangular quern. The plaster surface around the grinding stone platforms was also well-preserved.[266] A circular hearth is located off center roughly in line with the third post holes down from the North end of the building.

North of the grinding bench, approximately in line with the fifth saddle quern from the west end, a cluster of 27 loom weights sat on the floor, including two very large ones. A pile of 90 loom weights in the East corner was removed.[267] Three weights in this pile are cone-shaped pyramidal types and are larger, 0.14 m. high, 0.12 m. wide, and pierced horizontally. The excavator suggests that these weights must have belonged to special places on the loom. It is probable that these were used to reinforce the edges of the textile, giving heavier tension to the warp threads on the exterior to prevent an incurving edge.

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[266]This clay platform was removed. The pottery from this sounding is 95% Hittite. Good bowl fragments, thin red rimmed types are included. One fragmentary loom weight was also found, the crescent type described above.

[267]On July 25, 1955.

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Fifty-five spindle whorls in total were found. Five spindle whorls are found lying as if strung together, two large and three medium size, under a burnt brittle jar. The weights of these whorls falls within the usual range (chart 13). The average of all whorls from CC 2 was 24 grams, fitting the same average weight for all of the TB and CC whorls. In addition to the spindle whorls and loom weights, several iron knives and a needle were also found in the NE corner of CC 2. McClellan notes also that an additional group of iron needles was found, but not catalogued, with several pieces of fabric in this room.

Clay Cut 3[268]

This room contained the usual post holes to support a balcony on three sides and a hearth in the center. The grinding stand seems to have been destroyed in a later disturbance, though grinding stones were still present. North of this stand was perhaps a bin for the flour produced at the grinding stand.[269] In a circular area of the floor, the grain found was 'the sort ... of tail-corn and wild seeds that is normally separated from the better grain during the final stages of cleaning the grain with sieves.'[270] A pile of sprouted barley may have been set aside for Phrygian beer production. Sams has suggested that beer was probably brewed at Gordion based on the side-spouted sieve jugs, referred to as beer mugs, probably used to filter floating husks from the brew and based on ancient references to Phrygian beer, such as Archilochus who compares a certain sexual act to a Phrygian drinking beer through a long spouted jug.[271] These Phrygian beer mugs are found in elite

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[268]K. DeVries (1990) 385-387; NB 160 (1973) 109-111; NB 161 (1973) 168-170; NB 164 (1973) 123-125, 128, 130-199; NB 167 (1973) 4-16, 19-20, 24-26, 35-49, 51-61, 64-120, plan in flap.

[269]K. DeVries (1990) 385-6.

[270]G. Hillman, quoted by DeVries (1990) 386.

[271]G.K. Sams, "Beer in the City of Midas," Archaeology 30 (1977) 108-115. For ancient references see Xenophon Anabasis IV, 5, 26.

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contexts at Gordion and in Phrygian levels at other Anatolian sites.[272] In form, however, the mugs show a Syro-Palestinian derivation.[273] Other indications of food production include the skeletons of two partially butchered calves and traces of hazel nuts found together with spindle whorls in a jug.[274]

A total of 213 spindle whorls were recorded from CC 3. More than half of these weighed less than twenty grams (chart 14). 509 loom weights were discovered in the main room, some with traces of thread still in their holes.[275] These threads could either be from the warp-threads, or, more likely, they were the remains of cords used to fasten the weights to the warp. The weights were found in large groups, up to 72 per pile. The find spots of the weights do not suggest that they were attached to a loom when the destruction occurred though it could be that they fell from the balcony above. This deposit of textile tools makes the main room of CC 3 the workshop with the largest concentration of textile equipment excavated at Gordion.

Five spindle whorls were found inside a carved, round, wooden box with the remains of braided thread.[276] Clumps of coarsely twisted fibers were also found which may be rove. A bone tool that may be a weft-bobbin was also found in CC 3.[277] In addition to the spindle whorls and loom weights, pieces of thread, needles, and metal knives were also found. At least eleven bronze knives were catalogued from this room,

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[272]G. K. Sams (1977) 67-70.

[273]G. K. Sams (1988) 11.

[274]The cattle bones are discussed by K. DeVries (1990) 386-87 based on a report by S. Payne.

[275]K. DeVries (1990) 387. These weights, however, were in no discernible row as if on a loom and may have fallen from the balcony above, as DeVries suggests.

[276]K. DeVries (1990) 387.

[277]P. Sheftel (1974) BI 583 Clay Cut building 3, in burned fill just above floor. L. 0.168; GW 0.011; GTh. 0.005. Complete as mended; some chips in breaks. Burned. Long piece of bone with natural curve. Whittled to a flat-oval section with sharp point at one end and flat triangular head.

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three straight-edged, the rest curved.[278] The knives, as will be discussed below, are of great significance and probably suggest carpet weaving for cutting pile knots.

Summary

A great deal about Phrygian political and social organization is unknown, including aspects of military organization, but we can assume from the Near Eastern documents that refer to the Mushki, and their King Mita, that military might was a necessity in the belligerent times of Early Iron Age Anatolia. Alliances and loyalties switched quickly among people facing the Assyrian war-machine. Power and politics were based on military strength. Historical records discussed on pp. 194 and following seem to indicate that smaller kingdoms of Anatolia appealed to the Phrygians for military aid against the Assyrians suggesting that a powerful Phrygian army existed.

Three aspects are vital to the strength of any fighting force: social organization, adequate supplies, and troop mobility. The standardized architectural features of the workshop units along the great terrace and the elaborate nature of the Phrygian gateway and Megaron buildings show a high degree of planning and social organization among the Phrygians. The political force that mobilized the labor, technology and resources necessary for building the capital of Gordion would certainly have been able to organize and maintain a standing army.

Supplies for an army would include weapons, food, and textiles. The Phrygians were skilled metal workers, crafting fine objects of precious metal, and also strong tools of bronze and iron. Among these metal goods would have been weapons, such as the long spears and shields shown in Phrygian art. The furniture fragments showing horse and

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[278]K. DeVries (1981) 40 notes that there were 'twenty-two separate occurrences in the main room of CC 3 of one or more knives'.

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riders from Megaron 3 depict mounted warriors equipped with such weapons.[279] We have looked at the grinding platforms, hearths, ovens and bins in the Terrace and Clay Cut units which could have supplied a fighting force as well as fed supporting craftsmen and women. The famed Phrygian beer may have been brewed in the Citadel Mound and supplied to the army as both a reward and a good source of calories. Finally, evidence for the mass production of textiles by attached, dependent workers has been adequately demonstrated from the loom weights, spindle whorls, needles, and other tools found in the Terrace and Clay Cut buildings. The heavy duty cloth made here could have been used both as protective armor and to provide temporary shelter to an army.

Another possibility for the textiles at Gordion is that they were a form of payment or medium of exchange. This would be in keeping with a long tradition of textiles in the Near East and Mesopotamia. Since the mid-third millennium textiles, metal objects, and semi-precious stones were the most widely-recognized prestige goods in the Mediterranean region.[280] Comparing the organization of craft activities in Mesopotamia to that of textile production at Early Phrygian Gordion nearly two millennia later may seem a great stretch, especially since Gordion lacks the archival evidence of earlier Near Eastern centers. However, the workshops within the fortified citadel in close proximity to the elite quarters, artistic depictions of Phrygian garments on royal figures, dedications of cloth-associated goods in Greek sanctuaries, and finds of well-made cloth in elite burial chambers at Gordion, all suggest that cloth was both a prestige item and a principal medium of exchange in Early Iron Age Anatolia. This is paralleled in the evidence from third and second millennium Anatolia, at centers such as Ebla, Mari, and Nuzi.

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[279]K. DeVries (1980) 47, fig. 7 and 8.

[280]See G. Stein and M. Blackman, "The Organizational Context of Specialized Craft Production in Early Mesopotamian States," Research in Economic Anthropology 14 (1993) 29-59.

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The actual fragments of cloth found at Gordion show that the Phrygians valued high quality textiles. Along with these fine textiles we know that Phrygian craftsmen excelled at wooden furniture, ivory carving and fine metal work.[281] These goods of high quality were most likely used in elite settings, such as the funerary gifts of the man buried in the great Midas Mound. They could also have functioned as items of exchange with other rulers or officials, bestowing Phrygian authority or approval. One recalls the Phrygian fibula and cloth shown on the Ivriz relief discussed above. This tradition of gift exchange also extended to the sphere of religious dedications and we are reminded that King Midas is cited by Herodotus as the first foreigner to send dedications to the Greek sanctuary of Delphi.

In addition to elaborate garments shown in sculpture or possibly dedicated in religious contexts, it also seems likely that the Phrygians at Gordion were producing pile-carpets.[282] The large numbers of iron and bronze knives found in the Gordion excavations strongly suggest that they were used to cut the knotted threads of carpet weaves (fig. 4.6). The shape and size of these blades is appropriate to a hand-held tool used for cutting wool thread. The earliest indications for pile-carpet technology come from a special type of bronze knife found in women's graves in the Sumbar Valley, east of the Caspian Sea, dating to the late third millennium B.C.[283] The knives from Gordion have their inner blades sharpen in contrast to those found in S. Russia but their association with warp-

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[281]See, for example, E. Kohler, A Study of Wood and Ivory Carvings at Gordion, Phrygia (ca. 750-700 B.C.) Ph. D. Dissertation, Bryn Mawr College (1958); E. Simpson, *et al.* Gordion: Wooden Furniture (1992); *Eadem*, "The Phrygian Artistic Intellect," Source: Notes in the History of Art VII.3-4 (1988) 24-42.

[282]For more on pile-carpets see E. Barber (1991) 171; S. Dalley, "Ancient Assyrian Textiles and the Origins of Carpet Design," Iran: Journal of Persian Studies 29 (1991) 117-135; I. Khlopin, "The Manufacture of Pile Carpets in Bronze Age Central Asia," Hali 5.2 (1982) 116-118.

[283]See discussion by E. Barber (1991) 171 n. 10. And see I. Khlopin (1982) 116-118; H. Böhmer and J. Thompson, "The Pazyryk Carpet: A Technical Discussion," Source: Notes on the History of Art 10.4 (1991).

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woven textiles and their hand-held size suggests carpet production in the Terrace and Clay Cut units.

The earliest known examples of pile-carpets come from the astounding burials in Pazyryk Siberia. The origins of these carpets, dating to *ca.* 500 B.C., with Achaemenid and Greek designs have been traced to Lydia by Greenewalt and Majewski.[284] That the true origin for this type of textile might be found in Phrygia is another possibility. As Young suggests, the decorative patterns found in the mosaic of Megaron 2 imitates thrown Anatolian rugs, as seen in mosques today (fig. 4.9).[285] The small, multi-colored patterns are suggestive of pile carpets. The hand-tied knots of thread on pile rugs can easily vary in color and design, much more so than on tapestry. In addition, the bronze and iron blades found in the excavation can almost certainly be associated with carpet weaving. This is the only weaving technology where each weft thread is cut by hand once it is knotted to the warp threads.

Conclusion

Archaeological evidence from Gordion demonstrates that cloth production and food preparation were the major activities in the Terrace and Clay Cut Buildings at the time of the Early Phrygian destruction. As in many Aegean palaces, craft residues of industries such as bronze working and pottery production are lacking at Gordion.[286] The location of

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[284]C. Greenewalt, jr. and L. Majewski, "Lydian Textiles," in K. DeVries (ed.) From Athens to Gordion (1980) 133-147.

[285]R. Young, "Early Mosaics at Gordion," Expedition 7 (1965) 4-13. The pebble by pebble drawing by J. Last on page 11 is reproduced as figure 4.9.

[286]One should recall that traces of ivory carving, such as flakes and unfinished pieces of ivory, were found in the area of Megaron 4. For pottery production at Gordion, see R. Henrickson and M. Blackman, "Large-scale production of pottery at Gordion: comparison of the Late Bronze and Early Phrygian industries," Paléorient 22.1 (1996) 67-87; and R. Henrickson, "Continuity and discontinuity in the ceramic tradition at Gordion during the Iron Age," in A. Çilingiroğlu and D. French (eds.) Anatolian Iron Ages 3: The Proceedings of the Third Anatolian Iron Ages Colloquium held at Van, 6-12 August 1990 (1994) 95-129.

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textile production and food (including beer) preparation in standardized workshop units within the massive Phrygian fortification walls suggests that these were the two most important products of the Phrygian political-economy.

As Mellink states, 'The organization reflected in the citadel plan is one of a privileged society preparing for war and siege...The Phrygians show themselves heirs to a long tradition, to which Trojans of Troy II and VI belonged and to a branch of which, on the other side of the Aegean, the Mycenaeans were also indebted.'[287] As with cloth in the Minoan and Mycenaean kingdoms, Phrygian textiles probably functioned as both a kind of currency or medium of exchange, and as a prestige good. The research presented here suggests different qualities of cloth were woven in the Terrace and Clay Cut Buildings. The large numbers of spindle whorls and loom weights of fairly similar shape and standardized size indicate cloth made on a massive scale, for distribution directed by the ruling elite. The heavy weight of the doughnut-shaped loom weights might imply that the woven textiles were strong and durable. Heavy-duty textiles could have been used to equip a standing army of the Phrygians or at least as some means of payment or compensation to the soldiers.[288] Alternatively, some of the cloth woven just before the great conflagration may have been meant for tribute to assuage an aggressive neighbor.

Rodney Young, director of Gordion from 1950-1974, stated in his 1959 preliminary report, 'We can have little admiration for the Phrygians as neat and orderly housekeepers. In a palace, as at Knossos for example, one would expect many storerooms for stowing of goods and equipment of all kinds, one would expect also workrooms in which chores of daily living were performed. But one would expect at least a modicum of

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[287]M. Mellink (1991) 629.

[288]G. K. Sams alludes to this possibility in (1995) 1152.

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neatness and order: one for the storage of bronzes,... perhaps one for the archives....'[289]. While the organization of the Terrace Buildings may not have conformed to Rodney Young's original conception of a palace, his later researches and those of others have broadened our understanding of Phrygian economy and culture.

From this investigation of textile production and related craft activities we can see that the Phrygians had a highly developed system of cloth manufacture. The masses of spindle whorls and loom weights contained in the destruction level of the Terrace and Clay Cut buildings suggest a great deal of importance on textile production to the Phrygian economy and political structure. Cloth can serve many purposes in a state-level economy. It can function as a kind of currency in tribute demands placed upon commoners. Cloth of various qualities and grades bestowed by a ruler can signify certain levels of authority to peers and subordinate officials. Textiles can also function as protection and payment for members of a fighting force.

From surveys and excavations in central Anatolia the geographical extent of Phrygia is estimated to have covered a large portion of central Turkey. It is possible that some amount of tribute was demanded by the center, some of this may have been in the form of textiles sent to the center. Artistic representations far from the capital of Gordion, such as the Ivriz relief, show cloth that appears to be Phrygian in style. These garments with traditional Phrygian designs, fastened with fibulae of well-known Phrygian types, do suggest that they were exchanged between rulers following a long tradition of royal gift-exchange. Finally, we also know that the Phrygian citadel of Gordion was sacked by a destructive force, probably setting fire purposefully in different parts of the city mound at the same time. From diplomatic records we also hear of shifting alliances between the various rulers of Anatolian city-states and the Assyrian kingdoms. Whether it was the

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[289]R. Young (1960) 242-3.

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Cimmerians as traditionally believed, or the Assyrians, some military force destroyed the Phrygian capital. That there was such an intense effort to produce textiles within the center at the exact same time, suggests that this industry might be associated with a Phrygian military force. These fighters would have needed both equipment and payment. As it seems, though, the Phrygian military efforts were unsuccessful.

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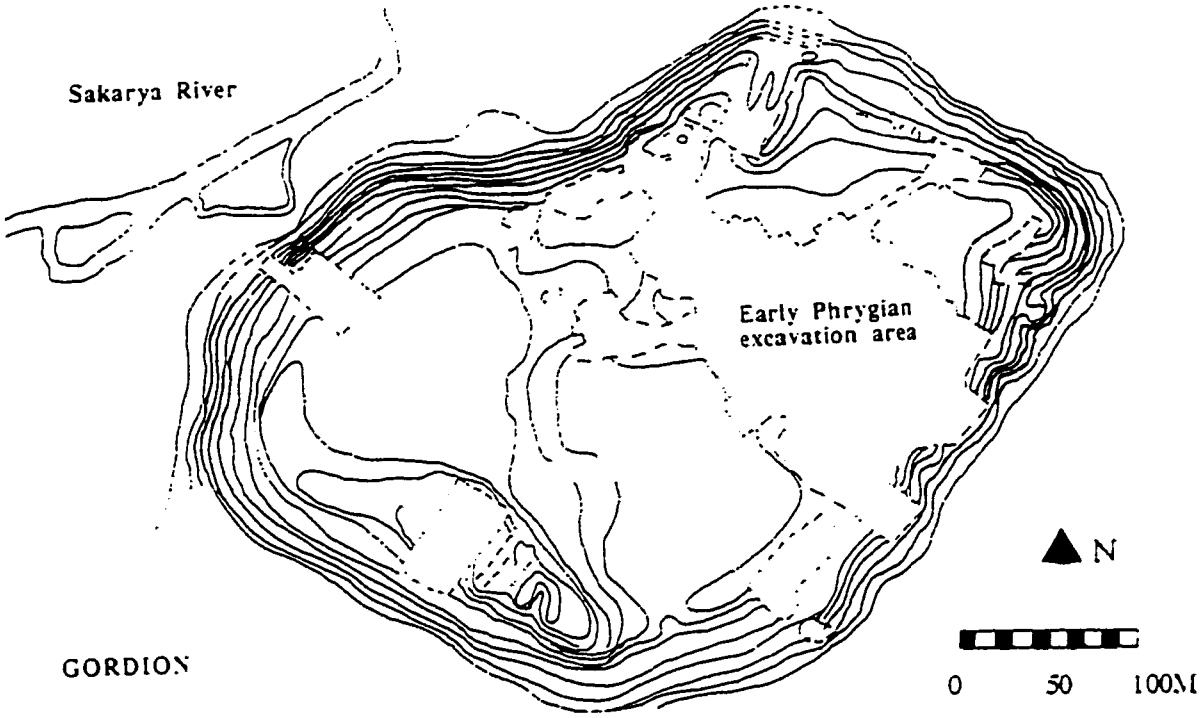


Fig. 4.1 Citadel Mound, Gordion

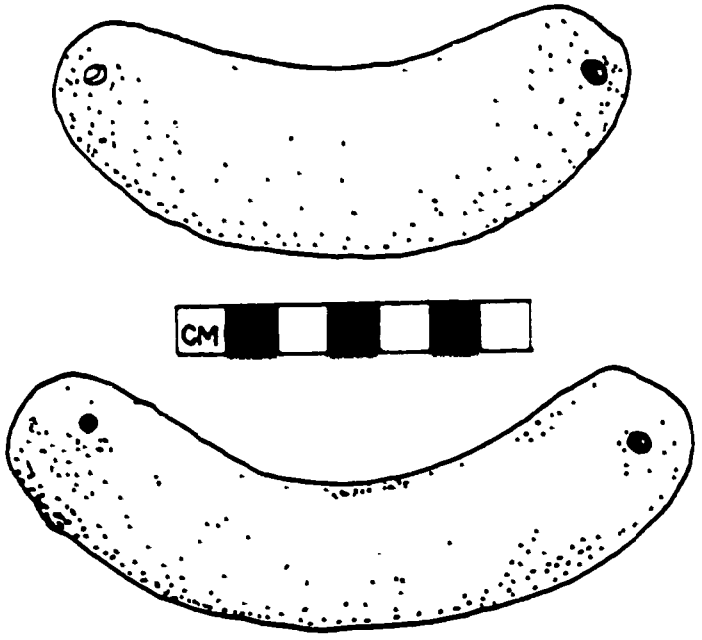


Fig. 4.2 'Hittite loom weights'

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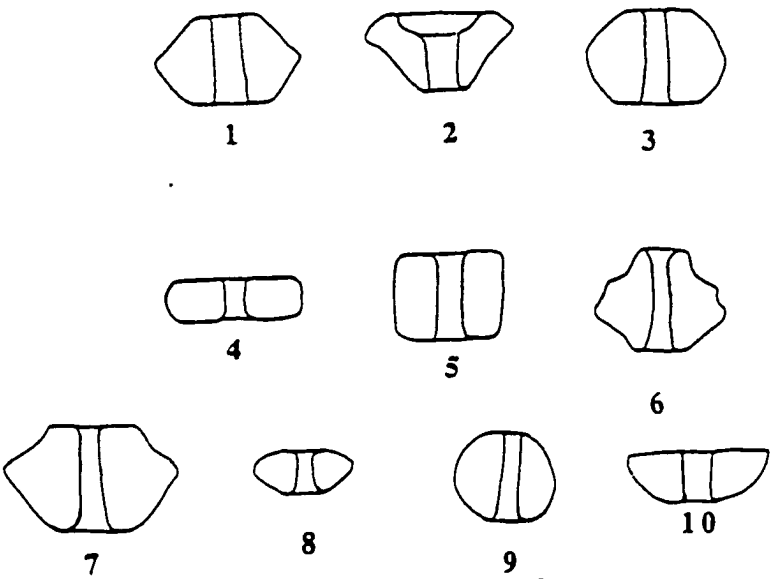


Fig. 4.3 spindle whorl types

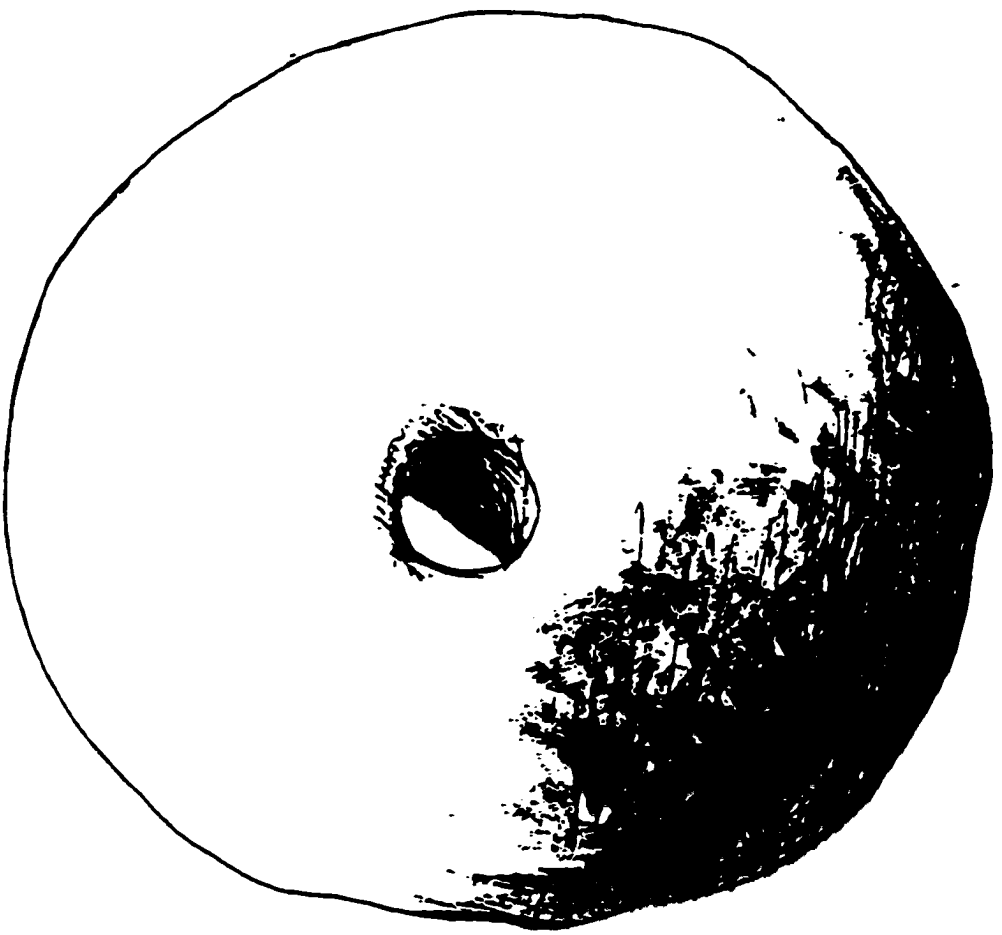
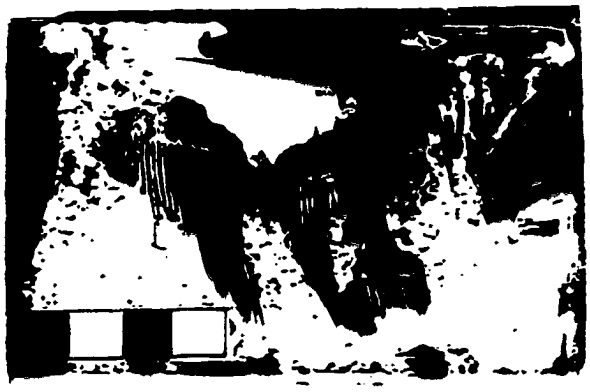
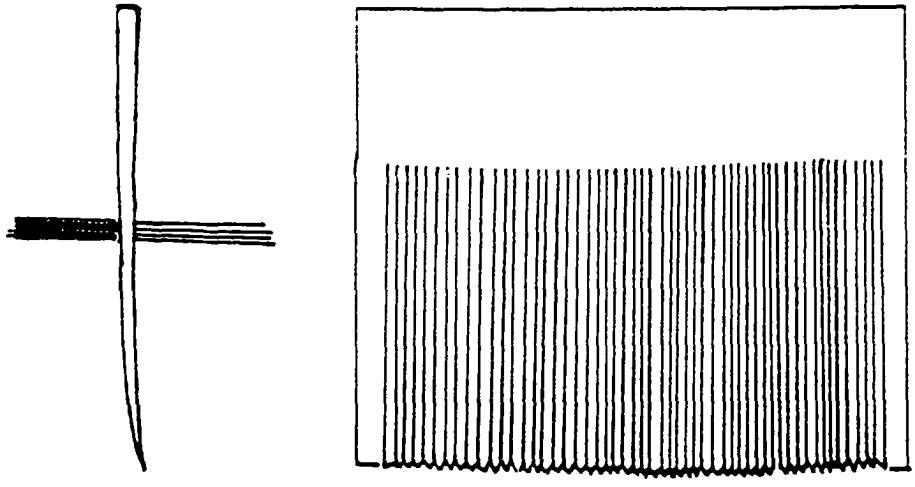


Fig. 4.4 typical loom weight, actual size

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a) photograph of comb with scale



b) restored view of comb, actual size

Fig. 4.5 a and b comb/reed from TB 2A

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Fig. 4.6 Bronze knife from TB 8

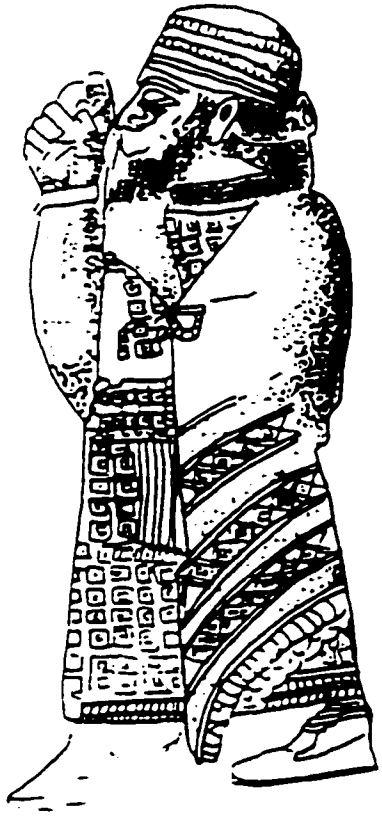


Fig. 4.7 King Warpalawas of Tyana, relief sculpture

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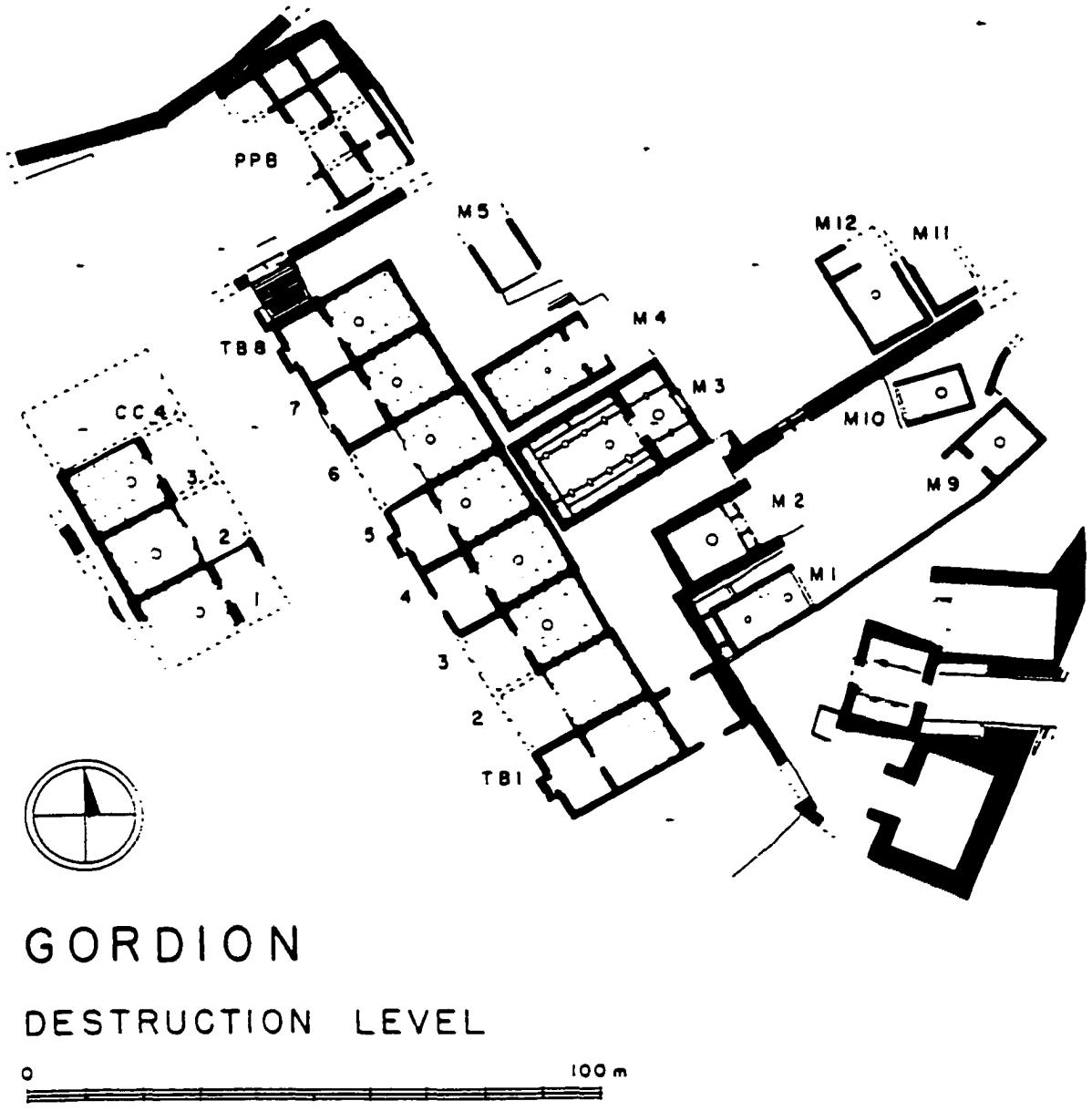


Fig. 4.8 Early Phrygian Gordion

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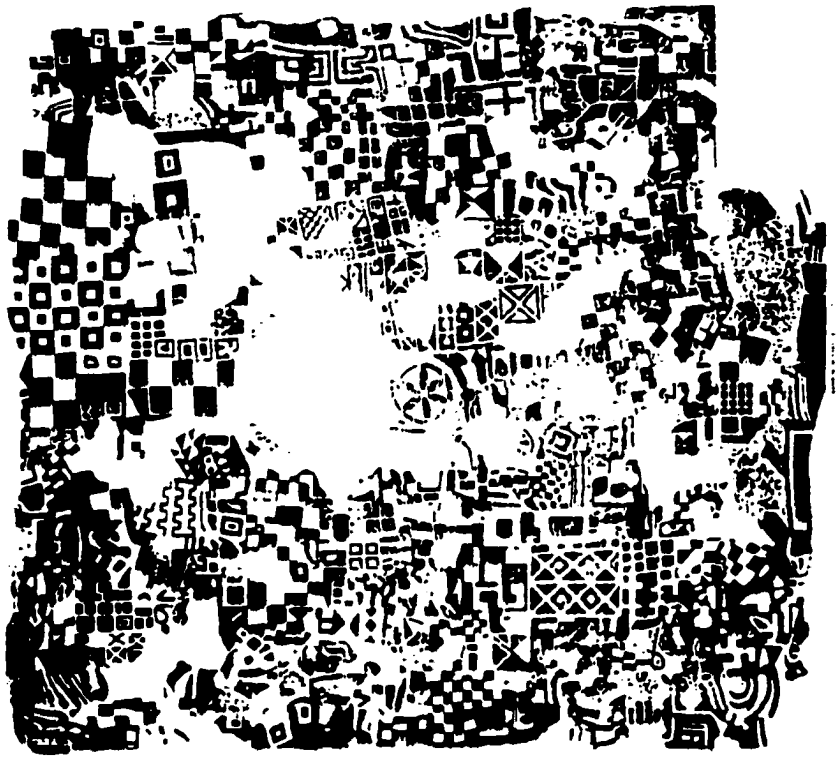


Fig. 4.9 Mosaic from Megaron 2

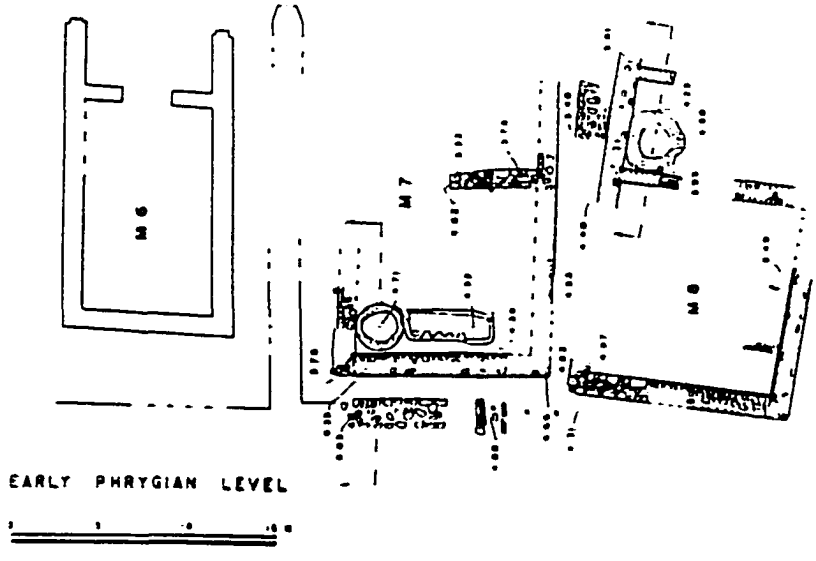


Fig. 4.10 Megara 6, 7, and 8

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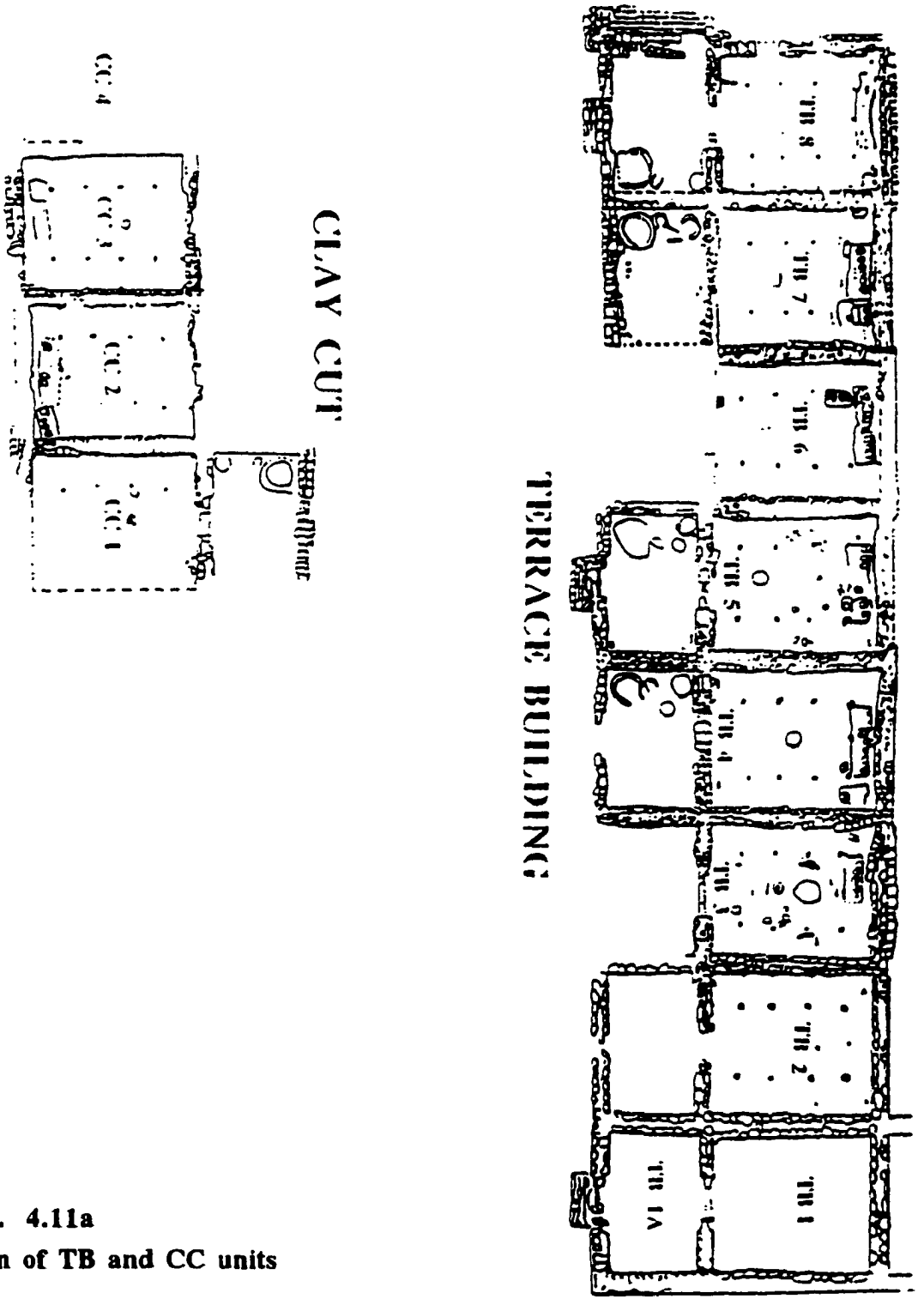


Fig. 4.11a
Plan of TB and CC units

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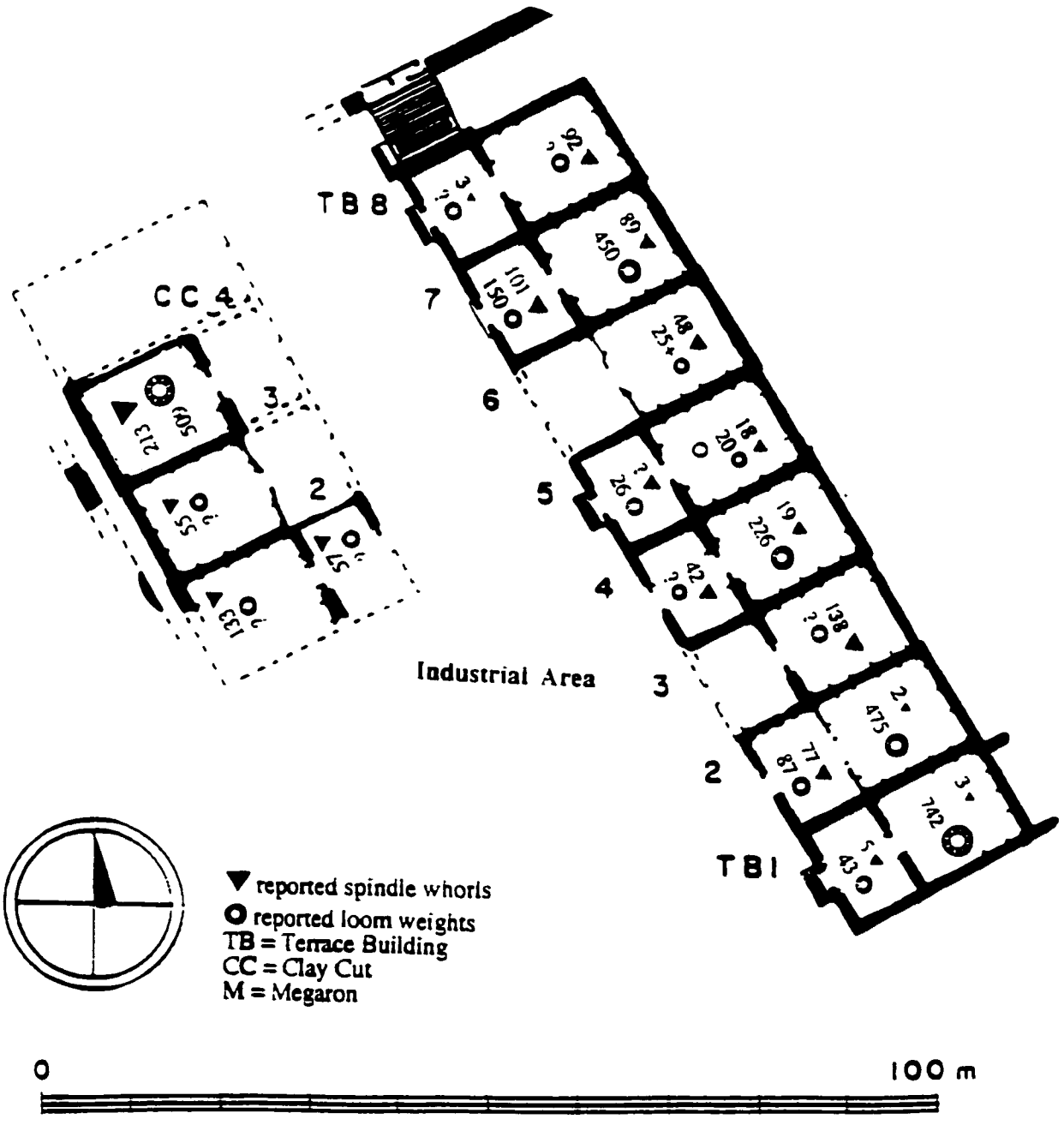


Fig. 11.b Industrial area of Gordion showing textile equipment

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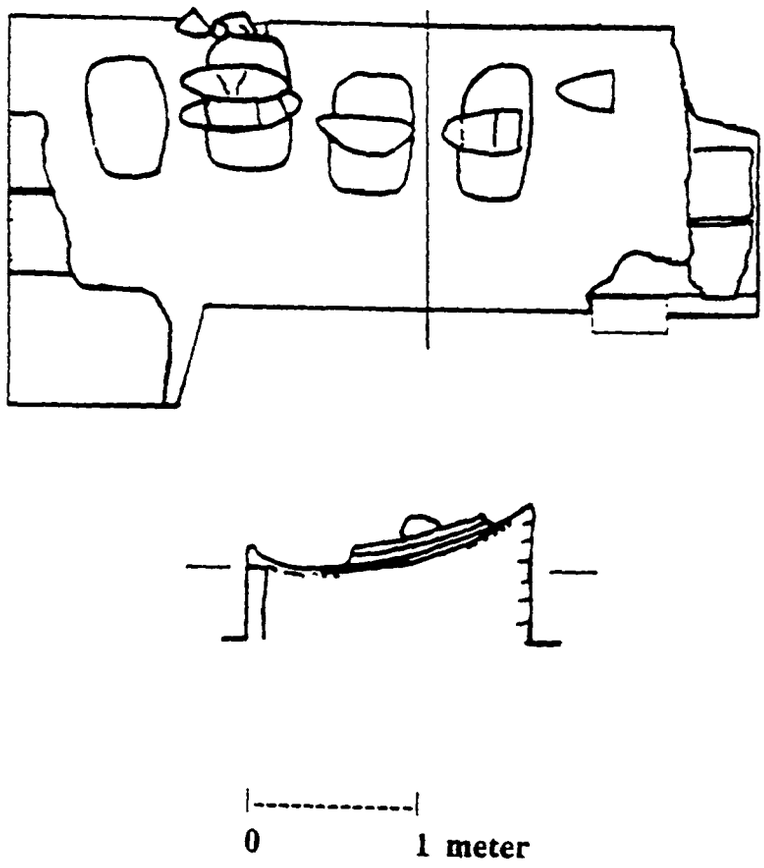


Fig. 4.12 Grinding stand from TB 3

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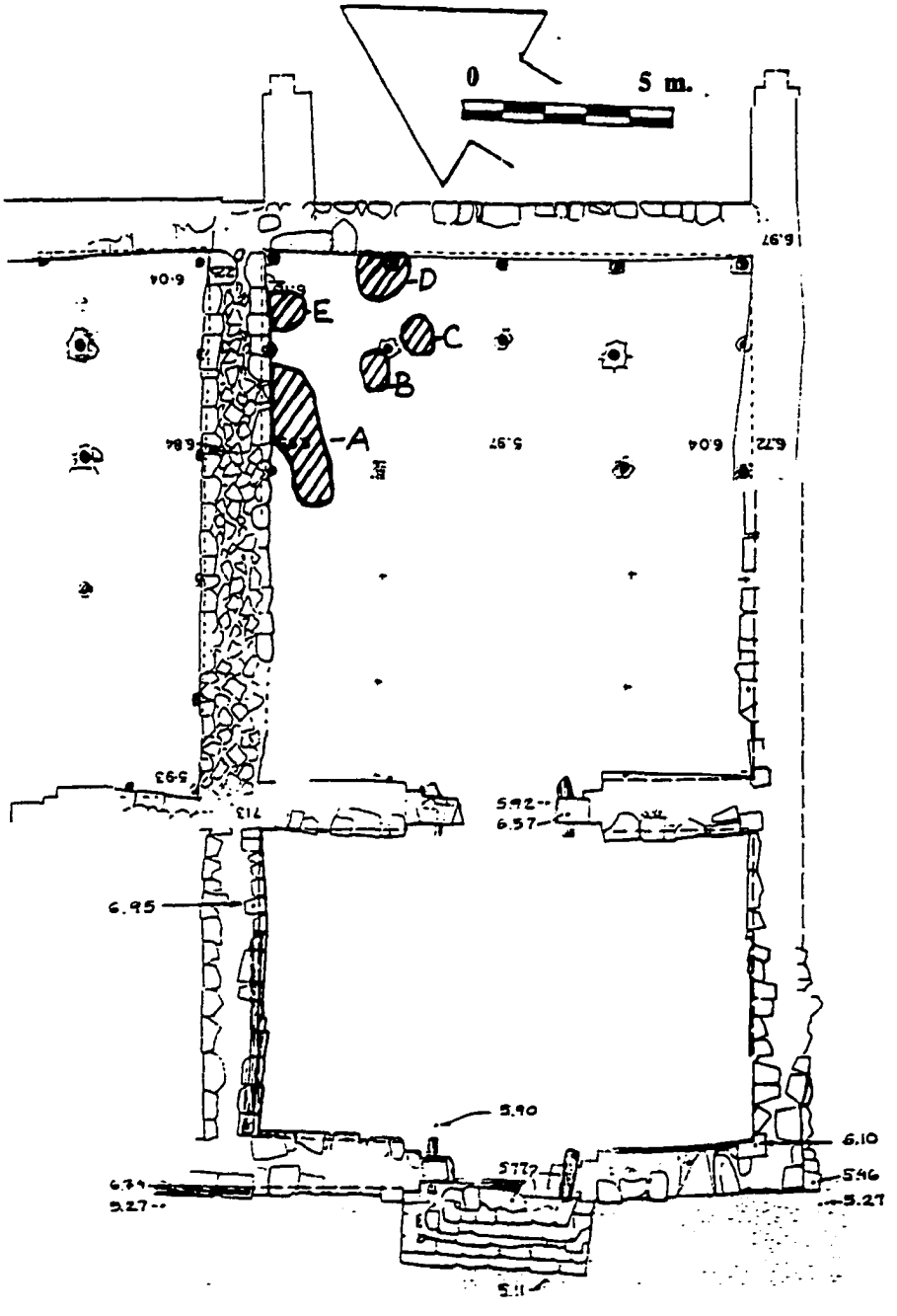


Fig. 4.13 Terrace Building 1

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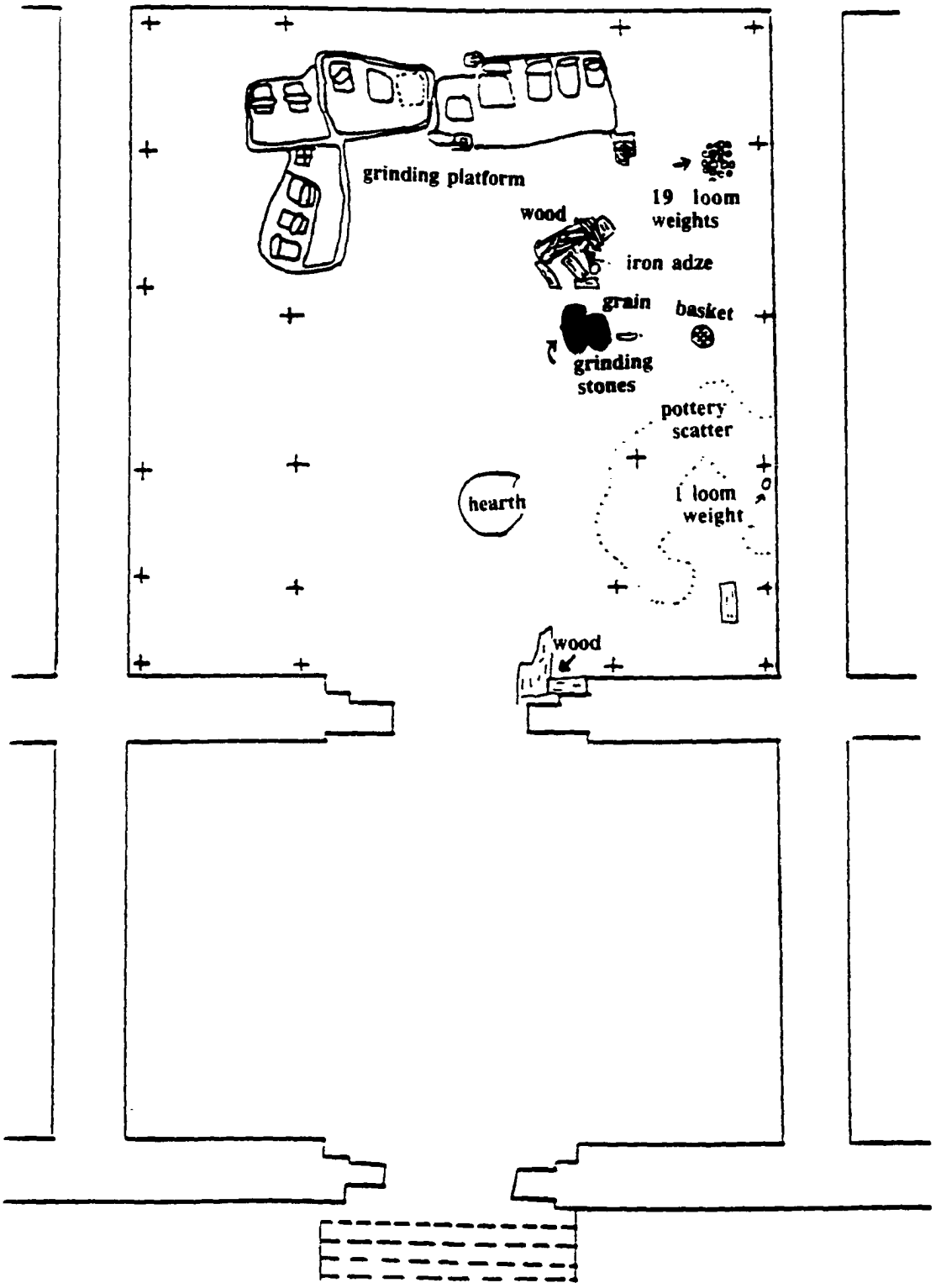


Fig. 4.14 Terrace Building 6

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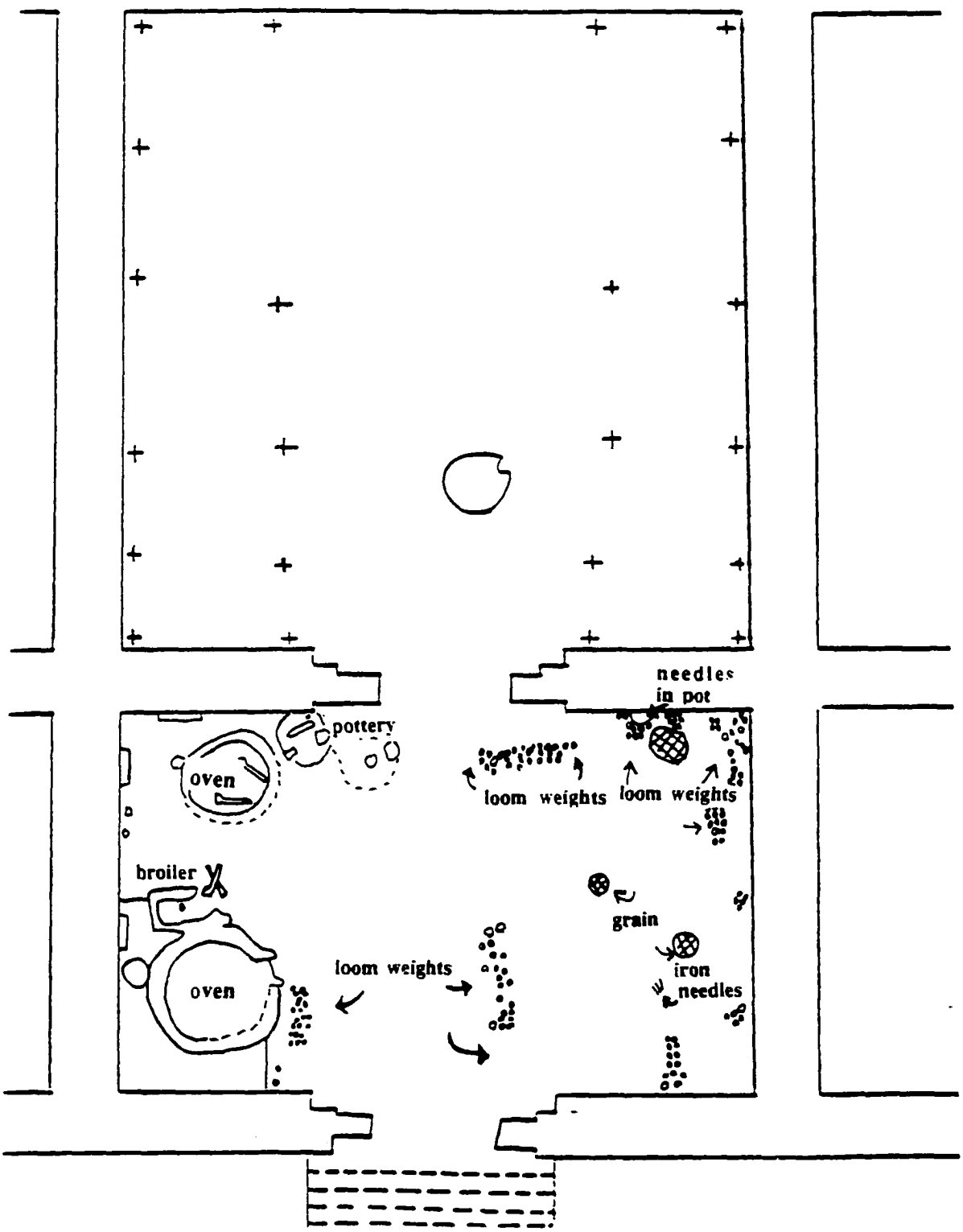


Fig. 4.15 Anteroom Terrace Building 7

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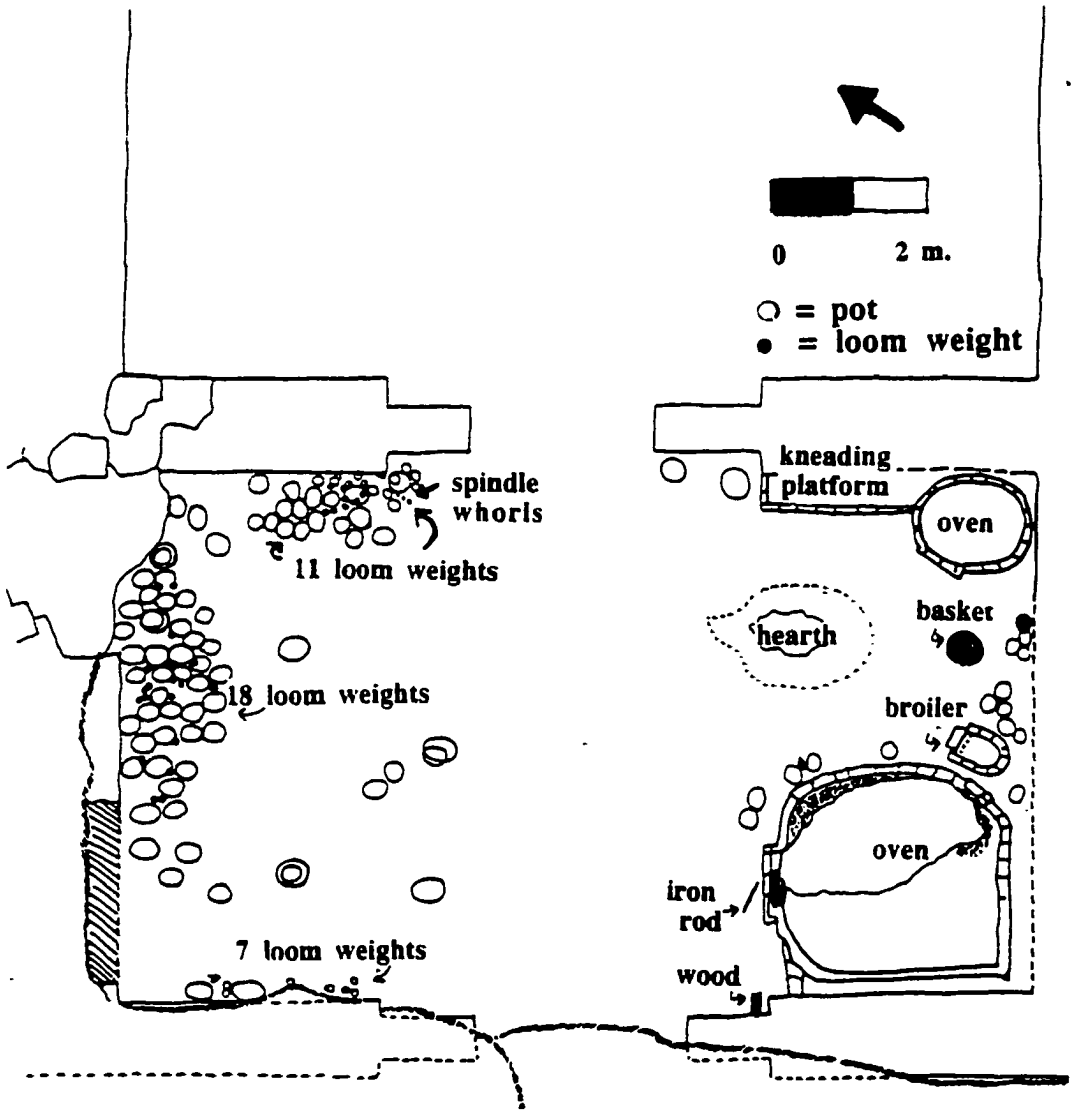
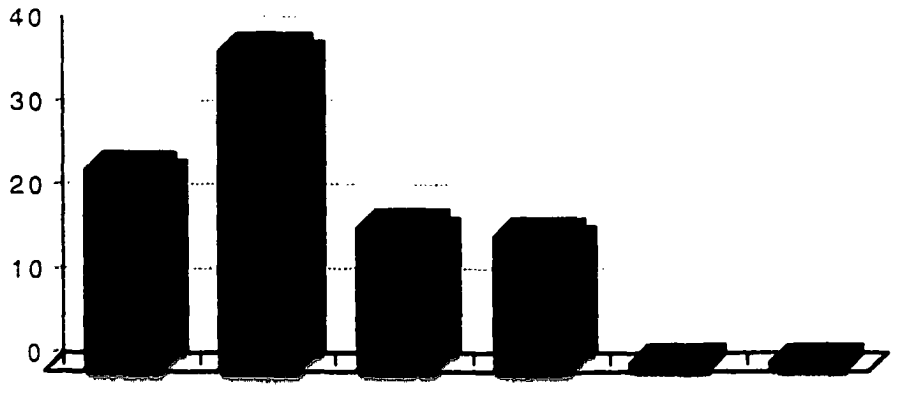


Fig. 4.16 Anteroom of Terrace Building 8

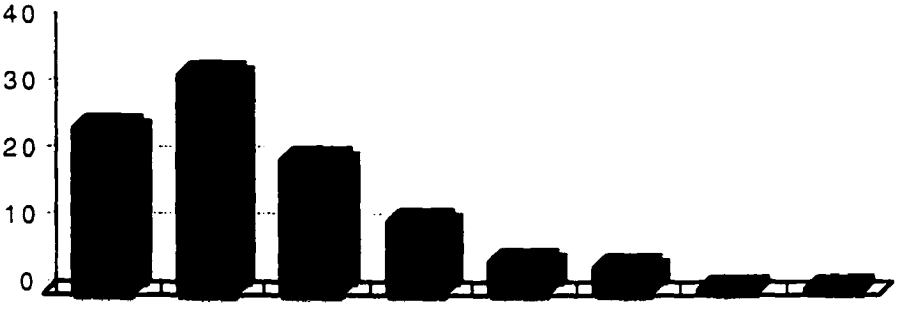
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Charts 1-3



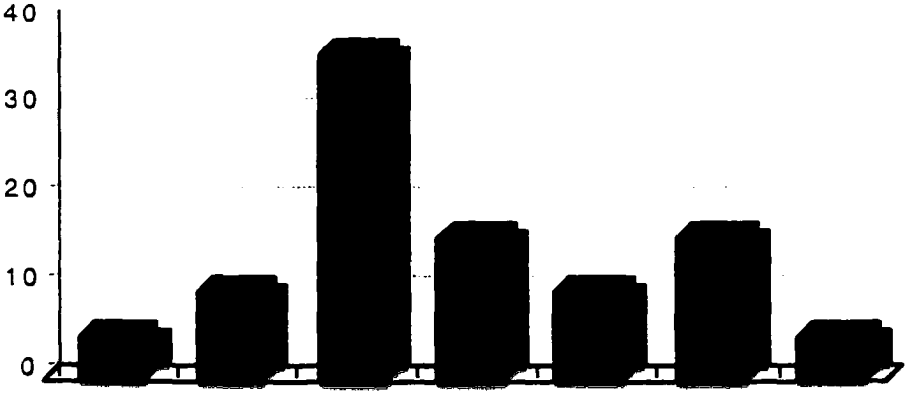
% of TB 2A Whorls by 10 gram groups

1



% of TB 3 Whorls by 10 gram groups

2

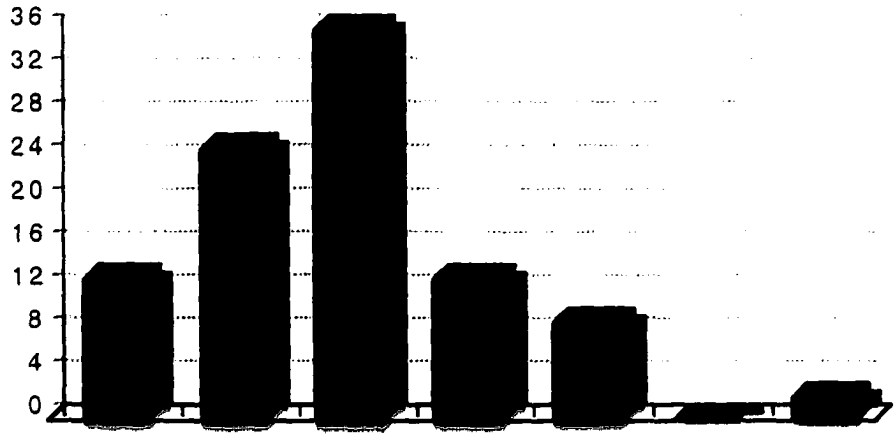


% of TB 4 whorls by 10 gram groups

3

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Charts 4-6



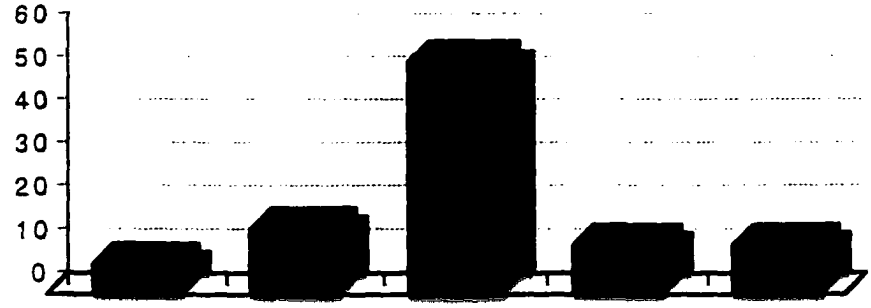
% of TB 4A whorls in 10 gram groups

4



% of TB 5 Whorls in 10 gram groups

5



% of TB 5A Whorls in 10 gram groups

6

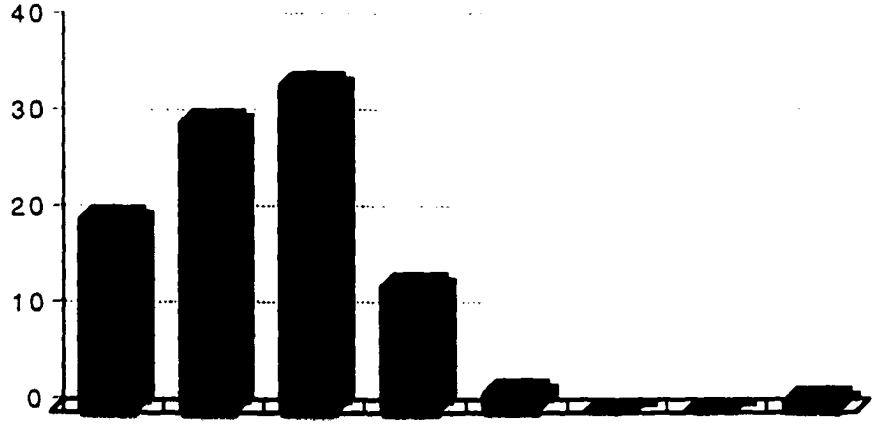
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Charts 7-9



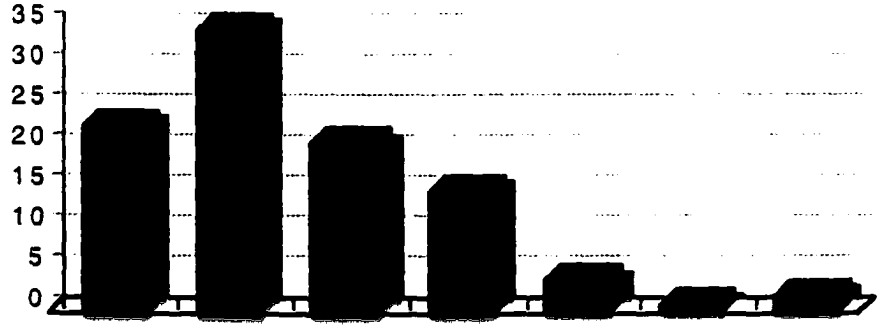
% of TB 6 Whorls in 10 gram groups

7



% of TB 7 Whorls in 10 gram groups

8

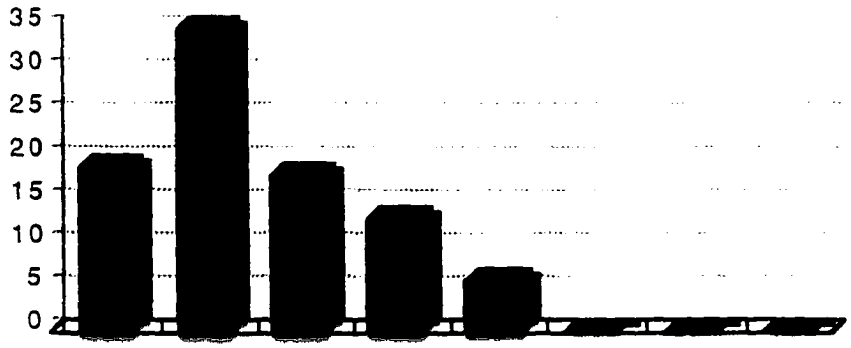


% of TB 7A Whorls in 10 gram groups

9

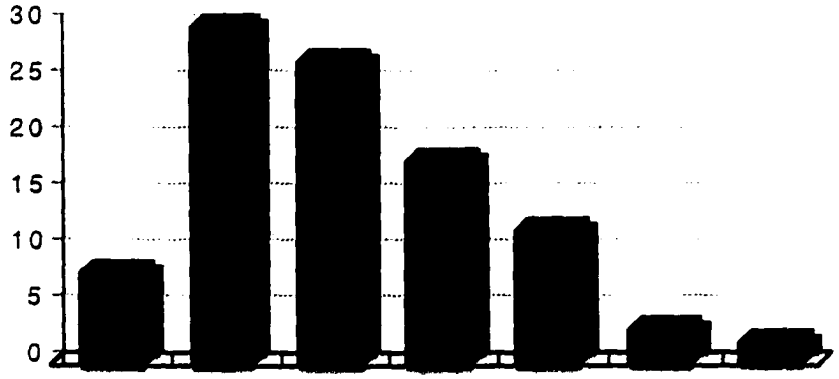
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Charts 10-12



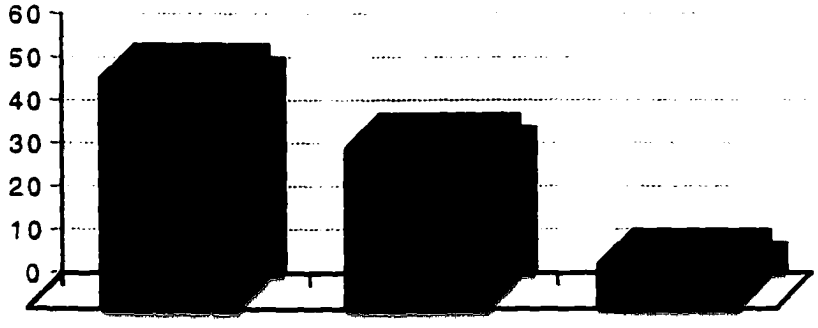
% of TB 8 Whorls in 10 gram groups

10



% of CC 1 Whorls in 10 gram groups

11

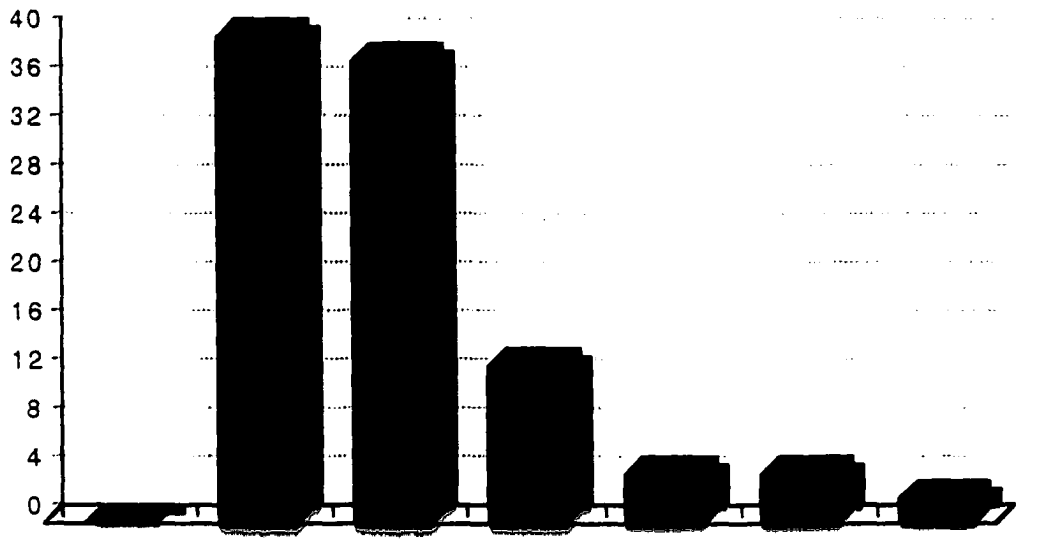


% of CC 1A spindle whorls 10 gram groups

12

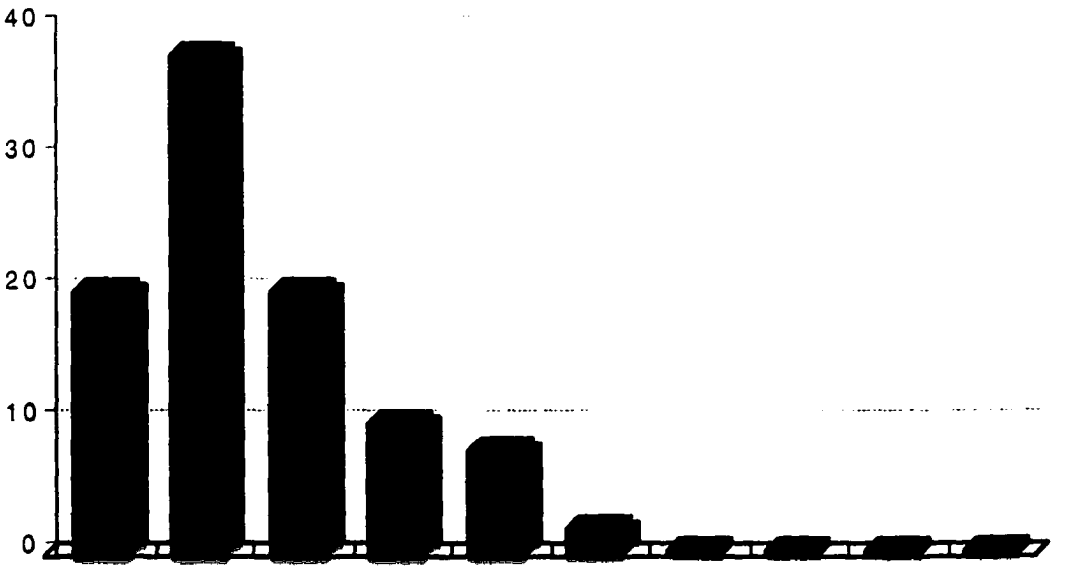
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Charts 13-14



% of CC 2 Whorls in 10 gram groups

13



% of CC 3 Whorls in 10 gram groups

14

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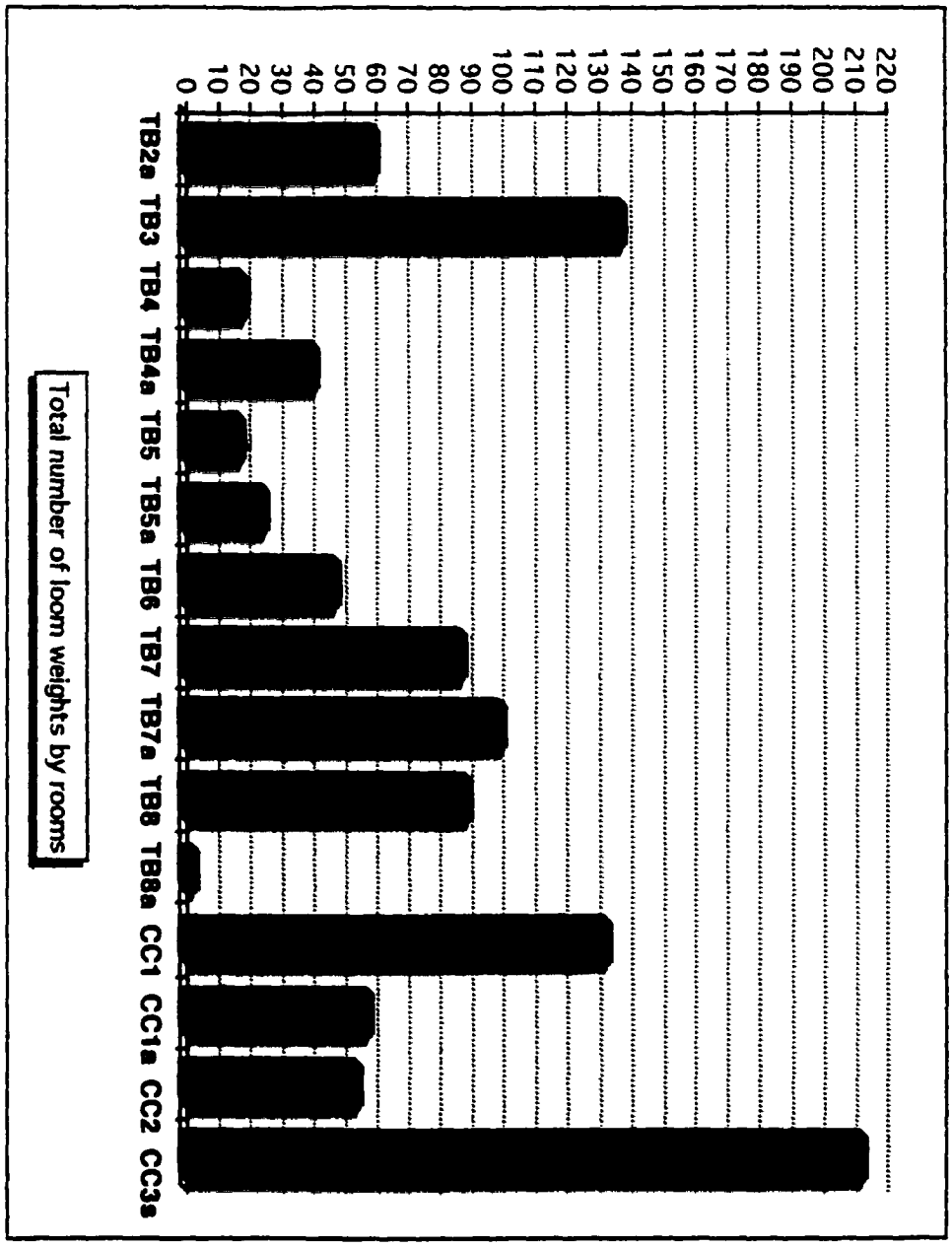


Chart 15

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Chapter Five - Comparative Systems and Conclusions

In many different parts of the world the many stages of fiber processing are often organized along the same general principles. One gap in our knowledge of textile production among the Minoans, Mycenaeans and Phrygians lies in understanding the function of so much cloth in these early states. Some evidence has been cited that connects intensive textile production with military service. The *ke-se-nu-wi-ja* cloth recorded in the Linear B tablets is most likely associated with mercenary soldiers. This cloth was either a kind of uniform supplied by the palace or a payment for services rendered.[1] Since the tablets that record this cloth were preserved by violent means, it is not unlikely that the Mycenaeans were facing some military threat. The citadel at Gordion was perhaps under a similar siege just prior to the great conflagration that destroyed the Phrygian capital and preserved so much of the equipment presented in Chapter four. Perhaps the textile activity within the palace walls of Pylos, Knossos and Gordion was intensified during their respective times of crisis to supply a standing army or to pay mercenaries. Concurrently more slave labor may have been employed to fulfill this increased need for cloth.

To supplement our understanding of the function of cloth we can also look to Mesopotamia, the Levant and empires of the New World to see how textiles operated as a means of finance in these areas.[2] Evidence from each region provides interesting comparative examples for Aegean and Anatolian textile production and demonstrates the

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[1]J. Driessen, "Mercenaries at Mycenaean Knossos," BSA 79 (1984) 49-56.

[2]See M. Van de Mieroop, The Ancient Mesopotamian City (1997) 176.

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important role of cloth in financing an elite power base. These forays through time and space are not meant to prove a preconceived notion of what I think all textile industries should be like, but rather this comparative evidence better directs my own line of inquiry in the Aegean and Anatolian spheres.

Mesopotamia

We learn a great deal about the organization of textile production during the third millennium from palace and temple records at Mesopotamian centers such as Ebla, Ur and Lagash.[3] Mortuary and iconographic evidence also indicate that textiles were a marker of status and prestige. Generally speaking (and each center differed somewhat) once each year subordinate city-states and professional herders under palatial administration sent enormous amounts of wool, goat hair, and finished textiles to a 'wool office'. Mari often sent wool to Ebla, for example.[4] Raw wool could either be sent raw to an urban center or spun locally and then disbursed to weaving workshops. The 'wool houses' were a distribution center for weaving, storage and disbursement and are perhaps comparable to the structure at Thebes discussed on page 162-163. Finished garments were also sent on to wool houses. One inventory at Ebla lists various qualities of 7,000 garments in one wool house. In exchange for the yearly tribute of wool, herders would presumably be granted

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[3]See Th. Jacobsen, "On the Textile Industry of Ur under Ibbi-Sin," Toward the Image of Tammuz and Other Essays on Mesopotamian History and Culture (ed.) W. Moran (1970) 216-230; H. Waetzoldt, Untersuchungen zur neusumerischen Textilindustrie (1972); K. Maekawa, "Female Weavers and their Children in Lagash- Pre-Sargonic and Ur III. Acta Sumerologica 2 (1980) 82-125; A. Archi, "About the Organization of the Eblaite State," Studi Eblaiti 5 (1982) 201-220; F. Pinnock, "Trade at Ebla," Bulletin of the Society of Mesopotamian Studies 7 (1984) 19-36; E. Sollberger, Administrative Texts Chiefly Concerning Textiles. Archivi Reali di Ebla Testi VIII (1986); K. Szarzynska, "Records of Garments and Cloth in Archaic Uruk/Warka," Altorientalische Forschungen 15 (1988) 220-230; G. Stein and M. Blackman, "Specialized Craft Production in Mesopotamia," Research in Economic Anthropology 14 (1993) 29-59; M. Van de Mieroop (1997).

[4]F. Pinnock (1984) 27.

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land of their own to raise palace sheep along with private live-stock and/or crops of their choice.

Regarding the organization of labor, almost no craft specialists in Mesopotamia attached to administrative centers were slaves. Interestingly, the only exceptions were male and female textile workers.[5] It took at least five years for an apprentice weaver to learn the necessary weaving skills. An estimate of 13,200 weavers were employed around Ur on a full-time basis, entirely dependent on the palace for their food and clothing.[6] Most of these workers were female and lived at home each day since they are known to have had children. These children would be taken to work each day, both because they would need to be breast-fed and for additional labor. The keen eyesight and small hands of older children made them especially desirable for fine textile manufacture, just as child labor is still exploited today. After the textiles were woven they were sent on to fullers and finishers also located in the nearby villages around Ur. Less is known about these workers but their tasks were obviously arduous and required a great amount of water. Van de Mieroop reproduces one texts that fully records the phases of a single three and a half kilogram garment of second rate quality.[7] The total amount of time spent on this piece is well over one year by laborers twisting, cleaning, trimming, sewing, pressing, beating, picking, and blowing the garment. This example gives a good idea of the attention and effort that went into making cloth under the Mesopotamian temple-state.

Textiles and metals were manufactured by attached specialists in Mesopotamia, under strict control of the palace-temple institutions. Access to raw materials was the major way centers controlled the production of metal goods. Since natural deposits of precious

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[5]C. Zaccagnini, "Patterns of Mobility among Ancient Near Eastern Craftsmen," JNES 42 (1983) 24; G. Stein and J. Blackman (1993) 52.

[6]M. Van de Mieroop (1997) 186.

[7]M. Van de Mieroop (1997) 187.

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metals were lacking in Mesopotamia, raw materials had to be imported from Anatolia, Oman, and Iran.[8] These imported materials were strictly controlled by the ruling centers. Textual evidence from Assyrian trading colonies in Anatolia, primarily from cuneiform documents at Kültepe Karum Kanesh, provide information on the two main items of exchange at the trading colonies of Anatolia: metals and textiles. Wool and linen cloth woven in communities surrounding Assur were sent by caravan northward. In Anatolia these garments would be exchange for valuable metals lacking in the home cities.

With textiles, however, access to the raw material wool was much more difficult to control. As Stein and Blackman suggest, 'The centralized institutions might have maintained their control over high-prestige, luxury textiles through a strategy of close control over the knowledge and skills of the specialists themselves.'[9] The absence of metal sources on Crete has already been noted. One wonders if the Minoans organized their metal and textile industries in a way similar to the Mesopotamian palace-temple states, by controlling access to certain textile technologies, such as purple dyeing or whatever kind of weaving was done with the spherical 'melon'-weights, and trading with the Near East for metals.

The Levant and Tell es-Sa'idiyeh

The University Museum excavations at Tell es-Sa'idiyeh, between 1964 and 1967 under the direction of J. Pritchard, provides a striking comparison to textile production at Gordion. Located in modern Jordan, the mound yielded evidence for occupation throughout the first millennium B.C.[10]

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[8]G. Stein and J. Blackman (1993) 52.

[9]G. Stein and J. Blackman (1993) 52.

[10]For more on textile production in the Levant, see D. Browning, The Textile Industry of Iron Age Timnah and Its Regional and Socioeconomic Contexts: A Literary and Artifactual Analysis Ph. D. dissertation,

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Because of modern political turbulence in the region, the excavation has never been completed. The ancient name of the site remains a mystery as does the character of the occupation sequence. Traces of fortification walls and a dominating presence in the landscape demonstrate that the tell was obviously an important center in the region. From the published findings interesting questions about the organization of this site and its industrial potential are raised in comparison to the remains from Gordion.

A substantial phase roughly contemporaneous with the Early Phrygian period is called Stratum V by the excavators. This stratum provided the most substantial information from the excavations and shows a group of building units that are very similar to the Terrace and Clay Cut buildings at Gordion. These structures are identified by the excavator as domestic houses. The standardized plan and shared walls show that all of the units were planned and constructed at the same time. Pritchard states that, 'every indication is that these were private dwellings, with the furniture and artifacts appropriate to residences.'[11] The finds, however, and the architectural units, when viewed in comparison to the Terrace and Clay Cut buildings, imply an industrial quarter of specialized workshops. The major classes of artifacts from these units are ceramic bowls and storage vessels, doughnut-shaped clay loom weights, grinding stones, and clay bake-ovens. The standard architectural unit consists of a main room and a smaller room built in a line with shared side walls. The one feature of comparison found at Gordion and not at Tell es-Sa'idiyeh is a large number of clay spindle whorls. The other evidence, however, strongly suggests that the main activities in these buildings were grinding grain, baking, and weaving. These buildings suggest a nucleated workshop aggregated within a single community.

Southwestern Baptist Seminary (1988); "Loom weights in Iron Age Israel: Evidence of production for Assyrian tribute?" (n.d., personal communication).

[11] J. Pritchard, Tell es-Sa'idiyeh: Excavations on the Tell 1964-1967 (1985) 30.

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Six identical units are built in a line with their entrances facing a north-south street. Through these thresholds one enters the main room of the unit, rather than an anteroom. A comparable smaller chamber is found behind the main room through an internal doorway, similar to the Gordion megaron plan but inverted with the smaller chamber behind the main room. The back wall of these rear chambers is shared by a parallel set of units built directly behind this first set of six. This second set of rooms corresponds to the facing Clay Cut building at Gordion in that the plan is a mirror image of the units across the way. The buildings at Tel es-Sa'idiyeh are located on an eroded ridge of the mound that has been detrimental to the preservation of some of the buildings, particularly the first two. Their organized plan and consistent features of cooking facilities and textile equipment are highly reminiscent of the remains from Gordion.

New World Evidence

Comparative evidence for organized textile production is also found in kingdoms of the New World. Evidence from the Inca center of Huanuco Pampa in Peru and at textile production sites among the Aztecs in Mexico provide interesting data for understanding the role of cloth in early states. While the complexity and scope of the Inca and Aztec empire are greater than that of what we might reconstruct for the Aegean or Phrygia, similarities in the organization of craft activities do provide useful evidence.

Aztec

In the early part of the second millennium A.D., the Aztec empire expanded throughout large portions of Mesoamerica. Aztec culture, like Bronze Age Greece, was an amalgamation of many native and subjugated groups. Tribute, trade, and war financed a

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large empire.[12] Aztec textile production played a large part in all of these spheres. Like linen and wool in the Aegean and Anatolia, two fibers dominate Aztec textiles: cotton was the finer fabric used for elites and in ceremonial contexts. Maguey fabrics were coarser and less expensive and could be worn by commoners. This distinction appears in the archaeological record with two different types of spindle whorls, light ones for cotton and heavier ones for maguey.[13]

Aztec tribute lists from the 38 provinces are preserved in codices recording tallies of various goods required by the ruling elites at Tenochtitlan. Among these are assessments of cloth that compare to the Mycenaean records. The Aztec pictographic script depicts images of various types of cloth with identifying markers inside the sign. The Aztec (Nawat) words used to identify the textile do not describe what the cloth looks like but they inform the reader what technology was used for making the cloth.[14] This representational system compares well with the ligatured ideograms of TELA in Mycenaean. Like the Mycenaean ideograms, the signs are in the shape of cloth with distinguishing features drawn inside the textile unit. The scribes use a generic form for any textile and then within the sign they give it a distinguishing design. Some Aztec ideograms indicate extra thick cloth (*cf. pu-ka-ta-ri-ja*?), fabric from certain areas (*cf. ku-do-ni-ja*?), specially dyed cloths, or finely woven textiles, by the marker inside. This is the same scribal method used by the

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[12]See, for example, F. Berdan and P. Anawalt, The Essential Codex Mendoza (1997). See also the essays and bibliography in M. Schevill, J. Berlo, and E. Dwyer (eds.) Textile Traditions of Mesoamerica and the Andes (1991).

[13]M. Smith and K. Hirth, "The Development of Prehispanic Cotton-Spinning Technology in Western Morelos, Mexico," JFA 15 (1988) 349-358; E. Brumfiel, "Weaving and Cooking: Women's Production in Aztec Mexico" in Engendering Archaeology: Women and Prehistory (eds.) J. Gero and M. Conkey (1991) 224-251; M. Parsons, "The Distribution of Late Postclassic Spindle Whorls in the Valley of Mexico", American Antiquity 40 (1975) 209-215; J. Parsons and M. Parsons, Maguey Utilization in Highland Central Mexico: An Archaeological Ethnography (1990).

[14]P. Anawalt, "Severed Heads and Thigh Bones: The Meanings Encoded in Aztec Textile Motifs," UCLA Institute of Archaeology Lecture, April 23, 1998.

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Mycenaeans and the Minoans as described above. This graphic representation might also suggest that those required to pay tribute in textiles were illiterate but could recognize a cloth representation in an ideogram.

Among others, Elizabeth Brumfiel has looked at weaving and cooking as gendered activities in Mesoamerica and examined the changes over time in tribute demands when the Aztecs rose to power.[15] Because the Aztecs placed increasing demands on subject for textiles in their tribute payments, it was expected that many spindle whorls would be found in surveys of urban Aztec centers. There were, however, fewer spindle whorls than expected. This dearth in evidence compares well with the absence of spindle whorls noted at Minoan palaces. Brumfiel explains that in urban areas, Aztec rulers gave cloth to commoners who served the state. These people could then go to the market to exchange the textiles for food. The farmers would then take the cloth and give it back to the state as tribute. Textiles had become a kind of currency in this finance system. A similar explanation for the Minoan system may not be unlikely.

Another important work regarding Mesoamerican textile production is the study by Mary Parsons of the spindle whorls collected from survey in the valley of Mexico. She was able to suggest that certain areas predominated in cotton textile production even though they most likely did not wear what they made but sent the goods on as tribute. She also did ethnographic work with local spinners asking them to test various weights of spindle whorls to see which kind of thread they could spin. While the spinners said they could make most thicknesses of thread with any weight, they preferred a certain range, 16 to 74.8 grams. This interesting work has applications to the assemblage of spindle whorls from Gordion which have paralllel weight ranges.

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[15]E. Brumfiel (1991) 224-251. See also B. Hamann, "Weaving and the Iconography of Prestige," in Women in Prehistory: North America and Mesoamerica, (eds.) C. Claassen and R. Joyce (1997) 153-172.

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Inca

In the Andean highlands the Inca empire also financed itself with many different industries. While the complexity and scope of the Inca empire is greater than that of anything known from the Bronze Age Aegean or Anatolia there are similarities in the administration of labor, craft activities and storage. The Inca empire was divided into four provinces, each with its own provincial capital, but the imperial capital was at Cuzco. To this city only were the most valuable wealth goods sent. An early 16th century traveler recounts that only two kinds of goods were sent here, 'gold and cloth', because they weighed little, were easily transported, and highly valuable.[16]

In each region of the empire, the Inca built store houses with a full range of goods, supplying the local population and thus eliminating the need for one area to be in contact with any other center but the capital. These secondary centers are comparable to the outlying places known in the Minoan and Mycenaean kingdoms.[17] Another possible similarity with Aegean palaces is that the Inca were known for ceremonial feasts and public hospitality at the state centers. Part of the royal gifts at such ceremonies were elaborately woven textiles.[18] These cloths would bestow authority on loyal chiefs and legitimize the dominant Inca system. Drinking vessels, open spaces, and quantities of goods recorded in the administrative documents at Minoan and Mycenaean centers have suggested to some

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[16]J. Murra and C. Morris, "Dynastic Oral Tradition, Administrative Records and Archaeology in the Andes," World Archaeology 7.3 (1976) 270-79.

[17]For example, see essays on the Minoan 'Villa' in R. Hägg (1997).

[18]See J. Murra, "Cloth and its Function in the Inca State," in Cloth and Human Experience (eds.) A. Weiner and J. Schneider (1989) 275-302; C. Costin, "Textiles, Women, and Political Economy in Late Prehispanic Peru", Research in Economic Anthropology 14 (1993) 3-28.; "Craft Production and Mobilization Strategies in the Inka Empire" in Craft Specialization and Social Evolution: In Memory of V. Gordon Childe (1996) 211-228; A. Rowe, "Inca Weaving and Costume," Textile Museum Journal 34-35 (1995-1996) 5-54.

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that these centers also functioned as communal feasting areas, as a way of 'redistributing' the kingdoms' wealth, possibly including finely woven textiles.[19]

Two major types of Incan cloth are recorded in ethnohistoric documents. The first type was a thick, rough cloth called *'awasqa*. Women weavers in nearly every provincial household would receive wool or cotton from local storage facilities, administered by a subordinate Incan lord called the *kuraka*.[20] They could use this fiber for their own household needs but they were also required by the Incan state to make a specific amount of state cloth each year.[21] Officials would inspect each community to account for the raw materials and finished goods. One source states that one blanket and one shirt was the annual tax required of a household, although other sources disagree.[22]

The other type of cloth was a finer textile called *qompi* or *kumpi*. This was a colorful, soft textile often decorated with shell beads or feathers. These garments seem to have been made by male weavers, *qompi-kamayoq*, and have particular associations with soldiers and warfare.[23] One settlement near Lake Titicaca is reported to have housed one-thousand full-time male weavers of this feather-cloth.[24]

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[19]See J. W. Graham, "The Minoan Banquet Hall," AJA 65 (1961) 165-172; "A Banquet Hall at Mycenaean Pylos," AJA 71 (1967) 353-360; J. Killen "Thebes Sealings, Knossos Tablets and Mycenaean State Banquets", BICS 39 (1994) 65-84.

[20]For an analysis of Inca textile production along gender lines, see C. Costin, "Exploring the relationship between Gender and Crafts," Gender and Archaeology (ed.) R. Wright (1997) 111-140. I am very grateful to C. Costin for sending me an off-print of this article prior to its publication.

[21]J. Murra (1989) 282 citing the Spanish chronicler and lawyer Juan Polo de Ondegardo (1571).

[22]C. Costin (1993) 4, citing the 16th century chronicler Cieza.

[23]J. Murra (1989) 278.

[24]See J. Murra (1989) 289 and C. Costin (1993) 5 for the questionable authority of this source.

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Another Inca center of particular interest to textile production is Huanuco Pampa.[25] Excavations within one part of this site revealed a large concentration of textile equipment and food processing. This area, Unit 5, was built in a standardized architectural idiom, with secured and controlled access. From sixteenth-century accounts, it is believed that this unit housed the *aqlla*, the chosen or, more accurately, 'abducted' women who were specialists of cloth manufacture and may have had a role in supplying food for military personnel.[26] The high density of liquid storage jars in Complex 5 also suggests that liquids were being prepared here, most probably *chicha*, corn beer.[27] It is most significant that no other craft activities have been found at Huanuco Pampa, such as pottery kilns or evidence for metal working. Only textile production and food processing were visible crafts in the archaeological record.

All Andean households had both obligations and claims to the stores of community fibers and textiles. As a valued commodity, textiles were required from peasants as payment to the crown. In exchange they would receive the right to work state lands. As with the Aztecs, textiles given by the king were used as gifts to other kings, members of the royal family, or to administrators as a way of ensuring loyalty. Finally, textiles were used in ceremonies and as funerary gifts.[28] The role of state textiles supplying the military recalls some of the uses of cloth suggested for the Mycenaeans and Phrygians.

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[25]See C. Morris, "Reconstructing Patterns of Non-agricultural Production in the Inca Economy: Archaeology and Documents in Institutional Analysis", in Reconstructing Complex Societies: An Archaeological Colloquium, BASOR Suppl. 20, (ed.) C. Moore (1974) 49-68; J. Murra (1989) 275-302; J. Murra and C. Morris, "Dynastic Oral Tradition, Administrative Records and Archaeology in the Andes," World Archaeology 7.3 (1976) 270-79.

[26]C. Morris (1974) 52: C. Costin (1997) 123-127.

[27]C. Morris (1974) 53.

[28]These uses of cloth are discussed in J. Murra (1989).

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Conclusions

Textile production of the type described above and in the preceding chapters is a highly specialized craft with attached laborers participating in staple/wealth finance economies. Most probably these industries were administered by centralized bureaucracies, although no such evidence from Phrygian Gordion has yet been found. This kind of organization could only occur at urban centers that had close ties to rural hinterlands. The evidence from the Minoan, Mycenaean and Phrygian kingdoms shows that there was a close interaction between these two spheres. The development of surplus agriculture may have been the result of increased trade and commerce in Minoan Crete, including the production of valuable textiles. Food rations are recorded in the Linear B records as payment to the textile workers. The evidence from Gordion shows that textile and food production were the two major activities the Phrygians chose to locate within their citadel walls, close to the ruling elites.

Evidence presented above demonstrates that differences in social complexity and productive strategies translate into differential distributions of textile equipment. For this dissertation, the unit of analysis has been, for lack of a better term, the Aegean and Anatolian palace. Rather than seeing specialization as a 'presence or absence' at palatial centers, we have looked at textile production in terms of degrees. As cited in Chapter one, Costin proposes an eight-part typology of increasing complexity for the organization of specialized production. Applying these criteria to examples presented in this dissertation provides a summary of organized textile production in the Bronze and Iron Ages:[29]

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[29]C. Costin (1991) 1-56.

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Table 1. Types of specialization

Costin's Specialized Production Types	Archaeological Examples
Specialization by autonomous individuals for unrestricted local consumption	Little evidence in examples studied
Dispersed specialists working for unrestricted local consumption	-Middle Neolithic Knossos, -Early Minoan Knossos (?)
Community specialization by autonomous individual or household-based production units, aggregated within a single community, producing for unrestricted regional consumption	-Later Neolithic Knossos -Early Minoan II A-B Myrtos (Fournou Korifi)
Large, nucleated workshops aggregated within a single community, producing for unrestricted regional consumption.	-Perhaps at Early Minoan II A-B Myrtos (Fournou Korifi) -Vasiliki
Dispersesd corveé labor, working part-time for elite and government institutions within households or local community settings.	-Mycenaean Nichoria -evidenced by the production of textile *146
Individual retainers and individual artisans usually working full-time for elite patrons or governments	Little evidence in examples studied
Nucleated corveé laborers working part-time, recruited by government for special purposes.	Perhaps some evidence at Mycenaean palaces.
Retainer workshops reflecting large-scale operations with full-time artisans working for an elite patron or government institution within a highly segregated, specialized setting or facility.	-Old Palace at Knossos (Loom weight Basement) -Houses around Knossos (MUM, North House, South House) -Gordion.

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Having looked at many examples of centralized textile production, both with textual and archaeological sources, what can we say about palace production? From the Mycenaean period we have documentary records of cloth manufacture in archives at palace centers. Many archaeologists have focused a great deal of attention on the Mycenaean palace while not investigating smaller settlements where activities such as pottery, metallurgy and textile production are more likely to be located. It is highly probable that the production of cloth occurred in smaller Mycenaean villages located in the hinterlands of the palaces, at settlements like Nichoria. It seems that large scale production of textiles was primarily the work of women dependent upon the central authority, possibly against their will. Similar to the Inca *aqlla*, some of the women laborers at the Mycenaean palaces may have been prisoners captured in war, from areas across the Aegean, like Miletus, Lemnos, and Halikarnassos. Palace-made cloth was probably used to appease potentially hostile neighbors, impress valuable trading partners, or to legitimize the elite power base. Textiles produced by commoners in household settings were often assessed and collected as tribute by the center and then redistributed as a form of currency to non-food producers. This circulation of textiles would have continued as long as the system was working.

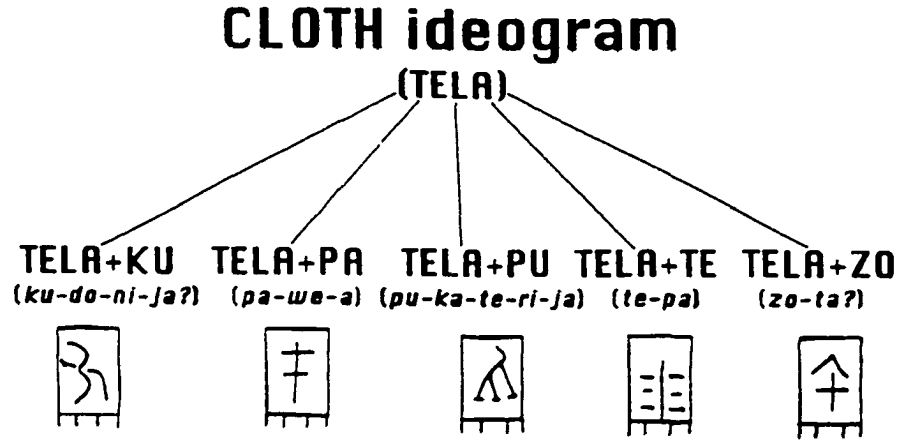
The Mycenaean system contrasts to the earlier Minoan period on Crete where we have both administrative documents and artifactual evidence for textile production at different centers throughout the Crete. The Mycenaeans may have adopted some of the administrative aspects of the Minoan system, but they do not seem to have employed the same production system. The Minoans might be closer in organization to the Phrygians, who, as we saw, located their textile production within the citadel walls of their capital. The parallels with the Inca and their cloistered women laborers, the *aqlla*, provide a suggestive explanation for such intensive textile production. Is it possible that the Phrygians were manufacturing so much cloth and preparing so much food, including beer,

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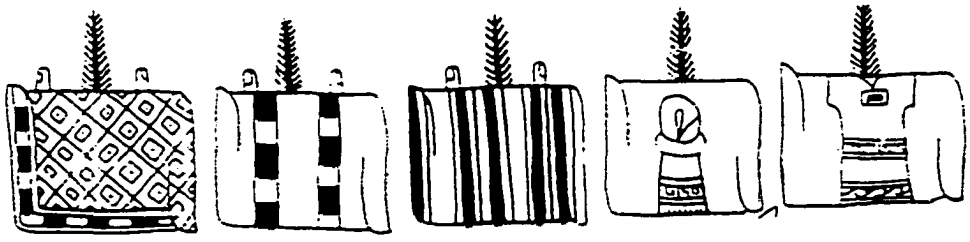
to supply an army? At the time of destruction at Gordion, sometime around 700 B.C., we know there was a violent military invasion that destroyed the citadel. The Mycenaeans, at the time of the Linear B tablets were written, were also facing a violent threat that eventually over-ran the palaces on the Greek mainland and on Crete.

After the Bronze Age the organization of textile production changes dramatically in the archaic and classical periods in Greece. The contribution of women's labor to Classical household economics, as described in Xenophon's Oeconomicus 7, or the role of textiles in Greek religion, are subjects for further investigation. This dissertation has examined the many different sources of information for understanding the role of controlled textile production at large palatial centers of the Bronze Age Aegean and the Anatolian Iron Age. This investigation shows that cloth manufacture in the eastern Mediterranean is an industry of wealth-based economies. Changes in textile technology and organization reflect developments in both complexity and craft specialization. The rich archaeological record demonstrates that humbler artifactual evidence can inform us about larger issues of emerging complexity.

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a) Mycenaean ideograms



b) Aztec ideograms (after Berdan and Anawalt, 1997)

Fig. 5.1 a and b Cloth ideograms

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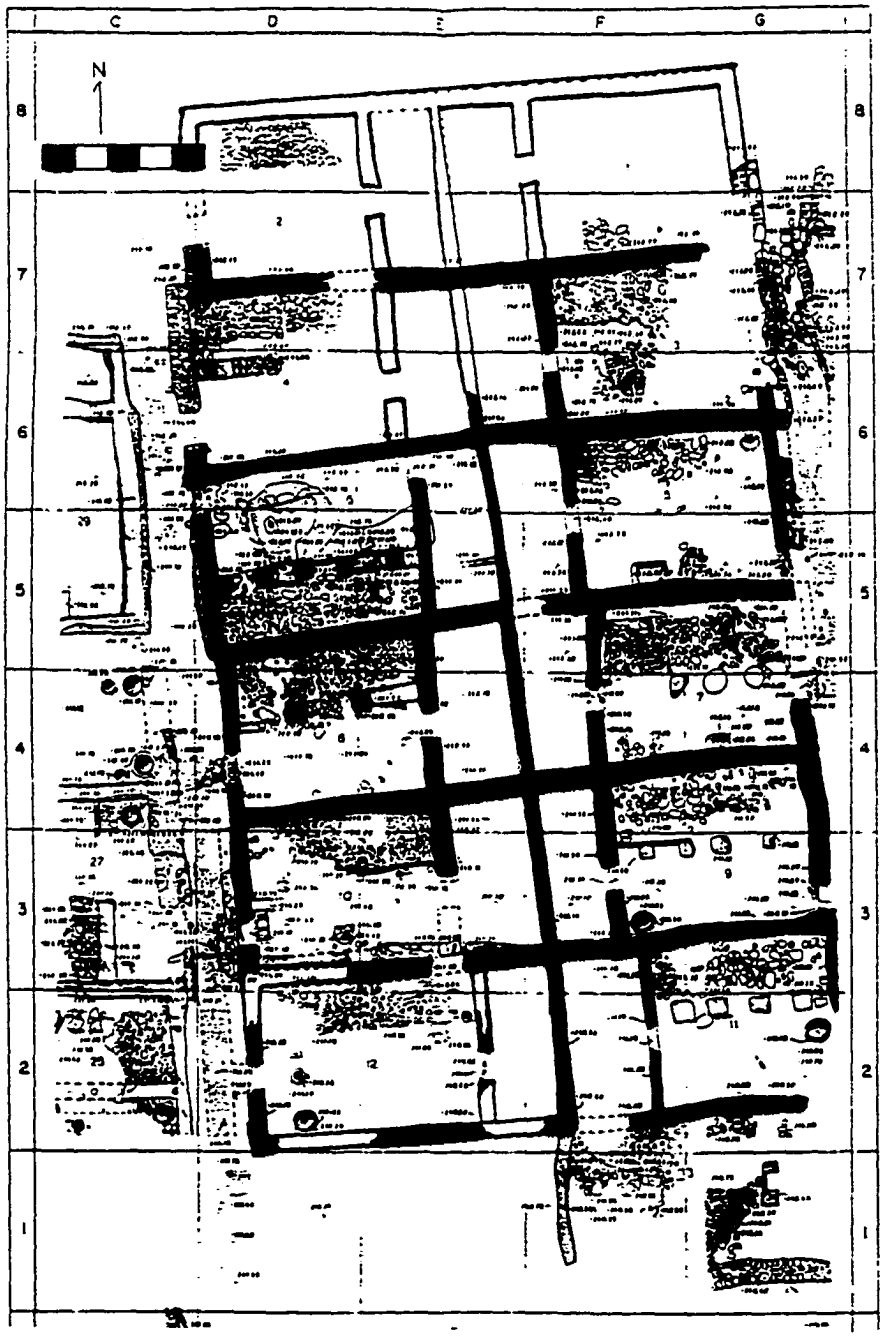


Figure 5.2 plan of Tel es-Sa'idiyeh

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Abbreviations follow the usual conventions in the American Journal of Archaeology. The abbreviation *CMS* is used for the Corpus der minioschen und mykenischen Siegel.

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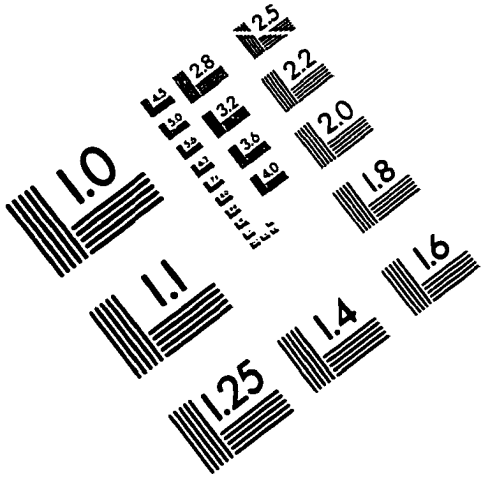
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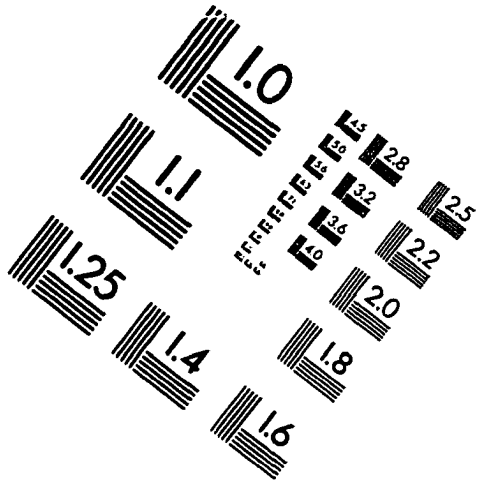
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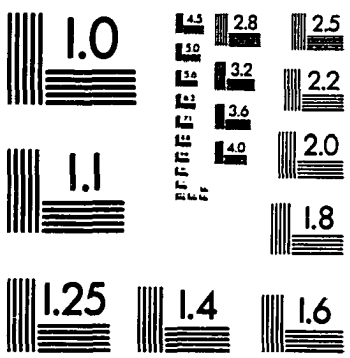
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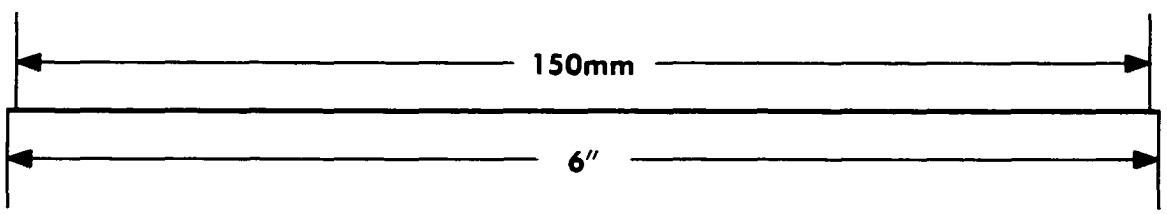
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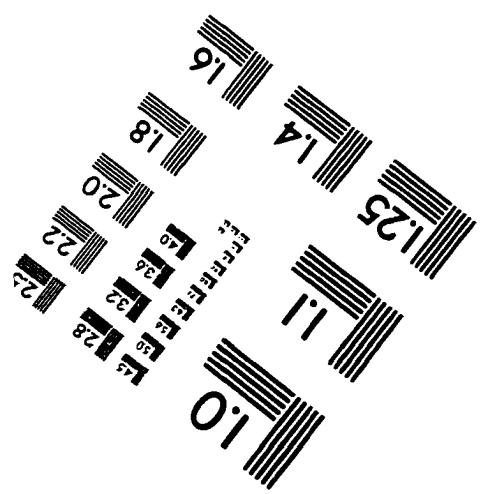
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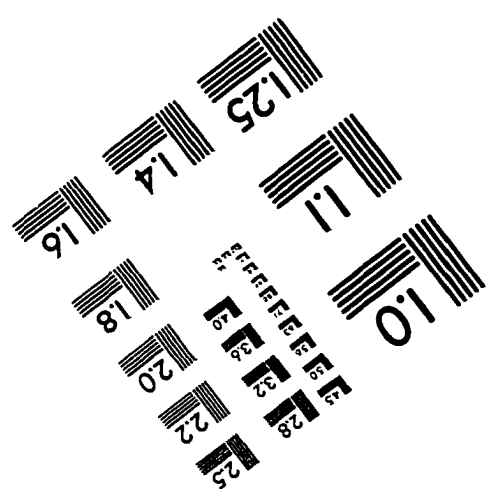












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